Investor Tax Heterogeneity and the Term Structure: Endogenous Segmentation and the Municipal Bond Puzzle*

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Abstract

I develop a term-structure model that jointly prices taxable and tax-exempt bonds when investors face different taxes under arbitrage frictions. Differences in tax benefits and balance-sheet exposures determine investors' participation across markets and maturities, generating endogenous partial segmentation. I apply the model to the municipal bond puzzle. Tax exemption is most valuable to high-tax households, since they are subject to highest tax. Literature use their after-tax yields on comparable taxable bonds as benchmark for municipal yields. Short municipal yields closely match this benchmark, but long yields exceed it. Municipal bond ownership have investors with lower tax participate municipal bonds at longer maturity when their balance sheets make interest-rate risk less costly to bear. Their participation makes long yields reflect a broader mix of tax rates. Calibrated to U.S. bond markets without targeting municipal yields, the model explains 83 percent of the 117-basis-point gap at thirty years. Empirically, household tax effects weaken with maturity while bank- and insurer-specific tax effects strengthen, consistent with the model mechanism.

Keywords: Term Structure, Investor Heterogeneity, Taxes, Endogenous Segmentation, Asset Ownership, Municipal Bonds
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1 Introduction

Under the U.S. tax code, investors face different tax treatment of the same asset. How does this tax heterogeneity shape bond yield curves? I develop an equilibrium term-structure model in which investors with heterogeneous tax treatment choose portfolios jointly across bond markets and maturities, subject to limits on short selling. Their taxes and balance sheets determine which bonds they hold and the returns they require, creating partial segmentation.

I apply the model to the joint pricing of taxable bonds and tax-exempt municipal bonds issued by U.S. state and local governments. Households pay their marginal income tax rate on taxable bond interest but no federal income tax on exempt municipal interest. The municipal exemption is most valuable to top-bracket households because they face higher tax rates on taxable interest than lower-tax households and institutional investors. Thus, top-bracket households can accept the lowest municipal yield. Together with limits on short selling, this tax-sorting logic motivates the representative-tax benchmark used in the literature: at every maturity, municipal yields should equal the household benchmark, taxable yields multiplied by one minus the top-bracket household's tax rate (Longstaff, 2011; Schwert, 2017; Fleckenstein and Longstaff, 2025).

However, the ownership data suggest that top-bracket households hold only about half the municipal market. Lower-tax households, banks and insurers hold substantial portfolios despite smaller net exemption benefits. Yields also depart from the literature's benchmark. Short municipal yields closely match the household benchmark, but long municipal yields exceed it. This is the municipal bond puzzle, documented in the literature for over thirty years (Feenberg and Poterba, 1991). The gap between the observed municipal yield and the household benchmark increases with maturity, reaching 117 basis points at thirty years. Equivalently, the implied tax, the tax rate that equates the after-tax taxable yield with the municipal yield, falls with maturity.

The heterogeneity in municipal ownership is central to the model's explanation of the municipal bond puzzle. Intuitively, when investors with different tax rates participate in the market, equilibrium yields reflect the tax treatment of participating investors, including those with lower tax rates than top-bracket households. Their participation explains why the implied tax rate falls below the household rate and municipal yields exceed the household's after-tax taxable benchmark. The exact equilibrium pattern across maturities depends on investors' endogenous participation and maturity sorting discussed below. The detailed mechanism proceeds in two steps. First, why do lower-tax investors hold municipal bonds despite their smaller exemption benefits? Second, how does this heterogeneous ownership affect equilibrium bond yields? In the model, investors differ in tax treatment and exogenous liability duration exposure. They choose portfolios jointly across maturities in taxable and tax-exempt bond markets, subject to short-sale constraints.

For the first question on participation, some lower-tax investors hold municipal bonds despite smaller exemption benefits because they bear interest-rate risk at a lower cost than high-tax households. Liability duration exposures on investors' balance sheets provide a microfoundation for this advantage. When interest rates rise, the value of long-term liabilities falls, offsetting losses on long bonds. Household mortgage debt, insurers' obligations and banks' deposit liabilities give investors

different net liability exposures to rate movements. The resulting duration benefit grows with a bond's rate exposure, while the difference in exemption benefits is constant across maturities. Lower-tax investors participate where their duration benefit outweighs the difference in exemption benefits, beginning at longer maturities. In the two-investor illustration, the lower-tax investor has a lower unit cost of rate risk. The high-tax household holds shorter municipal bonds, the lower-tax investor holds risky taxable bonds, and both hold municipal bonds beyond the maturity threshold.

For the second question, three channels explain how these holdings affect the relative equilibrium returns of taxable and tax-exempt bonds. First, a lower-tax investor keeps more taxable interest than the high-tax household, making exemption less valuable. It therefore accepts a smaller reduction in return for the exemption. At long maturities where it holds both municipal and taxable bonds, this difference contributes to municipal returns above the household benchmark. Second, the cost of bearing rate risk differs: the high-tax household requires more return per unit of rate exposure than the lower-tax investor. At risky maturities below the participation threshold, the household alone holds municipal bonds, while the lower-tax investor holds comparable taxable bonds. Municipal returns therefore exceed the household's after-tax taxable benchmark even where the lower-tax investor holds no municipal bonds. Third, taxes scale bonds' rate loadings. The household benchmark scales taxable loadings by the household's after-tax share. Long municipal bonds also reflect lower-tax holders' larger after-tax shares, making them more sensitive to rates than this benchmark predicts. Household portfolio substitution carries this effect to shorter bonds held only by households. Holders require additional return for this extra rate risk, widening the gap from the household benchmark. All three channels vanish at the riskless shortest maturity. Since a bond's yield reflects the equilibrium one-period returns over its remaining life, these return gaps accumulate into yield differences across maturities.

Liability hedging provides a natural foundation for the quantitative application. Insurers' long-term obligations create demand for duration, while households' outside assets and debt give them different exposures to rate movements. Observed balance sheets and maturity holdings therefore provide a concrete basis for differences in rate-risk costs and data with which to discipline them. The mechanism also accommodates other motives for participation, such as lower trading costs relative to households or a desire for default diversification. When the resulting advantages grow with maturity and apply equally to taxable and tax-exempt bonds, the relative-pricing results hold (Section 4.7). Thus, the pricing mechanism does not rely on a particular microfoundation for lower-tax investors' participation.

I calibrate the model to U.S. bond markets, with multiple investor types, municipal markets across U.S. states and a taxable bond market. Investors include high- and lower-tax households residing in each state, banks, property-and-casualty insurers, life insurers and tax-neutral investors. They differ in tax treatment, state of residence (for households), resources and outside rate exposure. Observed tax rules, balance sheets, bond ownership and the process for taxable interest rates discipline the quantitative model. Without targeting municipal yields, the model explains 83% of

the thirty-year puzzle. Measured differences in default risk, liquidity and capital-gains taxation explain almost all of the remainder.

Three empirical tests, untargeted in the calibration, distinguish the model from the representative-tax benchmark. The model predicts that variation in top-bracket households' exemption benefits is fully reflected in implied taxes at the shortest maturity and less strongly at longer maturities. Changes in lower-tax investors' exemption benefits have no effect at the shortest maturity and matter more at longer maturities. First, states exempt residents' interest on their own municipal bonds from state income tax. Cross-state variation in state tax rates is reflected almost one-for-one in implied state taxes at short maturities and progressively less at longer maturities. Second, bank-qualified designation gives banks an additional tax benefit on a subset of municipal bonds issued by small borrowers. This bank-only benefit is barely reflected in short municipal yields and increasingly reflected at longer maturities. Third, I compare municipal yield changes with changes in the treasury reference yield at the same maturity. This relative response rises with maturity, consistent with the model's rate-loading channel.

Two counterfactual tax reforms measure the fiscal and financing effects of changing investor tax benefits. The first removes a state's home-state exemption for resident households, taxing own-state interest as Illinois, Iowa and Wisconsin generally do. Portfolio reallocation changes the tax base and raises financing costs for issuers in other states and in the taxable market. The total increase in annual financing costs exceeds added state tax receipts. The second makes municipal bonds taxable, equalizing investors' tax treatment of municipal and taxable interest. Taxable municipal issuance has a precedent in the 2009–2010 Build America Bonds program, which allowed state and local governments to issue taxable bonds with federal interest subsidies. Averaged over the six 2000–2017 economies, issuers' replacement-financing costs rise by \$1.20 per dollar of added federal and state receipts, including \$0.13 borne by taxable issuers. The excess is an increase in net investor interest income, distributed unevenly: high-tax households, banks and property-and-casualty insurers lose on average, while lower-tax households, life insurers and tax-neutral investors gain. Evaluating either reform at the old allocation misses both the changing tax base and financing effects outside the targeted market.

I apply the framework to taxable and tax-exempt bonds because their tax rules provide clean variation for testing the mechanism and their relative yields present a longstanding puzzle. The framework applies more broadly wherever investors face different tax treatment across assets, including state-tax differences between treasury and corporate bonds, tax-favored accounts and cross-border withholding. Its central contribution is to show how investor tax heterogeneity shapes the entire yield curve through endogenous participation and joint portfolio choice. Whose tax treatment is reflected in yields varies with maturity, and portfolio substitution transmits tax effects across markets, even to bonds the affected investor never holds. Accounting for this heterogeneity is therefore essential both to understanding relative bond yields and to evaluating the financing and revenue effects of tax policy.

Sections 2–3 review related work and present the tax, ownership and yield evidence. Section 4 develops the model, Sections 5–6 evaluate its quantitative and empirical implications, and Section 7 studies tax policy. Formal results, proofs and supporting evidence are in the appendix.

2 Related Literature

Investor taxes shape portfolio choice, capital structure and equilibrium asset prices (Miller, 1977; Auerbach and King, 1983; Sialm, 2009). Tax-clientele theory distinguishes taxes' effects on holdings from their effects on prices, while tax-arbitrage models characterize when equilibrium exists under unrestricted short sales (Dybvig and Ross, 1986; Dammon and Green, 1987). I extend this literature to the term structure: heterogeneous taxes and rate-risk costs determine participation across maturities and jointly shape taxable and tax-exempt yield curves.

Segmented-market term-structure models link maturity-specific demand and bond supply to yields through risk-averse investors (Vayanos and Vila, 2021; Greenwood and Vayanos, 2014; Guibaud et al., 2013). Related research studies cross-market transmission, demand systems and investor composition (Greenwood et al., 2018, 2023; Koijen and Yogo, 2019; Coppola, 2025; Gabaix et al., 2026). Municipal fund flows and fragility provide further evidence on how investor demand affects pricing and financing (Li et al., 2024; Adelino et al., 2026). Limits to tax arbitrage and cross-market risk taking can sustain segmentation (Hines and Horton, 2024; Gromb and Vayanos, 2018; Siriwardane et al., 2025). I derive participation from heterogeneous taxes and outside rate exposures, making partial segmentation an equilibrium outcome across jointly priced yield curves.

Institutional balance sheets shape the cost of bearing risk. Intermediary asset-pricing research links risk premia to equity capital and banks' rate exposure (He and Krishnamurthy, 2013; Haddad and Sraer, 2020). Kekre et al. (2024) show how monetary tightening can raise term premia through wealth losses on arbitrageurs' positive-duration portfolios. Insurers' duration matching and duration mismatch affect long yields and credit spreads (Domanski et al., 2017; Li, 2026), while banks' rate exposure and funds' liquidity needs are reflected in municipal offering spreads (Pati, 2023). I show how a balance-sheet advantage in bearing rate risk can offset smaller exemption benefits and sustain lower-tax participation in tax-exempt bonds.

The municipal puzzle has motivated both clientele and bond-side explanations. Early segmentation accounts placed banks at short maturities and households at long maturities (Mussa and Kormendi, 1979; Hendershott and Koch, 1980; Kidwell and Koch, 1983; Skelton, 1983). The puzzle persisted after banks lost their general carrying-cost deduction in 1986, challenging fixed-clientele accounts (Green, 1993; Mankiw and Poterba, 1996). Subsequent bond-side analyses largely retain the top-bracket household benchmark while examining default and liquidity (Chalmers, 1998; Longstaff, 2011; Schwert, 2017), convenience premia (Fleckenstein and Longstaff, 2025), and tax timing and callability (Green, 1993; Ang et al., 2010; Brown, 2011). Municipal swaps imply declining tax rates with maturity, while the measured convenience premium is concentrated at short maturities (Longstaff, 2011; Fleckenstein and Longstaff, 2025). Both patterns are consistent with

an increasing gap above the household benchmark. Schwert's credit-risk evidence comes primarily from bonds maturing within ten years. I measure these bond-side adjustments and show that investor heterogeneity explains most of the municipal puzzle. Endogenous participation frees the explanation from fixed habitats, allowing maturity sorting to change with tax policy.[1]

Investor-specific tax rules provide evidence on municipal segmentation and financing. State exemptions shape home-state ownership, risk sharing and financing conditions (Hendershott and Kidwell, 1978; Pirinsky and Wang, 2011; Babina et al., 2021; Moussawi, 2022, 2024). Bank financing incentives and changes in the bank-qualified advantage likewise affect municipal borrowing (Dagostino, 2025; Benson and Marks, 2024). I contribute a maturity-based test of this investor heterogeneity: household state-tax pass-through declines with maturity while the bank-qualified yield reduction rises, the opposite gradients implied by the model's participation pattern.

Finally, research on tax exemption links investor substitution and competition to revenue costs and municipal borrowing (Poterba and Ramírez Verdugo, 2011; Garrett et al., 2023). I jointly solve for portfolios, taxable and tax-exempt yield curves and tax receipts after reform, accounting for financing effects on taxable issuers as well as municipal borrowers.

3 Taxes, Ownership and the Municipal Puzzle

This section establishes the empirical setting for the model. I describe the data and the tax benefits available to different investors, then show that municipal ownership extends well beyond high-tax households. I measure the municipal puzzle and examine whether bond characteristics or the choice of taxable benchmark explain its maturity pattern.

3.1 Data

I combine data on bond issuance, investor holdings and taxes. Mergent provides issuance records, ratings, bond characteristics and call schedules. Insurers' filings in Schedule D of the National Association of Insurance Commissioners (NAIC) and a pension holdings panel collected through Freedom of Information Act requests allow me to examine institutional portfolios at the level of individual bond holdings. I use the Financial Accounts to measure aggregate holdings, the Survey of Consumer Finances to measure household balance sheets, and IRS Statistics of Income to document household tax-exempt interest income across income brackets.

The main yield comparison covers 2000–2020 and uses FRED's A-or-above corporate zero curve as the taxable benchmark. The treasury curves (Gürkaynak et al., 2007) obtained from the Federal Reserve provide the benchmark for the interest-rate response test. To examine trading and liquidity, I use municipal transactions from the MSRB and corporate transactions from TRACE Enhanced. Markit CDS quotes provide credit default swap data for municipal and taxable bond issuers, and

[1] Related municipal pricing, tax-policy and maturity-clientele analyses include Fama (1977); Livingston (1982); Poterba (1986); Brown and Sirmans (2011). Trading-cost and dealer-intermediation evidence informs the liquidity adjustment (Harris and Piwowar, 2006; Green et al., 2007; Bagley et al., 2023). For surveys, see Cestau et al. (2019); Spreen and Gerrish (2022). Appendix D.5 compares the model with Green's tax-timing benchmark.

Tax Foundation schedules identify state income-tax rates. Appendix E.7 documents the sources and their coverage.

3.2 Municipal issuance

Table 1: Municipal issuance: descriptive statistics (sample means).

Variable	Full sample	Tax-exempt, non-AMT	
		AA and above (score ≤ 3)	A− and above (score ≤ 7)
Offering yield (%)	3.11	2.82	2.96
Corporate benchmark yield (%)	4.47	4.33	4.41
Maturity at issuance (years)	10.15	10.76	10.86
Offering price (per \$100 par)	102.88	104.68	103.85
Issue rating (1 = AAA, 3 = AA, 7 = A−)	3.80	2.18	3.55
Implied tax rate $1 - y^M/y^C$	0.29	0.35	0.33
Share callable	0.46	0.51	0.51
Share tax-exempt, non-AMT	0.92	1.00	1.00
Sample size (CUSIPs)	3,014,420	984,210	1,990,791

Notes. The table reports bond-level means before the call and issuance-premium adjustments, with one observation per municipal issue (CUSIP). The full sample contains all issues. The AA-and-above and A-or-above columns restrict to federally tax-exempt, non-AMT issues with average agency score ≤ 3 and ≤ 7, the latter matching the A-or-above corporate credit band. The implied-tax row averages the bond-level ratio $1 - y^M/y^C$, winsorized at the first and ninety-ninth percentiles, so it need not equal one minus the ratio of the displayed yield means. Each row uses non-missing observations: the full-sample rating mean uses the 2,292,092 rated issues, and the restricted columns exclude unrated issues.

Table 1 describes municipal bonds issued during 2000–2020, with one observation per bond identifier (CUSIP). The full sample contains more than three million issues, with an average maturity of about ten years. Most pay federally exempt interest: taxable municipal bonds account for 5.8% of issues, bonds subject to the alternative minimum tax (AMT) account for 1.9%, and the remaining issues are tax-exempt, non-AMT bonds. Call provisions allow issuers to redeem bonds before maturity and are common in the data. Callable bonds account for 46% of all issues and 51% of the high-grade, tax-exempt samples, making callability important for comparing yields.

The table also reports two high-grade samples of tax-exempt, non-AMT bonds. The broader sample includes A− and above issues (average agency score ≤ 7 on the twenty-two-notch scale) and matches the credit band of the corporate benchmark used for the main yield curves. The main empirical specifications use the AA-and-above sample (≤ 3) to minimize the impact of default risk.

3.3 Bond taxation and the representative-tax benchmark

For a taxable household, interest on taxable bonds is ordinary income, while interest on tax-exempt municipal bonds is federally exempt. Interest includes both coupon payments and annual original-issue-discount accrual. Institutions generally do not receive the full exemption benefit from holding municipal bonds. For example, property-and-casualty (P&C) insurers pay an effective federal tax of 5.25% on municipal interest. Let τ_i^C and τ_i^M denote investor i's effective tax rates on taxable and

municipal interest. I define the relative after-tax share as the fraction of taxable interest retained divided by the fraction of municipal interest retained:

$$\xi_i \equiv \frac{1-\tau_i^C}{1-\tau_i^M}. \tag{1}$$

Investor i's relative municipal tax benefit, the exemption benefit, is $1-\xi_i = (\tau_i^C - \tau_i^M)/(1-\tau_i^M)$. A smaller ξ_i means a larger benefit. When municipal interest is fully exempt, $\xi_i = 1-\tau_i^C$, so the benefit equals the tax rate on taxable interest. Appendix E explains the underlying tax rules.

The representative-tax benchmark uses the top-bracket household's tax treatment at every maturity. Write y_T^M and y_T^C for municipal and taxable yields at maturity T, and H for this household. Under the benchmark, the two bonds offer the household the same after-tax yield, $y_T^M = \xi_H y_T^C$. I call $\xi_H y_T^C$ the household benchmark and $1-\xi_H$ the household wedge. The benchmark's yield-implied tax is therefore $1 - y_T^M/y_T^C = 1-\xi_H$. With fully exempt municipal interest, it equals τ_H^C. Investors with different ξ_i cannot all be indifferent at this yield ratio. Their participation in the municipal bond market is the central question of the model.

3.4 Investor taxes and ownership

The tax benefit of municipal interest varies substantially across investors in the roughly $4 trillion municipal market. Table 2 summarizes these differences before the 2017 Act. For households and insurers, the net exemption benefit is $1-\xi_i$ in (1). Households in higher tax brackets receive a larger benefit because they pay more tax on taxable interest. Exemption from home-state income tax adds to the benefit on in-state municipal bonds. Appendix E gives the detailed tax treatment of each investor group and the changes after the 2017 Act.

Insurers receive smaller benefits than high-tax households because exempt interest reduces their deductions. The reduction is less severe for P&C insurers than for life insurers, giving P&C insurers the larger benefit. Banks generally lose the interest deduction allocated to municipal holdings, with partial relief for bank-qualified bonds, so the table reports their benefit net of this financing-tax cost.[2] Tax-exempt institutions such as qualified pension funds and endowment funds already receive taxable interest free of current income tax and gain no additional benefit from the municipal exemption. They hold almost no municipal bonds (Appendix D.9).

Figure 1 shows that municipal ownership spans these tax groups. Panel (a) reports aggregate holdings: households and municipal funds hold the largest amounts, while banks and P&C insurers also hold substantial portfolios despite smaller exemption benefits. P&C insurers allocate about 40% of their bond portfolios to municipals and maintain a larger municipal portfolio share than life insurers throughout the sample (Appendix D.9). Panel (b) documents household tax-exempt interest income across income brackets. Assigning municipal-fund holdings to households and allocating household holdings using tax-exempt interest shares puts roughly half the market with high-tax

[2]The quantitative model charges the bank financing-tax cost separately. Table 3 therefore reports the bank's exemption benefit $1-\xi_i$ before that cost.

Table 2: Investor heterogeneity in the municipal market before the 2017 Tax Cuts and Jobs Act

Investor type	Federal treatment	State treatment	Net exemption benefit
High-tax household	Interest exempt	Exempt on in-state bonds only	0.35–0.43
Middle-tax household	Interest exempt	Exempt on in-state bonds only	0.25–0.33
Low-tax household	Interest exempt	Exempt on in-state bonds only	0.10–0.15
P&C insurer	Exempt, less proration	Premium taxes, no distinction	0.31
Life insurer	Exempt, less proration	Premium taxes, no distinction	0.21
Bank	Exempt, funding cost disallowed	Varies by state	0.15–0.33
Other (pensions, etc.)	Varies	Varies	< 0.05

Notes. For households and insurers, the net exemption benefit is $1 - \xi_i$, from statutory rates and proration over 2000–2017. For banks it also deducts the financing-tax cost described below. Appendix Table 50 gives the values after the 2017 Act. Household values are federal marginal rates in three groups (35% and above, 25% to 33%, below 25%), and the top value includes the 3.8% net investment income tax from 2013. The home-state exemption adds to the benefit on in-state bonds. The quantitative model pools middle- and low-tax households into one lower-tax group, whose benefit with home-state taxes is 0.21 to 0.27 across states in 2015–2017, against 0.43 to 0.50 for the high-tax group (Table 3). Proration cuts a P&C insurer's losses-incurred deduction by 15% of its exempt interest and a life insurer's reserve deduction by the policyholders' share. Section 265(b) disallows the share of a bank's interest expense allocable to exempt holdings, of which 80% stays deductible for bank-qualified bonds. The bank value, net of that disallowance, is the break-even implied tax, equal to the bank's tax rate minus its annual financing-tax cost per dollar of municipal holdings divided by the taxable yield. That cost is 1.02% in 2000–2002 and 0.11% in 2015–2017, and the calculation covers the six three-year periods of 2000–2017 and maturities of two to thirty years. States generally tax insurers through premium taxes, which do not distinguish municipal from taxable interest, and banks through franchise or income taxes.

households.[3] The remaining holdings by investors with smaller exemption benefits motivate a model in which investors with different exemption benefits choose municipal and taxable bonds jointly. The analytical benchmark has two investors. The quantitative model includes household groups by state of residence, insurers, banks and a residual group (Section 5.1).

3.5 Limits to arbitrage

Tax rules and trading frictions limit the trades that could exploit yield differences or equalize investors' costs of bearing rate risk. Interest and short-sale payments incurred to carry exempt bonds generally lose their deduction. Consider a household that shorts a long-maturity corporate bond to hedge its municipal holdings. With the deduction denied, an equal-principal position earns coupon carry equal to the pretax municipal-minus-corporate yield difference. This carry is negative along the sample-average par curves and about -95 basis points a year at thirty years. Municipal shorts are scarce and costly because substitute payments to the lender do not retain the interest exemption. Off-exchange swaps face eligibility restrictions, while futures losses can face deduction limits and deferral. The model summarizes these tax asymmetries and trading frictions through nonnegative bond positions and cash balances, ruling out short sales and borrowing. Appendix E.6 examines the tax treatment, carry and residual risk of these trades.

[3] I use IRS Statistics of Income shares of tax-exempt interest on taxable returns, including fund distributions, to approximate household holdings by income group in 2010–2017. Adjusted gross income of at least $200,000 proxies for the high-tax group.

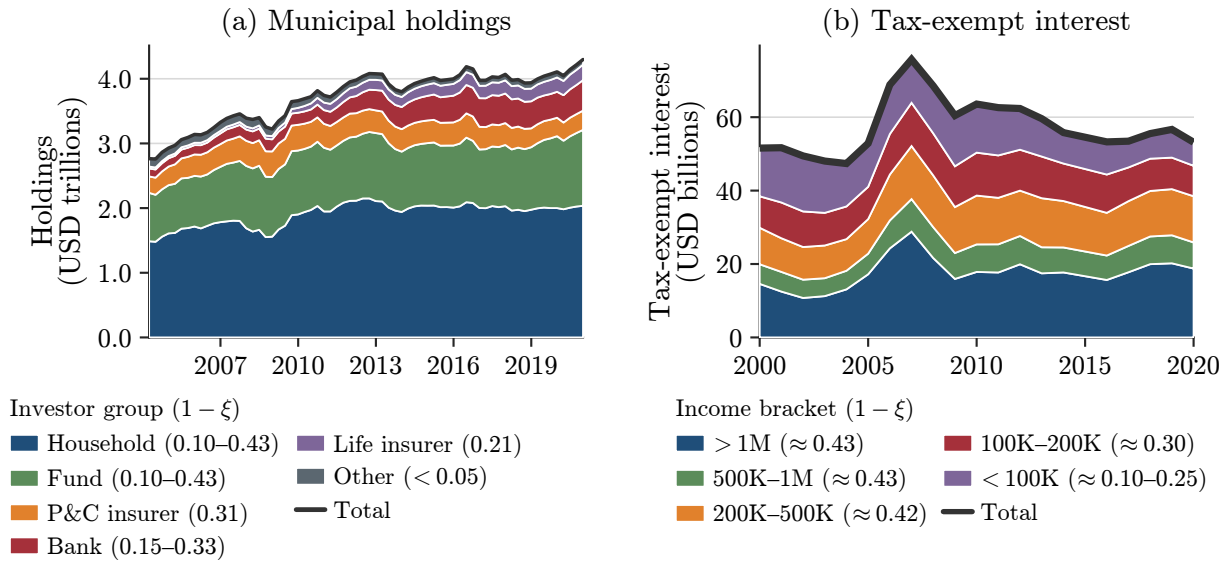


Figure 1: Municipal holdings and tax-exempt interest by investor group. Panel (a): direct holdings of taxable and tax-exempt municipal securities, 2005–2021 (USD trillions, Financial Accounts, with households including nonprofits). Panel (b): tax-exempt interest, an income flow, on taxable individual returns by adjusted-gross-income bracket, 2000–2020 (USD billions, IRS Statistics of Income). 2015 lacks comparable bracket detail and is omitted. Legend values in parentheses are net exemption benefits: in panel (a), the values before the 2017 Act from Table 2, with funds assigned their household shareholders' range, and in panel (b), approximate values by bracket.

3.6 The municipal puzzle

The municipal puzzle is a maturity pattern: short municipal yields are close to the household benchmark, while long municipal yields substantially exceed it. To measure this pattern, I use the seasoned, option-free A-or-above corporate zero-coupon curve as the taxable comparison. High-grade corporate bonds are closer to municipals in credit and liquidity than treasuries. Treasury yields also contain a premium for safety and liquidity (Krishnamurthy and Vissing-Jorgensen, 2012), which would enter the measured gap if treasuries supplied the benchmark. Previous studies using treasuries as the taxable benchmark also find that the municipal puzzle persists after adjusting for these differences (Chalmers, 1998; Longstaff, 2011).

I construct municipal zero-coupon yields in the same credit band from offering prices and contractual cash flows. Issuance data provide broad coverage and a common measurement date. Offering yields also avoid the direct tax effects of capital gains and market discounts arising from prior secondary-market trades. I remove callability and the issuance premium to make municipal yields comparable with the seasoned, option-free corporate curve.[4] Let $y_t^C(T)$ denote the taxable yield and $y_{s,t}^M(T)$ the municipal yield after these two adjustments, where T is maturity, s is the issuer's state and t is the issue date. The municipal puzzle is the gap between $y_{s,t}^M(T)$ and the household benchmark $\xi_H y_t^C(T)$. The yield-implied tax expresses the same comparison in tax-rate

[4]The market-discount tax applies to later secondary purchasers. Its expected effect on resale values can also affect offering prices. The capital-gains adjustment includes this effect as well as taxes on realized gains.

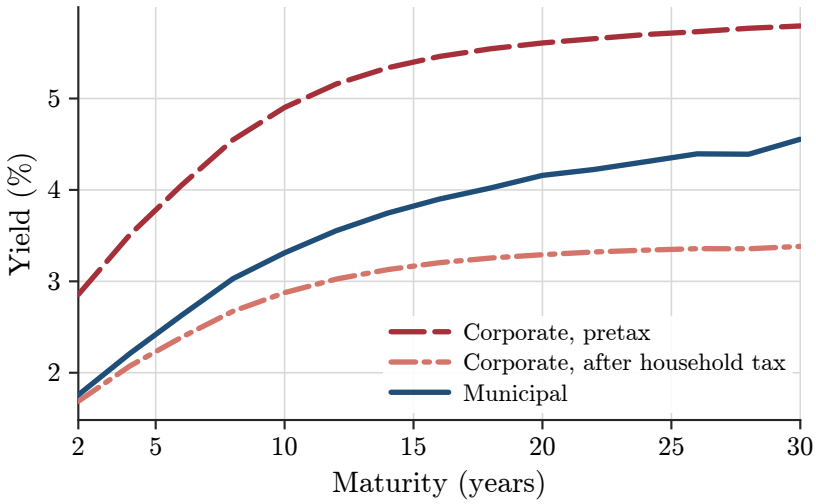


Figure 2: The municipal puzzle: A-or-above zero-coupon yields, 2000–2020. Municipal yields remove callability and the issuance premium. Averaging follows Appendix D.1.1. Appendix D.12.1 gives the AA-and-above counterpart.

units:

$$ImpliedTax_{s,t}(T) \equiv 1 - \frac{y^M_{s,t}(T)}{y^C_t(T)}. \tag{2}$$

For a given household's after-tax share ξ_H, the two measures satisfy

$$\underbrace{y^M_{s,t}(T) - \xi_H y^C_t(T)}_{\text{municipal yield gap}} = \underbrace{\left[(1-\xi_H) - ImpliedTax_{s,t}(T)\right]}_{\text{shortfall from the household wedge}} y^C_t(T). \tag{3}$$

For positive taxable yields, a positive gap means an implied tax below the household wedge $1-\xi_H$. In Figure 2, the municipal yield is near the household benchmark at the short end but about 117 basis points above it at thirty years. The implied tax falls from about 38% to 21% while the household wedge remains near 41% (Appendix Figure 14).

The declining implied-tax pattern also appears against treasuries (Chalmers, 1998), within issuer-matched tax-exempt and taxable municipals (Section 3.8), and in the AA-and-above and unlimited-tax general-obligation samples (Appendices D.12.1 and D.4).

3.7 Bond-side adjustments

Studies such as Fleckenstein and Longstaff (2025) use secondary-market prices of option-free bonds to examine the municipal puzzle. Consistent with that comparison, I remove callability and the issuance premium before defining the municipal puzzle as the gap over the household benchmark. I then assess default risk, liquidity and capital-gains taxation as explanations of this gap. Each adjustment measures the difference in compensation between municipal and corporate bonds, expressed in the household's after-tax yield units. Compensation common to both markets therefore cancels. The household wedge, one minus the ratio of averaged after-tax to pretax corporate yields,

averages 41.3% and is nearly flat across maturities. Appendix D explains the adjustments, and Appendix Table 26 reports their contributions by maturity. Appendix D.5 discusses the tax timing of taxable bonds in Green (1993).

At thirty years, the call and issuance-premium adjustments reduce the 159.8-basis-point spread over the household benchmark to the 117.2-basis-point municipal puzzle. The default, liquidity and capital-gains differentials explain 51% of the puzzle at ten years but only 16% at thirty. Subtracting them leaves 98.4 basis points at thirty years for comparison with the model in Section 5.4. At two years, their sum exceeds the small puzzle, leaving a residual of −9.1 basis points, consistent with the small convenience premium discussed in Fleckenstein and Longstaff (2025). The adjustments leave the central pattern intact: an unexplained municipal yield gap that widens with maturity.

3.8 Issuer-matched evidence

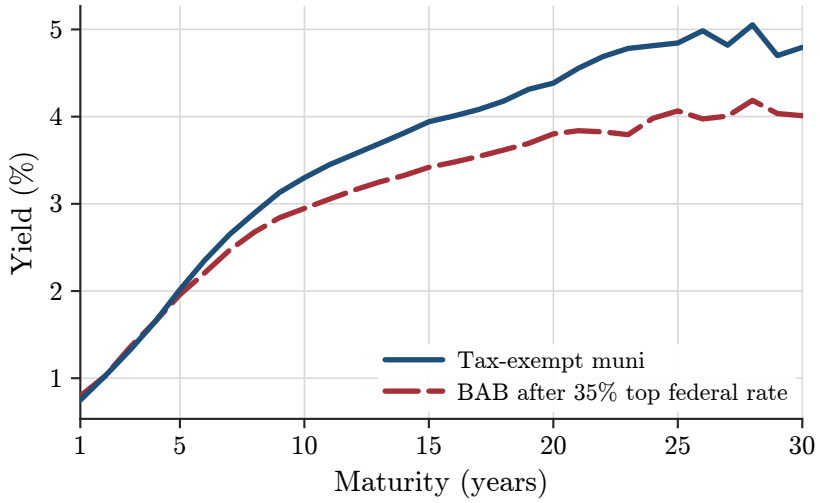


Figure 3: Tax-exempt and Build America Bond offering yields by maturity, AA-and-above issues, April 2009–December 2010. BAB yields are multiplied by 0.65, one minus the 35% top federal rate in 2009–2010. Appendix Table 29 reports regressions with and without call-status matching. Appendix D.3 describes the sample.

Taxable and tax-exempt bonds issued by the same municipal borrower provide a further comparison that does not rely on corporate bonds. During 2009–2010, the Build America Bonds (BAB) program allowed municipal issuers to sell taxable bonds with a federal subsidy equal to 35% of their interest payments. The top federal household income-tax rate was also 35%. I leave tax-exempt offering yields unchanged and multiply BAB yields by 0.65 to put both on the household's after-tax scale. I then compare bonds from the same issuer and estimate how their yield difference varies with maturity:

$$y_b \;=\; \alpha_1\, EXMP_b \;+\; \beta\,(EXMP_b \times m_b) \;+\; \gamma_C\, y^{C}_{m_b,t_b} \;+\; \lambda_{j(b)\times m_b\times q_b} \;+\; \varepsilon_b, \tag{4}$$

where y_b is the offering yield for a tax-exempt bond and 0.65 times the offering yield for a BAB. The indicator $EXMP_b$ equals one for a tax-exempt bond and zero for a BAB, and m_b is maturity in years. The fixed effect $\lambda_{j(b)\times m_b\times q_b}$ compares bonds from issuer $j(b)$ at the same maturity and in the same issue quarter q_b. The call-matched specification also requires the same call status. The corporate yield $y^C_{m_b,t_b}$, measured at the bond's maturity and issue month t_b, controls for market yield movements within that quarter. The coefficient β therefore measures how the tax-exempt yield minus the household after-tax BAB yield changes with maturity. Standard errors are clustered by state. Comparing bonds from the same issuer at the same maturity and with matched call status holds borrower identity and these bond characteristics fixed. The widening gap persists in this comparison.

The estimated gradient β is 3.4 basis points per year in the AA-and-above sample and 3.3 in the A-or-above sample. Matching call status gives gradients of 3.1 and 3.0 basis points. The binned estimates also rise monotonically: in the AA-and-above sample with call matching, the $[25, 30]$-year coefficient is 63.9 basis points above the one-to-five-year bin. Figure 3 shows mean yields by maturity before issuer matching. Appendix Table 29 reports the issuer-matched estimates. Both show a widening gap, and the regressions establish that it persists within the same issuer, using offering yields for both bond types and matching call status.

4 Model

I develop a term-structure model in which investors differ in tax treatment and balance-sheet exposure to interest rates. They choose bonds jointly across markets and maturities, subject to short-sale constraints. The model answers two questions: why investors with smaller exemption benefits hold tax-exempt bonds, and how their participation changes yields relative to the household benchmark.

I begin with a general economy with multiple asset markets and investors whose tax treatment differs across markets. This is the model used for quantitative calibration. A two-investor, two-market affine illustration provides analytical insights, with formal results and proofs in Appendices A and B.

4.1 Bonds, investors and portfolio choice

Time is discrete. Each investor solves an after-tax mean–variance portfolio choice problem across all assets and maturities, subject to short-sale constraints.

Interest rates. A single rate state z_t drives the short rate r, the pretax return on cash, which is subject to a zero lower bound:

$$z_{t+1} = \rho z_t + \sigma_r \varepsilon_{t+1}, \qquad r(z_t) = \max\{\bar{r} + z_t, 0\}, \qquad 0 < \rho < 1. \tag{5}$$

Here ε_{t+1} is standard normal, ρ measures persistence, σ_r the size of rate shocks and $\bar{r}$ the mean rate before the floor. Taxable cash earns this known log gross return $r(z_t)$ over the next period.

Bonds and returns. Bonds are zero-coupon and grouped into markets a. Taxes are investor-market specific. For the same investor, all the bonds in one market across maturities receive the same tax treatment. A bond of market a with T periods to maturity has price $P_T^a(z)$, pays one unit at maturity and becomes a $(T-1)$-period bond next period. Cash is the one-period taxable asset. Supplies Q_T^a are fixed.

An investor who buys a T-period bond today and sells it next period earns

$$\text{gross holding return} = \frac{P_{T-1}^a(z_{t+1})}{P_T^a(z_t)} \exp\big(\sqrt{v_T^a}\,\eta_{T,t+1}^a\big), \qquad P_0^a = 1.$$

The price ratio is the resale value per dollar invested. It carries rate risk because next period's price depends on next period's rate state. The standard normal shock $\eta_{T,t+1}^a$ is independent across bonds and of the rate shock, adding bond-specific log-return variance v_T^a. This small idiosyncratic shock represents residual return variation. Without it, bonds of different maturities can become perfect substitutes, leaving their separate holdings indeterminate. Rate risk drives the mechanism, while the bond-specific component has a small effect on the yield gap (Appendix C.8).

At maturity one, the payment is known, so the bond is fully riskless. A zero-coupon bond's yield is $y_T^a = -\log P_T^a/T$. The holding return above covers one period, from purchase to resale next period. Investors optimize over this one-period horizon.

Following Greenwood et al. (2024), I use a log-return approximation for portfolio choice. Write that return as

$$\mathcal{R}_{T,t+1}^a = \mu_T^a(z_t) + u_{T,t+1}^a, \tag{6}$$

where $\mu_T^a(z_t)$ is its conditional mean and $u_{T,t+1}^a$ is the unexpected part, with $\mathbb{E}_t[u_{T,t+1}^a] = 0$. Investors care about both: μ adds to expected wealth, while u changes its risk. The approximation includes the Jensen adjustment and becomes exact in the continuous-time limit.[5]

Investors. Investor i has funding W_i that can be invested in any bonds or cash, risk aversion γ_i and outside rate exposure ℓ_i from assets and liabilities beyond its bond portfolio. A positive ℓ_i raises net wealth when rates rise, as does a long-term liability whose value falls. A negative ℓ_i adds to bond losses. Investor i keeps the share $\xi_i^a = 1 - \tau_i^a$ of returns in market a, where τ_i^a is its tax rate on bonds in market a. Cash is taxed like taxable bonds.

The investor chooses dollar positions $x_{i,T}^a \geq 0$ across all markets a and all maturities T, keeps the rest of its funding in cash, $x_i^0 = W_i - \sum_{a,T} x_{i,T}^a \geq 0$, and maximizes expected next-period wealth less a penalty for its variance. With pretax returns measured by $\mathcal{R}$, next-period after-tax

[5] Let R denote the gross holding return above minus one. Precisely, $\mathcal{R} = \mu + u$ approximates R, with $\mu_T^a = \log \mathbb{E}_t[1 + R_{T,t+1}^a]$ and $u_{T,t+1}^a = \log(1 + R_{T,t+1}^a) - \mathbb{E}_t[\log(1 + R_{T,t+1}^a)]$. Thus μ includes the Jensen adjustment, and u is the centered log return. Conditional means and variances scale with the time step, as in equations (3)–(8) of Greenwood et al. (2024). Cash uses r in place of $e^r - 1$. Appendix B.1 gives the exact wealth equation and the simple-return comparison.

wealth is $\widehat{W}_i'$, and the problem is

$$\begin{aligned} \max_{x_i \geq 0,\, x_i^0 \geq 0} \quad & \mathbb{E}_t[\widehat{W}_i'] - \frac{\gamma_i}{2} \operatorname{Var}_t(\widehat{W}_i'), \\ \text{subject to} \quad & \widehat{W}_i' = W_i + \xi_i^C r(z_t)\, x_i^0 + \sum_{a,T} \xi_i^a x_{i,T}^a \mathcal{R}_{T,t+1}^a + \ell_i \sigma_r \varepsilon_{t+1}. \end{aligned} \tag{7}$$

Next-period wealth adds after-tax cash income, after-tax bond returns and the change in outside wealth to current funding. Investors choose portfolios each period, taking prices as given and conditioning on today's rate state. Nonnegative positions rule out short sales and borrowing (Section 3.5).

Required returns. The portfolio problem gives the Kuhn–Tucker conditions

$$\underbrace{\xi_i^a \mu_T^a - \xi_i^C r}_{\text{after-tax return over cash}} \quad \leq \quad \underbrace{\gamma_i \xi_i^a \operatorname{Cov}_z\big(u_{T,t+1}^a, \widehat{W}_i'\big)}_{\text{cost of the risk the bond adds}} + \underbrace{\beta_i}_{\text{value of funding}}, \tag{8}$$

with equality whenever $x_{i,T}^a > 0$. The covariance measures the risk the bond adds to total wealth, conditional on today's state. The funding multiplier $\beta_i \geq 0$ is the Lagrange multiplier on the cash constraint and is zero when the investor holds cash and can invest another dollar without borrowing. A held bond pays for both its added risk and any scarcity of funding. If it offers less than an investor's required return in equilibrium, that investor holds none. Differences in taxes and balance sheets therefore allow the same bond to suit one investor but not another in equilibrium.

Equilibrium. Prices clear every market at every rate state:

$$\textstyle\sum_i m_i x_{i,T}^a(z) = Q_T^a, \tag{9}$$

where m_i is the mass of type i, converting its positions and funding into aggregate holdings $m_i x_{i,T}^a$ and funding $m_i W_i$. In the calibration, positions are measured per dollar of resources, so m_i is the type's total resources and W_i the share available for cash and bonds. The rate process, supplies, taxes, funding, risk aversion and outside exposures are exogenous inputs. Equilibrium prices imply holding returns for which investors satisfy (8) and all markets clear. Who holds each bond is determined by these conditions.

4.2 Taxes and the cost of bearing rate risk

To isolate the mechanism, I use a special case with two investors and two markets. Consider a high-tax household H and a lower-tax investor I choosing municipal bonds M, taxable bonds C and cash. Municipal maturities begin at $T = 1$, while risky taxable bonds begin at $T = 2$. The two determinants of required returns are the investor's tax treatment and the cost of adding risk to its portfolio.

The relative after-tax share of Section 3.3 summarizes the tax comparison:

$$\xi_i = \frac{\xi_i^C}{\xi_i^M} = \frac{1-\tau_i^C}{1-\tau_i^M}, \qquad \xi_H < \xi_I.$$

The household's smaller ξ_H means a larger exemption benefit, $1-\xi_H$. I measure each investor's payoffs in units of its after-tax municipal income. Municipal returns are then kept in full, and taxable returns and cash income are multiplied by ξ_i. This change of units preserves portfolio choices.[6]

The full model is nonlinear because participation can change with rates and the rate floor can bind. Today's prices reflect these future nonlinearities. For an analytical solution, I use slack funding, $\beta_i = 0$, and a floorless short rate $r = \bar{r} + z$. Prices are solved with the same investors holding each bond across rate states, although their positions adjust. This gives log prices linear in z, the analytical affine solution.

Define a bond's rate loading as $A_T^a \equiv -\partial \log P_T^a / \partial z$, which measures its proportional price decline when the rate state rises. It is determined by equilibrium prices, as Section 4.5 shows. Rate shocks affect the holding return through next period's resale price, when the bond has $T-1$ periods remaining. Define the investor's net wealth exposure to rate risk as

$$\widehat{D}_i \equiv -\frac{\partial \widehat{W}_i'}{\partial z_{t+1}} = \sum_T x_{i,T}^M A_{T-1}^M + \xi_i \sum_T x_{i,T}^C A_{T-1}^C - \ell_i. \tag{10}$$

The derivative measures how much next-period wealth falls when the rate state rises, holding today's portfolio fixed. Liabilities whose value also falls offset losses on bonds, so a larger liability hedge ℓ_i lowers this net wealth exposure.

The investor's cost per unit of additional rate exposure is $\gamma_i \sigma_r^2 \widehat{D}_i$, which falls with the liability hedge at given bond holdings. The portfolio first-order condition is:

$$\begin{aligned} \mu_T^M - \xi_i r &\le \gamma_i \sigma_r^2 \widehat{D}_i \, A_{T-1}^M + \gamma_i v_T^M x_{i,T}^M, \\ \mu_T^C - r &\le \gamma_i \sigma_r^2 \widehat{D}_i \, A_{T-1}^C + \gamma_i \xi_i v_T^C x_{i,T}^C, \end{aligned} \tag{11}$$

Equality holds for a positive position. The municipal condition compares μ_T^M with after-tax cash, $\xi_i r$. Dividing the taxable condition by ξ_i puts it on a pretax scale, so it compares μ_T^C with r. In both lines the right side is rate-risk compensation plus a small bond-specific compensation, which grows with the position. The quantitative model uses common risk aversion, so its investors differ in tax treatment ξ_i and net rate exposure $\widehat{D}_i$, which depends on portfolio choice and outside exposures. The analytical benchmark allows γ_i to differ.

[6] Divide payoffs by ξ_i^M, and replace risk aversion by $\gamma_i \xi_i^M$ and outside exposure by ℓ_i / ξ_i^M. The objective becomes $1/\xi_i^M$ times the original. Below, γ_i and ℓ_i denote these normalized values.

These conditions separate the two forces behind participation. The household's larger exemption benefit lowers its required municipal return. A lower unit cost of rate risk also lowers the required return, increasingly so for a bond with a larger loading.

4.3 Participation at longer maturities

At the riskless maturity, only the cash alternative matters. With $A_0^M = v_1^M = 0$, a municipal holder requires $\mu_1^M = \xi_i r$. The household accepts the lowest return because it keeps the smallest share of taxable interest. With a positive short rate and slack household funding, it holds the municipal supply and

$$y_1^M = \mu_1^M = \xi_H r, \qquad y_1^C = r. \tag{12}$$

At the riskless short end the municipal yield equals the household benchmark exactly.

At a risky maturity, subtract the two investors' municipal conditions in (11). The common return μ_T^M cancels, leaving a comparison between their exemption benefits and their costs of bearing the same bond's rate exposure. The lower-tax investor participates whenever[7]

$$\underbrace{\sigma_r^2 A_{T-1}^M (\gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I)}_{\text{duration benefit}} > \underbrace{(\xi_I - \xi_H) r}_{\text{difference in exemption benefits}}. \tag{13}$$

All quantities are evaluated at equilibrium. A positive $\gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I$ means that the lower-tax investor requires less return per unit of rate exposure. With the same γ_i, this happens when the lower-tax investor's outside liability exposure ℓ_i is large enough to give it a lower equilibrium $\widehat{D}_i$. When the duration benefit exceeds the difference in exemption benefits, it is willing to accept a lower municipal return than the household, so the household cannot hold the supply alone.

On this affine branch, the bonds' rate loading A_{T-1}^M is endogenous and increases with maturity. Section 4.5 derives it from primitives. At a given equilibrium, the difference in unit risk costs is common across maturities, while A_{T-1}^M is increasing in maturity. The duration benefit therefore grows with maturity, while $(\xi_I - \xi_H)r$ stays fixed. Lower-tax participation can begin at the long end. For taxable bonds, bond returns and cash income receive the same tax treatment, so there is no difference in exemption benefits to overcome. With common risk aversion, the investor with lower $\widehat{D}_i$ has a lower rate-risk cost at every risky maturity.

Figure 4 shows the resulting allocation in the illustration. The household alone holds shorter municipal bonds, both investors hold longer municipal bonds, and the lower-tax investor holds risky taxable bonds. Bond-specific risk allows the investors to share a range of municipal maturities. This allocation underlies the return comparisons below.

[7] With $v_T^M > 0$, subtraction and market clearing give the exact criterion $x_{I,T}^M > 0 \iff \sigma_r^2 A_{T-1}^M (\gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I) + \gamma_H v_T^M Q_T^M / m_H > (\xi_I - \xi_H) r$. The additional term is the household's bond-specific risk cost if it holds the supply alone. It makes participation easier. Appendix Theorem 1 proves the equivalence.

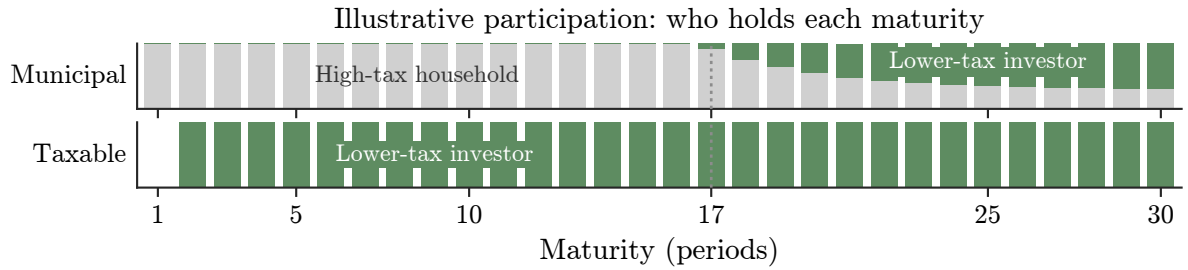


Figure 4: Participation in the two-investor illustration. The bars show each investor's share of the supply at the mean rate state. The lower-tax investor holds municipal bonds from $T^* = 17$ (dotted line). The one-period taxable claim is cash. Appendix B.9 gives the illustrative parameters. Section 5 reports the calibrated allocation.

4.4 Three channels in required returns

Given this allocation, I compare municipal and taxable bonds of the same maturity, the relevant comparison for the municipal puzzle. The municipal return above the household's after-tax taxable return is

$$g_T = \mu_T^M - \xi_H \mu_T^C. \tag{14}$$

The representative-tax benchmark sets $g_T = 0$ at every maturity. The portfolio conditions explain why this equality fails when holders have different taxes and risk costs.

The lower-tax investor holds risky taxable bonds. For municipal bonds, use the household's condition ($i = H$) below T^* and the lower-tax investor's condition ($i = I$) from T^* onward, where both investors hold them. Subtracting ξ_H times the taxable holder's condition from the municipal holder's condition in (11) gives

$$\begin{aligned} g_T = \; & \underbrace{(\xi_i - \xi_H) r}_{\text{after-tax cash return}} + \underbrace{\sigma_r^2 A_{T-1}^M (\gamma_i \widehat{D}_i - \gamma_I \widehat{D}_I)}_{\text{duration cost}} \\ & + \underbrace{\gamma_I \sigma_r^2 \widehat{D}_I (A_{T-1}^M - \xi_H A_{T-1}^C)}_{\text{rate loading}} + \underbrace{\gamma_i v_T^M x_{i,T}^M - \xi_H \gamma_I \xi_I v_T^C x_{I,T}^C}_{\text{bond-specific compensation}}. \end{aligned} \tag{15}$$

The three channels compare cash alternatives, costs per unit of rate exposure, and the exposures themselves. The cash-return term is zero below the threshold, the duration-cost term is zero from the threshold onward, and the loading term applies to both ranges. The final term accounts for independent bond-specific risk and is small in the illustration. Appendix Proposition 2 gives the derivation.

After-tax cash return. At long maturities, the lower-tax investor holds both municipal and taxable bonds. Because it pays less tax on taxable interest, tax exemption saves it less than it saves the high-tax household. It therefore requires a higher municipal return than the household benchmark predicts. The difference between their after-tax cash returns, $(\xi_I - \xi_H) r$, contributes to this gap.

Duration cost. At shorter risky maturities, the household alone holds municipal bonds, while the lower-tax investor holds taxable bonds. The household needs more return to bear the same rate exposure because rate risk is more costly for it. This difference contributes $\sigma_r^2 A_{T-1}^M(\gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I)$ to municipal returns above the household benchmark, even where the lower-tax investor holds no municipal bonds.

Rate loading. When taxes scale returns, they also scale risk. The household benchmark scales taxable returns and rate exposure by ξ_H, predicting $A_{T-1}^M = \xi_H A_{T-1}^C$. With lower-tax participation and $\xi_I > \xi_H$, municipal rate loadings satisfy $A_{T-1}^M \geq \xi_H A_{T-1}^C$. The extra exposure contributes positively to the gap when the lower-tax investor's net rate exposure is positive, as in the illustration. The contribution equals the extra exposure times $\gamma_I \sigma_r^2 \widehat{D}_I$, the lower-tax investor's unit risk cost. Section 4.5 derives the extra rate loading $A_{T-1}^M - \xi_H A_{T-1}^C$.

All three channels vanish at the riskless maturity, and $g_1 = 0$. Since $A_0 = 0$, the duration-cost and rate-loading terms vanish. The after-tax cash-return term is also zero because the household holds the riskless municipal bond. Figure 5 shows how the channels combine across maturities. It also decomposes the rate-loading channel into three effects discussed in the next section.

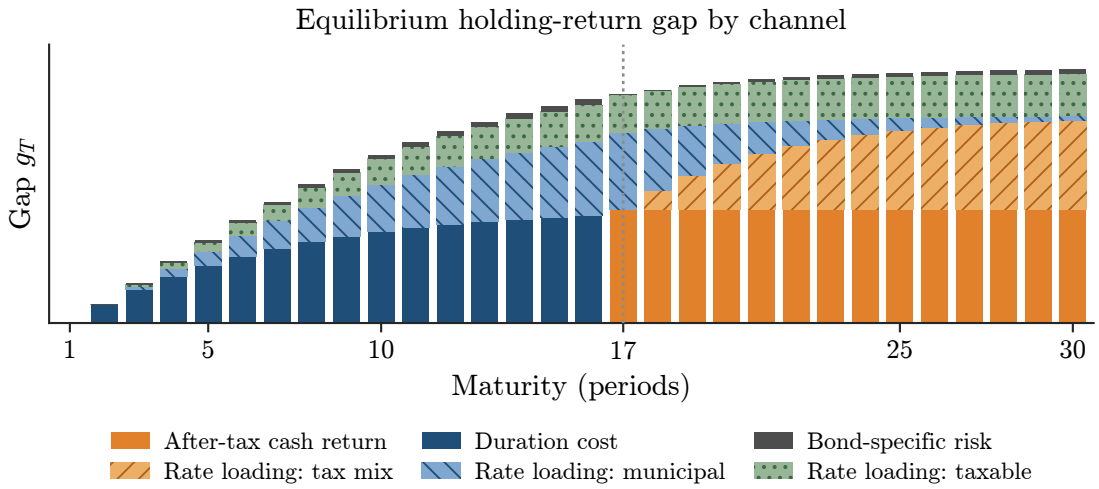


Figure 5: Required-return gap by channel. The allocation is that of Figure 4. The bars decompose $g_T = \mu_T^M - \xi_H \mu_T^C$ using (15). Within the rate-loading channel, the hatching separates the tax mix and the municipal and taxable portfolio responses. The small bond-specific term is on top. The vertical scale is omitted because levels depend on the illustrative parameters.

4.5 Rate loadings and the yield gap

A bond's rate loading depends on how its expected holding return responds to rates. Let $s_T^a = \partial \mu_T^a / \partial z$ denote the expected return's response to the current rate state. Since taxes returns, it scales risk accordingly, so the household benchmark scales taxable returns and rate loadings by ξ_H. Its return responses are ξ_H for municipal bonds and 1 for taxable bonds. The lower-tax investor keeps a larger share, $\xi_I > \xi_H$. Where both investors hold municipal bonds, the return response

reflects the tax mix larger than household benchmark prediction $\xi^* = \omega_H \xi_H + \omega_I \xi_I > \xi_H$, where ω_i is each type's share of total risk tolerance.[8]

Portfolio substitution also changes return responses elsewhere in both markets. As rates rise, the difference in exemption benefits, $(\xi_I - \xi_H)r$, grows, making municipal bonds relatively less attractive to the lower-tax investor. As a result, insurers will sell muni bonds when interest rise, and the household have to absorb more rate exposure in high-rate states when bond prices are lower. Its higher portfolio exposure raises its unit cost of rate risk, so it requires even more compensation when rate increases, making return rate sensitivity of bonds held along by households increase. On the other side, lower-tax investor sheds exposure when rate rises and requires less risk compensation on taxable bonds. Municipal returns therefore respond more to rates and taxable returns less than the household benchmark predicts.

Let $\bar{\xi}_T = \xi_H$ below T^* and $\bar{\xi}_T = \xi^*$ from T^* onward denote the holders' tax mix. Differentiating (11) gives the return responses and their gap from the household benchmark:

$$
\begin{aligned}
s_T^M &= \begin{cases} \xi_H + \gamma_H \sigma_r^2 A_{T-1}^M \frac{\partial \widehat{D}_H}{\partial z}, & \text{household alone,} \\ \xi^*, & \text{both investors,} \end{cases} \\
s_T^C &= 1 + \gamma_I \sigma_r^2 A_{T-1}^C \frac{\partial \widehat{D}_I}{\partial z}, \\
s_T^M - \xi_H s_T^C &= \underbrace{\bar{\xi}_T - \xi_H}_{\text{tax mix}} + \underbrace{s_T^M - \bar{\xi}_T}_{\text{municipal portfolio}} + \underbrace{\xi_H (1 - s_T^C)}_{\text{taxable portfolio}}.
\end{aligned}
\tag{16}
$$

Here $\frac{\partial \widehat{D}_i}{\partial z}$ is the investor's portfolio rate exposure change as rates rise. The exemption benefit gap between two investors $(\xi_I - \xi_H)r$ increases as rate state r increases. On this allocation, $\frac{\partial \widehat{D}_H}{\partial z} > 0 > \frac{\partial \widehat{D}_I}{\partial z}$ since lower-tax investors find municipal bonds less attractive and will sell it when rate rises, and households will clear what the lower-tax investors sell (Appendix Proposition 10). The tax-mix term is positive where both investors hold municipal bonds. The municipal portfolio term is positive at shorter risky maturities held only by the household, whose exposure rises. The lower-tax investor's falling exposure lowers the risky taxable response, making the taxable portfolio term positive as well. Figure 6(a) shows these three contributions. Here the tax mix concerns sensitivity to rates, while the cash-return channel in (15) concerns the return level.

The definition $\mu_T^a(z) = \log \mathbb{E}_z[1 + R_{T,t+1}^a]$ links expected holding returns to prices. Taking logs of the price recursion gives

$$
\log P_T^a(z) = -\mu_T^a(z) + \log \mathbb{E}_z\left[P_{T-1}^a(z')\right] + \frac{v_T^a}{2}.
$$

Today's log price equals the log expected resale value less the required holding return, with the bond-specific variance correction $v_T^a/2$. The rate loading $A_T^a = -\partial \log P_T^a / \partial z$ measures the proportional

[8] The weights are $\omega_i = (m_i/\gamma_i)/(m_H/\gamma_H + m_I/\gamma_I)$ and sum to one. The rate derivative of bond-specific risk cost is zero for a sole holder, whose position is fixed by supply, and cancels across joint municipal holders by market clearing.

price decline per unit rise in z. With affine log prices and $\mathbb{E}_z[z'] = \rho z$, the log expected resale value has loading ρA^a_{T-1}. Since v^a_T is independent of z, taking the negative state derivative of the log-price recursion gives

$$A^a_T = \rho A^a_{T-1} + s^a_T, \qquad A^a_0 = 0. \tag{17}$$

The first term carries forward the shorter bond's loading, weighted by rate persistence, and the second adds the current return response. Starting from the riskless holding return, this relation carries the response differences into $A^M_T > \xi_H A^C_T$ for $T \geq 2$. Municipal bonds consequently carry more rate exposure than the household benchmark implies. Figure 6(b) accumulates the three return-response contributions into this loading gap.

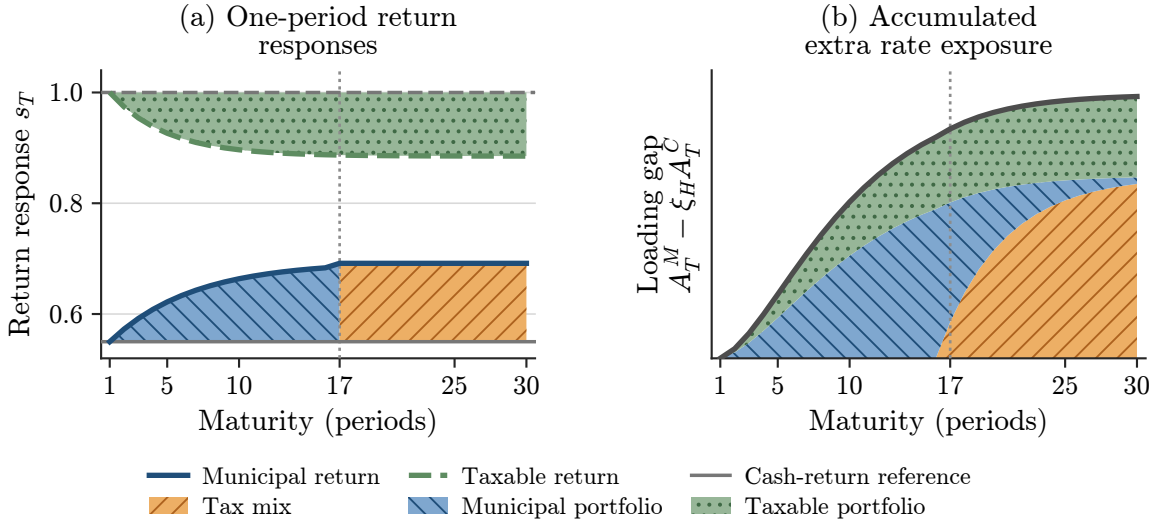


Figure 6: Rate responses in the illustration of Figure 4. Panel (a) shows municipal and taxable holding-return responses. Gray lines show the household municipal and pretax taxable cash responses, ξ_H and 1. Shading identifies the tax mix (orange), municipal portfolio response (blue) and taxable portfolio response (green). Panel (b) accumulates the three contributions into $A^M_T - \xi_H A^C_T$, scaling the taxable contribution by ξ_H. The dotted line marks lower-tax municipal participation from $T^* = 17$.

From return gaps to yield gaps. A long earns a sequence of holding returns before it matures. Its yield averages the expected returns along this path, less convexity (Appendix equation (39)). Applying this relation to both markets gives

$$y^M_T - \xi_H y^C_T = \frac{1}{T}\sum_{j=0}^{T-1} \mathbb{E}_t[g_{T-j}(z_{t+j})] - \text{convexity difference}. \tag{18}$$

The yield gap accumulates the three return-gap channels over the bond's remaining life. As a result, the municipal bond equilibrium yield is above the household benchmark at all maturities above shortest riskless one.

4.6 Empirical predictions

The household benchmark predicts full pass-through of household tax changes at every maturity and no response to tax benefits specific to lower-tax institutions. In the model, household taxes determine the riskless short municipal yield, while longer yields also reflect lower-tax participants' tax treatment and portfolio responses. I first use tax variation specific to households to test whether household tax pass-through declines with maturity (P1). I then use tax variation specific to banks and insurers to test whether lower-tax institutions' benefits affect longer yields while leaving the riskless short yield unchanged (P2). Together, these tests distinguish the participation mechanism from a single household tax rate applied across the yield curve. P3 tests the associated prediction that municipal rate sensitivity rises relative to the taxable benchmark with maturity.

The outcome of interest is the yield-implied tax $\tau_T^{Y,\text{impl}} = 1 - y_T^M / y_T^C$ from (2). Investor i's pass-through measures how much this equilibrium implied tax changes when investor i's exemption benefit changes:

$$\mathcal{P}_i^Y(T) = \frac{d\tau_T^{Y,\text{impl}}}{d(1-\xi_i)}, \qquad i \in \{H, I\}. \tag{19}$$

Here, pass-through of one means full transmission and zero means no effect. Household benchmark predict a pass through of 1 for household tax by definition.

In my model, at the riskless short end, the household alone holds the municipal bond, whose yield equals its after-tax return on taxable cash. Since $y_1^C = r$, this gives $y_1^M = \xi_H r$, the implied tax at riskless shortest maturity is $1 - \xi_H$. Therefore

$$y_1^M = \xi_H y_1^C \quad \Longrightarrow \quad \mathcal{P}_H^Y(1) = 1, \qquad \mathcal{P}_I^Y(1) = 0. \tag{20}$$

Only the household's benefit enters this pricing relation.

At long maturities, both investors hold municipal bonds. A larger exemption benefit makes these bonds more attractive to its recipient and induces portfolio adjustment. Equilibrium returns reflect both holders' tax treatment and the risk they bear. The household's benefit therefore no longer has the exclusive role it has at the riskless short end. The lower-tax investor's benefit also affects long municipal yields, and portfolio substitution transmits tax effects to other risky maturities. As holding returns accumulate into yields, longer bonds yields reflect both investors' tax benefits.

Portfolio adjustment reinforces this contrast in the affine benchmark. A larger household benefit widens the difference in exemption benefits and strengthens the rate-loading effect of Section 4.5, offsetting the increase in implied taxes, so household pass-through is smaller than 1 at long maturities. A larger lower-tax-investor benefit narrows that difference and weakens the rate-loading effect, reinforcing the increase in implied taxes, making their larger than 0 at long maturities. Under a moderate tax difference and a loose condition on mean rate, Appendix Proposition 6 gives

$$\mathcal{P}_H^Y(\infty) < 1, \qquad \mathcal{P}_I^Y(\infty) > 0. \tag{21}$$

Appendix B.8 derives the full yield-ratio derivatives and these conditions. In the solved illustration, household pass-through falls and institutional pass-through rises with maturity. In the empirical tests, we test the following three model predictions P1 to P3.

Household taxes (P1). A home-state exemption gives resident households an additional benefit on their state's bonds. That benefit is fully reflected in implied state taxes at the riskless short end and less strongly at longer maturities in the solved model. The representative-tax benchmark predicts full pass-through at every maturity.

Institutional taxes (P2). A benefit specific to a lower-tax institution has no effect at the riskless short end and matters more at longer maturities. It affects returns where the institution holds bonds and spreads to other risky maturities through portfolio substitution. Bank qualification and the 2017 corporate tax change for insurers test this prediction. For banks, the measure is the share of the bank-only benefit reflected in yields (Section 6.2). The representative-tax benchmark gives these benefits no role.

Interest-rate responses (P3). Section 4.5 shows why municipal bonds carry more rate exposure than the household benchmark predicts. Since $y_T^a = -\log P_T^a/T$, yield sensitivity is A_T^a/T. Comparing bonds at the same maturity cancels this division by T, so the yield-response ratio is the price-loading ratio:

$$\frac{\partial y_T^M/\partial z}{\partial y_T^C/\partial z} = \frac{A_T^M}{A_T^C}, \qquad \frac{A_1^M}{A_1^C} = \xi_H, \qquad \lim_{T\to\infty} \frac{A_T^M}{A_T^C} = \frac{\xi^*}{s_\infty^C} > \xi_H.$$

The larger long-end tax mix $\xi^* > \xi_H$ raises relative municipal sensitivity, and the taxable portfolio response $0 < s_\infty^C < 1$ reinforces it. P3 tests whether municipal yield sensitivity relative to same-maturity taxable bonds increases with maturity. The household benchmark predicts the constant ratio ξ_H.

4.7 Liability hedging and other motives

Liability hedging is one source of the participation advantage. Lower trading costs or better default diversification can also make municipal bonds less costly for a lower-tax investor to hold. An investor may trade less often or at lower cost than households, or diversify default risk across markets more effectively than a household concentrated in its home state's bonds. When these advantages grow with maturity, they can outweigh the maturity-invariant difference in exemption benefits and sustain participation in long municipal bonds.

The pricing mechanism extends to these motives when the relevant advantage is neutral across taxable and tax-exempt bonds. Neutrality means that the characteristic creates no return gap relative to the household benchmark when the household holds both bonds. Lower-tax participation then brings the participant's smaller exemption benefit into the return comparison. Appendix A.7 states the result, and Appendix B.12.4 gives solved trading-cost and default-diversification examples, with the default comparison net of default compensation. A preference for municipal bonds

as such, including warm glow from financing local projects, lowers the return an investor requires on them. Such a preference alone does not explain yields above the household benchmark.

I use liability hedging in the quantitative application because rate risk is common to both markets and balance sheets provide a measurable source of differences in investors' risk-bearing costs. Observed differences in maturity holdings help discipline those exposures. The broader pricing mechanism does not depend on this particular microfoundation. The trading-cost and diversification examples show how lower-tax participation of the kind observed in the data generates municipal yields above the household benchmark under the neutrality condition.

5 Quantitative Results

5.1 The quantitative model

The quantitative model asks how much of the municipal puzzle measured taxes, balance sheets and ownership can explain. It solves the general model of Section 4.1 with municipal markets across the 50 U.S. states and the District of Columbia, a taxable market and six investor groups. Tax rules, quantities and household balance sheets are measured directly. Seventeen common coefficients governing risk aversion, bond-specific risk, outside exposures and within-group resource shares are estimated from twenty-two moment conditions on ownership, maturities, portfolios and the taxable yield curve. Municipal yields are untargeted.

States with a personal income tax have separate municipal markets. The seven states with no state income tax are combined into one market because they are symmetric in the model. Residents value their home-state exemption, while institutions do not receive this state-level benefit under the tax code. To reduce the computational burden, the resulting 45 municipal markets have maturities from two to thirty years in two-year steps. The taxable market pools treasury and corporate supply. Cash and the two-year municipal claim are riskless. Prices in both markets are determined in equilibrium.

Investors solve (7) with market-specific after-tax shares, and prices satisfy the portfolio conditions (8) and clearing (9) in every market and rate state. Banks' measured financing-tax cost is subtracted from municipal returns according to their tax treatment (Appendix C.7). The model retains the rate floor in (5) and allows funding constraints and participation to change across rate states. I solve $P_T^a(z)$ numerically through the resale relation in Appendix equation (35), including bond-specific risk and integrating over next-period states. Returns, portfolio risks, holdings and both yield curves are determined jointly. Prices can be nonlinear in the rate state because future participation, binding funding constraints and the rate floor affect resale values.

The measured inputs cover six three-year blocks, from 2000–2002 through 2015–2017, centered on Survey of Consumer Finances (SCF) waves. Financial Accounts stocks give sector resources and bond quantities. Mergent gives maturity profiles and municipal state shares. The SCF supplies household balance sheets and the income and filing status needed to apply statutory taxes. Histor-

Table 3: Investor groups in the quantitative model, 2015–2017

Group	Types	Types differ by	Exemption benefit $1-\xi_i$	Resources m_i
High-tax households	$7 \times 45 = 315$	Rate exposure, home state	0.43–0.50	6.97
Lower-tax households	$7 \times 45 = 315$	Rate exposure, home state	0.21–0.27	8.09
P&C insurers	2	Rate exposure	0.31	1.11
Life insurers	3	Rate exposure	0.21	3.05
Banks	1	None	0.35	4.83
Residual group	7	Rate exposure	0	26.47
Total	643			50.52

Notes. Each household group has seven types, differing in rate exposure, in each of the 45 markets of the computation, and also seven types in the one-market economy used to estimate the coefficients (Section 5.2). Household exposures are set by each type's measured outside assets and debt; institutional exposures are estimated coefficients (Appendix Table 10). The exemption benefit $1-\xi_i$, with $\xi_i = (1-\tau_i^C)/(1-\tau_i^M)$, follows from tax law. Households multiply the federal and home-state after-tax shares of each SCF tax unit, with the range running across states, and institutions face the 35% corporate rate, with proration for insurers. Banks also pay a financing-tax cost of 0.11% a year on municipal holdings, which lowers their net benefit (Table 2). Resources m_i, in trillions of dollars, are all of a group's debt securities and cash in the Financial Accounts. Holdings are chosen per dollar of resources and clear as $\sum_i m_i x_i = Q$, so m_i scales the group's risk-bearing capacity.

ical short rates determine the rate process. Appendix C.7 details the construction, and Table 15 lists the sources.

Table 3 lists the groups. The setup is implemented per dollar of resources: m_i is a type's total resources, W_i the share available for bonds and cash, and $x^a_{i,T}$ its position per dollar. Total investment is $m_i W_i$ and aggregate bond holdings are $m_i x^a_{i,T}$. Households differ by federal tax bracket and outside assets and debt. Outside assets add rate risk and liabilities offset it. Their mapping into exposures is estimated. Insurers differ in exposures and resource shares, banks face municipal financing-tax costs, and the residual group is tax-neutral. Section 3.4 and Appendix E.1 distinguish gross exemption benefits from these financing costs.

5.2 Estimation and fit

I measure directly the inputs available in the data: investor tax treatment, sector resources, bond quantities and maturity profiles, household assets and debt, and the short-rate process. These inputs vary across the six blocks of 2000–2017. The remaining seventeen coefficients govern risk aversion, bond-specific risk and the mapping from balance sheets into outside rate exposures. I estimate one common set of coefficients across all six time blocks.

Observed ownership provides the main discipline on outside rate exposure, which balance-sheet amounts alone cannot measure. Given each group's tax treatment and resources, its municipal and taxable holdings reveal how much rate risk it is willing to bear. For households, three coefficients translate measured outside assets and debt into rate exposure. For institutions, the estimated exposures and within-sector resource shares account for both the amounts held and their maturity composition. The taxable yield curve and long-bond allocations discipline risk aversion and bond-

specific risk. The coefficients are estimated jointly because portfolio choices and yields adjust together in equilibrium.

The estimation uses twenty-two targets. Nine are ownership stocks: municipal holdings of the two household groups, P&C insurers, life insurers and banks, and taxable holdings of households and the same three institutional sectors. Four are the mean maturities of insurers' municipal and taxable portfolios. Six describe the distribution of insurer holdings. Two measure whether municipal holdings are spread across insurers or concentrated in portfolios with high municipal shares, giving larger portfolios more weight. Four measure the fractions of each sector's municipal and taxable portfolios maturing beyond twenty years. The remaining three are taxable yields at ten, twenty and thirty years. I evaluate these conditions at observed rate states, average them across the six blocks and minimize their weighted squared deviations from the data.[9]

For computational tractability, I estimate the coefficients in an economy with one national municipal market, using municipal supply weights to average each household type's home-state after-tax shares. I then restore the 45 municipal markets and their measured tax differences, holding every coefficient fixed. No coefficient jointly calibrated to moments is state-specific. [10]

The one-market and full economies give similar relative municipal yields. Their thirty-year gaps average 102.6 and 102.4 basis points across the six blocks, differing by at most 2.5 basis points in any block. Thirty-year taxable yields differ by at most 0.5 basis points. Changing each coefficient group in turn moves the one-market gap by at most about 4 basis points, except for the common risk aversion.[11] Section 5.4 uses the 2000–2020 bond sample's weights and expresses the model's relative municipal yields on the observed corporate curve for the headline comparison.

Municipal yields, the municipal puzzle and the empirical tax-pass-through coefficients do not enter the estimation criterion. The municipal yield curve and cross-state tax pass-through are untargeted outcomes, evaluated against data in Sections 5.3–5.4 and 6.1. The calibration uses ownership to discipline outside rate exposure. Together with observed taxes, these exposures determine municipal yields in equilibrium, leaving the relative municipal yield curve as a test of the mechanism.[12]

5.3 Model and data yield curves

Figure 7 sets the calibrated model curve (top) against the data (bottom). To cover the full data sample, I keep the estimated coefficients fixed and add a seventh economy for 2018–2020 after the TCJA, built from that period's taxes, rates and quantities. Both rows show a municipal yield near the household benchmark at the short end and a spread over it that widens with maturity.

[9] Each averaged deviation is scaled by 10% of its observed normalizer. Maturity and long-portfolio conditions include model-portfolio weights. Appendix equation (110) gives the criterion, and Tables 10 and 11 report the coefficients and target definitions.

[10] Fourteen of the twenty-two absolute scaled deviations are below one. Appendix C.6 reports the estimation procedure and fit.

[11] Changing risk aversion alone scales both curves and the spread, moving the taxable yield level away from its calibration targets.

[12] Appendix C describes the solution method, and Appendix C.3 reports accuracy.

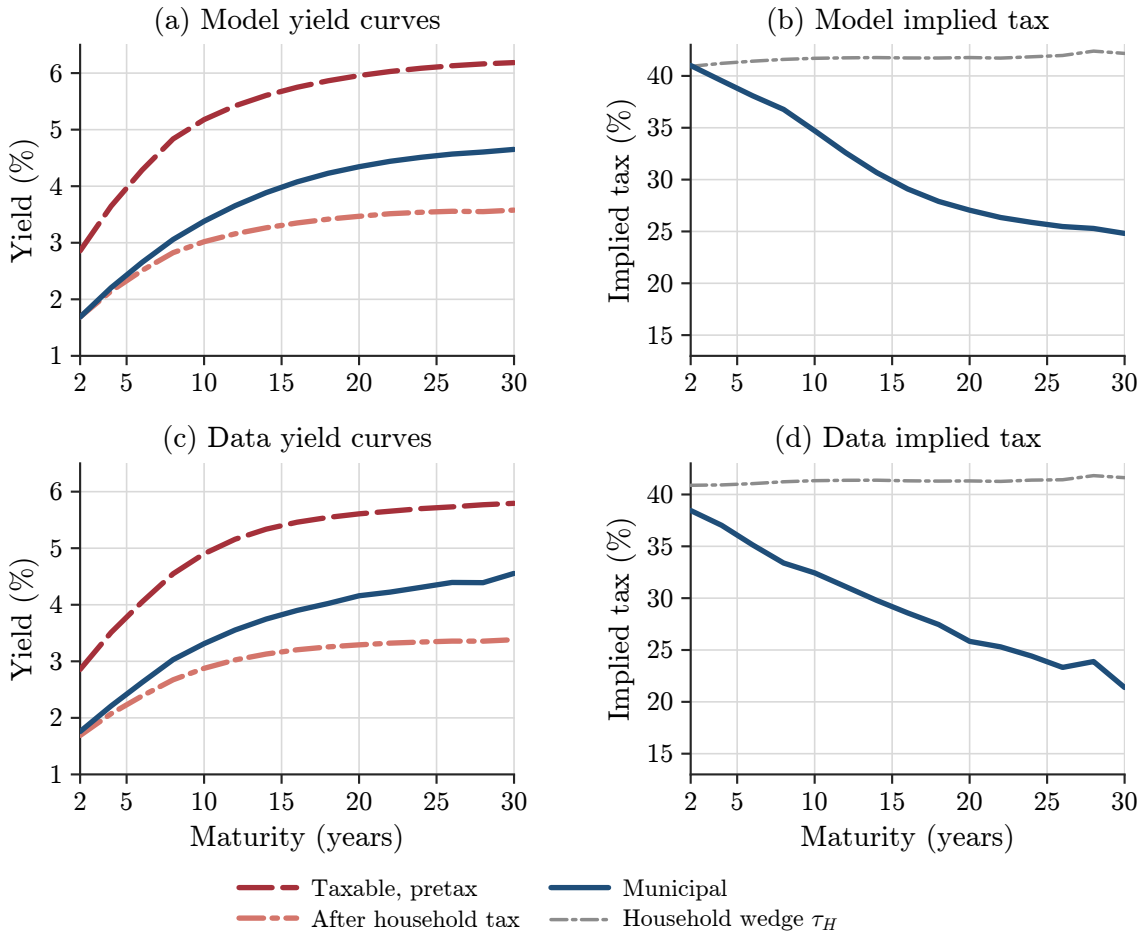


Figure 7: Model (top) and adjusted data (bottom), 2000–2020. Panels (a) and (c) plot the pretax taxable yield, the taxable yield after household tax (the household benchmark) and the municipal yield. Panels (b) and (d) plot the implied tax $1 - y_T^M / y_T^C$ and the household wedge. Corresponding panels share axis scales. The taxable curve is the model's pooled taxable market (top) and the observed corporate benchmark (bottom). The municipal data curve removes callability (Appendix D.1.2) and the issuance premium, and panel (d) uses this same adjusted curve. The model evaluates the estimated coefficients at observed issue-quarter rate states in the six calibration blocks and a 2018–2020 economy. Both rows use the empirical bond, rating and quarter weights and nominal semiannual yields. The household benchmark multiplies each observation's taxable yield by the high-tax household's relative after-tax share for that bond's state and date before averaging. Implied taxes and the household wedge are ratios of averaged yields.

In nominal semiannual yields, the model's implied tax $1 - y_T^M / y_T^C$ is 41.0% at two years, 34.7% at ten and 24.8% at thirty, a decline of 16.2 percentage points. The model's thirty-year municipal and taxable yields are 4.651% and 6.185%, against 4.555% and 5.794% in the adjusted data.

5.4 Accounting for the puzzle

The accounting asks how much of the municipal puzzle the model explains, alongside the measured capital-gains, liquidity and default differentials. To compare these contributions, I express them

on the same observed taxable yield curve. Otherwise, a higher taxable yield in the model would mechanically enlarge its gap in basis points, even with the same implied tax.

At each maturity, I keep the model's municipal-to-taxable yield ratio and multiply it by the observed taxable yield. I then subtract the household after-tax benchmark. The resulting gap, denoted $\widetilde{\mathcal{G}}_T$, measures the model's contribution on the data's taxable curve:

$$\begin{aligned} \text{Normalized municipal yield} &= \frac{\text{Model municipal yield}}{\text{Model taxable yield}} \times \text{Observed taxable yield}, \\ \widetilde{\mathcal{G}}_T &= \text{Normalized municipal yield} - \text{Household benchmark}. \end{aligned} \tag{22}$$

I call this the corporate normalization. It preserves the model's implied tax and changes only the scale of this comparison. The solved equilibrium and the curves in Figure 7 are unchanged. Appendix D.1.7 gives the averaging and normalization details.

The accounting has three steps at each maturity T:

$$\begin{aligned} \text{Spread}(T) - \underbrace{\big[\text{Call}(T) + \text{Prem}(T)\big]}_{\text{comparability adjustments}} &= \underbrace{\text{Puzzle}(T)}_{\text{municipal puzzle}}, \\ \text{Puzzle}(T) - \underbrace{\big[\text{CG}(T) + \text{Liq}(T) + \text{Def}(T)\big]}_{\text{bond-side adjustments}} &= \underbrace{\text{Puzzle}^{\text{adj}}(T)}_{\text{residual before the model}}, \\ \text{Puzzle}^{\text{adj}}(T) - \underbrace{\widetilde{\mathcal{G}}_T}_{\text{model}} &= \underbrace{\text{Residual}(T)}_{\text{unexplained}}. \end{aligned} \tag{23}$$

The spread, call adjustment and issuance premium follow Section 3.7. The terms CG, Liq and Def denote capital-gains, liquidity and default differentials. Since the model excludes these features, its gap is compared with $\text{Puzzle}^{\text{adj}}(T)$. The final residual equals the observed corporate yield times the implied-tax difference (model minus data), with the data reflecting all five adjustments.

At thirty years (Figure 8 and Table 4), the model accounts for 97.4 basis points of the 117.2-basis-point puzzle, or 83%, and the bond-side adjustments account for 18.8, or 16%, leaving 1.0 basis points unexplained. The model's contribution grows with maturity in basis points. Its share of the puzzle is 74% at ten years, 97% at eighteen and 83% at thirty. The unexplained residual is negative at every maturity except thirty years. The model gap exceeds $\text{Puzzle}^{\text{adj}}(T)$ by 11 to 15 basis points from twelve to twenty-four years, and the largest absolute residual is 14.9 basis points at sixteen years. This negative residual is consistent with the convenience premium interpretation in Fleckenstein and Longstaff (2025).

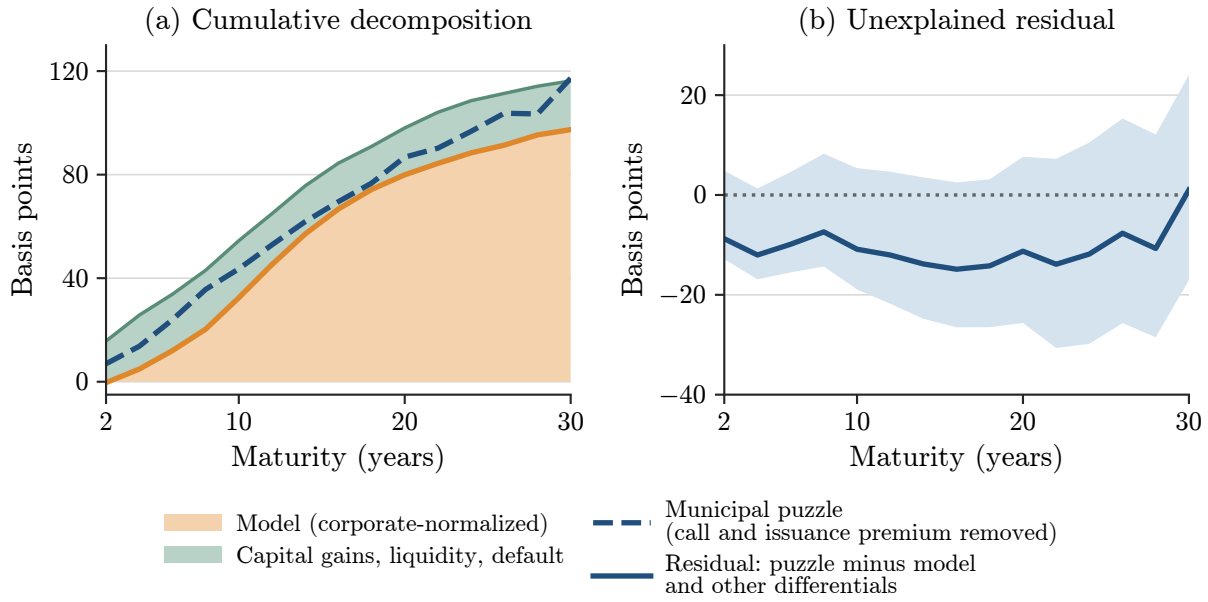


Figure 8: Cumulative decomposition of the municipal puzzle, 2000–2020, as in (23). The puzzle is the observed spread after removing callability and the issuance premium. Panel (a) stacks the model gap $\widetilde{\mathcal{G}}_T$ of (22) and the capital-gains, liquidity and default differentials against that puzzle. Panel (b) shows Residual(T), the puzzle less their sum. Shading spans the ranges of the five empirical adjustments (Appendix Table 27) and excludes model uncertainty. Weights and yield conventions match Figure 7.

Table 4: The model's share of the municipal puzzle. Removing callability and the issuance premium from the raw spread gives the puzzle. The normalized model gap applies the model's implied tax to observed corporate yields, as in equation (22). The unnormalized model gap is shown for comparison.

		comparability adjustments			model				
T	spread	call	premium	puzzle	unnormalized	normalized	model/puzzle	bond-side	unexplained
2	9.9	0.1	2.9	6.9	−0.2	−0.3		16.0	−8.8
6	27.9	0.7	3.3	23.9	14.3	12.0	50%	21.8	−9.9
10	52.2	4.6	4.0	43.6	36.2	32.4	74%	22.0	−10.9
14	87.0	18.0	7.2	61.8	62.1	57.1	92%	18.6	−13.8
18	110.3	24.0	9.6	76.7	81.1	74.3	97%	16.6	−14.2
22	128.8	27.2	11.4	90.2	92.7	84.3	93%	19.8	−13.9
26	145.0	28.7	12.5	103.7	101.1	91.4	88%	20.0	−7.7
30	159.8	29.8	12.8	117.2	107.4	97.4	83%	18.8	1.0

Notes. Basis points, except shares. The accounting is puzzle = normalized model + bond-side + unexplained, as in (23), where the bond-side adjustments sum the capital-gains, liquidity and default differentials. The model share divides the normalized model by the puzzle, using unrounded values, and is omitted where the puzzle is below 20 bp. Model and data both use the 2000–2020 sample, the averaging of Appendix D.1.1 and the nominal semiannual convention. The separately measured components are not jointly estimated. Rows show every second model maturity. Figure 8 shows all fifteen.

6 Empirical Tests

Table 5 summarizes three tests: household state-tax pass-through (P1), bank qualification and insurer exposure to the 2017 corporate tax cut (P2), and municipal responses to treasury-rate movements (P3). The estimates have the predicted maturity ordering. None is a calibration target or an input to the curve decomposition.

Table 5: Empirical predictions and sources of variation

	Source of variation	Model prediction	Representative-tax benchmark
P1	Residents' state income tax rates	Pass-through near one at the short end, declining with maturity	Pass-through of one at every maturity
P2	Bank qualification, 2017 insurer tax cut	Little effect at the short end, larger farther out	No effect at any maturity
P3	Treasury-rate changes	Long-end response above the coupon benchmark	Response equal to the coupon benchmark

The curve accounting in Sections 3 and 5 uses the A-or-above municipal sample (agency score seven or better), matching the corporate benchmark's credit band. The P1, TCJA and P3 tests use the narrower AA-and-above sample (agency score three or better) to limit credit heterogeneity that could affect both yields and trading patterns. Appendix D.12.2 reports each on the wider sample and shows that the point-estimate maturity ordering is not driven by the rating cutoff. The bank-qualified design uses all new issues with rating-by-year fixed effects. Below, m_b denotes remaining maturity in years, distinct from the investor masses m_i in the model.

6.1 P1. State-tax pass-through

P1 tests whether resident households' state-tax benefit is reflected less in municipal yields at longer maturities. Most states exempt residents' in-state municipal interest while taxing out-of-state interest. A household in state s thus keeps all in-state municipal interest and $1-\tau_s^{\text{st}}$ of out-of-state interest, with the same state tax applied to taxable bonds.

State rates move resident households' exemption benefits, a wedge institutions largely do not face. Pensions and tax-deferred accounts owe no current state interest tax. Insurer premium taxes generally do not distinguish municipal from taxable interest. Funds pass through the state benefit according to shareholder residence and portfolio eligibility: single-state funds preserve the home-state exemption, while national funds pass through only the eligible home-state share (Appendix E.3).

Under the conditions in Section 4.6, Proposition 6 gives household pass-through of one at the riskless boundary and below one in the long-end limit. Appendix Proposition 14 gives the state-market responses. The representative-tax benchmark predicts one at every maturity.

The ratio y^M/y^C contains both tax layers. Under the pre-2018 federal deduction convention used for taxable interest, the taxable after-tax share is $\xi_s^{\text{st}}\xi_t^{\text{fed}}$. Here $\xi_s^{\text{st}} = 1-\tau_s^{\text{st}}$ is the state

after-tax share and ξ_t^{fed} the federal share. Home-state municipal interest is exempt at both levels. Under the representative-tax benchmark, household indifference gives $y_{s,t}^M = \xi_t^{\text{fed}} \xi_s^{\text{st}} y_t^C$. Dividing out the federal after-tax share defines

$$ImpliedStateTax_{s,t}(T) \equiv 1 - \frac{y_{s,t}^M(T)/y_t^C(T)}{\xi_t^{\text{fed}}}. \tag{24}$$

Under the representative-tax benchmark, this outcome equals the statutory state rate τ_s^{st}, giving unit pass-through. Here ξ_t^{fed} is one minus the dated top federal ordinary-income rate: 0.604 in 2000, 0.609 in 2001, 0.614 in 2002, 0.65 in 2003–2012 and 0.604 in 2013–2017, excluding the net investment income tax. The model uses its own high-tax-household federal after-tax share in each block. The sample ends before the Tax Cuts and Jobs Act capped state and local tax deductions at \$10,000 from 2018. Appendix D.7 reports the later period under its tax mapping.

I partition bonds into five maturity bins by years to maturity ($[0,5)$, $[5,10)$, $[10,15)$, $[15,20)$, $[20,30]$ years) and estimate

$$\begin{aligned} ImpliedStateTax_{b,t} = &\sum_{g=1}^{5} \beta_g \big(StateTax_{s(b),t} \times \mathbf{1}\{m_b \in g\}\big) + \sum_{g=1}^{5} \delta_g' \big(Z_{b,t} \times \mathbf{1}\{m_b \in g\}\big) \\ &+ \lambda' V_b + \alpha_{t \times m \times callable} + \varepsilon_{b,t}, \end{aligned} \tag{25}$$

where $StateTax_{s,t}$ is the top statutory rate and (24) uses call-adjusted zeros. The coefficient β_g measures how much the implied state tax changes per percentage point of the statutory rate in maturity bin g. The representative-tax benchmark gives $\beta_g = 1$. Controls $Z_{b,t}$ are the rounded average rating and six lagged state variables: par-weighted outstanding maturity, preceding-year issuance maturity, outstanding par relative to personal income, log income, log preceding-year issuance and the outstanding share in the bond's maturity bin. The stock and issuance variables cover federally exempt bonds, and the stock and income variables are measured at the preceding quarter-end. Bin-specific coefficients allow supply, market size and maturity structure to affect each part of the curve differently. V_b includes log offering par and security-type indicators, and $\alpha_{t \times m \times callable}$ denotes issue-year–month–maturity–callability fixed effects.

The sample covers 2000–2017 and contains 681,839 observations with complete controls. The main estimates use unweighted OLS, so every bond counts once. Appendix D.7 reports equal-state and equal-state-year weighting. Standard errors are clustered by state, allowing dependence among a state's bonds across years and maturities.

For an IV comparison, I instrument each bin interaction of the current statutory rate with its 1990 counterpart, following the policy-history approach of Babina et al. (2021) and using the historical top-rate series of the Institute on Taxation and Economic Policy. Appendix D.7 reports alternative vintages with the same controls and sample.

Figure 9 compares OLS estimates of (25) for 2000–2017 with the full-state model's cross-state slope. The model uses (24), its own high-tax-household federal share ξ^{fed} and nominal-semiannual

yields. At each maturity, I take the slope on the state's top rate, with an intercept and equal weights across the 43 income-tax states and the District of Columbia. Within each economy, states are evaluated at a common rate state near the stationary mean. Slopes are averaged across the six pre-2018 blocks.

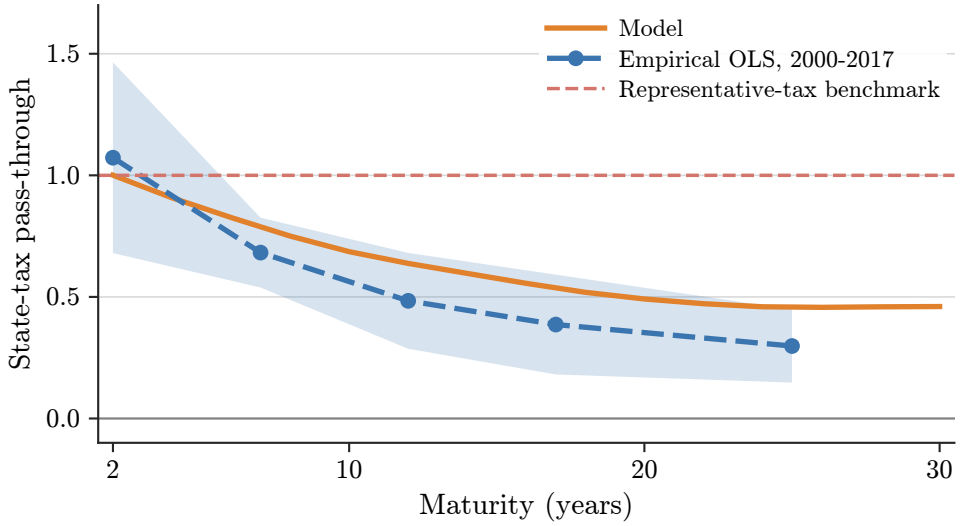


Figure 9: State-tax pass-through by maturity. Blue: controlled, unweighted OLS estimates for 2000–2017 call-adjusted zeros, with state-clustered 95% intervals using $t(46)$. Orange: the model's cross-state slope with its own federal after-tax share and nominal-semiannual yields, using a common rate state across states within each economy and averaging slopes over the six pre-2018 economies. Both use (24). The red dashed line is the representative-tax benchmark.

The OLS profile declines across all five bins, from 1.072 below five years to 0.298 beyond twenty (Appendix Table 30). A ten-percentage-point difference in a state's top rate corresponds to about 10.7 points in the implied tax at the short end and 3.0 points at the long end. The short-to-long decline is 0.774 (s.e. 0.234, $p = 0.002$). Unit pass-through cannot be rejected in the shortest bin and is strongly rejected in the longest. Every bin coefficient is significantly positive at one percent. The IV comparison also declines, from 1.092 to 0.227, with an endpoint-equality p-value below 0.001.

The untargeted model profile declines from 1 at two years to 0.69 at ten and 0.46 at thirty. Appendices D.6 and D.7 retain the main controls and pre-2018 sample while varying yield construction, credit controls, weights and instruments. They also report state fixed effects, single-state omissions and a separate post-2018 comparison under the changed tax mapping.

6.2 P2. Bank qualification

Bank qualification tests the maturity profile of a bank-only tax benefit. Banks ordinarily cannot deduct interest expense allocated to tax-exempt holdings (Section 265(b) of the Internal Revenue Code). For bank-qualified (BQ) bonds, Sections 265(b)(3) and 291 restore 80% of that deduction. Issuers qualify if reasonably anticipated annual tax-exempt issuance does not exceed $10 million ($30 million in 2009–10 under the American Recovery and Reinvestment Act) and they designate

the bonds. Designation changes neither payments nor other investors' taxes. Dagostino (2025) studies how the expanded cap affected issuance and employment.

A bank allocates interest expense across assets in proportion to tax basis, so a dollar of municipal basis carries the same annual funding cost c at every maturity. Designation saves $X = 0.8\,\tau_B c$ a year, where τ_B is the federal bank tax rate. Using Call Report interest expense, X is about 65 basis points in 2005–08 and 9 to 14 from 2011 to 2019 (Appendix D.8.3). I also scale premia by X_{tm}, the expected annual benefit over the life of a bond issued at t with maturity m (Appendix D.8.3). Its median is 45 basis points.

Three hypotheses differ in how much of X the designation premium $\varpi_k = y_k^N - y_k^Q$, the non-designated minus the designated yield at maturity k, reflects. Under the representative-tax benchmark, households require $(1-\tau_H)y_k^C$ on both bonds. When their exemption benefit exceeds banks' net benefit, banks prefer taxable bonds at these yields and $\varpi_k = 0$. Under a bank-only benchmark, indifference to taxable bonds gives full capitalization, $\varpi_k = X$, flat in maturity. In the model, the premium is the benefit times the bank's capitalization share θ_k, the average bank share of the total demand response to the bond's yield (Appendix D.8):

$$\varpi_k = \theta_k\, X, \qquad \underbrace{\theta_k = 0}_{\text{representative-tax}}, \qquad \underbrace{\theta_k = 1}_{\text{bank-only}}, \qquad \underbrace{\theta_{\text{short}} \simeq 0 < \theta_{\text{long}}}_{\text{model}}. \tag{26}$$

The capitalization share is the bank counterpart of $\mathcal{P}_I^Y(T)$ in Section 4.6. Household after-tax cash returns anchor short yields and banks hold few short bonds, making θ_k near zero. In the calibrated model, banks hold intermediate maturities where their duration benefit offsets their smaller exemption benefit (Section 5.3). Yields there reflect more of the bank benefit. A rising premium in expected-benefit units distinguishes this mechanism from both benchmarks.

The design compares issuers on either side of the cap. The running variable R_{jt} is issuer j's annual tax-exempt issuance minus that year's cap, in millions of dollars. The indicator $E_{jt} = \mathbf{1}\{R_{jt} \le 0\}$ marks below-cap issuer-years, and B_b marks designation. Measured issuance differs from the statutory count, giving a fuzzy discontinuity: below-cap status instruments designation. The Mergent sample covers tax-exempt new issues from 2000 to early 2021 with $0.5 \le |R_{jt}| \le 4$. The half-million donut removes managed bunching just below the cap and issuer-years exactly at it. With $\mathbf{1}_k$ indicating maturity at issuance (1–2, 2–5, 5–10, 10–20 or more than 20 years), the first and second stages are

$$\begin{aligned} B_b\mathbf{1}_k &= \sum_\ell \alpha_{k\ell}\, E_{jt}\mathbf{1}_\ell + V_b'\delta_k + e_{bk}, \qquad k = 1,\dots,5, \\ y_b &= \sum_k \beta_k\, B_b\mathbf{1}_k + V_b'\delta + u_b, \qquad B_b\mathbf{1}_k \text{ instrumented by } E_{jt}\mathbf{1}_k, \end{aligned} \tag{27}$$

where V_b contains local linear controls in R_{jt} on each side of the cap, log tranche par, and fixed effects for maturity, state, offering month, rating, security type, use of proceeds, coupon and callability, with the interactions listed in Appendix D.8.2. The outcome is the offering yield in basis points,

or that yield divided by X_{tm}, so $-\beta_k$ is the premium in basis points or in expected-benefit units. Standard errors are clustered by issuer. Identification requires that, conditional on these controls, crossing the cap moves offering yields only through designation. Callability and the number of maturities per issue are balanced at the cap. The first-stage designation jump is 0.42 to 0.45 across bins, too similar to explain the much larger long-maturity yield premium. Placebo cutoffs show essentially no designation effects, and no systematic yield effects at \$5 million and \$15 million (Appendix D.8).

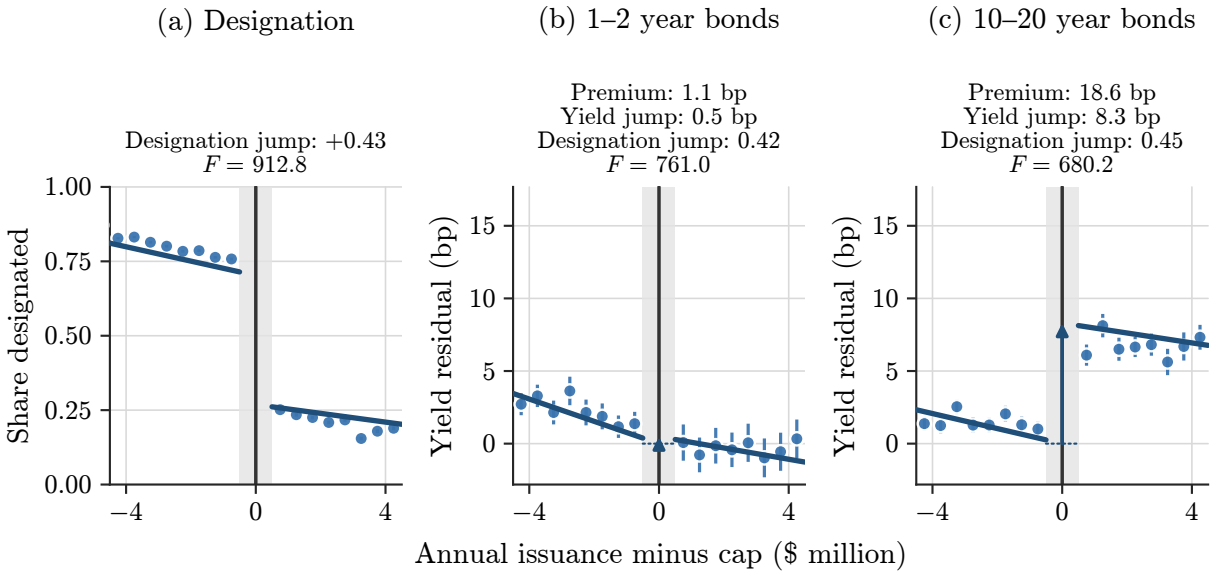


Figure 10: The bank-qualified discontinuity. Panel (a): designation around the cap. Panels (b)–(c): offering-yield discontinuities for 1–2 and 10–20 year bonds. Local linear fits exclude the shaded donut. Bars are 95% intervals. Jumps are differences between the local linear fits at the cap, below-cap minus above-cap for designation and above-cap minus below-cap for yields. First-stage F statistics test the below-cap coefficient with issuer-clustered standard errors. Dividing the bin-specific yield jumps by their first stages gives premia of 1.1 and 18.6 basis points. Appendix Figure 24 shows the other maturities.

Figure 10 shows unconditional designation rates of 0.79 below the cap and 0.21 above. The annotated first stages condition on the regression controls. The yield discontinuity is far larger for 10–20 year bonds than for 1–2 year bonds. Figure 11 reports the joint 2SLS estimates (Appendix Table 36). The premium is 1.8 basis points at 1–2 years, indistinguishable from zero, 8.7 at 2–5 years, 13.8 at 5–10, 18.4 at 10–20 and 18.2 beyond twenty, and the 10–20 year premium exceeds the 1–2 year premium by 16.5 basis points (s.e. 2.7). In expected-benefit units, the premium rises from 0.17 at 1–2 years to 0.52 at 10–20 years: the bank-only benefit is barely priced in short municipal yields, and about half of it is priced at 10–20 years. The positive premia at 5–20 years reject the representative-tax benchmark, with t statistics of 6 to 8, and the rise in expected-benefit units, 0.34 (s.e. 0.11), rejects the flat premium of full capitalization. The gradient appears in every rate regime and survives alternative donuts and bandwidths, credit-enhancement controls, a general-obligation sample, issuer effects, year clustering and the binned estimator of Dagostino (2025) (Appendix D.8).

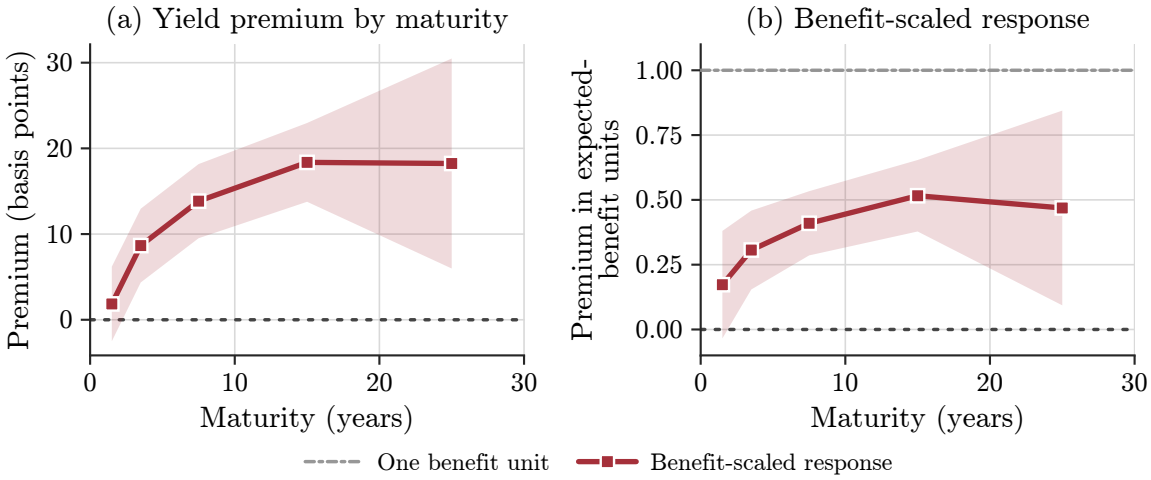


Figure 11: The bank-only benefit and municipal yields. Panel (a): 2SLS yield premium in basis points. Panel (b): the designation response when yield is divided by the expected annual benefit X_{tm}, with the same controls. Funding costs are municipal-holdings-weighted. Expected rates use the Kim–Wright expected-short-rate component, capped at ten years (Appendix D.8). Shading is the issuer-clustered 95% interval. The dashed line marks one benefit unit.

The 2018 cut in the bank tax rate, from 35% to 21%, lowered X by 40% at a given funding cost and lowered the value of the municipal exemption to banks. Relative to 2014–17, the premium fell most at 5–10 years, by 19.2 basis points (s.e. 7.6), while a substantial premium remained at 10–20 years and the first stage did not change (Appendix D.8.6). The participation margin moves when a clientele's tax treatment changes.

The rising bank profile contrasts with household pass-through, which falls from 1.07 to 0.30 in the controlled pre-2018 OLS estimates (Figure 9). Together, the designs show the opposite maturity gradients predicted for household and institutional benefits. The 2017 corporate tax cut provides a second institutional test: it reduced P&C insurers' exemption benefit, and implied taxes subsequently fell more at longer maturities in states where these insurers held more municipal debt. Appendix D.10 presents the design, estimates and event study.

6.3 P3. Rate responses by maturity

P3 tests whether municipal yields respond more to a given treasury yield change at longer maturities. Each municipal bond is compared with a treasury yield at the same maturity. Under the representative-tax benchmark, municipal zero yields are a constant fraction ξ_H of treasury zero yields, so their relative response is constant across maturities. In the model, long municipal bonds reflect their holders' smaller exemption benefits and have greater rate loadings relative to this benchmark. The model therefore predicts a municipal-to-taxable zero-coupon response ratio that rises with maturity (Section 4.6 and Proposition 6(iii)). I test this maturity pattern.[13]

[13]The observed response level, about 0.35 at one year, also reflects treasury safety and liquidity demand (Krishnamurthy and Vissing-Jorgensen, 2012) and transaction-outcome trimming. The empirical test therefore concerns how the relative response varies with maturity.

I compare consecutive trades in the same municipal bond. Let $\Delta y^{\mathrm{M}}_{b,t}$ be its yield change between the two trades, $\Delta y^{\mathrm{Tsy}}_{m_b,t}$ the corresponding change in the maturity-matched treasury yield, and m_b its remaining maturity in years. I estimate

$$\Delta y^{\mathrm{M}}_{b,t} = \beta_0 \, \Delta y^{\mathrm{Tsy}}_{m_b,t} + \beta_1 \big(\Delta y^{\mathrm{Tsy}}_{m_b,t} \times m_b \big) + \beta_2 \, m_b + \varepsilon_{b,t}. \tag{28}$$

The response to a treasury yield change is $\beta_0 + \beta_1 m_b$, so a positive β_1 means that this response is larger for longer bonds. Comparing trades within a bond removes its time-invariant characteristics.

The sample covers January 2005 to August 2020 and includes non-callable, federally tax-exempt municipal bonds rated AA and above. Trades are at most 90 days apart, and the treasury yield must move by at least 10 basis points in either direction to limit transaction noise. Standard errors are clustered by the second trade's date. The treasury comparison follows the bond's remaining maturity at each trade time. Appendix D.11 details the construction.

The estimates give $\hat{\beta}_0 = 0.346$ and $\hat{\beta}_1 = 0.007$ (Appendix Table 42). A 100-basis-point treasury increase is associated with a municipal increase of about 35 basis points at one year and 42 at ten years. Grouping bonds below five years and at five years or more gives the same ordering, with responses of 0.351 and 0.426.

The result survives changes in the minimum treasury move, the time between trades and the credit sample. All thirty combinations of treasury-move cutoffs and trade windows give a positive maturity interaction, from 0.0047 to 0.0104, significant at five percent (Appendix Table 44). Using the monthly corporate curve to replace the daily treasury curve also leave positive adjusted gradients (Appendix D.11.1). On the wider rating sample, the observed gradient rises from 0.007 to 0.013 in the continuous specification and from 0.075 to 0.113 in the binned specification (Appendix Tables 48 and 49).

7 Policy Counterfactuals

The municipal interest exemption is intended to lower borrowing costs for state and local governments. In the history of the U.S., investor-specific tax code changes have repeatedly reshaped municipal demand. The Tax Reform Act of 1986 generally removed banks' funding-cost deduction for new municipal purchases and was followed by a sharp decline in their holdings. The 2017 Tax Cuts and Jobs Act again changed banks' and insurers' relative exemption benefits. Evaluating such reforms requires accounting for changes in the investor base and the allocation of rate risk across markets.

I study two important reforms that extend existing state tax policies and federal experiments. Their broader adoption could substantially affect borrowing costs for state and local governments and taxable issuers, as well as redistributing investors' after-tax interest income. The first removes a state's exemption for residents' interest on its own municipal bonds. Illinois, Iowa and Wisconsin already generally tax this interest. The second gives municipal interest the same tax treatment as taxable interest for every investor. The federal government has also experimented with taxable

municipal finance through the 2009–2010 Build America Bonds program, and subsidized issuers' interest payments on taxable bonds. For each reform, I solve for holdings and yields in both markets and measure the resulting financing costs, tax receipts and distribution of after-tax interest income.

7.1 Design and fiscal accounting

The home-state experiments start from the calibrated 2015–2017 economy. For equal municipal and taxable treatment, I solve the reform separately in each of the six economies covering 2000–2017 and report their calendar-year-weighted average. These economies share the pre-2018 tax regime. In each case, I change the relevant tax rules and solve again for holdings and both yield curves, keeping bond supplies, investor resources, risk preferences and outside exposures at their dated baseline values. The comparison captures how a tax change reallocates the same debt and changes its financing cost.[14]

I measure annual financing costs by multiplying the amount of debt outstanding at each maturity by its yield and summing across maturities. The increase after a reform is the replacement-financing cost, the additional annual cost of financing the same debt at the new yields.[15]

I divide each group's bond and taxable cash interest into tax receipts and after-tax interest income. Household groups use federal and state rates by residence, and institutions use their own tax assignments. Aggregate cash and its pretax interest are unchanged because total investor funding, bond supplies and the short rate are fixed. Reallocation can still change the taxes paid on cash and the income its holders retain. The additional interest paid by bond issuers must therefore go either to governments or to investors. Summing across both markets and all governments gives

$$\Delta(\text{Financing costs}) = \Delta(\text{Tax receipts}) + \Delta(\text{After-tax interest income}). \tag{29}$$

Financing costs can thus exceed added receipts because investors retain part of the additional interest after tax. I report costs per dollar of added receipts. For home-state repeal, the denominator is the reforming state's added revenue. Other governments' receipts can change after home-state repeal, but do not enter that reform's denominator. For equal treatment, it is the combined increase in federal and state revenue. [16]

7.2 The home-state exemption

Most income-tax states exempt residents' interest on their own municipal bonds while taxing interest on other states' bonds. This home-state preference encourages residents to hold local debt.

[14] Tax changes are applied in each investor's baseline payoff units, preserving preferences and liability hedges. Yield changes average each economy's 36 monthly rate states. Dollar amounts and the California rate experiment in Figure 12(a) use $z = 0$. Ratios for 2000–2017 divide average dollar financing costs by average dollar receipts.

[15] The calculation uses continuously compounded yields to construct coupon-equivalent flows. Existing contractual payments and original-issue-discount schedules remain unchanged. The taxable market pools treasury and corporate debt. Other liability flows, issuer deductions, refinancing timing and fiscal feedback are outside this interest-income accounting (Appendix E.2).

[16] These ratios compare financing and interest-income flows. Welfare would also depend on changes in risk bearing, investment opportunities and public spending.

However, Illinois, Iowa and Wisconsin generally tax interest on their own municipal bonds as well, providing existing examples of the policy I study.[17] I first vary California's home-state benefit, then remove the exemption in seven separate state experiments to compare the effects on borrowing costs and tax receipts. These states are taken as example of the counterfactual experiment, and the messages are similar across states.

Panel (a) of Figure 12 shows how California's yield curve responds to a smaller home-state tax benefit. I lower household groups' state tax rates proportionally, keeping federal taxes and household composition fixed. A lower state rate makes the exemption less valuable. I measure its reflection in yields by comparing California with the pooled municipal market of states without an income tax. The implied state tax is the percentage by which California's yield is below this comparison yield.[18].

At the riskless two-year maturity, the implied state tax changes one-for-one with the high-tax household rate. At thirty years, each percentage-point reduction in that rate reduces the implied state tax by 0.377 to 0.525 percentage points across the four reductions considered. The household's state tax benefit is therefore fully reflected at the short end and only partly reflected at the long end, as P1 predicts.

Panel (b) instead removes the exemption while holding state tax rates fixed. I do so separately in California, New York, New Jersey, Massachusetts, Ohio, Pennsylvania and Minnesota, seven states with high supply-weighted tax benefits. Residents' home-state municipal interest becomes subject to the same tax as their out-of-state municipal interest. The change applies to all resident household types in the state. The tax treatment of institutions, nonresidents, cash and taxable bonds stays fixed.

The reforming state's municipal financing cost rises by \$0.83–\$1.02 per dollar of added state receipts. Including the costs borne by other municipal and taxable issuers raises the ratio to \$1.16–\$1.52. Both ratios use the same state receipts, so their difference measures the financing cost transmitted to other markets.

Portfolio substitution transmits the reform across markets. Residents sell home-state bonds and buy other municipal bonds, taxable bonds and cash. Nonresident high-tax households absorb short bonds, and banks absorb intermediate bonds. In California at $z = 0$, residents' share of the state's bonds at two to eighteen years falls from over 90% to about 35%, while banks' intermediate-maturity share rises from 10% to 30%. These trades change investors' portfolio rate exposure and the returns they require on their other bonds. Only residents owe the new tax, but all holders receive the new market yield. California's monthly-average yields rise 12.0, 26.4 and 24.7 basis points at two, ten and thirty years, and the thirty-year taxable yield rises 0.9 basis points.

[17] Each of these three states exempts particular issues. The Supreme Court upheld the home-state preference in *Department of Revenue of Kentucky v. Davis* (2008). Oklahoma exempts qualifying local obligations issued after July 1, 2001, while Utah exempts its own municipal interest. Since 2018, the federal state-and-local-tax deduction cap has also increased the exemption's value for residents whose deductions it limits.

[18] The ratio is $1 - y^M_{\mathrm{CA},T}/y^0_T$, where y^0_T is the pooled no-income-tax-state municipal yield. Both yields are solved at rate state $z = 0$

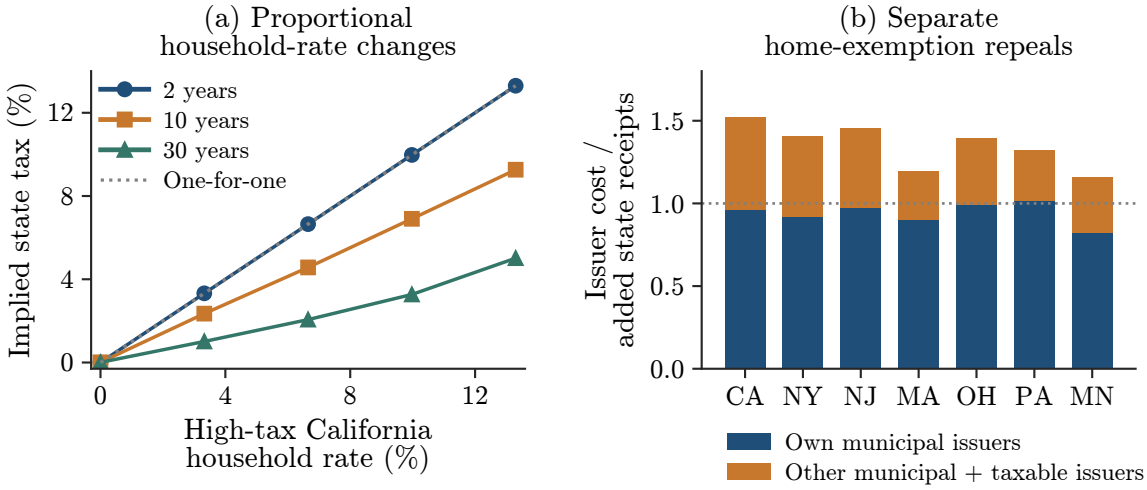


Figure 12: State taxes, municipal yields and financing costs in the 2015–2017 economy at $z = 0$. Panel (a) scales both California household groups' state rates, keeping the home-state exemption and federal taxes fixed. It plots $1 - y^M_{\text{CA},T}/y^0_T$ at two, ten and thirty years against the high-tax group's rate. The comparison yield y^0_T pools the seven states without an income tax. Markers are solved equilibria, and the dotted line is one-for-one. Panel (b) removes each listed state's home-state exemption at fixed tax rates. It reports replacement-financing costs for own-state municipal issuers and other municipal and pooled taxable issuers per dollar of added state receipts from residents' bond and taxable cash interest.

State receipts depend on residents' whole portfolios, including taxable cash. In California at $z = 0$, municipal financing costs rise by \$1.38 billion and state receipts by \$1.43 billion. Of the added receipts, \$0.51 billion comes from newly taxed home-state bonds, \$0.68 billion from other municipal and taxable bonds, and \$0.24 billion from cash interest. The own-state cost is \$0.96 per dollar of receipts, but the cost across all issuers is \$1.52.

Reallocation changes both the source and amount of taxable interest. Across states, residents' total riskless share rises by 0.03–0.56 percentage points of financial assets (Appendix Table 52). Moving from bonds into cash reduces bond-interest receipts but adds taxable cash interest. Together with higher yields, these changes raise California's added state receipts to \$1.43 billion, above the \$1.25 billion obtained at unchanged prices and holdings (Appendix E.2).

7.3 Equal municipal and taxable treatment

The second reform gives each investor's municipal interest the tax treatment of its taxable interest. It removes federal and state exemption benefits and banks' financing-tax cost across all investor groups.[19] I implement it by moving each municipal after-tax share 99.9% toward its taxable share and removing 99.9% of the bank financing-tax cost.[20] Cash and taxable-interest treatment remain fixed. Bond-specific risk and differences in supply continue to distinguish the two markets.

[19] The reform is broader than the investor-specific changes in the 2017 Tax Cuts and Jobs Act (Appendix D.10).

[20] Exact equality makes the riskless two-year municipal bond and cash perfect substitutes, leaving their holdings split indeterminate. The 99.9% endpoint avoids this indeterminacy.

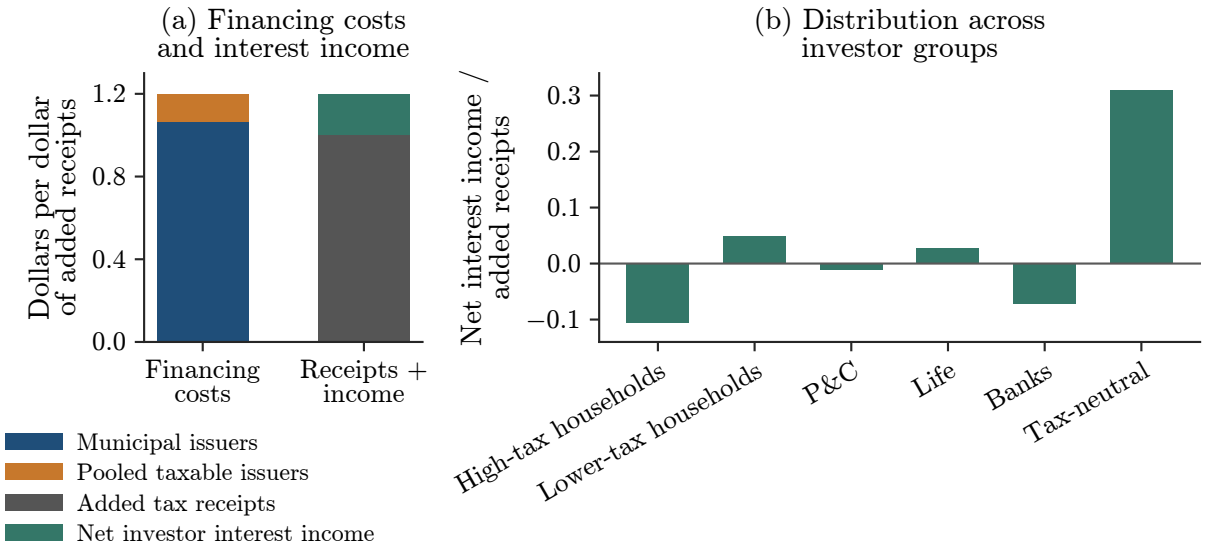


Figure 13: Average annual flows after removing 99.9% of each tax distinction in the six 2000–2017 economies. Panel (a) reports identity (29), dividing average financing costs and net investor interest income by average added federal and state receipts. Panel (b) allocates the income change across investor groups. Receipts and income include taxable cash interest and the bank financing-tax cost. Each economy uses continuous yields and equilibrium holdings at $z = 0$. Taxable issuers pool treasury and corporate quantities.

Table 6: Average fiscal effects of equal municipal and taxable treatment, 2000–2017

Average annual change	$bn/year	Per receipt dollar
Municipal issuer financing cost	41.16	1.063
Pooled taxable issuer financing cost	5.15	0.133
Total issuer financing cost	46.30	1.196
Federal receipts, including bank financing-tax cost	34.29	0.886
Residence-state receipts	4.42	0.114
Total receipts	38.71	1.000
Net investor interest income	7.59	0.196

Notes. Calendar-year-weighted changes across six three-year economies, each relative to its own baseline, with 99.9% of the tax distinction removed. Dollar amounts use dated-period quantities without inflation adjustment and continuous yields at $z = 0$. Ratios divide average dollar changes. Receipts and income include taxable cash interest. Federal receipts include the average $0.55 billion decline in the bank financing-tax cost. Before rounding, total issuer costs equal receipts plus net investor interest income. Totals may differ because of rounding.

Average annual financing costs rise by $46.30 billion and federal and state receipts by $38.71 billion. Total financing costs therefore rise by $1.20 per dollar of added receipts, including $0.13 for taxable issuers (Table 6 and Figure 13(a)). The ratio exceeds one in every period (Appendix Table 53). Taxable issuers bear 11.1% of the financing-cost increase even though their bonds' tax treatment has not changed. Of the added receipts, 88.6% accrue federally and 11.4% accrue to investors' states of residence.

Joint portfolio choice explains the spillover to taxable issuers. Averaging each economy's market shares at $z = 0$, households, banks and P&C insurers sell municipal bonds to life insurers and the tax-neutral group. The tax-neutral group's municipal share rises from zero to 37%, while high-tax

households' share falls from 46% to 17% and banks' from 12% to 7%. Taking on municipal rate exposure raises the tax-neutral group's required return on taxable bonds as well. Its taxable share falls from 69% to 60%, with households and banks absorbing the sales. Appendix Table 52 reports the corresponding changes in each group's own allocation.

Thirty-year municipal and taxable yields rise by 170.6 and 23.1 basis points, averaging across the six economies' monthly rate states. After the reform, municipal yields are below taxable yields by only 0.21 basis points at two years and 2.0 at thirty.

The $0.20 of financing costs above each dollar of added receipts becomes additional after-tax investor interest income, by (29). This aggregate gain is unevenly distributed. High-tax households, banks and P&C insurers lose on average, while lower-tax households and life insurers gain (Figure 13(b)). The tax-neutral group gains most, benefiting from higher yields and larger municipal holdings without losing a tax exemption.

Under equal treatment, applying new tax rates to old prices and holdings gives average added receipts of $26.5 billion and leaves issuer costs unchanged. Portfolio and yield changes generate a further $12.2 billion of receipts, while financing costs rise by $46.3 billion. In both reforms, fiscal outcomes therefore depend on who holds the debt after reform and the yields those investors require.

8 Conclusion

I develop an equilibrium term-structure model in which heterogeneous investor taxes create endogenous segmentation across bond markets and maturities. Investors choose portfolios jointly, balancing tax benefits against the cost of bearing interest-rate risk. Under limits on short selling, these choices determine which investors hold each bond and how their tax treatment is reflected in yields. Portfolio substitution links yields across maturities and markets, so an investor's tax treatment can affect even bonds it never holds.

The municipal application shows why the tax treatment of a single representative investor is insufficient to understand an entire yield curve. Top-bracket households value exemption most, yet investors with smaller exemption benefits hold substantial municipal portfolios. The model connects this heterogeneous ownership to relative yields: household taxes anchor the riskless short end, while longer yields also reflect lower-tax investors' valuations and the allocation of rate risk. The quantitative and empirical evidence supports this explanation of the longstanding municipal bond puzzle.

The tax heterogeneity model also has implications for taxable-bond pricing. Investors retain different shares of taxable interest, so heterogeneous taxes shape required returns in the taxable market as well. Joint portfolio choice transmits tax-policy changes across taxable and tax-exempt markets, as the counterfactuals demonstrate. Evaluating a tax reform therefore requires accounting for financing costs in both markets and the redistribution of investor income.

I apply the framework to municipal bonds because their tax rules provide clean variation for testing the mechanism and their yields present a longstanding puzzle. More broadly, the model applies wherever investors differ in their tax treatment across assets and choose portfolios jointly under arbitrage frictions. Investor tax heterogeneity is a first-order determinant of bond yield curves and should be considered in term structure models.

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A Analytical framework and results

This appendix is the formal companion to Section 4. Appendices A.1 and A.2 state the general economy, which the quantitative model solves, and show how equilibrium one-period returns determine prices and yields. Appendix A.3 specializes this economy to the two-investor affine benchmark of Section 4.2. The formal results then follow the main text: participation (Appendix A.4, for Section 4.3), the three channels of the one-period return gap and the rate-loading decomposition (Appendix A.5, for Sections 4.4 and 4.5), the yield-implied tax (Appendix A.6), other participation motives (Appendix A.7, for Section 4.7) and the empirical predictions (Appendix A.8, for Section 4.6). Appendix B gives the proofs and extensions.

A.1 The general economy

The quantitative model of Section 5 solves this economy with several investor groups, state municipal markets, binding funding constraints and the rate floor.

Assets and investors

Time is discrete and indexed by t. One aggregate rate state z_t follows a stationary first-order autoregression, AR(1), and the short rate, the pretax log gross return on cash, is bounded below at zero:

$$z_{t+1} \;=\; \rho z_t \;+\; \sigma_r \varepsilon_{t+1}, \qquad r_t = r(z_t) := \max\{\bar{r} + z_t, 0\}, \qquad \rho \in (0,1), \quad \varepsilon_{t+1} \sim \mathcal{N}(0,1). \tag{30}$$

Here ρ measures the persistence of the rate state and σ_r the size of its shocks. The mean rate $\bar{r}$ is the cash rate at the stationary mean $z = 0$. The floor affects only the cash payoff, and the state z_{t+1} remains Gaussian. Appendix B.2 explains how the floor makes prices nonlinear.

Let $\mathcal{A}$ be a finite set of bond markets, indexed by a and distinguished by asset class or issuer. In the two-investor benchmark, $\mathcal{A}$ contains a taxable market C and a municipal market M; the quantitative model has one municipal market for each issuer state. Each market contains zero-coupon bonds indexed by their remaining maturity $T \in \{1, \ldots, T_{\max}\}$, the number of periods until the bond pays one unit. The one-period claim of the taxable market C is the cash account, with return $r(z)$, so separately held taxable bonds begin at $T = 2$. A bond with T periods remaining has price $P^a_T(z_t)$ and holding-period return

$$1 + R^a_{T,t+1} \;=\; \frac{P^a_{T-1}(z_{t+1})}{P^a_T(z_t)} \exp\!\big(\sigma^a_{\eta,T}\, \eta^a_{T,t+1}\big). \tag{31}$$

The standard normal shocks $\eta^a_{T,t+1}$ are independent of each other and of ε_{t+1}, and they add bond-specific log-return variance $v^a_T := (\sigma^a_{\eta,T})^2$, the notation of Section 4.1. Every market has $\sigma^a_{\eta,1} = 0$, so a one-period claim is fully riskless. Bond-specific risk represents residual variation in holding returns, such as an issue-specific change in trading demand. It makes longer bonds imperfect substitutes without adding a persistent price state. Appendix C.8 explains why this shock is specified in holding returns and shows that normalizing its mean moves the calibrated thirty-year gap by 0.20 basis points.

Investors belong to finitely many types, indexed by i. Type i has mass $m_i > 0$, risk aversion $\gamma_i > 0$, funding W_i and outside exposure ℓ_i to the rate shock from assets and liabilities beyond its bonds. It keeps the absolute after-tax share $\xi^a_i = 1 - \tau^a_i > 0$ of returns in market a, where τ^a_i is its tax rate on those returns. Because after-tax shares differ across both investors and markets, the same structure covers municipal and taxable bonds, state taxes on treasury and corporate interest, tax-favored accounts and cross-border withholding (Appendix B.13).

Portfolio choice and equilibrium

Investor i holds bonds $x^a_{i,T,t}$ and cash $x^0_{i,t} = W_i - \sum_{a,T} x^a_{i,T,t}$, in dollars per unit of mass. Aggregate holdings are $m_i x^a_{i,T,t}$, in the same units as supply. Portfolio choice uses log returns plus a Jensen adjustment, $\mathcal{R}^a_{T,t+1} = \mu^a_T(z_t) + u^a_{T,t+1}$. Here $\mu^a_T(z) := \log \mathbb{E}_t[1 + R^a_{T,t+1}]$ is the log of the conditional expected gross holding return. It is the one-period return used in the portfolio problem and, in equilibrium, equals the return its holders require under this convention. The term $u^a_{T,t+1} := \log(1 + R^a_{T,t+1}) - \mathbb{E}_t[\log(1 + R^a_{T,t+1})]$ is the centered

log-return innovation. Substituting $R \approx \mathcal{R}$ for bonds and $e^r - 1 \approx r$ for cash gives approximated next-period wealth $\widehat{W}_i'$ and the mean–variance problem:[21]

$$\begin{aligned} \max_{x_i \geq 0,\, x^0_{i,t} \geq 0} \quad & \mathbb{E}_t[\widehat{W}_i'] - \frac{\gamma_i}{2} \operatorname{Var}_t(\widehat{W}_i'), \\ \text{subject to} \quad & \widehat{W}_i' = W_i + \xi_i^C r(z_t) x^0_{i,t} \\ & \qquad + \sum_{a,T} \xi_i^a x^a_{i,T,t} \mathcal{R}^a_{T,t+1} + \ell_i \sigma_r \varepsilon_{t+1}. \end{aligned} \tag{32}$$

Positions and cash are nonnegative, which rules out short sales and borrowing (Section 3.5). Funding and masses are fixed across states. A positive ℓ_i raises net wealth when rates rise and so offsets the rate exposure of the investor's bonds.

Investor i's after-tax excess return on bond (a, T) over cash is $\widetilde{\mu}^a_{i,T} = \xi_i^a \mu_T^a - \xi_i^C r$. Let $\widetilde{\Pi}_i(t+1)$ be the after-tax innovation of its whole portfolio, including the outside exposure. Appendix B.1 gives the exact wealth equation, derives the reduced problem and compares the return approximation with simple returns. Let $\beta_i(z) \geq 0$ be the shadow value of funding, the multiplier on the funding constraint $\sum_{a,T} x^a_{i,T} \leq W_i$. It is zero whenever the investor holds cash. The portfolio, or Karush–Kuhn–Tucker (KKT), conditions are

$$\begin{aligned} x^a_{i,T}(z) > 0 \quad &\Longrightarrow \quad \widetilde{\mu}^a_{i,T}(z) = \underbrace{\gamma_i \xi_i^a \operatorname{Cov}_z\Big(u^a_{T,t+1}, \widetilde{\Pi}_i(t+1)\Big)}_{\text{marginal portfolio-risk cost}} + \underbrace{\beta_i(z)}_{\text{shadow value of funding}}, \\ x^a_{i,T}(z) = 0 \quad &\Longrightarrow \quad \widetilde{\mu}^a_{i,T}(z) \leq \gamma_i \xi_i^a \operatorname{Cov}_z\Big(u^a_{T,t+1}, \widetilde{\Pi}_i(t+1)\Big) + \beta_i(z). \end{aligned} \tag{33}$$

The shadow value of funding also satisfies the complementary-slackness condition $\beta_i(z)[W_i - \sum_{a,T} x^a_{i,T}(z)] = 0$.

Every bond other than cash clears at every rate state:

$$x^a_{i,T}(z) \geq 0, \qquad \sum_i m_i \, x^a_{i,T}(z) = Q^a_T, \tag{34}$$

where Q^a_T is the fixed dollar supply. An equilibrium consists of prices and holdings satisfying (31), (33) and (34). Participation, which investors hold each bond, is determined by these conditions.

A.2 From holding returns to yields

Taking expectations of the gross holding return (31), including bond-specific risk with variance $v^a_T = (\sigma^a_{\eta,T})^2$, gives

$$P^a_T(z) = e^{-\mu^a_T(z) + v^a_T/2} \, \mathbb{E}_z\big[P^a_{T-1}(z')\big], \qquad P^a_0 = 1. \tag{35}$$

A higher required return lowers today's price for a given expected resale value. Iterating from the maturity payment links yields to the one-period returns along the bond's life. The factor $e^{v^a_T/2}$ accounts for the mean of the bond-specific shock (Appendix B.2), whose yield effect is small (Appendix C.8).

Write $p^a_T = \log P^a_T$ for the log price and $y^a_T = -p^a_T/T$ for the yield. In the affine benchmark of Appendix A.3,

$$\log P^a_T(z) = -A^a_T z - C^a_T, \qquad y^a_T(z) = \frac{A^a_T z + C^a_T}{T}. \tag{36}$$

The coefficient A^a_T is the bond's rate loading, the model's counterpart of duration: the proportional fall in its price per unit rise in the rate state. Thus A^a_T/T is the yield's response to the rate state and C^a_T/T is its level at the mean state. The response of the one-period return to the rate state is

$$s^a_T(z) := \frac{\partial \mu^a_T(z)}{\partial z}. \tag{37}$$

[21] Appendices D.1.4 and E.5 give the capital-income tax correction.

Substituting affine prices into the resale equation and matching its state and constant terms, with the bond-specific variance $(\sigma^a_{\eta,T})^2 = v^a_T$ retained, gives

$$A^a_T = \underbrace{\rho A^a_{T-1}}_{\text{inherited loading}} + \underbrace{s^a_T}_{\text{current response}}, \qquad C^a_T = C^a_{T-1} + \mu^a_T(0) - \tfrac{1}{2}\left[\sigma_r^2\left(A^a_{T-1}\right)^2 + \left(\sigma^a_{\eta,T}\right)^2\right]. \tag{38}$$

The first relation adds today's return response to the loading inherited through the expected resale price. The second accumulates the mean-state one-period returns less half-variance terms. Starting from $A^a_0 = C^a_0 = 0$, the loading $A^a_T = \sum_{j=1}^{T} \rho^{T-j} s^a_j$ accumulates the responses, weighted by rate persistence. With a riskless one-period claim, $\sigma^a_{\eta,1} = 0$, the same recursion gives the yield as the average of the one-period returns along the bond's life less convexity:

$$y^a_T(z_t) = \underbrace{\frac{1}{T}\sum_{j=0}^{T-1} \mathbb{E}_t\left[\mu^a_{T-j}(z_{t+j})\right]}_{\text{one-period returns as the bond ages}} \underbrace{- \frac{1}{2T}\sum_{k=2}^{T}\left[\sigma_r^2\left(A^a_{k-1}\right)^2 + \left(\sigma^a_{\eta,k}\right)^2\right]}_{\text{convexity}}. \tag{39}$$

In the two-investor benchmark of Appendix A.3, risky municipal bonds have positive bond-specific variance, while taxable bonds may have none, as in the illustration.

A.3 The two-investor affine benchmark across maturities

The two-investor benchmark yields the closed forms behind Sections 4.3–4.6.

The high-tax household H and the lower-tax investor I choose among municipal maturities $T = 1, \ldots, T_{\max}$, taxable maturities $T = 2, \ldots, T_{\max}$ and cash, the one-period taxable claim, with fixed supplies Q^a_T. The benchmark retains bond-specific risk, allows different risk aversions γ_i, and keeps the relative after-tax share $\xi_i = \xi^C_i/\xi^M_i$ of Section 4.2, with $\xi_H < \xi_I$. Payoffs are in units of after-tax municipal income, so taxable returns and cash income are multiplied by ξ_i, and γ_i and ℓ_i are normalized correspondingly. I fix a participation branch, which specifies which investors hold each bond, solve its portfolio conditions and market clearing, and check that no investor benefits from changing its holdings. Closed forms use three restrictions:

(i) funding constraints are nonbinding;

(ii) the floored cash rate is replaced by the linear rule $r(z) = \bar{r} + z$; and

(iii) prices are computed as if the same participation branch applied at every state, which makes log prices affine, and the branch is verified at the states studied.

On a fixed branch, holdings, one-period returns and log prices are affine in z, and the price coefficients solve a fixed point. Appendix B.4 states the feasibility and uniqueness conditions. The quantitative model lets participation and funding change across states and keeps the rate floor.

Log prices then have constant coefficients,

$$p^{a,\text{aff}}_T(z) = -A^{a,\text{aff}}_T z - C^{a,\text{aff}}_T, \tag{40}$$

and one-period returns are affine as well, $\mu^{a,\text{aff}}_T(z) = \mu^{a,\text{aff}}_T(0) + s^{a,\text{aff}}_T z$, so $s^{a,\text{aff}}_T$ and $A^{a,\text{aff}}_T$ are the state coefficients of the one-period return and of the negative log price. The affine recursions (38) and the yield average (39) of Appendix A.2 hold exactly. Below and in the proofs, loadings, level terms and responses refer to this affine branch, which the superscript "aff", where shown, marks explicitly.

Investor i's net rate exposure, the after-tax exposure of its whole balance sheet to the rate shock, is

$$\widehat{D}_i(z) := \underbrace{\sum_j x^M_{i,j}(z) A^M_{j-1} + \xi_i \sum_j x^C_{i,j}(z) A^C_{j-1}}_{\text{tax-weighted asset exposure}} - \underbrace{\ell_i}_{\text{outside liability exposure}}. \tag{41}$$

Municipal positions count at their loadings and taxable positions at ξ_i times theirs in these payoff units. With today's positions fixed, $\widehat{D}_i = -\partial \widehat{W}_i'/\partial z_{t+1}$, as in (10). It is dollar exposure before division by wealth. The wealth innovation is $-\widehat{D}_i \sigma_r \varepsilon_{t+1}$, so a larger liability hedge ℓ_i lowers this exposure. A bond's rate-risk compensation equals its loading times the investor's unit risk cost, $\gamma_i \sigma_r^2 \widehat{D}_i$. Loadings are constant on an affine branch, so changes in $\widehat{D}_i$ across rate states reflect trading alone.

Bond-specific risk allows both investors to hold several of the same municipal maturities. Without it, their municipal equalities hold at the same maturity only where the difference in their rate-risk compensation on that bond equals the difference in exemption benefits.

At one period, zero risk and nonbinding funding give $\mu_1^M = \xi_H r$ whenever $r > 0$ (Appendix Proposition 8). The household, which keeps the smallest share of taxable interest, then holds the riskless municipal claim. At longer maturities, participation determines whose taxes and risk costs shape equilibrium returns. Appendix B.4 gives the branch construction. Appendix Proposition 7 gives the uniqueness conditions for current-state clearing, and Appendix B.6 those for the loading solution.

The thirty-maturity illustration of Figures 4, 5 and 6 is a branch with joint municipal participation from maturity 17. Appendix B.9 gives its parameters, verifies its KKT conditions at the mean state and establishes a unique loading solution satisfying condition (S), which (56) in Appendix A.6 states. In the household-only benchmark, the household holds every municipal bond and the lower-tax investor every taxable bond, giving $A_T^M/A_T^C = \xi_H$ at every maturity (Appendix Proposition 11). Level panels omit the vertical scale because the illustration is not a calibration.

A.4 The participation margin

The comparison below first isolates the duration benefit at the household-only allocation, marked by the superscript hh, at which the high-tax household holds all municipal supply and the lower-tax investor holds all risky taxable supply. The lower-tax investor's net rate exposure there already reflects its taxable book, which adds exposure, and its outside exposure, which offsets it when positive.

Proposition 1 sets bond-specific risk to zero, $v_T^M = 0$, to show the duration comparison alone, whereas the equilibrium criterion of Section 4.3 retains $v_T^M > 0$. In the zero-variance case, the high-tax household's active KKT equality and the lower-tax investor's condition for a nonnegative first-order purchase incentive are

$$\underbrace{\mu_T^M - \xi_H r}_{\text{offered to the household}} = \underbrace{\gamma_H \sigma_r^2 A_{T-1}^M \widehat{D}_H^{\text{hh}}}_{\text{household marginal risk cost}}, \tag{42}$$

$$\underbrace{\mu_T^M - \xi_I r}_{\text{offered to the lower-tax investor}} \geq \underbrace{\gamma_I \sigma_r^2 A_{T-1}^M \widehat{D}_I^{\text{hh}}}_{\text{lower-tax investor's marginal risk cost}}. \tag{43}$$

If the second inequality fails, the lower-tax investor holds none of the bond. Substituting the household's equality cancels the common return μ_T^M and leaves one condition:

$$\underbrace{\sigma_r^2 A_{T-1}^M \Lambda}_{\text{duration benefit (grows with maturity)}} \geq \underbrace{(\xi_I - \xi_H) r}_{\text{difference in exemption benefits (flat in maturity)}}, \quad \Lambda := \gamma_H \widehat{D}_H^{\text{hh}} - \gamma_I \widehat{D}_I^{\text{hh}}. \tag{44}$$

The duration benefit on the left, the lower-tax investor's saving in rate-risk cost on the bond, equals the bond's loading times $\sigma_r^2 \Lambda$, the difference between the two investors' unit costs of rate risk. Appendix Assumption 1 requires $\Lambda > 0$: risk-weighted net rate exposure is larger for the household at this allocation. At the household-only allocation, clearing fixes every holding, so $s_T^M = \xi_H$ and (38) gives $A_{T-1}^M = \xi_H \sum_{j=0}^{T-2} \rho^j$. Starting from $A_0^M = 0$, the loading rises strictly with maturity, so the duration benefit grows from zero and crosses the difference in exemption benefits at most once.

Proposition 1 (Participation from the long end)**.** *In the two-investor affine benchmark without bond-specific risk, suppose $\Lambda > 0$ at the household-only allocation and $r_t > 0$. At that allocation, the maturities with a*

nonnegative first-order purchase incentive for the lower-tax investor form the upper interval $\{T : T \geq T^*(z_t)\}$, *where*

$$T^*(z_t) \;=\; \min\Big\{T \geq 1 :\; A^M_{T-1} \geq \frac{(\xi_I - \xi_H)\, r_t}{\sigma_r^2\, \Lambda}\Big\}, \tag{45}$$

with $\min \varnothing = +\infty$. *A strict inequality makes participation strictly profitable, and equality need not imply a positive holding.*

Proof. See Appendix B (proof of Proposition 1). □

The threshold T^* is set by taxes, balance sheets and loadings, and no maturity habitat is assigned. At fixed benchmark loadings, a larger difference in exemption benefits moves the threshold out and a larger Λ moves it in. With positive bond-specific risk, the household's required return also includes compensation for the bond-specific risk of its whole holding, while a new buyer's bond-specific risk cost is zero at a zero position. This additional participation incentive appears explicitly in the equilibrium index (77) of Appendix Theorem 1.

In equilibrium, prices and both portfolios are re-solved. Appendix Theorem 1 gives conditions under which the lower-tax investor's municipal holdings form an upper maturity interval, using the equilibrium difference $\widehat{\Lambda} := \gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I$. Section 4.3 states this equilibrium criterion in (13). From here on, T^* denotes the start of that interval.

On the branches studied below, including the thirty-maturity illustration, the household holds municipal bonds alone below T^* and both investors hold them from T^* on. The calibrated economy has several longer-maturity clienteles (Section 5).

A.5 The pricing channels

This subsection gives the formal results behind Sections 4.4 and 4.5. It first derives the three channels of the one-period return gap, the after-tax cash return, the duration cost and the rate loading, together with a small bond-specific term. It then derives the tax-mix contribution and the municipal and taxable portfolio responses within the rate-loading channel.

The three channels of the one-period return gap

Fix maturity T, and let the active set $\mathcal{U}^M_T \subseteq \{H, I\}$ denote the investors with positive holdings of that municipal bond. Each active investor satisfies its municipal condition in (11) with equality:

$$\underbrace{\mu^M_T - \xi_i r}_{\text{after-tax excess return offered to } i} = \gamma_i \Bigg[\underbrace{\sigma_r^2 A^M_{T-1} \widehat{D}_i}_{\text{whole-portfolio rate risk}} + \underbrace{(\sigma^M_{\eta,T})^2 x^M_{i,T}}_{\text{bond-specific risk}} \Bigg]. \tag{46}$$

On the branches studied here the lower-tax investor alone holds the risky taxable bonds. Dividing its taxable condition by its after-tax share gives, for $T \geq 2$,

$$\mu^C_T - r = \gamma_I \big[\sigma_r^2 A^C_{T-1} \widehat{D}_I + \xi_I (\sigma^C_{\eta,T})^2 x^C_{I,T}\big], \tag{47}$$

and $\mu^C_1 = r$ for cash. Subtracting ξ_H times the taxable equality from the equality of a municipal holder separates the one-period return gap $g_T = \mu^M_T - \xi_H \mu^C_T$ into the three channels plus bond-specific compensation. Proposition 2, the formal counterpart of (15) in Section 4.4, states this decomposition.

Proposition 2 (Three channels of the one-period return gap)**.** *In the two-investor affine benchmark, consider a state at which the lower-tax investor alone holds the risky taxable bonds, and let i be an investor holding the maturity-T municipal bond, the lower-tax investor whenever it does. Then*

$$g_T = \underbrace{(\xi_i - \xi_H)r}_{\text{after-tax cash return}} + \underbrace{\sigma_r^2 A_{T-1}^M\big(\gamma_i \widehat{D}_i - \gamma_I \widehat{D}_I\big)}_{\text{duration cost}} + \underbrace{\gamma_I \sigma_r^2 \widehat{D}_I\big(A_{T-1}^M - \xi_H A_{T-1}^C\big)}_{\text{rate loading}} + \underbrace{\gamma_i(\sigma_{\eta,T}^M)^2 x_{i,T}^M - \xi_H \gamma_I \xi_I (\sigma_{\eta,T}^C)^2 x_{I,T}^C}_{\text{bond-specific risk}}. \tag{48}$$

At $T = 1$ the duration-cost, rate-loading and bond-specific contributions are zero, and so is the cash-return contribution when $r > 0$. Where the household holds the bond alone, the cash-return contribution is zero and the duration-cost contribution is at most $(\xi_I - \xi_H)r$. Where the lower-tax investor holds it, the cash-return contribution is $(\xi_I - \xi_H)r$ and the duration-cost contribution is zero.

Proof. See Appendix B.6. □

Below the participation threshold, where $i = H$, the cash-return contribution is zero and the duration-cost contribution is $\sigma_r^2 A_{T-1}^M \widehat{\Lambda}$. From the threshold on, where $i = I$, the cash-return contribution is $(\xi_I - \xi_H)r$ and the duration-cost contribution is zero. At jointly held maturities either investor's equality gives the same total gap. The tax mix and portfolio responses derived below determine the loadings in the third channel.

The rate-loading channel links maturities. Iterating the loading recursion (38) from $A_0^a = 0$,

$$A_{T-1}^M - \xi_H A_{T-1}^C = \sum_{j=1}^{T-1} \rho^{T-1-j}\big(s_j^M - \xi_H s_j^C\big),$$

so the rate loading accumulates the response gaps $s_j^M - \xi_H s_j^C$ of the shorter maturities the bond passes through. It is zero at two periods, since the one-period responses are $s_1^M = \xi_H$ and $s_1^C = 1$ when $r > 0$.

Figure 5 shows the decomposition for the thirty-maturity illustration. The bond-specific term is below 5 percent of the gap. At thirty periods the rate loading accounts for just over half of the gap, two thirds of it inherited from the tax mix at jointly held maturities.

Tax mix and portfolio responses

The rate-loading channel depends on how one-period returns respond to the rate state. A bond's one-period return responds for two reasons. Each holder's after-tax return on cash rises by its after-tax share per unit of the rate state. Trading also changes how much net rate exposure each holder carries, and with it the compensation each requires. Aggregating the holders' equalities separates the tax-mix and portfolio responses. Multiply investor i's equality (46) by m_i/γ_i, sum over the active holders and differentiate with respect to z. The bond-specific terms cancel because fixed supply implies $\sum_i m_i\, \partial x_{i,T}^M(z)/\partial z = 0$. What remains is

$$\left(\sum_{i \in \mathcal{U}_T^M} \frac{m_i}{\gamma_i}\right) s_T^M - \sum_{i \in \mathcal{U}_T^M} \frac{m_i \xi_i}{\gamma_i} = \sigma_r^2 A_{T-1}^M \sum_{i \in \mathcal{U}_T^M} m_i \frac{\partial \widehat{D}_i(z)}{\partial z}. \tag{49}$$

Define the active holders' risk-tolerance shares and aggregate exposure response as

$$\omega_{i,T}^M := \frac{m_i/\gamma_i}{\sum_{j \in \mathcal{U}_T^M} m_j/\gamma_j}, \qquad E_T^M := \frac{\sum_{i \in \mathcal{U}_T^M} m_i\, \partial \widehat{D}_i(z)/\partial z}{\sum_{i \in \mathcal{U}_T^M} m_i/\gamma_i}.$$

The weight $\omega_{i,T}^M$ is type i's risk tolerance, m_i/γ_i, as a share of the active holders' total. The term E_T^M is the response of their aggregate net rate exposure to the rate state, divided by that total risk tolerance, and E_T^C is defined in the same way from the holders of the taxable bond. Dividing (49) by total risk tolerance gives Proposition 3, the formal counterpart of the response formulas (16) in Section 4.5.

Proposition 3 (Holding-return response to the rate state)**.** *Consider a regular state of the two-investor affine benchmark: price moments are differentiable, active risky holdings and inactive KKT slacks are strictly positive, and the stacked active KKT-and-clearing Jacobian is nonsingular, so the active set* $\mathcal{U}_T^M \subseteq \{H, I\}$ *is locally fixed. Then*

$$s_T^M = \underbrace{\sum_{i \in \mathcal{U}_T^M} \omega_{i,T}^M \xi_i}_{\text{holders' tax mix}} + \underbrace{\sigma_r^2 A_{T-1}^M E_T^M}_{\text{portfolio-risk response}} \tag{50}$$

exactly. The tax-mix term depends on who holds maturity T*. The portfolio-risk response depends on how equilibrium trading changes those holders' exposure across their whole portfolios. This decomposes one market's return response. Comparing the two markets below separates the portfolio response into its municipal and taxable contributions to the rate-loading channel.*

Proof. See Appendix B.6. □

The tax term is ξ_H under household-only holdings and, under joint holdings, the tax mix

$$\xi^* := \omega_H \xi_H + \omega_I \xi_I, \qquad \omega_i := \frac{m_i/\gamma_i}{m_H/\gamma_H + m_I/\gamma_I},$$

where ω_i is investor i's share of total risk tolerance. Thus the lower-tax investor's after-tax share enters this term only at maturities it holds.

The sign of the portfolio-risk response follows from reallocation on a joint-participation branch, on which the lower-tax investor alone holds risky taxable bonds, both investors hold the municipal maturities in a nonempty set $J \subseteq \{2, \ldots, T_{\max}\}$, and the household alone holds every other municipal maturity. Appendix Proposition 10 shows that at a regular state with $r(z) > 0$, positive loadings and positive bond-specific variance on J,

$$d := \frac{\partial \widehat{D}_H(z)}{\partial z} > 0, \qquad \frac{\partial \widehat{D}_I(z)}{\partial z} = -\frac{m_H}{m_I}\, d < 0.$$

A higher rate state raises the difference in exemption benefits, $(\xi_I - \xi_H)r$, and trading in jointly held bonds shifts net rate exposure to the household. Its unit cost of rate risk rises, raising its required compensation on all its risky municipal holdings, including maturities the lower-tax investor never holds. The sign holds for exposure-weighted portfolios, and individual jointly held positions can move either way. At a household-only risky maturity,

$$s_T^M - \xi_H = \gamma_H \sigma_r^2 A_{T-1}^M \frac{\partial \widehat{D}_H(z)}{\partial z} > 0. \tag{51}$$

The taxable leg moves the other way. Because the lower-tax investor's net rate exposure falls as the rate state rises, the response of its taxable return is

$$s_T^C = 1 - b A_{T-1}^C, \qquad b := \gamma_I \sigma_r^2 \frac{m_H}{m_I}\, d > 0,$$

which lies below one at every risky maturity. The coefficient b measures the feedback of portfolio reallocation into taxable returns.

If joint municipal participation begins at T^* and continues to the long end, the municipal responses are

$$s_T^M = \begin{cases} \xi_H + \gamma_H \sigma_r^2 A_{T-1}^M d, & T < T^*, \\ \xi^* = \omega_H \xi_H + \omega_I \xi_I, & T \geq T^*. \end{cases} \tag{52}$$

Below the threshold, the lower-tax investor's after-tax share affects the municipal return response through the household's portfolio adjustment. From the threshold on, both investors hold the bond. Clearing cancels their aggregate exposure response, so the tax mix ξ^* remains. These are the responses (16) of Section 4.5.

Figure 6(a) plots these municipal and taxable responses.

Combining the two legs gives the response gap $s_j^M - \xi_H s_j^C = \partial_z g_j$, whose persistence-weighted sum over $j < T$ is the loading difference $A_{T-1}^M - \xi_H A_{T-1}^C$ in the rate-loading channel of (48). Applying the

decomposition of Proposition 3 in both markets, with the lower-tax investor the sole holder of risky taxable bonds, gives

$$s_T^M - \xi_H s_T^C = \underbrace{\left(\bar{\xi}_T^M - \xi_H\right)}_{\text{difference in tax mix}} + \underbrace{\sigma_r^2\left(A_{T-1}^M E_T^M - \xi_H A_{T-1}^C E_T^C\right)}_{\text{difference in portfolio-risk responses}}, \tag{53}$$

where $\bar{\xi}_T^M := \sum_i \omega_{i,T}^M \xi_i$, equal to ξ_H or ξ^*, is the tax mix of the municipal bond's active holders. The second term combines the municipal and taxable portfolio responses shown separately in Figure 6. All three contributions are zero at one period, where the bond has no rate exposure and the high-tax household alone holds it. Below the participation threshold, portfolio adjustment raises the household's municipal response and lowers the lower-tax investor's taxable response, so the two legs reinforce each other. From the threshold on, the tax-mix term adds $\xi^* - \xi_H = \omega_I(\xi_I - \xi_H)$. Appendix equation (80) gives the exact counterpart with nonlinear covariances.

On an upper-interval branch, the rate-loading channel of Proposition 2 incorporates portfolio-risk responses from three periods on and the change in tax mix from $T^* + 1$ on.

For the empirical treasury comparison in Section 6.3, divide municipal loadings by $G_T := \sum_{j=1}^T \rho^{T-j} = A_T^{\text{Tsy}}$. This reference has a one-period return that responds one-for-one to the rate state. Since $A_T^M/G_T = \sum_{j=1}^T (\rho^{T-j}/G_T) s_j^M$, the relative yield sensitivity averages municipal responses with positive weights. It equals ξ_H in the household-only benchmark. Its excess over ξ_H accumulates portfolio-risk responses below T^* and tax-mix differences from T^* onward. Both vanish when the after-tax shares coincide. Figure 6(b) shows the loading gap relative to the equilibrium taxable market C, whose own portfolio response also contributes to $A_T^M - \xi_H A_T^C$.

A per-dollar benefit or cost that does not vary with the rate state, such as a trading-cost advantage or the calibration's fixed bank financing-tax cost, drops out of these response formulas on the same participation branch. Such a benefit or cost still changes which maturities an investor holds and the levels of one-period returns (Appendix B.6).

A.6 From one-period returns to the yield-implied tax

The channels act on one-period returns, while the municipal puzzle concerns yields to maturity. This subsection carries the channels into yields, the formal counterpart of the yield relation (18) in Section 4.5.

Applying (39) to both markets turns the one-period gap g_T of Proposition 2 into the yield gap:

$$\begin{aligned} y_T^M - \xi_H y_T^C = {} & \underbrace{\frac{1}{T}\sum_{j=0}^{T-1} \mathbb{E}_t\left[g_{T-j}(z_{t+j})\right]}_{\text{average gap along the bond's life}} \\ & \underbrace{- \frac{1}{2T}\sum_{k=2}^{T}\left[\sigma_r^2\left(\left(A_{k-1}^M\right)^2 - \xi_H\left(A_{k-1}^C\right)^2\right) + \left(\sigma_{\eta,k}^M\right)^2 - \xi_H\left(\sigma_{\eta,k}^C\right)^2\right]}_{\text{convexity difference}}. \end{aligned} \tag{54}$$

The riskless one-period gap is zero. The yield-implied tax is

$$\tau_T^{Y,\text{impl}} := 1 - \frac{y_T^M}{y_T^C}, \qquad \text{the model counterpart of } \textit{ImpliedTax} \text{ in (2).}$$

It equals the household wedge less the yield gap per unit of taxable yield, $\tau_T^{Y,\text{impl}} = (1-\xi_H) - (y_T^M - \xi_H y_T^C)/y_T^C$, as in the identity (3) of Section 3.6. A positive gap accumulated in (54) pulls the implied tax below the household wedge, by more at longer maturities when the gap grows faster in proportion than the taxable yield (Proposition 4(ii)).

The relative yield response sums the response gaps in (53). On a joint-participation branch the response gap is positive at every risky maturity, through portfolio responses below the participation threshold and

through both the tax mix and portfolio responses from that threshold on (Appendix Proposition 10), so for $T \geq 2$

$$\underbrace{\frac{\partial y_T^M}{\partial z} - \xi_H \frac{\partial y_T^C}{\partial z}}_{\text{relative yield response}} = \frac{1}{T}\sum_{j=2}^{T} \rho^{T-j} \underbrace{\left(\frac{\partial \mu_j^M}{\partial z} - \xi_H \frac{\partial \mu_j^C}{\partial z}\right)}_{\text{holding-return response gap}} > 0, \tag{55}$$

in model-period units, and both sides are zero for the riskless one-period bond. Long municipal yields therefore move more with the short rate than the household benchmark implies. The lower-tax investor's tax affects these responses even at maturities it does not hold.

Proposition 4 uses one restriction on taxes, risk-tolerance shares and rate persistence:

$$\text{(S)} \qquad \omega_H \frac{\xi_I - \xi_H}{\xi_H} \leq \rho. \tag{56}$$

Appendix equation (86) writes the same condition in terms of masses and risk aversions. This condition keeps the feedback b below ρ, so taxable loadings rise with maturity. Municipal loadings also rise beyond the participation threshold because the loading there lies below its long-end value, $A^M_{T^*-1} < \xi^*/(1-\rho)$ (Appendix equation (87)). The long-end taxable response $s^{C,\text{aff}}_\infty := \lim_{T\to\infty} s^{C,\text{aff}}_T$ lies in $(0,1)$, because the lower-tax investor's net rate exposure falls as the rate state rises.

For the yield levels in Proposition 4(ii), define the forward rate implied by today's prices for the period from $t+T-1$ to $t+T$:

$$f^a_T(z) = \log \frac{P^a_{T-1}(z)}{P^a_T(z)}, \qquad y^a_T(z) = \frac{1}{T}\sum_{j=1}^{T} f^a_j(z). \tag{57}$$

With positive taxable forwards, the yield ratio is a weighted average of municipal-to-taxable forward-rate ratios. The forward rate compares adjacent-maturity prices at the same date, whereas the holding return uses next period's resale price. The affine recursions (38) give their relation:

$$f^a_T(z) = \mu^a_T(z) - (1-\rho)A^a_{T-1}z - \tfrac{1}{2}\left[\sigma_r^2 (A^a_{T-1})^2 + (\sigma^a_{\eta,T})^2\right]. \tag{58}$$

For taxable bonds, the term premium is the yield less the average expected future short rate. Equation (39) expresses it as average expected bond risk premia less convexity. The municipal–taxable yield gap also reflects differences in holders' tax benefits.

Proposition 4 (Maturity ordering of the loading ratio and the yield-implied tax)**.** *In the affine two-investor benchmark, consider the equilibrium of Appendix Proposition 10 in which the high-tax household alone holds municipal bonds shorter than T^*, both investors hold municipal bonds from T^* to $T_{\max}$, and the lower-tax investor holds the taxable bonds. Assume condition (S) in (56).*

(i) Rate sensitivity. *The municipal-to-taxable loading ratio is non-decreasing in maturity. It starts at the household's after-tax share and, when the recursions are extended to infinite maturity with the household's exposure response d held at its solved value, converges to the long-end tax mix relative to the taxable response:*

$$\frac{A^{M,\text{aff}}_1}{A^{C,\text{aff}}_1} = \xi_H \qquad \text{and} \qquad \lim_{T\to\infty} \frac{A^{M,\text{aff}}_T}{A^{C,\text{aff}}_T} = \frac{\xi^*}{s^{C,\text{aff}}_\infty}. \tag{59}$$

(ii) Yield levels. *Suppose also that taxable forward rates are positive and the municipal-to-taxable forward-rate ratio is non-decreasing in maturity at the stationary mean $z = 0$. Then the yield ratio y^M_T/y^C_T is non-decreasing at $z = 0$ (Appendix Proposition 12), so the yield-implied tax $\tau^{Y,\text{impl,aff}}_T$ is non-increasing.*

(iii) Long end. *Extend the recursions to infinite maturity and suppose the loadings, the risk-premium coefficients π^a_T and the bond-specific compensation terms ι^a_T converge, with limits marked by the subscript ∞ and a positive limiting taxable forward. Then*

$$\lim_{T\to\infty} \tau^{Y,\text{impl,aff}}_T = 1 - \frac{\xi^*\bar{r} + A^M_\infty \pi^M_\infty - \frac{1}{2}\sigma_r^2 (A^M_\infty)^2 + \iota^M_\infty}{\bar{r} + A^C_\infty \pi^C_\infty - \frac{1}{2}\sigma_r^2 (A^C_\infty)^2 + \iota^C_\infty}, \tag{60}$$

where $\pi_T^a = \sum_i \omega_{i,T}^a \gamma_i \sigma_r^2 \widehat{D}_i(0)$ *is the holders' cost of rate risk per unit of loading, weighted by risk tolerance, and* ι_T^a *is their compensation for bond-specific risk net of its convexity term (Appendix equation (88)).*

Proof. See Appendix B.7. □

Loading monotonicity alone does not imply the yield-level ordering of part (ii) (Appendix B.7). Part (iii) adds relative term premia, convexity and bond-specific compensation to the long-end tax mix. With $\widehat{\Lambda} > 0$, small taxable feedback b and negligible convexity and bond-specific terms, the long-end implied tax is, to first order, below $1 - \xi^*$ and hence below $1 - \xi_H$ (Appendix equations (90) and (91)).

A.7 Participation from other motives

This subsection states the formal neutrality result behind Section 4.7. The link between participation and the return gap carries over to other motives when the participation advantage is neutral across the two markets. Lower trading costs or default diversification, for example, can induce long-end participation without an advantage in bearing rate risk.

An advantage that grows with maturity competes with the maturity-invariant difference in exemption benefits. Proposition 5 separates the resulting gap from the motive for participation.

The proposition sets aside every other source of a gap by making the remaining characteristics neutral. A characteristic is neutral when, for a household that bears it, it adds ξ_H times as much to the required municipal return as to the required taxable return, so it creates no gap when the household holds both bonds (Appendix Definition 1). Rate risk is neutral when the household holds both bonds, since municipal loadings are then ξ_H times taxable loadings.

Proposition 5 (Participation creates the gap)**.** *In the two-investor affine benchmark without bond-specific risk, suppose every non-tax characteristic other than rate risk is neutral and zero at the one-period claim: a risk common to both markets, independent of rate risk, with municipal exposure* ξ_H *times the taxable exposure, or a cost that reduces after-tax wealth by the same amount in both markets.*

(i) If the household held every municipal and taxable bond, $g_T = 0$ *at every maturity.*

(ii) Wherever the lower-tax investor holds the maturity-T municipal bond,

$$g_T \;\geq\; \underbrace{(\xi_I - \xi_H) r}_{\text{after-tax cash return}} \;+ \underbrace{\gamma_I \sigma_r^2 \widehat{D}_I \left(A_{T-1}^M - \xi_H A_{T-1}^C \right)}_{\text{rate loading}}.$$

The rate-loading term in part (ii) is positive on the branch of Appendix Proposition 10 when the lower-tax investor's net rate exposure is positive, but it can have either sign under other motives. Appendix B.12.2 proves part (i), and Appendix Proposition 15 proves part (ii).

Appendix Proposition 17 solves two three-maturity economies with constant cash rates, one for each motive. Lower trading costs give long-end participation and a municipal-to-taxable return ratio that increases with maturity. Municipal round-trip costs rise with maturity and fall with trade size (Harris and Piwowar, 2006; Green et al., 2007). Appendix Figure 19 reports the matched-market evidence.

In the diversification example, the state-tax exemption tilts households toward their own state's bonds, which limits their risk sharing (Babina et al., 2021). A lower-tax investor whose book is mostly taxable bonds holds little municipal default risk and bears it more cheaply. Default exposure rises with maturity, as do the municipal credit default swap spreads in Appendix Figure 20, so this advantage also grows with maturity and the lower-tax investor again holds only the long municipal bond. With separate municipal and taxable default factors, each gap taken net of the default compensation its holders require is zero at the riskless and middle bonds and equals the difference in exemption benefits at the long bond (Appendix Propositions 17 and 16). The raw gap at the middle bond also contains a default-premium difference, which reflects diversification.

With rate risk added back, numerical examples under both motives show participation only at the long end whenever it occurs. Their long gaps, net of default compensation in the diversification case, are at least the difference in exemption benefits, or fall short of it by less than 0.1 basis points where the rate-loading term is slightly negative (Appendix B.12.4).

A preference for municipal bonds as an asset class has a different pricing implication. In Appendix Proposition 18, such a preference supports long-end municipal participation while keeping municipal yields at or below the household benchmark. Explaining the puzzle through participation thus requires motives that are neutral across the two markets.

The quantitative model uses the duration motive of Appendix A.4 because rate risk is common to both markets, equilibrium prices determine the bonds' rate loadings, and ownership and the maturity of insurers' portfolios discipline the investors' outside exposures. Rate risk also plays the central quantitative role. Lowering risk aversion by 20% lowers the thirty-year gap by 13.6 basis points and the thirty-year taxable yield by 70 basis points, while a 20% change in the bond-specific variance moves each by less than 0.1 basis points (Appendix C.6).

A.8 From mechanism to empirical predictions

The mechanism predicts whose taxes are priced at each maturity. Of the three tests, P1 and P2 concern how tax benefits are priced in yield levels, and P3 concerns rate sensitivities, where the theory is sharpest because the loadings link maturities. I pair the yield-implied tax of Appendix A.6 with a loading-implied tax:

$$\tau_T^{Y,\text{impl}} = 1 - \frac{y_T^M}{y_T^C}, \qquad \tau_T^A = 1 - \frac{A_T^M}{A_T^C}.$$

A pass-through varies one investor's after-tax share at a fixed rate state and re-solves the equilibrium, letting holdings, risk costs and both yield curves adjust while non-tax primitives stay fixed. The pass-throughs of investor i's exemption benefit $1-\xi_i$ into the two implied taxes are

$$\mathcal{P}_i^Y(T) := \frac{d\tau_T^{Y,\text{impl}}}{d(1-\xi_i)}, \qquad \mathcal{P}_i^A(T) := \frac{d\tau_T^A}{d(1-\xi_i)}, \qquad i \in \{H, I\},$$

where an argument ∞ denotes the limit as $T \to \infty$. Under the representative-tax benchmark of Section 3.3, $y_T^M = \xi_H y_T^C$ and $A_T^M = \xi_H A_T^C$ at every maturity, so in both measures household pass-through is one and the lower-tax investor's pass-through is zero at every maturity. With heterogeneous taxes, the household alone holds the riskless short bond, whose return is $\xi_H r$. Both investors hold long bonds, whose returns depend on both after-tax shares through the tax mix ξ^*. For the empirical treasury comparison of Section 6.3, I also use a taxable reference whose one-period return moves one-for-one with the rate state, giving loading $A_T^{\text{Tsy}} = G_T$ (Appendix A.5).

Section 4.6 reports the long-end yield inequalities of Proposition 6 in (21).

Proposition 6 (Tax pass-through at the short and long ends)**.** *Consider the equilibrium of Proposition 4 (Appendix Proposition 10), with $0 < \rho < 1$, $\sigma_r > 0$, a positive cash rate and slack household funding. Extend its recursions to infinite maturity, and suppose long yields and their tax derivatives converge locally uniformly, with a positive long taxable yield.*

(i) (Short end) Both implied taxes are the household's own, $\tau_1^{Y,\text{impl}} = \tau_1^A = 1 - \xi_H$, so $\mathcal{P}_H^Y(1) = \mathcal{P}_H^A(1) = 1$ and $\mathcal{P}_I^Y(1) = \mathcal{P}_I^A(1) = 0$.

(ii) (Long end, P1 and P2) Suppose the after-tax shares satisfy the tax-difference condition

$$2\omega_H e + \omega_I e^2 \le 1, \qquad e := \frac{\xi_I - \xi_H}{\xi_H}.$$

Then

$$\mathcal{P}_H^A(\infty) < \omega_H, \qquad \mathcal{P}_I^A(\infty) > \omega_I.$$

The difference $\mathcal{P}_i^Y(\infty) - \mathcal{P}_i^A(\infty)$ shrinks to zero as the mean rate $\bar{r}$ rises, so for every $\bar{r}$ above a bound set by the other primitives, and at every feasible participation threshold,

$$\mathcal{P}_H^Y(\infty) < \omega_H < 1, \qquad \mathcal{P}_I^Y(\infty) > \omega_I > 0.$$

(iii) (Response ratios, P3) The municipal-to-taxable zero-coupon response ratio A_T^M/A_T^C is non-decreasing in maturity, equals ξ_H at the short end and converges to ξ^*/s_∞^C, with $0 < s_\infty^C < 1$. For the unit-response reference, A_T^M/A_T^{Tsy} has the same ordering and short endpoint, and converges to $\xi^* > \xi_H$.

Proof. See Appendix B.8. Appendix B.7 proves the monotonicity in part (iii). □

Part (ii) is proved first for the loading-implied tax. At the long end both investors hold municipal bonds, and Appendix B.8 solves the long-end loading ratio $R_\infty := A_\infty^M/A_\infty^C$ as

$$R_\infty = \xi^* + \varkappa\omega_H(\xi_I - \xi_H),$$

where $\varkappa > 0$ measures the feedback from the taxable response.[22]

The first term is the tax-mix contribution. It passes each investor's after-tax share through with its risk-tolerance share, ω_H for the household and ω_I for the lower-tax investor. The second term is the portfolio-risk response, which reflects the investors' reallocation of rate risk as the rate state changes. Under the stated tax-difference condition, a larger household exemption benefit strengthens this response, while a larger exemption benefit for the lower-tax investor weakens it. The resulting feedback offsets the household's tax-mix effect and adds to the lower-tax investor's, pushing $\mathcal{P}_H^A(\infty)$ below ω_H and $\mathcal{P}_I^A(\infty)$ above ω_I.

Raising ξ_H narrows the tax difference, which weakens the rate-induced transfer of net rate exposure to the household. It also raises the tax mix and bond loadings, which strengthens the risk feedback. The first effect dominates when the tax difference is not too large relative to ξ_H. The condition bounds this relative difference, e, and holds whenever $\xi_I < \frac{3}{2}\xi_H$, whatever the risk-tolerance shares or the participation threshold. Appendix Lemma 1 proves the derivative bounds with both effects included.

For yield levels, Appendix B.8 shows that $\mathcal{P}_i^Y(\infty)$ equals $\mathcal{P}_i^A(\infty)$ plus two corrections divided by the long taxable yield. The first is the tax response of $y_\infty^M - R_\infty y_\infty^C$, which contains only relative convexity and risk costs and does not depend on $\bar{r}$. The second reflects the tax response of the taxable yield itself. The taxable yield rises with $\bar{r}$, so both corrections shrink as $1/\bar{r}$, which gives the yield-level inequalities in part (ii) when the mean rate is high enough.

The tax-difference condition accommodates the tax difference between a top-bracket household and a corporate investor. The mean-rate condition limits how much risk compensation can move relative to interest income. The tax-difference condition and the yield-level inequalities hold with wide margins in the thirty-maturity illustration, where convexity accounts for most of the difference between the lower-tax investor's two pass-throughs. In the illustration, the yield-level inequalities also hold for every nonnegative value of the taxable holders' cost per unit of rate loading, π_∞^C. Stronger hedging demand or lower mean rates can enlarge the correction as the long taxable yield in its denominator approaches zero (Appendix B.9).

The proposition establishes the short-end and long-end contrasts that the empirical tests use. At intermediate maturities, the part of $\mathcal{P}_i^A(T)$ that comes from the tax mix alone shifts weight from the household to the lower-tax investor as more of a bond's remaining life falls at jointly held maturities (Appendix Proposition 13). In the thirty-maturity illustration, $\mathcal{P}_H^Y(T)$ falls and $\mathcal{P}_I^Y(T)$ rises strictly with maturity. With more clienteles, each has its own participation threshold and risk-tolerance share (Appendix B.11). These contrasts and the solved profiles give the three predictions below, which Section 4.6 states in words and Section 6 tests.

Prediction (P1: household state-tax pass-through declines with maturity)**.** A household's home-state tax is part of its exemption benefit on home-state bonds. Its pass-through into the yield-implied state tax of equation (24) is one at the riskless short end and declines with maturity in the solved illustration and the calibrated model as other clienteles hold municipal bonds. The empirical test compares pass-through across maturity bins. The representative-tax benchmark predicts one at every maturity.

Prediction (P2: institutional tax benefits affect longer maturities)**.** A change in a lower-tax institution's exemption benefit has no effect at the riskless short end. At longer maturities, the institution's own municipal holdings and the trading it induces make its tax treatment matter for municipal yields. Proposition 6 and the solved profiles predict a larger response at the long end, and portfolio adjustment also carries the change

[22] The identity $R_\infty = \xi^*[1 + b/(1-\rho)]$ gives $\varkappa = \xi^* b/[(1-\rho)\omega_H(\xi_I - \xi_H)]$. The coefficient and b adjust when an after-tax share changes.

to shorter risky bonds the institution does not hold. The representative-tax benchmark gives institutional tax treatment no role at any maturity.

Prediction (P3: the municipal-to-taxable zero-coupon response rises with maturity)**.** The municipal zero-coupon yield's response to the rate state, relative to the same-maturity taxable yield's response,

$$\frac{\partial y_T^M(z)/\partial z}{\partial y_T^C(z)/\partial z} = \frac{A_T^M}{A_T^C},$$

rises with maturity, from ξ_H at the short end toward ξ^*/s_∞^C. For the unit-response treasury reference used in Section 6.3, the long endpoint is ξ^*. The representative-tax benchmark predicts the constant ratio ξ_H in either comparison.

B Proofs and model extensions

This appendix proves the formal results of Appendix A, which support Sections 4.3–4.6, and extends the model of Section 4. Unless stated otherwise, funding is nonbinding ($\beta_i = 0$). These solutions also solve the funded problem wherever they leave cash nonnegative. Appendix C gives the funded system, and Table 7 collects the notation.

Table 7: Notation for the model and the proofs

Symbol	Meaning
z, $r(z) = \max\{\bar{r} + z, 0\}$	Latent rate state and the one-period log gross cash return
τ_i^a, ξ_i^a, $\xi_i = \xi_i^C/\xi_i^M$	Investor i's asset-specific tax rate and after-tax share. In the two-market benchmark, ξ_i is its relative after-tax share and $1-\xi_i$ its relative municipal tax benefit
$\xi_H < \xi_I$, ξ^*	Benchmark after-tax shares of the high-tax household and lower-tax investor, and their tax mix $\omega_H\xi_H + \omega_I\xi_I$ with risk-tolerance shares ω_i
$p_T^a(z)$, $A_T^a = -\partial_z p_T^a$, $y_T^a = -p_T^a/T$	Log price of the maturity-T bond in market a, its rate loading (the model's duration), and its yield, with $A_0^a = 0$
$C_T^a(z) = -p_T^a(z) - A_T^a(z)z$	Residual price-level term, the constant intercept in the affine benchmark
$\mu_T^a(z) = \log \mathbb{E}_z[1 + R_T^a]$, $s_T^a = \partial_z \mu_T^a$	Mean adjusted holding return, equal to its holders' required return in the KKT conditions, and its response to the state
$R_{T,t+1}^a$, $\mathcal{R}_{T,t+1}^a = \mu_T^a + u_{T,t+1}^a$	Simple holding-period return and the return used in portfolio choice, with mean μ_T^a and centered log-return innovation $u_{T,t+1}^a$
$\sigma_{\eta,T}^a$	Loading on bond-specific risk, zero for the one-period claim
$\nu_{i,T}^a$	KKT slack: marginal risk cost minus excess return, zero for a positive holding
$x_{i,T}^a$, Q_T^a, m_i, γ_i	Holdings per unit of mass, total supply, investor mass, risk aversion
W_i, W_i', $\widehat{W}_i'$, β_i	Funding per unit of mass, next-period net wealth, its discrete-time approximation, and the multiplier on the funding constraint
D_i, $\widehat{D}_i$, ℓ_i	Tax-weighted asset rate exposure, net rate exposure, and the outside liability loading. Appendix C.7 writes $h_i = -\ell_i$
Λ, $\widehat{\Lambda}$	Duration gap $\gamma_H\widehat{D}_H - \gamma_I\widehat{D}_I$ at the household-only allocation and at the equilibrium allocation
$\omega_{i,T}^a$, $\mathcal{U}_T^a$	Risk-tolerance share of an active holder and the set of active holders of the maturity-T bond in market a
$g_T = \mu_T^M - \xi_H\mu_T^C$	Gap between the municipal one-period return and the high-tax household's after-tax taxable return
$\tau_T^{Y,\text{impl}} = 1 - y_T^M/y_T^C$	Model yield-implied tax, the counterpart of *ImpliedTax* in (2)

B.1 The general economy: the reduced problem and conventions

The conventions here underlie Section 4.1 and the proofs below.

Portfolio choice uses the adjusted return (6), with mean $\mu_T^a = \log \mathbb{E}_t[1 + R_{T,t+1}^a]$ and centered log-return innovation $u_{T,t+1}^a$ as defined in Appendix A.1. Collecting the cash and bond terms in approximated wealth (32) gives each bond's expected after-tax excess return over cash

$$\widetilde{\mu}_{i,T}^a(z_t) := \underbrace{\xi_i^a \mu_T^a(z_t)}_{\text{after-tax bond return}} - \underbrace{\xi_i^C r(z_t)}_{\text{after-tax cash return}} . \tag{61}$$

The bond and outside innovations have conditional mean zero, so expected approximated wealth separates into the return on the initial funding and the portfolio's excess returns:

$$\mathbb{E}_t[\widehat{W}_i'] = W_i\big(1 + \xi_i^C r(z_t)\big) + \sum_{a,T} x_{i,T,t}^a \widetilde{\mu}_{i,T}^a(z_t).$$

The variance of wealth is that of the after-tax portfolio innovation, the random part of (32):

$$\widetilde{\Pi}_i(t+1) = \sum_{a \in \mathcal{A}} \sum_T \xi_i^a x_{i,T,t}^a u_{T,t+1}^a + \underbrace{\ell_i \sigma_r \varepsilon_{t+1}}_{\text{liability hedge}} . \tag{62}$$

Dropping the portfolio-independent term $W_i(1 + \xi_i^C r(z_t))$ and eliminating cash gives the equivalent reduced problem:

$$\max_{\substack{x_i \geq 0 \\ \sum_{a,T} x_{i,T,t}^a \leq W_i}} \underbrace{\sum_{a \in \mathcal{A}} \sum_T x_{i,T,t}^a \widetilde{\mu}_{i,T}^a(z_t)}_{\text{after-tax excess return}} - \underbrace{\frac{\gamma_i}{2} \operatorname{Var}_t\Big(\widetilde{\Pi}_i(t+1)\Big)}_{\text{whole-portfolio risk cost}} . \tag{63}$$

The problem's first-order conditions are (33).

Rate risk comes from the next-period resale price. With log price $p_T^a := \log P_T^a$ and rate loading $A_T^a := -\partial_z p_T^a$, a bond with T periods remaining has rate innovation $-A_{T-1}^a(\rho z_t)\sigma_r \varepsilon_{t+1}$ to first order, at the conditional mean resale state. A one-period bond carries no rate innovation, since $A_0^a = 0$, although its price can respond to today's state. A positive ℓ_i offsets the portfolio's rate exposure, and ℓ_i can represent net outside exposure of either sign.

The discrete-time return approximation

With simple holding-period returns $R_{T,t+1}^a$, exact next-period net wealth is

$$W_i' = W_i + \xi_i^C x_i^0 \big(e^{r(z_t)} - 1\big) + \sum_{a,T} \xi_i^a x_{i,T}^a R_{T,t+1}^a + \ell_i \sigma_r \varepsilon_{t+1}.$$

With rate risk alone, the log holding return is the change in log price, $\log(1 + R_{T,t+1}^a) = \log P_{T-1}^a(z_{t+1}) - \log P_T^a(z_t)$. Following Greenwood et al. (2024), I replace the simple bond return by the log return plus a Jensen adjustment, and $e^r - 1$ by r for cash. These substitutions give approximated wealth $\widehat{W}_i'$ in (32), with adjusted bond return

$$\mathcal{R}_{T,t+1}^a = \log(1 + R_{T,t+1}^a) + \underbrace{\log \mathbb{E}_t[1 + R_{T,t+1}^a] - \mathbb{E}_t[\log(1 + R_{T,t+1}^a)]}_{\text{Jensen adjustment}} .$$

The Jensen adjustment is known at time t. Taking the conditional expectation gives $\mathbb{E}_t[\mathcal{R}_{T,t+1}^a] = \mu_T^a(z_t) = \log \mathbb{E}_t[1 + R_{T,t+1}^a]$. Subtracting this mean cancels the adjustment:

$$u_{T,t+1}^a = \mathcal{R}_{T,t+1}^a - \mu_T^a(z_t) = \log(1 + R_{T,t+1}^a) - \mathbb{E}_t[\log(1 + R_{T,t+1}^a)].$$

In the affine benchmark, log returns are conditionally Gaussian. The Jensen term then equals $\frac{1}{2} \operatorname{Var}_t[\log(1 + R_{T,t+1}^a)]$, giving the same expected and unexpected returns as equations (7) and (8) of Greenwood et al. (2024). The nonlinear solver evaluates the exponential expectation directly, retaining the full Jensen term. Because the adjustment is conditionally known, it leaves return variances and covariances unchanged.

The price recursion is exact for the chosen mean $\mu = \log \mathbb{E}_t[1+R]$. Its use in the linear wealth equation approximates simple returns, whose conditional mean is $e^{\mu}-1$. Appendix E.5 compares the two conventions for a deterministic return. I re-solve the one-market economy of Section 5 with simple returns, taxed exactly, re-fitting risk aversion to restore the targeted taxable yields and holding the other coefficients fixed. The thirty-year gap in that economy rises from 102.6 to 107.7 basis points.

Funding and riskless claims

The funding constraint in (63) keeps cash nonnegative, so investors cannot borrow. Its multiplier $\beta_i(z)$ in (33), the shadow value of funding, is zero whenever the investor holds cash, since an extra dollar can then stay in cash. A positive shadow value requires the investor to be fully invested, so that buying more of one bond requires selling another, and it raises the required pretax return by β_i/ξ_i^a on each held bond.

Consider a riskless one-period claim in any market $a \neq C$. Its bond-specific variance $v_1^a = (\sigma_{\eta,1}^a)^2$ is zero and $p_0^a = 0$, so its return has no covariance with portfolios or liabilities. Retaining β_i in (33), each eligible investor's KKT condition becomes

$$\mu_1^a \leq \frac{\xi_i^C r + \beta_i}{\xi_i^a}, \qquad \text{with equality for a positive holding.}$$

With positive supply, at least one investor holds the claim, so its return equals the smallest of these bounds:

$$\mu_1^a(z) = \min_i \frac{\xi_i^C r(z) + \beta_i(z)}{\xi_i^a}. \tag{64}$$

The price recursion (66) then gives $p_1^a = -\mu_1^a$, hence $y_1^a = \mu_1^a$.

B.2 The general price recursion

The price recursion turns the equilibrium one-period returns μ_T^a, which the bonds' holders require, into prices and yields in two layers: levels accumulate one-period returns and rate sensitivities accumulate their responses to the rate state. Section 4.5 uses its affine form.

With fixed supplies and funding, I seek an equilibrium log-price function for each market and maturity:

$$p_{T,t}^a = p_T^a(z_t), \qquad p_0^a(z) \equiv 0, \qquad a \in \mathcal{A}. \tag{65}$$

Let $Z' = \rho z + \sigma_r \varepsilon'$. Since μ_T^a is the log of the bond's conditional expected gross return (31), solving for today's log price gives

$$p_T^a(z) = \underbrace{\log \mathbb{E}_z\left[\exp\left\{p_{T-1}^a(Z')\right\}\right]}_{\text{log expected resale price}} + \underbrace{\tfrac{1}{2}(\sigma_{\eta,T}^a)^2}_{\text{bond-specific payoff term}} - \underbrace{\mu_T^a(z)}_{\text{one-period return}}. \tag{66}$$

Equation (66) discounts the expected resale payoff at the equilibrium holding-period return, so a higher return lowers today's price. This recursion links adjacent maturities, while the portfolio conditions (33) link markets through investors' holdings. Prices and portfolios solve these equations and clearing (34) jointly at every state. Appendix C gives the numerical implementation. Equations (70)–(71) in Appendix B.3 write the portfolio system with nonbinding funding.

The rate floor, changes in participation and binding funding constraints make prices nonlinear in the state. Resale carries these features across maturities, because today's continuation value averages future prices over states with different participation and funding conditions.

Iterating (66) from $p_0^a = 0$ expresses the price through the one-period returns the bond earns as it ages through maturities $T, T-1, \ldots, 1$:

$$P_T^a(z_t) = e^{p_T^a(z_t)} = \mathbb{E}_t\Big[\exp\Big\{-\sum_{j=0}^{T-1}\Big(\mu_{T-j}^a(z_{t+j}) - \tfrac{1}{2}\big(\sigma_{\eta,T-j}^a\big)^2\Big)\Big\}\Big]. \tag{67}$$

Taking the negative log of (67) and dividing by T gives the yield $y_T^a = -p_T^a/T$. In the affine benchmark, this becomes the explicit average of expected one-period returns less convexity in (39). In the general economy, the yield's dependence on the state is summarized by writing the log price in loading form,

$$p_T^a(z) = -A_T^a(z)\, z - C_T^a(z), \qquad A_T^a(z) := -\frac{\partial p_T^a(z)}{\partial z}, \tag{68}$$

so that $P_T^a = \exp\{-A_T^a z - C_T^a\}$ and, at differentiability points,

$$y_T^a(z) = \frac{A_T^a(z) z + C_T^a(z)}{T}, \qquad \frac{\partial y_T^a(z)}{\partial z} = \frac{A_T^a(z)}{T}.$$

The level term C_T^a carries the price level, including accumulated one-period returns and convexity, and at the mean state the yield is $C_T^a(0)/T$. Both can vary with the state, with $\partial_z C_T^a = -z\, \partial_z A_T^a$ preserving the yield derivative.

The loading recursion links these coefficients to the one-period return response $s_T^a = \partial_z \mu_T^a$ of equation (37). Above the rate floor, cash has response one, and by (64) a riskless one-period claim in another market has response ξ_i^C/ξ_i^a for its active holder i when funding is slack. At longer maturities the response also includes the change in the holders' risk premia. For locally absolutely continuous continuation prices, an integrable envelope for the Gaussian exponential moment and its state derivative permits differentiation under the expectation in (66):

$$A_T^a(z) = \underbrace{s_T^a(z)}_{\text{current return response}} + \underbrace{\rho\, \frac{\mathbb{E}_z\left[\exp\{p_{T-1}^a(Z')\}\, A_{T-1}^a(Z')\right]}{\mathbb{E}_z\left[\exp\{p_{T-1}^a(Z')\}\right]}}_{\text{continuation price response}}. \tag{69}$$

The fraction averages the continuation loading, giving more weight to states where the resale bond is more valuable. In the affine benchmark, (69) reduces to the loading recursion (38).

B.3 Equilibrium conditions and the one-period boundary

The two-investor results assume $0 < \rho < 1$, $\sigma_r > 0$, $m_i, \gamma_i > 0$, $0 < \xi_H < \xi_I$ and positive traded supplies. The one-period taxable claim is riskless cash. An allocation's support is the set of investor–bond pairs with positive holdings. A participation branch fixes the support, and a bond's active set is its holders. Affine results use floorless rates and a fixed participation branch, whereas the numerical model retains the floor and funding constraints.

One KKT system for all markets. Index risky market–maturity coordinates by $j = (a, T)$ and write $v_j = (\sigma_{\eta,T}^a)^2$, $K_i = \mathrm{diag}(\xi_i^a)$ and the relative after-tax share $\xi_{ij} = \xi_i^C/\xi_i^a$, collected in the vector $\boldsymbol{\xi}_i$. In the normalized two-market benchmark, $\xi_{ij} = \xi_i$ at municipal coordinates and $\xi_{ij} = 1$ at taxable coordinates. With resale log price $p_{j,-} = p_{T-1}^a$, the exact covariance and liability credit are

$$\Sigma_{jk}(z) = \mathrm{Cov}_z(p_{j,-}(Z'), p_{k,-}(Z')) + \mathbf{1}_{\{j=k\}} v_j, \qquad c_{ij}^L(z) = -\mathrm{Cov}_z(p_{j,-}(Z'), \ell_i \sigma_r \varepsilon'). \tag{70}$$

Dividing the first-order condition by the positive asset after-tax share gives

$$\begin{gathered} \nu_{ij} := \xi_{ij} r + \gamma_i [\Sigma K_i x_i - c_i^L]_j - \mu_j \geq 0, \qquad x_{ij} \geq 0, \qquad x_{ij}\nu_{ij} = 0, \\ \sum_i m_i x_{ij} = Q_j. \end{gathered} \tag{71}$$

Concavity of (63) makes these conditions sufficient for investor optimality. Positive bond-specific variances give strict concavity on risky coordinates, but do not establish equilibrium uniqueness.

Proposition 7 (Current-state clearing)**.** *Fix the state and continuation prices, with finite $\Sigma \succeq 0$, finite liability credits and at least one eligible investor for every positive-supply coordinate.*

(i) The analytical system has a current-state KKT-and-clearing solution.

(ii) If, in addition, $\Sigma = V + \sigma_r^2 aa'$ with V positive diagonal and $a \gg 0$ on the risky coordinates, after-tax shares are positive and supported supplies are positive, the current-state clearing returns and risky holdings are unique, even with endogenous support changes. Deterministic one-period claims are treated separately. Payoff-equivalent holdings in tied boundary claims need not be unique.

Neither part establishes the dynamic fixed point of continuation prices.

Proof. (i) Let $X = \{x \geq 0 : \sum_i m_i x_i = Q\}$ be the set of nonnegative allocations that clear, with inner product $\langle x, y\rangle_m = \sum_i m_i x_i' y_i$. Positive masses and finite supplies make X nonempty, compact and convex. For $G_i(x_i) = \boldsymbol{\xi}_i r + \gamma_i(\Sigma K_i x_i - c_i^L)$, investor i's required return on each coordinate, Brouwer gives a fixed point of the continuous map $x \mapsto \mathrm{Proj}_X(x - G(x))$. The projection characterization implies

$$\sum_i m_i \langle G_i(x_i), y_i - x_i\rangle \geq 0 \qquad (y \in X).$$

Transfers from positive holders to other eligible investors show that each positive holder minimizes G_{ij}. Thus $\mu_j = \min_i G_{ij}$ and $\nu_{ij} = G_{ij} - \mu_j$ satisfy (71). Conversely, complementarity implies the displayed inequality for every feasible transfer. Predetermined trading exclusions restrict eligibility. (ii) Strict concavity makes individual demand $x_i(\mu)$ continuous, Lipschitz and piecewise affine. On a region where investor i holds exactly the coordinates in a set B_i, its derivative is

$$J_i = \gamma_i^{-1} E_{B_i} K_{i,B_i}^{-1} \Sigma_{B_i B_i}^{-1} E_{B_i}',$$

where E_B embeds the held coordinates. Sherman–Morrison gives

$$\Sigma_{BB}^{-1} = V_B^{-1} - \frac{\sigma_r^2 V_B^{-1} a_B a_B' V_B^{-1}}{1 + \sigma_r^2 a_B' V_B^{-1} a_B}, \qquad \Sigma_{BB}^{-1} a_B = \frac{V_B^{-1} a_B}{1 + \sigma_r^2 a_B' V_B^{-1} a_B} \gg 0.$$

Every selection of $J = \sum_i m_i J_i$ therefore has nonpositive off-diagonal entries and $Ja \geq 0$, strictly on held coordinates. If μ^0, μ^1 both cleared, piecewise-affine continuity would give $0 = \overline{J}(\mu^1 - \mu^0)$, with $\overline{J}$ the segment-average selection matrix. Positive supply at μ^0 and demand continuity imply positive holdings on an initial segment for each asset. Finitely many selections then give $(\overline{J}a)_j > 0$ everywhere. Hence $\overline{J}\,\mathrm{diag}(a)$ is strictly diagonally dominant and nonsingular, so returns coincide, and strict concavity gives identical risky holdings. On support-boundary segments, adjacent selections have the same directional derivative. □

Nonlinear continuation covariances need not have this sign structure.

Regularity for derivatives. Derivatives below are taken at regular states, at which the cash rate and price moments are continuously differentiable, complementarity is strict on risky coordinates and the active KKT-and-clearing Jacobian is nonsingular. The implicit-function theorem then gives a locally fixed participation branch and differentiable holdings and returns. Fixed supplies imply, for each bond's active set $\mathcal{U}_j$,

$$\sum_{i \in \mathcal{U}_j} m_i \frac{\partial x_{ij}}{\partial z} = 0. \tag{72}$$

Individual strict concavity does not imply nonsingularity, because with heterogeneous after-tax shares the normalized operator contains ΣK_i, which need not be symmetric or strongly monotone.

The one-period boundary. Specializing the riskless-claim return (64) to the normalized municipal–taxable benchmark gives the short-end anchor (12) and pass-throughs (20).

Proposition 8 (Deterministic one-period boundary)**.** *In the normalized municipal–taxable benchmark with nonbinding funding constraints, suppose the one-period municipal claim has $v_1^M = 0$ and positive supply, and*

$r(z) > 0$. *Suppose the high-tax household has the minimal after-tax share, $\xi_H = \min_i \xi_i$. Then only investors with the minimal after-tax share can hold it, and*

$$y_1^M = \mu_1^M = \xi_H r(z), \qquad y_1^C = r(z), \qquad \tau_1^{Y,\text{impl}} = 1 - \xi_H. \tag{73}$$

If the high-tax household is the unique after-tax share minimizer, its tax passes through one-for-one and the tax of an investor with a strictly higher after-tax share has zero effect at this boundary. At a tie the household derivative requires perturbing all tied minimal after-tax shares together. At $r = 0$ the yield ratio is undefined.

Proof. With $\xi_i^M = 1$, $\xi_i^C = \xi_i$ and $\beta_i = 0$, the riskless-claim KKT condition of Appendix B.1 is $\mu_1^M \le \xi_i r$, with equality for positive holdings. Since $r > 0$, only investors with the minimal after-tax share can satisfy it with equality. Positive supply gives $\mu_1^M = \min_i \xi_i r = \xi_H r$ and $p_1^M = -\mu_1^M$. Taxable cash has $p_1^C = -r$. Differentiating the yield ratio proves the tax statements. □

Bond-specific risk. Positive risky variances give strict concavity but no uniform bound relative to rate risk, because the cash floor compresses rate variation and bond-specific premia depend on holdings. Appendices C.3 and C.4 check the funded boundary, grid, barrier and floor knot.

B.4 Affine solutions

This subsection constructs the affine solutions of Appendix A.3. Fix a participation branch, set inactive positions to zero, set $r(z) = \bar{r} + z$ and use affine continuation prices on $\mathbb{R}$. Solve the active equalities jointly across maturities, retaining whole-portfolio risk. Then check nonnegative holdings, inactive slacks and, for finite budgets, $W_i - \sum_j x_{ij}(z) \ge 0$.

For affine candidates $p_T^a(z) = -A_T^a z - C_T^a$, Gaussian integration gives exactly

$$\log \mathbb{E}_z e^{p_{T-1}^a(Z')} = -\rho A_{T-1}^a z - C_{T-1}^a + \tfrac{1}{2}\sigma_r^2 (A_{T-1}^a)^2. \tag{74}$$

The current-state covariance $\Sigma = \sigma_r^2 AA' + \operatorname{diag}(v)$, with A collecting predecessor loadings, and liability credit $c_i^L = \ell_i \sigma_r^2 A$ are constant. If the active KKT-and-clearing matrix is nonsingular, holdings and μ are therefore affine in z, and the price recursion returns affine prices with updated loadings $\rho A_{T-1}^a + s_T^a$. At a coefficient fixed point, all active equalities, clearing equations and exact continuation equations hold identically, and, with $A_0^a = 0$,

$$A_T^a = \rho A_{T-1}^a + s_T^a = \sum_{j=1}^{T} \rho^{T-j} s_j^a, \qquad s_T^a = \frac{\partial \mu_T^a}{\partial z}. \tag{75}$$

Affine holdings and slacks make the KKT-valid states an interval, possibly empty, conditional on the global affine continuation. Gaussian transitions leave any proper interval, so interval feasibility is not global equilibrium feasibility. For example, at the household-only allocation the lower-tax investor's slack on the one-period municipal claim, $(\xi_I - \xi_H)(\bar{r} + z)$, turns negative at negative rates. The cash floor changes continuation values.

For a fixed participation branch, a continuous coefficient update on a nonempty compact convex invariant set has a fixed point by Brouwer. A contraction gives uniqueness within that set. Nonsingularity, invariance and reinstated KKT feasibility must be checked separately. Appendix B.9 verifies the reported solution of the illustration, and these checks supply no global existence argument.

B.5 Participation

This subsection proves the participation results of Appendix A.4 behind Section 4.3.

Assumption 1 (Household duration burden at the household-only allocation)**.** *At the household-only allocation, where the high-tax household holds all municipal bonds and the lower-tax investor all taxable bonds,*

$$\widehat{D}_H^{\text{hh}} = \sum_T A_{T-1}^M Q_T^M / m_H - \ell_H, \quad \widehat{D}_I^{\text{hh}} = \xi_I \sum_T A_{T-1}^C Q_T^C / m_I - \ell_I, \quad \Lambda := \gamma_H \widehat{D}_H^{\text{hh}} - \gamma_I \widehat{D}_I^{\text{hh}} > 0.$$

Under the assumption, the duration benefit $\sigma_r^2 A_{T-1}^M \Lambda$ is nonnegative and grows with the bond's loading.

Proof of Proposition 1. At the household-only allocation the household holds Q_T^M/m_H of each municipal bond, and its active municipal equality sets

$$\mu_T^{M,\mathrm{hh}} = \xi_H r + \gamma_H \sigma_r^2 A_{T-1}^M \widehat{D}_H^{\mathrm{hh}} + \gamma_H v_T^M Q_T^M/m_H.$$

Substituting it into the lower-tax investor's zero-holding condition gives its slack

$$\nu_{I,T}^{\mathrm{hh}} = (\xi_I - \xi_H) r - \sigma_r^2 A_{T-1}^M \Lambda - \gamma_H v_T^M Q_T^M/m_H. \tag{76}$$

Without bond-specific risk, the slack (76) satisfies $\nu_{I,T}^{\mathrm{hh}} \le 0$ exactly when (44) holds. Sole-holder quantities are state invariant, giving $s_T^M = \xi_H$ and strictly increasing $A_T^M = \xi_H \sum_{q=0}^{T-1} \rho^q$. If $v_T^M Q_T^M$ is nondecreasing, the slack decreases with maturity and profitable participation forms an upper interval. The bond-specific term in (76) weakly expands this interval at fixed benchmark loadings and exposures.

At an equilibrium allocation, Theorem 1 gives the exact criterion in the footnote to (13).

Theorem 1 (Upper-interval participation by the lower-tax investor on an affine branch)**.** *At a regular state of a two-investor affine solution, suppose $v_T^M > 0$ on the risky maturities considered. Define the duration gap at the equilibrium allocation, $\widehat{\Lambda} = \gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I$, and*

$$\Psi_T = (\xi_I - \xi_H) r - \sigma_r^2 A_{T-1}^M \widehat{\Lambda} - \gamma_H v_T^M Q_T^M/m_H. \tag{77}$$

Then $x_{I,T}^M > 0$ if and only if $\Psi_T < 0$. If $\widehat{\Lambda} > 0$ and the predecessor loadings and $v_T^M Q_T^M$ are nondecreasing, the lower-tax investor's positive support is an upper interval, possibly empty. This identifies the cutoff within a given equilibrium and does not establish uniqueness across equilibria.

Proof. If the lower-tax investor is inactive, $x_{H,T}^M = Q_T^M/m_H$. Substituting the household equality into the lower-tax investor's inequality gives $\Psi_T = \nu_{I,T} \ge 0$. If both investors hold the bond, subtract their equalities and use clearing:

$$\Psi_T = -v_T^M \left(\gamma_I + \gamma_H m_I/m_H\right) x_{I,T}^M < 0.$$

If the lower-tax investor alone is active, its equality and the household inequality imply $(\xi_I - \xi_H) r - \sigma_r^2 A_{T-1}^M \widehat{\Lambda} \le -\gamma_I v_T^M Q_T^M/m_I < 0$, which also gives $\Psi_T < 0$. These cases prove the equivalence. Finally, adjacent indices differ by

$$\Psi_{T+1} - \Psi_T = -\sigma_r^2 (A_T^M - A_{T-1}^M)\widehat{\Lambda} - \frac{\gamma_H}{m_H}(v_{T+1}^M Q_{T+1}^M - v_T^M Q_T^M) \le 0.$$

With $\widehat{\Lambda} > 0$ and nondecreasing predecessor loadings and $v_T^M Q_T^M$, the index is nonincreasing in maturity, and the first maturity with a strictly negative index is the participation threshold T^*. Equality is a zero-holding boundary. □

Joint holdings. For candidate net rate exposures, put $N_T = (\xi_I - \xi_H) r - \sigma_r^2 A_{T-1}^M \widehat{\Lambda}$, the difference in exemption benefits less the duration benefit. For $v_T := v_T^M > 0$, the difference of the two active KKT equalities and clearing give

$$x_{H,T}^M = \frac{\gamma_I Q_T^M + m_I N_T/v_T}{\gamma_H m_I + \gamma_I m_H}, \qquad x_{I,T}^M = \frac{\gamma_H Q_T^M - m_H N_T/v_T}{\gamma_H m_I + \gamma_I m_H}. \tag{78}$$

Both holdings are nonnegative exactly when $-\gamma_I Q_T^M/m_I \le N_T/v_T \le \gamma_H Q_T^M/m_H$. Each maturity's holdings depend on the net rate exposures, which are common to all maturities and must match the resulting portfolios. A nonzero tax difference $(\xi_I - \xi_H) r$ and distinct positive limiting loadings preclude a fixed block of two or more jointly held maturities as every $v_T \downarrow 0$, because bounded holdings require $N_T \to 0$ at each maturity, which assigns different values to the one common duration gap $\widehat{\Lambda}$. Formula (83) of Proposition 10, which couples the jointly held maturities through their positive variances, therefore describes no feasible equilibrium with such a block in this limit.

B.6 The pricing channels

This subsection proves the results of Appendix A.5 behind Sections 4.4 and 4.5.

Proof of Proposition 2. Investor i's active municipal equality (46) gives $\mu_T^M = \xi_i r + \gamma_i[\sigma_r^2 A_{T-1}^M \widehat{D}_i + (\sigma_{\eta,T}^M)^2 x_{i,T}^M]$. The lower-tax investor's taxable equality (47) gives $\mu_T^C = r + \gamma_I[\sigma_r^2 A_{T-1}^C \widehat{D}_I + \xi_I(\sigma_{\eta,T}^C)^2 x_{I,T}^C]$, also covering cash at $T=1$ because $A_0^C = 0$. Subtract ξ_H times the second equality from the first, then add and subtract $\gamma_I \sigma_r^2 A_{T-1}^M \widehat{D}_I$. This gives the three-channel identity (48).

At $T=1$, $A_0^a = 0$ and there is no bond-specific risk, so the duration-cost, rate-loading and bond-specific terms vanish. When $r > 0$, the household alone holds the claim (Proposition 8), so $i = H$ and the after-tax cash-return difference also vanishes.

If the household alone holds maturity T, then $i = H$ and $x_{H,T}^M = Q_T^M / m_H$. The lower-tax investor's inactive condition gives $\Psi_T \geq 0$ in (77), with $v_T^M = (\sigma_{\eta,T}^M)^2$, or $\sigma_r^2 A_{T-1}^M \widehat{\Lambda} + \gamma_H (\sigma_{\eta,T}^M)^2 Q_T^M / m_H \leq (\xi_I - \xi_H) r$. The duration-cost contribution is the first term on the left and is therefore at most $(\xi_I - \xi_H) r$. If the lower-tax investor holds the bond, use $i = I$ in the identity. The duration-cost contribution then vanishes, and the cash-return contribution is $(\xi_I - \xi_H) r$.

Proposition 9 is the exact, possibly nonlinear counterpart of Proposition 3.

Proposition 9 (Exact active-clientele aggregation)**.** *At a regular state, let* $\Gamma_j = \sum_{i \in \mathcal{U}_j} m_i / \gamma_i$, $\bar{\xi}_j = \Gamma_j^{-1} \sum_{i \in \mathcal{U}_j} m_i \xi_{ij} / \gamma_i$ *and* $\mathcal{C}_j = \Gamma_j^{-1} \sum_{i \in \mathcal{U}_j} m_i [\Sigma K_i x_i - c_i^L]_j$. *Then*

$$\begin{aligned} \mu_j &= \bar{\xi}_j r + \mathcal{C}_j, \\ s_j &= \bar{\xi}_j r' + \frac{1}{\Gamma_j} \sum_{i \in \mathcal{U}_j} m_i \left[(\partial_z \Sigma) K_i x_i + \Sigma K_i \partial_z x_i - \partial_z c_i^L \right]_j . \end{aligned} \tag{79}$$

In the normalized municipal–taxable benchmark, above the cash floor the response difference is

$$s_T^M - \xi_H s_T^C = (\bar{\xi}_T^M - \xi_H) + \partial_z \mathcal{C}_T^M - \xi_H \partial_z \mathcal{C}_T^C. \tag{80}$$

Proof. Weight the active equalities $\mu_j - \xi_{ij} r = \gamma_i [\Sigma K_i x_i - c_i^L]_j$ by m_i / γ_i, sum and divide by Γ_j. This gives the first line. Differentiating on the locally fixed participation branch gives the second line. In the two-market normalization, $\bar{\xi}_j = \bar{\xi}_T^M$ at municipal and $\bar{\xi}_j = 1$ at taxable coordinates. Subtraction gives the gap. □

With binding funding, each active equality also contains β_i / ξ_i^a by (33), so the return identity in (79) adds $\Gamma_j^{-1} \sum_{i \in \mathcal{U}_j} m_i \beta_i / (\gamma_i \xi_i^a)$ and the response identity adds its state derivative. A per-dollar charge such as the bank financing-tax cost appears in the return identity in the same units and, when it does not vary with the rate state, leaves the response identity unchanged on a given participation branch.

Proof of Proposition 3. With $\widehat{D}_i = A' K_i x_i - \ell_i$, constant affine covariance and liability credits reduce (79) to

$$s_j = \bar{\xi}_j + \frac{\sigma_r^2 A_j}{\Gamma_j} \sum_{i \in \mathcal{U}_j} m_i \frac{\partial \widehat{D}_i}{\partial z} + \frac{v_j}{\Gamma_j} \sum_{i \in \mathcal{U}_j} m_i \xi_i^a \frac{\partial x_{ij}}{\partial z}, \tag{81}$$

where $A_j = A_{T-1}^a$ is the predecessor loading. For municipal bonds, whose after-tax shares are normalized to one, (72) cancels the final term, leaving the tax-mix and portfolio-risk terms of (50) in the holding-return response.

Taxable counterpart. Proposition 10 also uses the taxable version of (81). With $\xi_{ij} = 1$ at taxable coordinates, its tax term is one. Its bond-specific term vanishes with a singleton taxable active set, since clearing fixes the sole holder's quantity, with equal active taxable after-tax shares, which factor out of (72),

or with $v_T^C = 0$. Heterogeneous after-tax shares with multiple holders prevent cancellation by unweighted clearing. With the lower-tax investor the sole holder of the taxable bond,

$$s_T^C = 1 + \gamma_I \sigma_r^2 A_{T-1}^C \partial_z \widehat{D}_I. \tag{82}$$

Proposition 10 supplies the signs in (16) and the positive response gap in (55).

Proposition 10 (Joint participation and propagation across maturities)**.** *At a regular state with $r(z) > 0$ in the two-investor affine benchmark, suppose the lower-tax investor alone holds risky taxable bonds. Let $\varnothing \neq J \subseteq \{2, \dots, T_{\max}\}$ be the fixed set of municipal maturities held by both investors, with the high-tax household alone holding every municipal maturity outside J, including the riskless one-period claim. Assume $A_{T-1}^M, A_{T-1}^C > 0$ for every $T \geq 2$ and $v_T^M > 0$ on J. Define*

$$P_J = \sum_{T \in J} \frac{A_{T-1}^M}{v_T^M}, \quad S_J = \sum_{T \in J} \frac{(A_{T-1}^M)^2}{v_T^M}, \quad d := \frac{\partial \widehat{D}_H}{\partial z}.$$

Then

$$d = \frac{(\xi_I - \xi_H) m_I P_J}{(\gamma_H m_I + \gamma_I m_H)(1 + \sigma_r^2 S_J)} > 0, \qquad \frac{\partial \widehat{D}_I}{\partial z} = -\frac{m_H}{m_I} d. \tag{83}$$

Writing $\omega_i = (m_i/\gamma_i)/(m_H/\gamma_H + m_I/\gamma_I)$, $\xi^* = \omega_H \xi_H + \omega_I \xi_I$ and $b = \gamma_I \sigma_r^2 (m_H/m_I) d$, the slopes are

$$s_T^M = \begin{cases} \xi_H + \gamma_H \sigma_r^2 A_{T-1}^M d, & T \notin J, \\ \xi^*, & T \in J, \end{cases} \qquad s_T^C = 1 - b A_{T-1}^C. \tag{84}$$

Hence $s_T^M - \xi_H s_T^C > 0$ and $A_T^M - \xi_H A_T^C > 0$ at every risky maturity $T \geq 2$, with equality in both comparisons at $T = 1$.

Proof. Clearing fixes sole-held quantities at Q_T^M/m_H for households outside J and Q_T^C/m_I for the lower-tax investor's risky taxable bonds. Only jointly held municipal positions vary. Differentiate clearing and sum against municipal predecessor loadings. Constant liabilities and the fixed taxable book give

$$m_H \partial_z \widehat{D}_H + m_I \partial_z \widehat{D}_I = 0. \tag{85}$$

At each $T \in J$, subtracting the lower-tax investor's differentiated municipal equality from the household's gives

$$\xi_I - \xi_H = \sigma_r^2 A_{T-1}^M (\gamma_H \partial_z \widehat{D}_H - \gamma_I \partial_z \widehat{D}_I) + v_T^M (\gamma_H \partial_z x_{H,T}^M - \gamma_I \partial_z x_{I,T}^M).$$

With $u_T = \partial_z x_{I,T}^M$ and $k = (\gamma_H m_I + \gamma_I m_H)/m_H$, clearing reduces this to $u_T = -(\xi_I - \xi_H)/(k v_T^M) - \sigma_r^2 A_{T-1}^M \partial_z \widehat{D}_I / v_T^M$. Since $\partial_z \widehat{D}_I = \sum_{T \in J} A_{T-1}^M u_T$, multiplying by A_{T-1}^M and summing over J gives

$$(1 + \sigma_r^2 S_J) \partial_z \widehat{D}_I = -\frac{\xi_I - \xi_H}{k} P_J < 0.$$

Equation (85) gives (83). The slopes follow from (81). Household-only quantities have zero derivatives, giving the first municipal slope. At jointly held maturities, risk-tolerance aggregation cancels the aggregate exposure and bond-specific derivatives and leaves ξ^*, and (82) gives the taxable formula.

Equation (84) gives the response gap $\partial_z g_T = s_T^M - \xi_H s_T^C$ as

$$\partial_z g_T = \begin{cases} \sigma_r^2 d \left[\gamma_H A_{T-1}^M + \xi_H \gamma_I \dfrac{m_H}{m_I} A_{T-1}^C \right], & T \notin J, \\ \omega_I (\xi_I - \xi_H) + \xi_H b A_{T-1}^C, & T \in J. \end{cases}$$

Thus $\partial_z g_T > 0$ for every $T \geq 2$, while $\partial_z g_1 = 0$ because both predecessor loadings vanish. Iterating (75), with $0 < \rho < 1$, yields

$$A_T^M - \xi_H A_T^C = \sum_{j=2}^{T} \rho^{T-j} \partial_z g_j > 0, \qquad T \geq 2.$$

□

The signed reallocation is $\partial_z \widehat{D}_I$, the change in the lower-tax investor's loading-weighted municipal portfolio. Individual jointly held positions can move in either direction.

State-independent holding benefits. Suppose an investor's objective includes $\sum_{a,T} b_{i,T}^a x_{i,T}^a$, with each $b_{i,T}^a$ independent of the rate state. Its active municipal condition becomes

$$\mu_T^M - \xi_i r + b_{i,T}^M = \gamma_i \left[\sigma_r^2 A_{T-1}^M \widehat{D}_i + v_T^M x_{i,T}^M \right],$$

and the taxable condition similarly gains $b_{i,T}^C / \xi_i$. With covariance, financing and constraints fixed, differentiation removes state-independent benefits and liabilities. The preceding proof and loading equations hold for the same feasible set J of jointly held maturities. Benefits and liabilities affect return intercepts, gap levels (Appendix B.12) and the feasible support. A given loading solution is therefore unchanged on the same participation branch. Equality across distinct coefficient fixed points additionally requires uniqueness. State-dependent benefits add their derivatives. Fixed participation fees affect global participation without a per-unit KKT term.

Uniqueness of the loading fixed point. Appendix B.9 verifies the following sufficient condition for a unique loading fixed point in the illustration. For given d, generate municipal loadings from (84) and the loading recursion (75), and let $\mathcal{F}_J(d)$ be the right side of (83) at those loadings. If $\mathcal{F}_J$ maps $[0, \infty)$ into $[0, d_{\max}]$ and $\sup_{[0,d_{\max}]} |\mathcal{F}_J'| < 1$, every nonnegative exposure fixed point lies in $[0, d_{\max}]$, and Banach gives exactly one, with its loading path on this branch. Holdings and inactive inequalities still require verification. Equilibria with other feasible participation thresholds remain possible. The derivative of $\mathcal{F}_J$ is

$$\mathcal{F}_J'(d) = c \frac{P_J'(1 + \sigma_r^2 S_J) - \sigma_r^2 P_J S_J'}{(1 + \sigma_r^2 S_J)^2}, \qquad c = \frac{(\xi_I - \xi_H) m_I}{\gamma_H m_I + \gamma_I m_H},$$

with the derivatives of P_J, S_J obtained by differentiating the same loading recursions.

The last result in this subsection gives the household-only benchmark. Without participation by the lower-tax investor, municipal loadings are exactly ξ_H times taxable loadings.

Proposition 11 (Household-only loadings)**.** *In the two-investor affine benchmark, if the high-tax household alone holds every municipal bond and the lower-tax investor alone holds every taxable bond, then* $s_T^M = \xi_H$, $s_T^C = 1$ *and* $A_T^M = \xi_H A_T^C$ *at every maturity.*

Proof. Clearing fixes every holding, so the derivatives of both net rate exposures are zero. Apply (81) and iterate (75). The identity concerns loadings, and different intercept risk premia can still affect the yield ratio. □

B.7 Loading and yield ratios

This subsection proves Proposition 4 of Appendix A.6 and the treasury-reference ordering in Proposition 6(iii).

The sufficient condition and the threshold loading. Specialize Proposition 10 to the upper interval $J = \{T^*, \dots, T_{\max}\}$. The maturity ordering uses condition (S) and the bound (M) on the loading at the threshold:

$$\textbf{(S)} \quad \frac{\gamma_I m_H(\xi_I - \xi_H)}{\xi_H(\gamma_H m_I + \gamma_I m_H)} \le \rho, \tag{86}$$

$$\textbf{(M)} \quad A^M_{T^*-1} < \frac{\xi^*}{1-\rho}. \tag{87}$$

Condition (S) is condition (56) of Appendix A.6 with ω_H written out and is assumed. The bound (M) holds at every solution of (83) on this branch. Put $n = T^* - 1$ and $\lambda = \gamma_H \sigma_r^2 d$, so that (83) reads $\lambda = \omega_I(\xi_I - \xi_H)\sigma_r^2 P_J/(1 + \sigma_r^2 S_J)$. The recursions at household-only and jointly held maturities give $A^M_n = \xi_H \sum_{q=0}^{n-1}(\rho+\lambda)^q$ and $A^M_{n+l} = \rho^l A^M_n + (1-\rho^l)\xi^*/(1-\rho)$ for $l \ge 0$. If $A^M_n \ge \xi^*/(1-\rho)$, every loading on J is at least $\xi^*/(1-\rho)$, so $S_J/P_J \ge \xi^*/(1-\rho)$ and $\lambda < (1-\rho)(\xi^* - \xi_H)/\xi^* < 1-\rho$, using $\xi^* - \xi_H = \omega_I(\xi_I - \xi_H)$. Then $A^M_n < \xi_H/(1-\rho-\lambda) < \xi^*/(1-\rho)$, a contradiction. Both statements concern the upper-interval branch of the two-investor affine benchmark and do not prove that this participation branch exists.

Proof of Proposition 4(i). Step 1 concerns municipal loadings. At household-only maturities, $A^M_T = (\rho + \gamma_H \sigma_r^2 d)A^M_{T-1} + \xi_H$, so A^M_T is increasing from zero. At jointly held maturities, $A^M_T = \rho A^M_{T-1} + \xi^*$, whose increments have the sign of $\xi^* - (1-\rho)A^M_{T^*-1}$. Condition (M) makes them positive. Step 2 concerns the taxable leg. Since $A^M_{T-1} \ge \xi_H$ on J, $P_J/S_J \le 1/\xi_H$, and (83) and (S) give $0 \le b \le \rho$. Therefore

$$A^C_T = \sum_{q=0}^{T-1}(\rho - b)^q, \qquad s^C_T = 1 - bA^C_{T-1} \downarrow \frac{1-\rho}{1-\rho+b} > 0.$$

Step 3 concerns the municipal response. The municipal slope rises at household-only maturities, is constant at jointly held maturities and does not jump down when the lower-tax investor begins to participate. For the last claim, the slope at $T^* - 1$ must not exceed ξ^*. Substituting (83) shows that the required inequality $\omega_I(\xi_I - \xi_H) \ge \gamma_H \sigma_r^2 A^M_{T^*-2} d$ is equivalent to $\sigma_r^2 A^M_{T^*-2} P_J \le 1 + \sigma_r^2 S_J$, which follows from $A^M_{T-1} > A^M_{T^*-2}$ for every $T \in J$, established above using (M). Since Step 2 makes s^C_j positive and nonincreasing, the response ratio s^M_j/s^C_j is nondecreasing in maturity.

Step 4 is an averaging argument. For any positive e_j and nondecreasing ratios $q_j = a_j/e_j$, the ratio weighted by rate persistence is

$$\bar{q}_T = \frac{\sum_{j \le T} \rho^{T-j} a_j}{\sum_{j \le T} \rho^{T-j} e_j}$$

is nondecreasing, because it is a positive-weight average of $q_1, \dots, q_T$, and

$$\bar{q}_{T+1} - \bar{q}_T = \frac{e_{T+1}(q_{T+1} - \bar{q}_T)}{\rho \sum_{j \le T} \rho^{T-j} e_j + e_{T+1}} \ge 0.$$

Applying this with $a_j = s^M_j$, $e_j = s^C_j$ and (75), which gives these weights in A^M_T/A^C_T, proves the loading-ratio ordering, starting from ξ_H.

Step 5 gives the limits and the treasury reference. Extending the same branch recursions to arbitrarily long maturities with fixed d and eventual joint participation gives the limits $A^M_\infty = \xi^*/(1-\rho)$ and $A^C_\infty = 1/(1-\rho+b)$, hence the endpoint in (59). This extends the branch and asserts no existence for an infinite-supply economy. Taking $e_j = 1$, for which (M) suffices, proves by the same argument that A^M_T/G_T is nondecreasing, where $G_T = \sum_{j \le T} \rho^{T-j}$ is the treasury-reference loading of Appendix A.5. Thus the same ordering holds for the empirical treasury comparison of Section 6.3.

Part (ii) of Proposition 4 rests on the following bridge.

Proposition 12 (Exact forward-ratio bridge)**.** *For any price functions, affine or nonlinear, let* $f^a_T(z) = -p^a_T(z) + p^a_{T-1}(z)$. *If* $f^C_T(z) > 0$ *and* $f^M_T(z)/f^C_T(z)$ *is nondecreasing over the maturities considered, then* $y^M_T(z)/y^C_T(z)$ *is nondecreasing and* $\tau^{Y,\text{impl}}_T(z)$ *is nonincreasing.*

Proof. Telescoping gives $Ty_T^a = -p_T^a = \sum_{j \le T} f_j^a$. The yield ratio averages forward ratios with positive weights $f_j^C > 0$. Adding a ratio at least as large as its predecessors cannot lower the average. On a two-year grid, T counts steps and annualization cancels. □

The premise concerns full forwards, including bond-specific and convexity terms, so loading monotonicity alone does not establish it. The nonlinear maturity profiles are checked numerically in Section 5.

Affine forward levels and Proposition 4(iii). For coordinate $j = (a, T)$, put $\Gamma_T^a := \sum_{i \in \mathcal{U}_j} m_i/\gamma_i$ and

$$\bar{\xi}_T^a = \bar{\xi}_j = \frac{\sum_i (m_i/\gamma_i)\xi_{ij}}{\Gamma_T^a}, \quad \pi_T^a = \frac{\sigma_r^2}{\Gamma_T^a} \sum_i m_i \widehat{D}_i(0), \quad \iota_T^a = v_j \left[\frac{\sum_i m_i \xi_i^a x_{i,T}^a(0)}{\Gamma_T^a} - \tfrac{1}{2} \right],$$

where sums run over active holders. Here π_T^a is the holders' cost of rate risk per unit of loading and ι_T^a their compensation for bond-specific risk net of its convexity term. Aggregating their level KKT equations at $z = 0$ and subtracting the two Jensen terms in the exact Gaussian mean gives

$$C_T^a - C_{T-1}^a = \bar{\xi}_T^a \bar{r} + A_{T-1}^a \pi_T^a - \tfrac{1}{2}\sigma_r^2 (A_{T-1}^a)^2 + \iota_T^a. \tag{88}$$

For municipal bonds the weighted holding sum is Q_T^M. For taxable bonds heterogeneous after-tax shares remain inside it. The increment $C_T^a - C_{T-1}^a$ is the forward rate at $z = 0$. If the displayed coefficients and loadings converge and the limiting taxable forward f_∞^C is positive, Cesàro convergence gives

$$y_T^a(z) \longrightarrow f_\infty^a = \bar{\xi}_\infty^a \bar{r} + A_\infty^a \pi_\infty^a - \tfrac{1}{2}\sigma_r^2 (A_\infty^a)^2 + \iota_\infty^a \tag{89}$$

at fixed z, because C_T^a/T converges to the forward limit and $A_T^a z/T \to 0$. At jointly held long maturities, $\bar{\xi}_\infty^M = \xi^*$, and $\bar{\xi}_\infty^C = 1$ because relative after-tax shares on taxable bonds equal one, so the implied-tax limit $1 - f_\infty^M/f_\infty^C$ is (60). This proves part (iii).

For the first-order approximation, aggregate by risk tolerance over both investors for long municipal bonds and over the lower-tax investor alone for long taxable bonds:

$$\pi_\infty^M - \pi_\infty^C = \omega_H \, \sigma_r^2 \, \widehat{\Lambda}, \qquad \widehat{\Lambda} = \gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I, \tag{90}$$

where $\widehat{D}_i = \widehat{D}_i(0)$ is stationary-mean net rate exposure. The premium difference alone cannot bound other level corrections. With small taxable loading feedback ($b \ll 1 - \rho$), negligible convexity and bond-specific terms and a positive denominator, substituting $A_\infty^M = \xi^*/(1-\rho)$, $A_\infty^C \approx 1/(1-\rho)$ and (90) into (89) gives the first-order endpoint

$$\tau_\infty^{Y,\text{impl,aff}} \approx \underbrace{(1 - \xi^*)}_{\text{clientele-mix tax}} - \underbrace{\frac{\xi^* \, \omega_H \, \sigma_r^2 \, \widehat{\Lambda}}{(1-\rho)\bar{r} + \pi_\infty^C}}_{\text{muni term-premium excess}}. \tag{91}$$

With $\widehat{\Lambda} > 0$, this first-order endpoint therefore lies below one minus the tax mix, $1 - \xi^*$, and hence below the one-period implied tax $1 - \xi_H$.

B.8 Tax pass-through

Pass-throughs (Appendix A.8) are the derivatives of y_T^M/y_T^C and A_T^M/A_T^C with respect to ξ_i.
Proof of Proposition 6.

(i) The riskless short end. The one-period claims have zero variance. With slack household funding, Proposition 8 gives $y_1^M = \xi_H r$ and $y_1^C = r$ at every rate state with $r > 0$. The one-period loadings equal the responses $s_1^M = \xi_H$ and $s_1^C = 1$ by (38), so $y_1^M/y_1^C = A_1^M/A_1^C = \xi_H$. Differentiating the two ratios gives $\mathcal{P}_H^Y(1) = \mathcal{P}_H^A(1) = 1$ and $\mathcal{P}_I^Y(1) = \mathcal{P}_I^A(1) = 0$.

(ii) The long end. Let d be the household's exposure response in (83) and $b = \gamma_I \sigma_r^2 (m_H/m_I) d$ the feedback into taxable responses, as in (84). The solved long responses and loadings satisfy

$$s_\infty^M = \xi^*, \qquad s_\infty^C = \frac{1-\rho}{1-\rho+b}, \qquad A_\infty^a = \frac{s_\infty^a}{1-\rho}.$$

Thus $R_\infty = A_\infty^M / A_\infty^C = \xi^* / s_\infty^C$. At the mean state, the solved yield formula (89) gives

$$\begin{aligned} y_\infty^M &= \xi^* \bar{r} + A_\infty^M \pi_\infty^M - \tfrac{1}{2}\sigma_r^2 (A_\infty^M)^2 + \iota_\infty^M, \\ y_\infty^C &= \bar{r} + A_\infty^C \pi_\infty^C - \tfrac{1}{2}\sigma_r^2 (A_\infty^C)^2 + \iota_\infty^C. \end{aligned}$$

The four terms are interest income, portfolio rate-risk compensation, convexity and bond-specific compensation. Dividing the two yields separates the loading ratio from the remaining yield-level difference:

$$\frac{y_\infty^M}{y_\infty^C} = R_\infty + \frac{y_\infty^M - R_\infty y_\infty^C}{y_\infty^C}.$$

All derivatives below are with respect to ξ_i, with holdings, loadings and both yields adjusting. The assumed convergence permits differentiation of the limiting ratio. The quotient rule gives

$$\begin{aligned} \mathcal{P}_i^Y(\infty) = \mathcal{P}_i^A(\infty) &+ \frac{1}{y_\infty^C} \frac{d(y_\infty^M - R_\infty y_\infty^C)}{d\xi_i} \\ &- \frac{y_\infty^M - R_\infty y_\infty^C}{(y_\infty^C)^2} \frac{dy_\infty^C}{d\xi_i}. \end{aligned} \tag{92}$$

The three terms are the loading-ratio response, the response of relative yield compensation and the response of the taxable denominator. The first retains the endogenous taxable response:

$$\mathcal{P}_i^A(\infty) = \frac{\omega_i}{s_\infty^C} - \frac{\xi^*}{(s_\infty^C)^2} \frac{ds_\infty^C}{d\xi_i}.$$

Lemma 1 below proves $\mathcal{P}_H^A(\infty) < \omega_H$ and $\mathcal{P}_I^A(\infty) > \omega_I$ under the tax-difference condition, with uniform positive margins.

The calculation after the lemma shows that $y_\infty^M - R_\infty y_\infty^C$ and its tax derivatives are independent of $\bar{r}$. Also, $y_\infty^C = A_\infty^C(1-\rho)\bar{r} + O(1)$ and $dy_\infty^C/d\xi_i = O(\bar{r})$. Both correction terms in (92) therefore vanish at rate $1/\bar{r}$. The loading and derivative bounds in the lemma are uniform over all feasible cutoffs, so a lower bound on $\bar{r}$ that depends only on the other primitives makes both yields positive and keeps the corrections smaller than the loading margins. This proves $\mathcal{P}_H^Y(\infty) < \omega_H < 1$ and $\mathcal{P}_I^Y(\infty) > \omega_I > 0$.

(iii) The yield-response ratios. At a common maturity, dividing both price loadings by T leaves their ratio unchanged. Proposition 4(i) gives the ordering and endpoints for the equilibrium taxable market C: the ratio starts at ξ_H and converges to ξ^*/s_∞^C. For the unit-response reference of Appendix A.5, $A_T^{\mathrm{Tsy}} = G_T$, with $A_1^{\mathrm{Tsy}} = 1$ and $A_\infty^{\mathrm{Tsy}} = 1/(1-\rho)$. Its ratio therefore starts at ξ_H and converges to ξ^*. Appendix B.7 proves monotonicity for both comparisons. □

Lemma 1 (The loading response)**.** *Under the tax-difference condition in Proposition 6, $\mathcal{P}_H^A(\infty) < \omega_H$ and $\mathcal{P}_I^A(\infty) > \omega_I$. With other primitives fixed, both strict margins have positive lower bounds over all feasible upper-interval cutoffs.*

Proof. For the finite loading calculation, put $n = T^* - 1$ and $\lambda = \gamma_H \sigma_r^2 d$, the response of the household's unit cost of rate risk to the rate state. Equation (83) gives

$$\lambda = \omega_I(\xi_I - \xi_H)\sigma_r^2 f, \qquad f = \frac{P_J}{1+\sigma_r^2 S_J}, \qquad R_\infty = \xi^* \left[1 + \frac{\omega_H \lambda}{\omega_I(1-\rho)}\right].$$

The sums P_J and S_J are those of Proposition 10. With $G_n(p) = \sum_{q=0}^{n-1} p^q$, the loading recursion gives

$$A_n^M = \xi_H G_n(\rho+\lambda), \qquad A_{n+l}^M = \rho^l A_n^M + (1-\rho^l)A_\infty^M, \quad l \geq 0.$$

By (87), $A_n^M < A_\infty^M$, so loadings rise along the jointly held block. Thus $S_J/P_J \geq A_n^M$ and

$$\lambda G_n(\rho+\lambda) < \omega_I e, \qquad \lambda \frac{G_n'(\rho+\lambda)}{G_n(\rho+\lambda)} < \omega_I e,$$

where e is the relative tax difference of Appendix Proposition 6. The first bound uses $\lambda < \omega_I(\xi_I - \xi_H)P_J/S_J$, and the second follows from $G_n' \leq G_n^2$. The tax-difference condition implies $e < 1$ and hence $b = (\omega_H/\omega_I)\lambda < \omega_H e < 1$, ensuring convergence of the taxable loadings.

To differentiate the loading sums, let $t_j = \rho^l A_n^M / A_{j-1}^M$ for $j = T^* + l$ and

$$X = \bar{t}_P - \frac{2\sigma_r^2 S_J}{1+\sigma_r^2 S_J}\bar{t}_S.$$

Here $\bar{t}_P$ and $\bar{t}_S$ are the means of t_j with weights proportional to A_{j-1}^M/v_j^M and $(A_{j-1}^M)^2/v_j^M$, respectively. Since $0 \leq t_j \leq 1$, $-2 < X \leq 1$. Let ∂_i denote differentiation with respect to ξ_i holding λ fixed. Differentiating the loadings gives

$$\partial_H \ln f = \left(\frac{1}{\xi_H} - \frac{\omega_H}{\xi^*}\right) X + \frac{\omega_H}{\xi^*}\frac{1-\sigma_r^2 S_J}{1+\sigma_r^2 S_J},$$
$$\partial_I \ln f = \frac{\omega_I}{\xi^*}\left(\frac{1-\sigma_r^2 S_J}{1+\sigma_r^2 S_J} - X\right).$$

Hereafter G_n and G_n' are evaluated at $\rho+\lambda$. Differentiating the fixed point for λ then gives

$$\frac{d\lambda}{d\xi_i} = \frac{\lambda\,\partial_i \ln[(\xi_I - \xi_H)f]}{D}, \qquad D = 1 - \lambda\frac{G_n'}{G_n}X > 1 - \omega_I e > 0.$$

Substitute into $dR_\infty/d\xi_i = \mathcal{P}_i^A(\infty)$. The coefficient of X is positive in the household derivative and negative in the lower-tax derivative. Using $X \leq 1$ and $\lambda G_n'/G_n < \omega_I e$ gives

$$\mathcal{P}_H^A(\infty) - \omega_H \leq \frac{\omega_H \lambda}{\omega_I(1-\rho)D}\left(\frac{2\omega_H}{1+\sigma_r^2 S_J} - \frac{1}{e} + \omega_I e\right) < 0,$$
$$\mathcal{P}_I^A(\infty) - \omega_I > \frac{\omega_H \lambda}{\omega_I(1-\rho)D}(1/e - \omega_I^2 e) > 0.$$

The tax-difference condition makes the household bracket at most $-2\omega_H\sigma_r^2 S_J/(1+\sigma_r^2 S_J)$, giving a positive margin for each investor.

For uniformity, consider $n \in \{1, \ldots, T_{\max}-1\}$ in the finite loading system and trial feedbacks $x \geq 0$ with $xG_n(\rho+x) \leq \omega_I e$. This compact primitive domain contains every actual pair. Its positive loading sums give positive minima for f and $\sigma_r^2 S_J/(1+\sigma_r^2 S_J)$, while $D \geq 1 - \omega_I e$ keeps full tax derivatives bounded. At a fixed point, $\lambda \geq \omega_I(\xi_I - \xi_H)\sigma_r^2 \min f > 0$ and $D < 1 + 2\omega_I e$. The displayed margins consequently have uniform positive lower bounds. □

The relative yield compensation. Subtract the two solved long yields, weighting the taxable yield by $R_\infty = A_\infty^M/A_\infty^C$. Equation (90) gives $\pi_\infty^M - \pi_\infty^C = \omega_H\sigma_r^2\widehat{\Lambda}$, while $\xi^* - R_\infty = -bA_\infty^M$. Therefore

$$y_\infty^M - R_\infty y_\infty^C = A_\infty^M(\omega_H\sigma_r^2\widehat{\Lambda} - b\bar{r}) + \tfrac{1}{2}\sigma_r^2 A_\infty^M(A_\infty^C - A_\infty^M) + \iota_\infty^M - R_\infty\iota_\infty^C.$$

The three terms are the difference in risk costs on the books each investor holds alone, relative convexity and bond-specific compensation. Subtract the lower-tax investor's municipal equality from the household's on the shared block and use clearing:

$$\omega_H \sigma_r^2 \widehat{\Lambda} - b\bar{r} = \frac{\omega_H \sigma_r^2}{1 + \sigma_r^2 S_J} \left[\gamma_H \left(\sum_{T \notin J} \frac{A_{T-1}^M Q_T^M}{m_H} - \ell_H \right) - \gamma_I \left(\frac{\xi_I}{m_I} \sum_T A_{T-1}^C Q_T^C - \ell_I \right) \right].$$

Every term is independent of $\bar{r}$ on the maintained branch, including its tax derivative. Common duration compensation per unit of loading cancels from $y_\infty^M - R_\infty y_\infty^C$.

The joint holders' risk-premium coefficient is also independent of $\bar{r}$. With $\Gamma := m_H/\gamma_H + m_I/\gamma_I$, the value of Γ_T^M on the jointly held block, clearing gives

$$\pi_\infty^M = \frac{\sigma_r^2}{\Gamma} \left(\sum_T A_{T-1}^M Q_T^M + \xi_I \sum_T A_{T-1}^C Q_T^C - m_H \ell_H - m_I \ell_I \right).$$

Since $1/A_\infty^C = 1 - \rho + b$, dividing the taxable yield by its loading cancels $b\bar{r}$ against the mean-rate term in π_∞^C, which by (90) and the two previous displays is $-b\bar{r}$. This establishes $y_\infty^C = A_\infty^C(1-\rho)\bar{r} + O(1)$. The bond-specific terms are $\iota_\infty^M = v_\infty^M(Q_\infty^M/\Gamma - \frac{1}{2})$ and $\iota_\infty^C = v_\infty^C(\gamma_I \xi_I Q_\infty^C/m_I - \frac{1}{2})$.

Convexity in the long-end correction. For the illustration (Appendix B.9), dividing by y_∞^C isolates convexity:

$$\Delta\tau_\infty := \frac{y_\infty^M}{y_\infty^C} - R_\infty = R_\infty(1-R_\infty)\chi_\infty + \psi_\infty,$$
$$\chi_\infty = \frac{\frac{1}{2}\sigma_r^2 (A_\infty^C)^2}{y_\infty^C}, \qquad \psi_\infty = \frac{A_\infty^M(\omega_H \sigma_r^2 \widehat{\Lambda} - b\bar{r}) + \iota_\infty^M - R_\infty \iota_\infty^C}{y_\infty^C}.$$

The signed convexity contribution is at most $\chi_\infty/4$, since $R_\infty(1-R_\infty) \le 1/4$. The derivatives of the two terms sum to $\mathcal{P}_i^Y(\infty) - \mathcal{P}_i^A(\infty)$ in (92).

The tax-mix pass-through across maturities. The following supporting result supplies the intermediate-maturity profile used in Section 4.6 and Appendix B.10.

Proposition 13 (Tax-mix pass-through)**.** *Switch off exposure-response terms in the affine pricing recursions, with support and risk-tolerance weights fixed. Put*

$$G_T = \sum_{j=1}^T \rho^{T-j}, \qquad \Theta_S(T) = \frac{\sum_{j=1}^{\min(T,T^*-1)} \rho^{T-j}}{G_T}, \qquad \Theta_L(T) = 1 - \Theta_S(T).$$

For the loading-implied tax τ_T^A of Appendix A.8, the tax-mix pass-throughs $\mathcal{D}_i(T) := -\partial_{\xi_i} \tau_T^A$, the part of $\mathcal{P}_i^A(T)$ that remains with exposure responses switched off, are

$$\mathcal{D}_H(T) = \Theta_S(T) + \omega_H \Theta_L(T), \qquad \mathcal{D}_I(T) = \omega_I \Theta_L(T). \tag{93}$$

They equal $(1,0)$ before participation and converge monotonically to (ω_H, ω_I).

Proof. The channel-off recursions give $A_T^C = G_T$ and

$$A_T^M = \xi_H \sum_{j=1}^{\min(T,T^*-1)} \rho^{T-j} + \xi^* \sum_{j=T^*}^T \rho^{T-j}.$$

Differentiating A_T^M/G_T gives (93). For $T \geq T^*$, $\Theta_S(T+1) = \Theta_S(T)\rho G_T/(\rho G_T+1) < \Theta_S(T)$ and $\Theta_S(T) \to 0$. □

The pass-throughs average $(1,0)$ and (ω_H, ω_I), using the shares of household-only and jointly held maturities in the bond's remaining life weighted by rate persistence.

Price derivatives in the nonlinear model. On a differentiable nonlinear equilibrium branch, let θ be a tax parameter, such as an after-tax share ξ_i, and put $q_T^a = \partial_\theta p_T^a$ and $\upsilon_T^a = -\partial_\theta \mu_T^a$, including portfolio and covariance responses. Hold the state process, cash rule and bond-specific variances fixed. Under a common integrable envelope for the exponential moments and their derivatives, differentiation of the price recursion (66) gives

$$q_T^a = \mathcal{K}_T^a q_{T-1}^a + \upsilon_T^a, \qquad q_0^a = 0, \qquad (\mathcal{K}_T^a \psi)(z) = \mathbb{E}^{\mathbb{Q}_{T,z}^a} \psi(Z'), \tag{94}$$

where F_z is the conditional law of Z' and $d\mathbb{Q}_{T,z}^a/dF_z = e^{p_{T-1}^a(Z')}/\mathbb{E}_z e^{p_{T-1}^a(Z')}$. The identity follows from $\partial_\theta \log \mathbb{E}_z e^{p_{T-1}^a} = \mathbb{E}^{\mathbb{Q}_{T,z}^a} q_{T-1}^a$. The full implied-tax response is

$$\partial_\theta \tau_T^{Y,\text{impl}} = \frac{q_T^M - (y_T^M/y_T^C) q_T^C}{T y_T^C}, \qquad y_T^C > 0. \tag{95}$$

Both price responses therefore enter the implied tax measured in the data.

B.9 The thirty-maturity illustration

This subsection verifies the two-investor illustration of Section 4.

Figures 4 and 6 use $\rho = 0.8$, $\sigma_r = 0.025$, $(\gamma_H, \gamma_I) = (8, 5)$, $(\xi_H, \xi_I) = (0.55, 0.78)$, equal investor masses and $T^* = 17$ on a thirty-period ladder. For $T \geq 2$, the municipal bond-specific standard deviation is $0.004[1 + 7\sqrt{(T-1)/29}]$. The one-period claim has zero variance. The exposure fixed point in (83) includes only the jointly held block's variances.

A unique loading solution. For every $d \geq 0$, the predecessor loadings on the joint block are at least $a_{\min} = \xi_H \sum_{q=0}^{15} \rho^q$, so $P_J/S_J \leq 1/a_{\min}$ and the continuous map $\mathcal{F}_J$ takes $[0, \infty)$ into $[0, d_{\max}]$ for

$$d_{\max} = \frac{\xi_I - \xi_H}{(\gamma_H + \gamma_I)\sigma_r^2 a_{\min}} < 10.591841.$$

Outward-rounded forty-digit decimal evaluation of $\mathcal{F}_J'$ (Appendix B.6) on 20,000 interval cells gives $\sup |\mathcal{F}_J'| < 0.080131$. Since this bound is below one, $\mathcal{F}_J$ has a unique fixed point d^*. Directed endpoint evaluations bracket it:

$$8.310258726885486 < d^* < 8.310258733114512.$$

The resulting A_{16}^M lies between 3.251488563972084 and 3.251488564481132, below $\xi^*/(1-\rho) = 3.457692307692308$, verifying condition (M) in (87). Condition (S) in (56) has left side $0.16084 < \rho$.

The three rate-loading contributions. Figure 6(b) accumulates the tax-mix, municipal portfolio and taxable portfolio responses of Section 4.5. With $\bar{\xi}_j = \xi_H$ below T^* and $\bar{\xi}_j = \xi^*$ from T^* onward, iterating the loading recursion gives

$$A_T^M - \xi_H A_T^C = \sum_{j=1}^{T} \rho^{T-j} \left[(\bar{\xi}_j - \xi_H) + (s_j^M - \bar{\xi}_j) + \xi_H (1 - s_j^C) \right].$$

The three terms in brackets give the orange, blue and green contributions. Their sum is the loading gap plotted in black. At maturity T, the return-gap decomposition uses this sum at $T-1$ multiplied by $\gamma_I \sigma_r^2 \widehat{D}_I$.

Levels and the KKT conditions. The level solution uses $\bar{r} = 0.04$, unit masses, $Q_T^M = 0.1$, $Q_T^C = 0.01$ for $T \geq 2$, no taxable bond-specific risk, $\ell_I = 0$ and $\ell_H = 2.985$, chosen so that the duration gap $\widehat{\Lambda} = 4.319$ sits at the midpoint of $(4.285, 4.353]$, the interval that selects $T^* = 17$ under Theorem 1. The threshold is therefore set for the illustration and then verified, and the exposure fixed point is solved given it. At $z = 0$, closed-form holdings clear both markets. Active KKT errors are below 10^{-15}. Inactive municipal slacks exceed 6.8×10^{-5} and household taxable slacks exceed 0.0026.

The return gap. Figure 4 reports supply shares. Figure 5 decomposes the one-period return gap into the three channels of (48), separating the tax-mix and portfolio-risk contributions to its rate-loading term. The identity error is below 10^{-15}. At $T = 16$, the last household-only maturity, the return gap of 181 basis points (bp) comprises 87 bp of duration cost, 89 of rate loading and 5 of bond-specific compensation. At $T = 30$ the gap of 205 bp comprises 92 of after-tax cash return, 109 of rate loading (72 from the tax mix and 37 from portfolio-risk responses) and 4 of bond-specific compensation.

Pass-through across maturities. Full yield-implied-tax derivatives re-solve the exposure fixed point, loadings and holdings, with supplies, masses and outside exposures fixed. Over thirty maturities, household pass-through falls from 1 to 0.24, and the lower-tax investor's pass-through rises from 0 to 0.73, including 0.06 at $T = 2$, which it does not hold. Between adjacent maturities, household pass-through falls by at least 0.0088 and the lower-tax investor's rises by at least 0.0083. At every maturity, $\mathcal{P}_H^Y(T) + \mathcal{P}_I^Y(T)$ lies between 0.97 and 1.00. Outward-rounded interval evaluation of the analytic derivatives verifies every adjacent comparison, positive holdings and inactive slacks, with $T^* = 17$ unchanged.

The long end. The long-end loadings are $A_\infty^M = 3.458$ and $A_\infty^C = 4.425$, with long taxable response $s_\infty^{C,\text{aff}} = 0.885$. The risk-premium coefficients are $\pi_\infty^M = 0.0118$ and $\pi_\infty^C = 0.0108$, and the long taxable yield is $y_\infty^C = 0.0816$. Loading-implied pass-throughs are $(\mathcal{P}_H^A, \mathcal{P}_I^A) = (-0.012, 1.012)$ and yield-implied pass-throughs $(-0.008, 0.956)$. The tax-difference condition of Appendix Proposition 6 has left side $0.429 < 1$. In the decomposition of Appendix B.8, convexity absorbs $\chi_\infty = 0.0750$ of the taxable yield, with $R_\infty(1 - R_\infty)\chi_\infty = 0.0128$, $\psi_\infty = -0.0024$ and $\Delta\tau_\infty = 0.0104$. The derivatives of the convexity term and of ψ_∞ contribute -0.057 and 0.002 to the lower-tax investor's pass-through and -0.001 and 0.006 to household pass-through, preserving the comparisons with $(\omega_H, \omega_I) = (0.385, 0.615)$.

Stronger hedging demand and lower rates. Raising $\gamma_H \ell_H$ and $\gamma_I \ell_I$ equally lowers both risk-premium coefficients without changing holdings, participation or $\widehat{\Lambda}$. Both endpoint comparisons hold for every nonnegative π_∞^C. At $\pi_\infty^C = 0$, convexity absorbs 0.18 of the taxable yield and contributes -0.168 to the lower-tax investor's pass-through, and yield-implied pass-throughs become $(-0.027, 0.853)$. The lower-tax investor's pass-through falls below ω_I only when hedging demand pushes π_∞^C below zero and lowers the long taxable yield to 0.017, against 0.0816 at baseline, or when, with $\pi_\infty^C = 0$, a fall in the mean rate lowers it to 0.0165.

Equilibrium selection. Enumerated affine configurations are retained only when their linear level KKT-and-clearing system has a unique feasible solution. The uniqueness check applies to each configuration separately, and other feasible cutoffs remain possible, as noted in Appendix B.6. Full-state nonlinear solutions pass the independent funded-grid checks in Appendix C.3, and their maturity patterns are numerical results.

B.10 State-specific exemptions and multiple municipal states

A household resident in state s' pays her home state's income tax on interest from a municipal bond issued by another state $s \neq s'$ but not on a home-state municipal bond. The quantitative institutional types have no modeled state-income-tax wedge on either, so state taxation is a household-specific channel (Appendix E.3). Statutory top state rates vary across states, with a supply-weighted mean of 5.8%. This cross-state variation is both a calibration input and P1's source of identification.

Multi-state aggregation. The multi-state model replaces ξ_i with $\xi_{i,s} = \xi_i^C/\xi_{i,s}^M$, which depends on the issuing state s and on the investor, whose type includes a household's state of residence. For state s's market M_s, put $\Gamma_T^{M_s} := \sum_{i\in\mathcal{U}_T^{M_s}} m_i/\gamma_i$ and $\omega_{i,T}^s := (m_i/\gamma_i)/\Gamma_T^{M_s}$, where $\mathcal{U}_T^{M_s}$ is the set of holders of the maturity-T bond. The exact KKT aggregation (79) applies state by state, with $\xi_{ij} = \xi_{i,s}$. Applying (81) coordinate by coordinate gives, in the affine construction,

$$\begin{aligned} s_T^{M_s} = &\sum_{i\in\mathcal{U}_T^{M_s}} \omega_{i,T}^s \xi_{i,s} + \frac{\sigma_r^2 A_{T-1}^{M_s}}{\Gamma_T^{M_s}} \sum_{i\in\mathcal{U}_T^{M_s}} m_i \partial_z \widehat{D}_i \\ &+ \frac{v_{M_s,T}}{\Gamma_T^{M_s}} \sum_{i\in\mathcal{U}_T^{M_s}} m_i \xi_{i,s}^M \partial_z x_{i,M_s}(T). \end{aligned} \tag{96}$$

As for the taxable counterpart before (82), the final term vanishes under common active municipal after-tax shares, a singleton active set or zero bond-specific variance. Unweighted clearing alone does not remove it.

Proposition 14 is the state-tax counterpart of Proposition 13. Its implied state tax is the model counterpart of (24), and T_s^* is the first maturity at which investors other than the resident household hold state s's bonds.

Proposition 14 (State-tax anchor and tax-mix pass-through)**.** *If the household H_s resident in state s uniquely minimizes $\xi_{i,s}$ and the one-period municipal boundary is deterministic with $r > 0$, then $y_1^{M_s}/y_1^C = \xi_{H_s,s}$. Under the multiplicative mapping $\xi_{H_s,s} = \xi^{\text{fed}}(1-\tau_s^{\text{st}})$, with federal after-tax share ξ^{fed} and state income-tax rate τ_s^{st}, define*

$$\tau_{s,T}^{\text{st,impl}} = 1 - \frac{y_T^{M_s}/y_T^C}{\xi^{\text{fed}}}, \qquad \tau_{s,T}^{A,\text{st}} = 1 - \frac{A_T^{M_s}/A_T^C}{\xi^{\text{fed}}}. \tag{97}$$

Then $\tau_{s,1}^{\text{st,impl}} = \tau_s^{\text{st}}$, with unit pass-through. In the affine channel-off decomposition, if only the resident household's effective after-tax share changes and its active weight is one below T_s^* and constant at $\omega_{H_s}^s$ thereafter, the tax-mix derivative of the loading-implied state tax $\tau_{s,T}^{A,\text{st}}$ is $\Theta_S^s(T) + \omega_{H_s}^s \Theta_L^s(T)$, where Θ_S^s, Θ_L^s are the weights in (93) with T_s^* in place of T^*.

Proof. The boundary KKT minimizes $\xi_{i,s} r$ (Proposition 8). Equation (97) gives the anchor. Applying (93) with T_s^* in place of T^* gives the tax-mix derivative, because $\partial\xi_{H_s,s}/\partial\tau_s^{\text{st}} = -\xi^{\text{fed}}$ cancels the factor $1/\xi^{\text{fed}}$ in (97). □

Changes to out-of-state after-tax shares or other clienteles add their derivatives. Other statutory mappings require a corresponding implied-tax transformation. The primitive boundary remains $y_1^{M_s}/y_1^C = \xi_{H_s,s}$. Reported state-tax responses use the full multi-state model, which supplies P1's full-yield pass-through profile, including level and exposure corrections.

B.11 Participation and pass-through with many clienteles

Extend the two-investor affine benchmark of Appendix A.3 to N investor types with slack funding. Order after-tax shares $\xi_1 < \cdots < \xi_N$, so type 1 is the household with the largest exemption benefit. Without bond-specific risk, subtract type 1's active municipal equality from type $j > 1$'s condition for a nonnegative first-order purchase incentive (Appendix A.4). Type j has that incentive at maturity T when

$$\underbrace{\sigma_r^2 A_{T-1}^M \Lambda_j}_{\text{duration benefit}} \geq \underbrace{(\xi_j - \xi_1) r}_{\text{difference in exemption benefits}}, \qquad \Lambda_j := \gamma_1 \widehat{D}_1 - \gamma_j \widehat{D}_j. \tag{98}$$

At fixed net rate exposures the duration benefit grows with the bond's rate loading, while the difference in exemption benefits is flat in maturity. Positive bond-specific risk adds the incumbent's bond-specific compensation to the comparison, as in the KKT system (71) and the participation index (77).

Assumption 2 (Long-end participation, clientele by clientele)**.** *Type 1 is active at every maturity, so* $T_1^* = 1$*. For each type* $j > 1$*,* $\Lambda_j > 0$*, and its active municipal set is an upper maturity interval* $[T_j^*, T_{\max}]$ *with* $T_j^* \geq 2$*. The thresholds need not be ordered by tax advantage.*

Many-clientele tax-mix pass-through. Under the nested supports of Assumption 2, fix the support and the risk-tolerance weights, as in Proposition 13. Let $\omega_{k,j}$ be type k's share of the risk tolerance of the holders of the maturity-j municipal bond, zero when type k does not hold it. Then $A_T^C = G_T$ and $A_T^M = \sum_{j \leq T} \rho^{T-j} \sum_k \omega_{k,j} \xi_k$, so differentiating $\tau_T^A = 1 - A_T^M / G_T$ in ξ_i gives the channel-off affine derivative for type i, $-\partial \tau_T^A / \partial \xi_i$,

$$\mathcal{D}_i(T) = \frac{\sum_{j=T_i^*}^{T} \rho^{T-j} \omega_{i,j}}{\sum_{j=1}^{T} \rho^{T-j}}, \tag{99}$$

with an empty numerator equal to zero. It is positive after participation begins and converges to ω_i if the active weight is eventually constant at ω_i, because the finite pre-participation block receives vanishing weight at the long end. Participation ordered by increasing after-tax shares makes the tax mix, the clientele-weighted after-tax share, nondecreasing in maturity, because adding an after-tax share no smaller than the incumbents' after-tax shares does not lower their risk-tolerance-weighted mean. Such ordered participation does not by itself order total slopes, loading ratios or yields, since other terms in (96) and the taxable leg remain unrestricted. Clientele-specific pass-through and endpoint comparisons need no tax ordering.

Under Assumption 2, the household anchors the riskless boundary, and (99) gives each clientele's tax-mix derivative. Full yield responses also include relative compensation and convexity (Appendix A.6). The portfolio-risk response can transmit a later clientele's tax to shorter risky bonds it never holds. P1 is the always-active household case and P2 the later-participating case. Assumption 2 restricts only this affine extension. The calibration does not impose it (Section 5).

B.12 Participation motives and the municipal gap

This subsection proves the neutrality results of Appendix A.7 behind Section 4.7. Specialize the economy of Appendix A.1 to one municipal market, the high-tax household H and the lower-tax investor I, with $\xi_H < \xi_I$. The one-period return gap is $g_T = \mu_T^M - \xi_H \mu_T^C$.

B.12.1 Non-tax required returns and neutral characteristics

Add to investor i's objective in (63) a differentiable concave term $\Phi_i(x_i)$ in the same after-tax units. Dividing each KKT condition (33) by its after-tax share gives, under the normalized municipal–taxable benchmark,

$$\mu_T^M \leq \xi_i r + K_{i,T}^M, \qquad \mu_T^C \leq r + K_{i,T}^C, \qquad \text{with equality for a positive holding,} \tag{100}$$

where

$$K_{i,T}^a := \gamma_i \operatorname{Cov}_z\big(u_{T,t+1}^a, \widetilde{\Pi}_i(t+1)\big) + \frac{\beta_i}{\xi_i^a} - \frac{1}{\xi_i^a} \frac{\partial \Phi_i}{\partial x_{i,T}^a}.$$

Additive shocks split $K_{i,T}^a$ into components $k_{ij,T}^a$ for rate risk, bond-specific risk and each additional shock. Additional shocks are independent of the rate and bond-specific shocks.

A common return characteristic is a unit-variance shock q with nonnegative maturity-T exposures e_T^M, e_T^C in both markets, taxed with returns. Its component is $k_{iq,T}^a = c_{iq} e_T^a$, where the cost per unit of exposure, $c_{iq} := \gamma_i \operatorname{Cov}_z(q_{t+1}, \widetilde{\Pi}_i(t+1))$, is the same in both markets. Affine rate risk has $q = -\varepsilon$, $e_T^a = \sigma_r A_{T-1}^a$, $c_{iq} = \gamma_i \sigma_r \widehat{D}_i$ and component $\gamma_i \sigma_r^2 A_{T-1}^a \widehat{D}_i$.

A wealth characteristic changes after-tax wealth by w_T^a per dollar at investor i's intensity $\lambda_i \geq 0$, as with trading costs. This intensity is unrelated to the feedback λ in the proof of Lemma 1. For $\Phi_i = -\lambda_i \sum_{a,T} w_T^a x_{i,T}^a$, the component is $k_{iw,T}^a = \lambda_i w_T^a / \xi_i^a$. Funding similarly gives β_i / ξ_i^a.

Definition 1 (Neutral characteristic)**.** *A common return characteristic is neutral if* $e_T^M = \xi_H e_T^C$*, and a wealth characteristic if* $w_T^M = w_T^C$*, at every maturity. For a household that bears it, a neutral characteristic satisfies* $k_{H,T}^M = \xi_H k_{H,T}^C$*, so it creates no gap when the household holds both bonds.*

Neutrality for each investor. A neutral common return characteristic gives $k^M_{iq,T} = c_{iq}e^M_T = c_{iq}\xi_H e^C_T = \xi_H k^C_{iq,T}$ for every investor i. A neutral wealth characteristic gives $k^M_{iw,T} = \lambda_i w_T = \xi_i\,(\lambda_i w_T/\xi_i) = \xi_i k^C_{iw,T}$, so

$$k^M_{Iw,T} - \xi_H k^C_{Iw,T} = (\xi_I - \xi_H)k^C_{Iw,T}. \tag{101}$$

In the affine benchmark in which the household alone holds both bonds, rate risk is neutral. Clearing fixes its exposure, so differentiation gives $s^M_T = \xi_H$ and $s^C_T = 1$, and recursion (38), starting at $A^M_1 = \xi_H$, $A^C_1 = 1$, gives $A^M_{T-1} = \xi_H A^C_{T-1}$ at every maturity by induction, as in Proposition 11.

Neutrality is stated for the household because the puzzle is defined against the household benchmark. A characteristic that weighs more on municipal bonds would add a premium even there.

B.12.2 The gap without and with participation

No gap when the household holds both bonds. Suppose the household alone holds both bonds, there is no bond-specific risk and every additional characteristic is neutral and zero at the one-period claim. Both household conditions in (100) bind, so $g_T = \sum_j (k^M_{Hj,T} - \xi_H k^C_{Hj,T})$, and its funding terms β_H and β_H/ξ_H cancel. Definition 1 removes the additional components. In the affine benchmark, the neutrality of rate risk makes the rate contribution $\gamma_H\sigma_r^2\widehat{D}_H(A^M_{T-1} - \xi_H A^C_{T-1})$ zero, and with a constant cash rate there is no rate component. At riskless $T = 1$, the same equalities give $\mu^M_1 = \xi_H\mu^C_1$, reducing to $\mu^M_1 = \xi_H r$ and $\mu^C_1 = r$ when funding is slack. Hence

$$g_T = 0 \qquad \text{at every maturity.} \tag{102}$$

Participation incentive for any motive. At the household-only allocation of Appendix A.4, with components evaluated there, household-only municipal holdings give $\mu^M_T = \xi_H r + K^M_{H,T}$. The lower-tax investor's first-order gain from buying the maturity-T municipal bond, $\mu^M_T - \xi_I r - K^M_{I,T}$, is then nonnegative if and only if

$$\Delta_T := K^M_{H,T} - K^M_{I,T} \;\geq\; (\xi_I - \xi_H)\,r. \tag{103}$$

If $r > 0$, $\Delta_1 = 0$ and Δ_T is nondecreasing in T, the maturities satisfying (103) form an upper interval, possibly empty, that excludes the one-period claim. A nondecreasing Δ_T crosses the maturity-invariant tax difference at most once, and $\Delta_1 = 0 < (\xi_I - \xi_H)r$.

Rate risk gives $\Delta_T = \sigma_r^2 A^M_{T-1}\Lambda$ (Appendix Proposition 1). At that allocation, the advantages from several motives add. This comparison holds prices and portfolios fixed. Equilibrium upper-interval participation follows from Proposition 16(i) for the default and trading-cost motives and under the conditions of Theorem 1 for rate risk.

Proposition 15 (The gap when the lower-tax investor participates)**.** *In the economy of Appendix A.1 with one municipal market, suppose every additional characteristic is a neutral common return or wealth characteristic. If the lower-tax investor holds the maturity-T municipal bond at state z, then*

$$g_T \;\geq\; (\xi_I - \xi_H)\,r + g^{\text{risk}}_T + (\xi_I - \xi_H)\Big(\sum_w k^C_{Iw,T} + \frac{\beta_I}{\xi_I}\Big),$$
$$g^{\text{risk}}_T := \gamma_I\Big[\operatorname{Cov}_z\big(u^M_{T,t+1}, \widetilde{\Pi}_I(t+1)\big) - \xi_H \operatorname{Cov}_z\big(u^C_{T,t+1}, \widetilde{\Pi}_I(t+1)\big)\Big],$$

where the covariances run over the rate and bond-specific shocks, with equality if it also holds the maturity-T taxable bond. In the affine benchmark without bond-specific risk, $g^{\text{risk}}_T = \gamma_I\sigma_r^2\widehat{D}_I(A^M_{T-1} - \xi_H A^C_{T-1})$ and funding is slack. With nonnegative costs, this gives Proposition 5(ii).

Proof. The active municipal condition gives $\mu^M_T = \xi_I r + K^M_{I,T}$. Taxable KKT gives $\mu^C_T \leq r + K^C_{I,T}$, with equality when taxable holdings are positive. Subtract ξ_H times the latter:

$$g_T \;\geq\; (\xi_I - \xi_H)r + \sum_j \big(k^M_{Ij,T} - \xi_H k^C_{Ij,T}\big).$$

Rate and bond-specific components give g_T^{risk}. Neutral common-return components cancel. Neutral wealth costs give $(\xi_I - \xi_H)k^C_{Iw,T}$ by (101). Funding terms β_I, β_I/ξ_I give $(\xi_I - \xi_H)\beta_I/\xi_I$. □

The third term prices wealth costs and scarce funding at the lower-tax investor's after-tax share. Nonnegative g_T^{risk} and nonnegative costs suffice for $g_T \geq (\xi_I - \xi_H)r$.

B.12.3 Three motives that produce the long-end gap

Proposition 16 (Three motives that produce the long-end gap)**.** *Let $r > 0$ and consider one of the three cases below. Cases (b) and (c) are set in the two-investor economy with a constant cash rate, no bond-specific risk and nonbinding funding, so the stated characteristic is the only non-tax component. In each case the characteristic is zero at the one-period claim. In (b), investors have no outside exposure to the default shocks.*

(a) Liability hedging and risk-bearing capacity: the two-investor affine benchmark with $\Lambda > 0$ at the household-only allocation, without bond-specific risk for parts (i) and (iv).

(b) Default diversification: municipal default risk loads on one unit-variance shock and taxable default risk on another, with correlation $\varrho \in [0,1]$ and exposures $\xi_H e_T$ and e_T, where e_T is strictly increasing for $T \geq 1$. Write $X_i^M := \xi_H \sum_T e_T x^M_{i,T}$ and $X_i^C := \xi_i \sum_T e_T x^C_{i,T}$ for investor i's municipal and taxable default exposures, and $k^M_{id,T} := \gamma_i \xi_H e_T(X_i^M + \varrho X_i^C)$ and $k^C_{id,T} := \gamma_i e_T(X_i^C + \varrho X_i^M)$ for its default components. Assume $\gamma_H X_H^{M,\text{hh}} > \varrho\, \gamma_I X_I^{C,\text{hh}}$.

(c) Trading costs: the same round-trip cost $w_T \geq 0$ in both markets, strictly increasing for $T \geq 1$, and trading intensities $\lambda_I < \lambda_H$.

Then:

(i) the participation advantage Δ_T in (103) is zero at $T = 1$ and nondecreasing in T. In (b) and (c), the municipal maturities the lower-tax investor holds in any equilibrium form an upper interval;

(ii) $g_1 = 0$: the household anchors the one-period claim;

(iii) at every maturity at which the lower-tax investor holds the municipal bond, $g_T \geq (\xi_I - \xi_H)r + g_T^{\text{risk}}$ in (a), with g_T^{risk} of Proposition 15, $g_T \geq (\xi_I - \xi_H)r$ in (c), and in (b)

$$g_T = (\xi_I - \xi_H)r + k^M_{Id,T} - \xi_H k^C_{d,T},$$

where $k^C_{d,T}$ is the default component of any investor who holds the taxable bond, so the gap net of these default components is $(\xi_I - \xi_H)r$;

(iv) with the household alone holding both bonds, $g_T = 0$ in (a) and (c), and $g_T = k^M_{Hd,T} - \xi_H k^C_{Hd,T}$ in (b).

Proof. (i) At household-only holdings, (a) gives $\Delta_T = \sigma_r^2 A^M_{T-1}\Lambda$ with $A^M_{T-1} = \xi_H \sum_{j=0}^{T-2} \rho^j$ (Proposition 11); (b) gives $X_H^C = X_I^M = 0$ and $\Delta_T = \xi_H e_T(\gamma_H X_H^{M,\text{hh}} - \varrho\, \gamma_I X_I^{C,\text{hh}})$; (c) gives $\Delta_T = (\lambda_H - \lambda_I)w_T$. All vanish at $T = 1$ and increase weakly. In equilibrium (b), define investor i's unit municipal-default cost $c_i := \gamma_i(X_i^M + \varrho X_i^C)$. The lower-tax investor's required municipal return is no greater than the household's exactly when $\xi_H e_T(c_H - c_I) \geq (\xi_I - \xi_H)r$. If the lower-tax investor holds maturity T, this comparison holds there and $c_H > c_I$. Strictly increasing e_T makes it strict beyond T, precluding household holdings and forcing lower-tax-investor holdings. Case (c) uses $(\lambda_H - \lambda_I)w_T$ identically.

(ii) Every non-tax component vanishes at the one-period claim, so the municipal conditions reduce to $\mu_1^M \leq \xi_i r$. Positive supply gives $\mu_1^M = \xi_H r$, while $\mu_1^C = r$, as in Proposition 8.

(iii) Apply Proposition 15 to (a), with no additional characteristic, and (c), with $g_T^{\text{risk}} = 0$ and cost $(\xi_I - \xi_H)\lambda_I w_T/\xi_I \geq 0$. In (b), active conditions give $\mu_T^M = \xi_I r + k^M_{Id,T}$ and $\mu_T^C = r + k^C_{d,T}$. With a constant rate, every taxable holder has the same default component. Subtraction proves the expression.

(iv) In (a) and (c) this is (102). In (b), the household's conditions give $\mu_T^M = \xi_H r + k^M_{Hd,T}$ and $\mu_T^C = r + k^C_{Hd,T}$. □

In (a), Λ is positive from liability hedges or lower duration costs, including lower risk aversion. In (b), households bear all municipal default risk at the household-only allocation, while the lower-tax investor's taxable book is exposed to it only through ϱ, so the lower-tax investor can add it more cheaply. Independent

shocks guarantee this advantage. Raw gaps also contain holder-specific default-premium differences driven by relative risk supplies. Appendix D.1.6 adjusts these with credit default swap (CDS) costs. In (c), trading costs reduce after-tax wealth (Appendix D.1.5). For matched long bonds at 26 to 30 years, institutional round-trip costs are 0.53% of price in both markets (Table 22), supporting equal costs. With combined motives, the decomposition of Proposition 15 still applies. Its tax-difference bound requires the remaining risk term g_T^{risk} to be nonnegative at the resulting equilibrium, and separate default shocks are treated as in (b).

B.12.4 Two solved examples without a duration motive

Consider two investors and three maturities with constant $r > 0$ and no bond-specific risk, eliminating rate-risk advantages. Both investors have risk aversion γ, masses m_H, m_I, after-tax shares $\xi_H < \xi_I$ and slack funding. Supplies are Q_T^M for $T = 1, 2, 3$ and Q_T^C for $T = 2, 3$. Write $M1$, $M2$, $M3$ for the municipal bonds and $C2$, $C3$ for the risky taxable bonds, indexed by maturity. Each example adds only its stated non-tax characteristic.

Proposition 17 (Participation for diversification or lower trading costs)**.** *Consider the economy just described.*

(a) Default diversification. Municipal and taxable default risk load on independent unit-variance shocks, with exposures $\xi_H e_T$ and e_T and $0 = e_1 < e_2 < e_3$. Let $X_H^M := \xi_H e_2 Q_2^M / m_H$ and $X_I^M := \xi_H e_3 Q_3^M / m_I$ be the investors' municipal default exposures when the household holds $M1$ and $M2$ and the lower-tax investor holds $M3$, and let

$$X^C := \frac{e_2 Q_2^C + e_3 Q_3^C}{m_H/\xi_H + m_I/\xi_I}.$$

There is an equilibrium with this municipal allocation if and only if

$$\gamma\,\xi_H e_2 (X_H^M - X_I^M) \;\le\; (\xi_I - \xi_H) r \;\le\; \gamma\,\xi_H e_3 (X_H^M - X_I^M).$$

In it both investors hold taxable bonds, each with taxable default exposure X^C, and

$$g_1 = 0, \qquad g_2 = \gamma\,\xi_H e_2 (X_H^M - X^C), \qquad g_3 = (\xi_I - \xi_H) r + \gamma\,\xi_H e_3 (X_I^M - X^C).$$

Net of the default components of the investors who hold each bond, the gaps are 0, 0 and $(\xi_I - \xi_H) r$.

(b) Trading costs. Each investor pays the after-tax cost $\lambda_i w_T$ per period on either bond of maturity T, with $0 = w_1 < w_2 < w_3$ and $0 \le \lambda_I < \lambda_H$. If $(\lambda_H - \lambda_I) w_2 < (\xi_I - \xi_H) r \le (\lambda_H - \lambda_I) w_3$, the household holds $M1$ and $M2$, the lower-tax investor holds $M3$, $C2$ and $C3$, and

$$g_1 = 0, \qquad g_2 = \Big(\lambda_H - \frac{\xi_H}{\xi_I}\lambda_I\Big) w_2, \qquad g_3 = (\xi_I - \xi_H)\Big(r + \frac{\lambda_I w_3}{\xi_I}\Big),$$

with $0 < g_2 < g_3$. The ratio μ_T^M / μ_T^C equals ξ_H at $T = 1$, lies strictly between ξ_H and ξ_I at $T = 2$ and equals ξ_I at $T = 3$.

Proof. (a) Independent default shocks separate market risks. Investor i's KKT conditions are $\mu_T^M \le \xi_i r + \gamma \xi_H e_T X_i^M$ and $\mu_T^C \le r + \gamma e_T X_i^C$, for municipal and taxable exposures X_i^M, X_i^C. Lower taxable exposure implies lower required return and profitable purchases, so both investors hold taxable bonds with equal exposures. Clearing $\sum_i m_i X_i^C / \xi_i = e_2 Q_2^C + e_3 Q_3^C$ gives $X_H^C = X_I^C = X^C$ and $\mu_T^C = r + \gamma e_T X^C$. Each investor's split of its taxable exposure across the two maturities need not be unique. Supply-proportional holdings implement these exposures and clear both maturities. Active municipal conditions give $\mu_1^M = \xi_H r$, $\mu_2^M = \xi_H r + \gamma \xi_H e_2 X_H^M$ and $\mu_3^M = \xi_I r + \gamma \xi_H e_3 X_I^M$. Inactive KKT requires $\xi_H r < \xi_I r$ at $M1$, the left threshold inequality at $M2$ and the right at $M3$. Subtracting returns gives the gaps. Remove the holder-specific municipal component $\gamma \xi_H e_T X_i^M$ and the taxable component $\gamma e_T X^C$ to leave $\xi_H r - \xi_H r$ at $T = 2$ and $\xi_I r - \xi_H r$ at $T = 3$.

(b) Investor i requires $\xi_i r + \lambda_i w_T$ on municipal bonds and $r + \lambda_i w_T / \xi_i$ on taxable bonds. With slack funding, each bond's return equals the minimum eligible requirement. Since $\lambda_I / \xi_I < \lambda_H / \xi_H$, the lower-tax

investor holds all taxable bonds. It holds the maturity-T municipal bond when $\xi_I r + \lambda_I w_T \le \xi_H r + \lambda_H w_T$, here only at $T = 3$. Subtract returns to obtain the gaps and

$$g_2 = (\lambda_H - \lambda_I)w_2 + (\xi_I - \xi_H)\frac{\lambda_I w_2}{\xi_I} < (\xi_I - \xi_H)\Big(r + \frac{\lambda_I w_2}{\xi_I}\Big) \le g_3.$$

At $T = 2$ the ratio is below ξ_I because the household's required municipal return is below the lower-tax investor's. At $T = 3$ it is $(\xi_I r + \lambda_I w_3)/(r + \lambda_I w_3/\xi_I) = \xi_I$. □

Case (a) implements default diversification (Proposition 16), requiring $X_H^M > X_I^M$. The intermediate raw gap $\gamma\xi_H e_2(X_H^M - X^C)$ is a default-premium difference. With $\varrho \in (0, 1)$, the same allocation, where it is an equilibrium, has default-net gaps 0, 0, $(\xi_I - \xi_H)r$ (Proposition 16(b)). Case (b)'s intermediate gap reflects higher household trading costs, independent of other holdings.

Examples with rate risk. Adding the affine rate process $z' = \rho z + \sigma_r \varepsilon$, the examples use $\xi_H = 0.6$, $\xi_I = 0.8$, $\rho = 0.8$, $\bar{r} = 4\%$ and small municipal bond-specific variance, with $\Lambda = 0$ at household-only holdings to remove the initial rate-risk advantage. For each motive, 120 economies with other parameters drawn randomly, including $\varrho \in [0, 0.9]$, are solved and screened for household-only participation incentives. With trading costs, the lower-tax investor holds only the longest municipal bond, with $0 = g_1 < g_2 \le g_3$ and $g_3 \ge (\xi_I - \xi_H)r$, in all 120. With default diversification, it does so in 72 economies and holds no municipal bond in 48. In the 72, the default-net gap is the tax difference plus Proposition 15's rate and bond-specific terms. The bond-specific term is below 0.01 bp. The default-net long gap is at least the tax difference in 68 economies. In the other four, negative rate terms lower it below the tax difference by at most 0.09 bp. Rate terms across both motives span −0.09 to 1.7 bp.

B.12.5 A municipal-specific benefit

A municipal-only benefit $\zeta_T > 0$, from a mandate, issuer relationship or favorable regulation at matched maturity and credit, violates neutrality, because a household that had it would accept municipal returns below the benchmark. For the lower-tax investor, it lowers the gap by ζ_T and gives $\Delta_T = \zeta_T$, so a ζ_T that rises with maturity induces long-end participation. Unlike default compensation, this benefit has no taxable counterpart, and no adjustment removes it.

Proposition 18 (Same participation, opposite pricing)**.** *Consider the two-investor economy without rate risk or bond-specific risk, with a constant cash rate $r > 0$, nonbinding funding and no additional household characteristic. A fixed per-dollar benefit lowers the lower-tax investor's required municipal return by $\zeta_T \ge 0$, with $\zeta_1 = 0$, ζ_T nondecreasing and $\zeta_T < \xi_H r$. Compare a neutral benefit, which also lowers the investor's required taxable return by the household-equivalent ζ_T/ξ_H, with a municipal-specific benefit that leaves the required taxable return unchanged.*

(i) In both cases the lower-tax investor holds the maturity-T municipal bond only if $\zeta_T \ge (\xi_I - \xi_H)r$, so participation begins at the same maturity.

(ii) With the neutral benefit, $\mu_T^C = r - \zeta_T/\xi_H$ and

$$g_T = \min\{\zeta_T,\ (\xi_I - \xi_H)r\} \ge 0,$$

so the gap equals the full difference in exemption benefits wherever the lower-tax investor participates. The ratio μ_T^M/μ_T^C rises with ζ_T.

(iii) With the municipal-specific benefit, $\mu_T^C = r$ and

$$g_T = \min\{0,\ (\xi_I - \xi_H)r - \zeta_T\} \le 0,$$

and the ratio $\mu_T^M/\mu_T^C = \min\{\xi_H,\ \xi_I - \zeta_T/r\}$ is weakly below ξ_H where the lower-tax investor participates and strictly below it above the participation threshold.

Because the rate is constant, each yield averages the one-period returns $\mu_1^a, \ldots, \mu_T^a$. The yield gap $y_T^M - \xi_H y_T^C$ therefore has the sign of the g_j, and the yield-implied tax is non-increasing in maturity in case (ii) and non-decreasing in case (iii).

Proof. As in Proposition 17(b), minimum required returns determine holdings. The household requires $\xi_H r$ and r. The lower-tax investor requires $\xi_I r - \zeta_T$ and either $r - \zeta_T/\xi_H$ or r. This proves (i). With neutrality, $g_T = \mu_T^M - \xi_H r + \zeta_T$ equals ζ_T when $\xi_H r$ is minimal, otherwise $(\xi_I - \xi_H)r$. The ratios $\xi_H r/(r - \zeta_T/\xi_H)$ below the threshold and $(\xi_I r - \zeta_T)/(r - \zeta_T/\xi_H)$ above increase in ζ_T since $\xi_I > \xi_H$. A municipal-only benefit gives $g_T = \mu_T^M - \xi_H r$. Since ζ_T is nondecreasing, μ_j^M/μ_j^C is nondecreasing in j in (ii) and nonincreasing in (iii). Adding a maturity whose ratio is at least (in (iii), at most) the running yield ratio cannot lower (raise) that ratio, which proves the yield statements. □

B.13 Other markets

The mechanism requires common portfolio risk, heterogeneous relative taxes and a shorting friction. For a tax-favored bond and a comparable fully taxed bond, $\xi_i = (1-\tau_i^{\text{tax}})/(1-\tau_i^{\text{fav}})$, with investor i's tax rates on the taxed and favored bonds. Symmetric taxation gives $\xi_i = 1$ and removes the relative tax incentive. The response decomposition (79) applies when after-tax shares differ. Positive taxable forwards and an increasing relative forward ratio imply a declining yield-implied tax (Proposition 12). Condition (S) orders the affine loading ratio (Appendix Proposition 4).

Applications include state-exempt treasury versus corporate interest; state-exempt Federal Home Loan Bank and Farm Credit issues versus Fannie Mae and Freddie Mac issues; treaty-dependent withholding on cross-border sovereign debt; and the inflation accrual on treasury inflation-protected securities (TIPS), taxed before payment in taxable accounts. Taxable accounts, individual retirement accounts (IRAs) and pensions also assign different after-tax payoffs to the same security, which retains one market price. In each application, the maturity profile depends on which clienteles hold the bonds. Municipal bonds combine a large federal wedge with tight shorting restrictions.

C Numerical solution and calibration

The quantitative economy of Section 5, with 643 investor types and 689 bond coordinates, is solved for municipal and taxable price functions of the rate state and funded portfolios, in which bonds and cash are financed from funding W_i and cash is nonnegative.

C.1 State-contingent moments and funded market clearing

The solver imposes the conditions of Section 4.1 and Appendix A.1 at each node of a grid of rate states.

Rate-state grid. Let $\{z_g\}_{g=1}^G$ be the rate-state grid. It combines fifteen equally spaced nodes on $[-6\sigma_z, 6\sigma_z]$, which include $z = 0$, with the block's three observed annual rate states, for eighteen nodes in all; $\sigma_z = \sigma_r/\sqrt{1-\rho^2}$ is the unconditional standard deviation of the rate state. An observed rate state is $z_t = r_t - \bar{r}$, where r_t is twice the continuously compounded two-year corporate zero yield in month t. The annual nodes use December of each block year, and yields are read at all 36 months of the block. The grid has no node at the rate floor $z = -\bar{r}$, the kink of the cash return, so the two-period taxable bond's continuation evaluates the cash price through the same piecewise-linear interpolant as other prices. No rate state is selected to match long corporate yields.

Continuation moments. Let j be a market–maturity coordinate, v_j its bond-specific return variance and $p_{j,-}(Z')$ its continuation log price, the log price function of the one-period-shorter bond in the same

market at next period's rate state. Conditional on the current rate state z, $Z' = \rho z + \sigma_r U$ with $U \sim N(0,1)$, the rate shock of (30). The continuation objects are

$$\mathcal{M}_j(z) = \log \mathbb{E}_z \exp\{p_{j,-}(Z')\} + \tfrac{1}{2} v_j, \tag{104}$$

$$\Sigma_{jk}(z) = \mathrm{Cov}_z(p_{j,-}(Z'), p_{k,-}(Z')) + \mathbf{1}_{\{j=k\}} v_j. \tag{105}$$

The one-period return is $\mu_j = \mathcal{M}_j - p_j = \log \mathbb{E}_z[1+R_j]$, including the full Jensen adjustment of Appendix B.1, and Σ is the covariance matrix of centered log-return innovations. Log prices are piecewise linear in the rate state and extrapolated beyond the end nodes along the end segments, so their exponential moments, covariance and rate-innovation covariance have analytic Gaussian integrals (Appendix C.3). Interpolation gives Σ a low-rank common component, and bond-specific variances add its diagonal component.

Investors and payoff units. Investor i has bond after-tax shares $K_i = \mathrm{diag}(\xi_i^{a(j)})$ ($a(j)$ is coordinate j's market), cash after-tax share ξ_i^C, funding W_i, cash $x_i^0 = W_i - \mathbf{1}' x_i$ and shadow value of funding β_i, the multiplier on nonnegative cash. After-tax shares are after-tax payoffs per dollar of pretax payoff under each block's tax law, in the normalized payoff units of Section 4.2, which set to one a household's absolute after-tax share on home-state municipal interest and an institution's on all municipal interest. Risk aversion γ_i and outside exposure ℓ_i are transformed as in Section 4.2. In these units ξ_i^C is the investor's relative after-tax share ξ_i, which includes proration for insurers, and a household's share on another state's municipal bonds is one minus its home-state rate.

The liability credit is $c_{ij}^L(z) = -\mathrm{Cov}_z(p_{j,-}(Z'), \ell_i \sigma_r U)$ as in (70). The two-year financing-tax vector τ_i^{fin} is zero for nonbanks and taxable bonds. For banks it equals twice the annual financing-tax cost of Appendix C.7 on every municipal coordinate and at every rate state.

Portfolio conditions and clearing. Stationarity is

$$\begin{aligned} g_i &= \gamma_i K_i[\Sigma K_i x_i - c_i^L] - K_i \mu + \xi_i^C r \mathbf{1} + \tau_i^{\mathrm{fin}} + \beta_i \mathbf{1}, \\ g_{ij} &= \frac{\epsilon w_j}{x_{ij}}, \qquad \beta_i = \frac{\epsilon}{x_i^0}, \qquad x_{ij} > 0, \quad x_i^0 > 0. \end{aligned} \tag{106}$$

Here g_{ij}, distinct from the return gap g_T, is investor i's net cost of holding coordinate j. The second line replaces exact complementarity in the Kuhn–Tucker conditions (33) (in normalized units, plus the bank's financing-tax cost) by a log barrier on holdings and cash with weight $\epsilon > 0$ (10^{-9} in the baseline). The two-year log gross cash return is $r = \max\{\bar{r} + z, 0\}$. A municipal coordinate's barrier weight w_j equals its issuer state's share of national municipal supply at every maturity, so splitting supply across issuer states does not multiply the barrier. Taxable coordinates have unit weight. Clearing requires

$$\sum_i m_i x_i(z_g) = Q. \tag{107}$$

Appendix C.2 describes how these equations are solved.

C.2 The solution algorithm

Given candidate log price functions, continuation moments (104)–(105) define a static funded-clearing problem at each grid node. Its one-period returns imply the price update

$$\widehat{p}_j(z_g) = \mathcal{M}_j(z_g) - \mu_j(z_g), \tag{108}$$

and the algorithm iterates on prices until $\widehat{p} = p$ (Algorithm 1). Two boundary conditions close the recursion. A one-period municipal claim pays one next period, so its continuation log price is zero and its return is riskless. Its current price is solved in equilibrium. Cash, the one-period taxable claim, is not one of the 689 coordinates. It earns $r = \max\{\bar{r} + z, 0\}$, so its log price is $-r$, which supplies the continuation price of the two-period taxable bond.

Consistent expectations. Moments integrate all segments and extrapolated tails of the candidate functions. A fixed point uses the same prices in expected resale values and current clearing at every grid node. Covariances and funding couple all maturities, so the entire price array is updated together.

Participation. Every investor may hold every bond. The barrier gives $x_{ij} = \epsilon w_j / g_{ij}$, so positions whose net cost is large relative to ϵ are small, of order 10^{-12} in the reported equilibria. Portfolios are re-solved at each grid node in every iteration, so participation adjusts with prices. At the three annual nodes, high-tax households hold more than 99.99% of each municipal maturity through six years and less than 0.01% beyond eighteen years.

Step 1: initial prices and portfolios. The block starts from the one-market equilibrium at the same coefficients (Appendix C.6), which is solved by the same algorithm with one municipal market and household after-tax shares averaged across issuer states by supply shares. Its municipal prices are copied to every issuer-state market. Each household's municipal holdings are placed in its home-state market, each institution's are split across markets by supply shares, and taxable holdings are unchanged. One-period returns start at $\mathcal{M} - p$. Before the first Newton solve, one-period returns and shadow values of funding are adjusted so that the mapped portfolios satisfy the market-level first-order conditions, and positions that the mapping leaves near zero are reset to their barrier values. Every position stays free, so the start does not restrict holdings.

Step 2: portfolios and clearing at each grid node, for given prices. At each grid node, the moments $\mathcal{M}(z_g)$ and $\Sigma(z_g)$ and the liability credits follow in closed form from the current price functions. During the solve, these moments trim singular values below 10^{-12} of the largest. Newton's method then solves stationarity and clearing, (106)–(107), jointly for holdings x_i, cash x_i^0 and one-period returns μ. Each investor's Hessian is diagonal plus low rank, so it is inverted with the Woodbury identity. Eliminating holdings leaves a 689×689 linear system in μ, the market Schur complement, which is solved by LU factorization and refined when needed to a relative residual of 10^{-12}. A fraction-to-boundary rule and backtracking keep bond and cash holdings positive. In the first price iteration, the barrier weight falls from 10^{-5} to 10^{-9} in factor-of-ten stages. Later iterations start from the node's previous solution at $\epsilon = 10^{-9}$, and a failed solve restarts from proportional holdings with a larger barrier. Newton iteration stops when the stationarity and clearing residuals are below 10^{-11} and the largest product of a holding and its stationarity residual is below 2.5×10^{-12}. The grid nodes are solved independently and in parallel.

Step 3: price update. All prices are updated together from (108), across markets, maturities and grid nodes. A damped step moves prices a fraction $\min\{0.65,\ 0.025/\|\widehat{p} - p\|_\infty\}$ of the way to $\widehat{p}$, so no log price changes by more than 0.025. Anderson acceleration combines the last seven price iterates with least-squares weights and mixes at the damped step size. It restarts when the price error more than doubles, is used only when every weight is below 50 in absolute value, and is scaled to the same 0.025 cap.

Step 4: stopping and acceptance. Iteration stops when $\|\widehat{p} - p\|_\infty < 10^{-10}$. All moments are then recomputed without the singular-value trimming, and the equilibrium is accepted only if the price, stationarity, clearing and budget residuals of Appendix C.3 meet their thresholds. Yields at the 36 monthly rate states of the block come from the solved log prices by linear interpolation in z, and the national municipal yield averages the issuer-state yields with supply weights.

Convergence. The price fixed point is the only outer loop. In the six calibration blocks, the price error $\|\widehat{p} - p\|_\infty$ starts near 10^{-2} and falls steadily, slowly for the first fifteen to twenty iterations and then by about a factor of ten every five iterations. It reaches 10^{-6} within forty iterations and 10^{-10} within seventy. The first iteration, which carries the barrier stages, is the most expensive. Later iterations need about two Newton steps per grid node.

Policy counterfactuals use the same algorithm, barrier and acceptance checks. The equal-treatment counterfactual (Section 7.3) uses price-update, inner, holding-weighted and linear-solve tolerances of 2×10^{-7}, 2×10^{-8}, 5×10^{-9} and 2×10^{-9}, respectively. Its 2015–2017 endpoint starts from the solution that removes

99% of each tax distinction. The other five pre-2018 economies use this accepted 99.9% endpoint as a warm start and independently pass the same checks. Near binding funding, the cash Newton direction is computed from the low-rank inverse to avoid cancellation. Other policy solves use the baseline tolerances or tolerances between these and the baseline.

Algorithm 1 Full-state equilibrium of one block

Require: after-tax shares, funding, masses, supplies, exposures, rate dynamics, state grid $\{z_g\}$

Initialize p and holdings from the one-market equilibrium; $\mu \leftarrow \mathcal{M} - p$ ▷ Step 1
repeat ▷ outer loop over price functions
  for all grid states z_g, in parallel **do** ▷ Step 2
    Compute $\mathcal{M}(z_g)$, $\Sigma(z_g)$ and liability credits from the current p
    Solve stationarity and clearing (106)–(107) for x, x^0 and μ by barrier Newton
  end for
  $\widehat{p} \leftarrow \mathcal{M} - \mu$ at every grid state ▷ Step 3
  Move p toward $\widehat{p}$ by a damped or Anderson step, with log-price changes capped at 0.025
until $\|\widehat{p} - p\|_\infty < 10^{-10}$ ▷ Step 4
Recompute the moments; accept only if the price, stationarity, clearing and budget checks pass
Read yields at the block's monthly states and holdings at its annual nodes

The local rate loadings $A^a_T = -\partial_z p^a_T$ satisfy the nonlinear recursion (69) under the differentiation conditions of Appendix B.2.

C.3 Equilibrium residuals

Table 8 reports four independently reconstructed equilibrium diagnostics for each of the six blocks. With g_i from (106),

$$\begin{aligned} R_p &= \|p - \mathcal{M} + \mu\|_\infty, \qquad R_c = \|Q - \textstyle\sum_i m_i x_i\|_\infty, \\ R_s &= \|g - \epsilon w/x\|_\infty, \qquad R_{\text{comp}} = \max\{\|x \odot g\|_\infty, \|\beta \odot x^0\|_\infty\}. \end{aligned} \tag{109}$$

All six equilibria have zero budget and negative-holding violations, and their riskless claims have zero return variance. Acceptance requires $R_p < 3 \times 10^{-7}$, $R_s < 10^{-7}$, $R_c < 10^{-8}$ and a budget violation $\max_i(\mathbf{1}' x_i - W_i)$ below 10^{-10}, which all six blocks meet. The complementarity measure R_{comp} is reported only, since it equals ϵ at an exact barrier solution.

Table 8: Full-state equilibrium diagnostics

Block	R_p	R_s	R_c	R_{comp}
2000–2002	7.511×10^{-11}	5.884×10^{-12}	5.551×10^{-17}	1.002×10^{-9}
2003–2005	4.456×10^{-11}	8.340×10^{-12}	1.110×10^{-16}	1.002×10^{-9}
2006–2008	8.156×10^{-11}	8.547×10^{-12}	1.110×10^{-16}	1.002×10^{-9}
2009–2011	5.498×10^{-11}	8.687×10^{-12}	2.220×10^{-16}	1.002×10^{-9}
2012–2014	9.289×10^{-11}	9.864×10^{-12}	2.220×10^{-16}	1.002×10^{-9}
2015–2017	8.495×10^{-11}	8.352×10^{-12}	2.220×10^{-16}	1.002×10^{-9}

Notes. Maxima over each block's eighteen state nodes and the applicable coordinates, with $\epsilon = 10^{-9}$. Complementarity is the largest unpenalized product of a bond or cash holding and its multiplier, which equals ϵ at an exact barrier solution. The diagnostics assess only the discretized funded equations.

The fixed-coefficient 2018–2020 extension (Appendix C.7) also satisfies these checks: $R_p = 4.519 \times 10^{-11}$, $R_s = 7.777 \times 10^{-12}$, $R_c = 2.220 \times 10^{-16}$ and $R_{\text{comp}} = 1.001 \times 10^{-9}$, with zero budget and negative-holding violations and zero variance on riskless claims.

Riskless boundary and the household benchmark. The 45 one-period municipal claims are priced in funded clearing like other bonds. The lowest return at which a household in the set $\mathcal{H}$ holds the claim of issuer state s, the household counterpart of the boundary (64), is

$$\min_{i\in\mathcal{H}\text{ eligible}} \frac{\xi_i^C r(z)+\beta_i(z)}{\xi_i^{M_s}},$$

which includes the shadow value of funding and so can exceed $r(z)$ times the smallest after-tax-share ratio. In every equilibrium, each issuer state's one-period municipal return is within 0.01 basis points of this frontier at every grid node. The household benchmark uses the after-tax share of the high-tax household resident in the issuing state, which can differ from that of the household attaining the minimum. At each observed month, the national one-period municipal yield differs from the household benchmark by the supply-weighted difference between the frontier and the benchmark, plus less than 0.01 basis points.

Analytic integration of the price interpolant. For a standard normal U and an affine segment $p = a + bU$ on $l < U < u$,

$$\mathbb{E}[e^{a+bU}\mathbf{1}_{\{l<U<u\}}] = e^{a+b^2/2}\{\Phi(u-b)-\Phi(l-b)\}.$$

Summing over segments and the extrapolated tails eliminates quadrature truncation, leaving interpolation and finite-precision error. The residuals above hold at the grid nodes and do not bound errors uniformly between them. Grid refinement assesses accuracy between nodes (Appendix C.4).

Weighted barriers and funding. Exact stationarity gives $x_{ij}g_{ij} = \epsilon w_j$ and $x_i^0\beta_i = \epsilon$. For concave preferences at fixed prices, the funded utility gap is bounded by $\epsilon(\sum_j w_j + 1)$, provided the stationarity error is controlled.

C.4 Sensitivity to the grid, the barrier and a node at the rate floor

Three comparisons fix the seventeen coefficients and measured inputs in 2015–2017. The first increases the regular nodes from fifteen to twenty-nine, retaining the original nodes and endpoints. The second lowers ϵ from 10^{-9} to 10^{-10} with unchanged inner tolerances. The third adds a grid node at the rate floor (row "Exact floor knot"). I check each re-solve independently, and Table 9 reports the results.

Table 9: Numerical sensitivity at fixed coefficients, 2015–2017

Comparison	Actual nodes	ϵ	Δ 30-year gap (bp)	Δ HH lower muni book (%)	Δ bank taxable book (%)
Nested grid	32	1.000×10^{-9}	+0.199365	-4.020	-2.585
Smaller barrier	18	1.000×10^{-10}	+0.000138	-0.004	+0.001
Exact floor knot	19	1.000×10^{-9}	-0.228439	-1.689	-2.797

Notes. Each row compares one change with the original eighteen-node, 10^{-9}-barrier economy. Rows are not a sequence. The yield gap is the national supply-weighted municipal yield minus the original resident-household after-tax taxable benchmark, in continuously compounded yields over 36 actual months. Its baseline is 104.270250 bp. Holdings use the three annual measurement nodes, and percentage changes use each group's original book as denominator. The rows are neither confidence intervals nor the seven-block empirical-weight-matched gap. The floor is economically active in every row.

Changes in the thirty-year gap are small relative to holding changes. Refinement reduces the lower-tax household municipal book by 4.02%, and the node at the rate floor reduces it by 1.69% and bank taxable holdings by 2.80%. Changes in municipal and taxable yields partly offset each other.

Lowering the barrier leaves risky holdings and gaps stable, while riskless and near-zero holdings remain sensitive to it. Individual positions in riskless claims change by at most 77.06% of budgets across grid nodes and 24.84% at annual holdings nodes, compared with maxima of 0.1179% and 0.0709% for risky positions.

Table 10: Common estimated coefficients, held fixed in the full-state evaluation

Symbol	Economic role	Value
Risk. Main information: taxable term spreads and the allocation of long bonds		
γ	Common risk-aversion curvature, per unit of balance-sheet mass	9.6969
v_{30}	Bond-specific variance of a thirty-year bond's two-year return	0.000722
p_v	Maturity exponent of the bond-specific variance	1.2597
Households. Main information: household municipal and taxable stocks		
a_H	Exposure intercept	1.3045
b_A	Loading on the outside-asset ratio	2.704
b_D	Loading on the debt ratio	0.8379
P&C insurers. Main information: P&C stocks, maturities, composition and long fractions		
a_P	Low-type exposure	−1.961
d_P	High-minus-low exposure	3.0311
w_P	Low-type resource fraction	0.0802
Life insurers. Main information: life-insurer stocks, maturities, composition and long fractions		
a_L	Low-type exposure	−5.3927
d_L	High-minus-low exposure	6.3007
w_L	Low-type resource fraction	0.2616
t_L	Middle type's location within the exposure span	0.8625
u_L	Middle type's fraction of non-low resources	0.0162
Banks and residual group. Main information: bank stocks and taxable term spreads		
h_B	Bank exposure	1.2045
μ_R	Residual-group mean exposure	1.0683
s_R	Residual-group exposure dispersion	1.7168

Notes. All seventeen coefficients are estimated jointly to the twenty-two averaged conditions of Table 11. Every coefficient affects every condition through equilibrium prices and clearing, and the headings name the conditions most informative about each block. Exposures are measured like net duration, in the model's two-year units. Resource fractions split each insurer sector's measured resources across its types. Appendix Table 12 reports the search intervals, starting values and step scales.

The largest riskless shift is at a positive cash rate. Complementarity falls from 1.0016×10^{-9} to 1.0134×10^{-10}, and the tiny residual-group municipal and high-tax household taxable books fall about 90%.

C.5 Risk specification

Appendix C.6 reports sensitivity to the coefficients below.

The bond-specific variance profile is $v_n = v_{30}(n/30)^{p_v}$ for maturities of $n = 4, 6, \ldots, 30$ years, with $v_2 = 0$ and the same variances for municipal and taxable bonds at each maturity. The coefficients v_{30} and p_v are fitted jointly with preferences and effective exposures (Table 10) and held fixed in the comparisons above.

Effective outside exposures summarize rate risk without matching every asset position to liability cash flows. The averaged ownership and maturity conditions assess them, and Section 5.4 assesses the fit to the yield curve.

C.6 Calibration conditions, parameter search and full-state evaluation

I estimate the seventeen common coefficients of Table 10 with the criterion (110) in six dated economies with one national municipal market. Table 11 lists the targeted conditions and untargeted tests. Exposure parameters use $h_i = -\ell_i$: positive values add rate risk, while negative values hedge it (Appendix C.7).

Measured inputs. The 1952–2024 annual rate estimates are a persistence of 0.853903 and an innovation standard deviation of 0.016266, and the mean 0.030206 is the 2000–2017 average of the two-year corporate-like yield. Two-year units give $(\rho, \sigma_r, \bar{r}) = (0.729151, 0.042778, 0.060412)$ with floor $r = \max\{\bar{r} + z, 0\}$. Monthly two-year yields set the observed states (Appendix C.2). The pooled treasury and corporate asset

Table 11: Targeted conditions and untargeted tests

A. Targeted: twenty-two conditions, each averaged over the six blocks of 2000–2017	
Ownership stocks (9)	Municipal holdings of high-tax households, lower-tax households, P&C insurers, life insurers and banks. Taxable holdings of households, P&C insurers, life insurers and banks
Insurer maturity (4)	Holdings-weighted mean maturity of the P&C and life-insurer municipal and taxable books
Taxable term spreads (3)	Ten-, twenty- and thirty-year taxable yields over the two-year yield
Insurer composition (2)	Book-weighted mean of squared municipal portfolio shares, P&C and life insurers
Insurer long fractions (4)	Share of each insurer sector's municipal and taxable book maturing beyond twenty years
B. Untargeted tests of the mechanism	
Municipal yield curve	Evaluated on the observed corporate curve, the model accounts for 83% of the 117.2-basis-point thirty-year puzzle after callability and the issuance premium (Figure 8)
State-tax pass-through (P1)	With the model's own federal after-tax share, its cross-state slope falls from 1 at two years to 0.46 at thirty and lies inside the controlled pre-2018 OLS 95% band in the first four maturity bins and 0.01 above its upper bound in the longest (Figure 9)

Notes. Each averaged condition is scaled by one tenth of its observed six-block mean magnitude. Appendix Table 16 reports every scaled deviation at the estimated coefficients.

uses corporate yields and household taxes, despite the different statutory treatment of treasuries. After-tax shares and bank financing-tax costs are distinct fixed inputs (Table 15), and after-tax shares alone do not identify tax receipts.

Condition construction and normalization. Table 16 lists the twenty-two conditions in evaluation order. Each condition j in block k is a signed model-minus-observed difference g_{kj} with a positive observed normalizer o_{kj}. For the nine ownership conditions, g_{kj} is the model minus the observed stock in national dollars and o_{kj} is the observed stock.

For each insurer sector and asset, the maturity condition normalizes $B(\overline{T} - \overline{T}^{\text{obs}})$ by $B^{\text{obs}}\overline{T}^{\text{obs}}$, with B its model book and $\overline{T}$ its holdings-weighted maturity. The three taxable-yield conditions average the errors in annual log yields at ten, twenty and thirty years over 36 monthly states. Because observed two-year yields set the states, these are term-spread errors, normalized by the observed block-mean spreads over the two-year yield.

The remaining six conditions use maturities above two through thirty years, excluding the riskless two-year municipal stock included in the ownership and maturity conditions. For investor i in each insurer sector, let M_i and C_i be municipal and taxable holdings in this window and $B_i = M_i + C_i$. The municipal-share second moment $\sum_i (M_i^2/B_i)/\sum_i B_i$ is the book-weighted average of squared municipal portfolio shares. It captures how unevenly insurers allocate their bond portfolios to municipals and is compared with its observed counterpart. For each sector–asset book B with holdings L beyond twenty years and observed long fraction f^{obs}, the long-maturity condition is

$$g^{\text{long}} = \frac{L - f^{\text{obs}}B}{B^{\text{obs}}_{\text{window}}}.$$

Observed window books equal national sector stocks times measured window coverage. These six conditions are normalized by observed second moments and long fractions.

Holdings conditions are evaluated at each block's three observed end-year states and then averaged, because averaging books first would change products and ratios. Let $g_{kj}(\theta)$ be signed condition j in block k after its equilibrium solve at the common coefficient vector θ, and $o_{kj} > 0$ its observed normalizer. The criterion is

$$f_j(\theta) = \frac{6^{-1}\sum_{k=1}^{6} g_{kj}(\theta)}{0.1\,(6^{-1}\sum_{k=1}^{6} o_{kj})}, \qquad \mathcal{L}(\theta) = \frac{1}{2}\sum_{j=1}^{22} f_j(\theta)^2. \tag{110}$$

The book-weighted residuals above enter $\mathcal{L}$ with fixed inverse-squared-normalizer weights, which supply no covariance estimate, parameter standard errors or overidentification test.

Table 12: Parameter-search intervals, initialization and coordinate scales

Coefficient	Search interval	Initial value θ_k^0	Step scale c_k
γ	$[1, 10]$	9	1
a_H	$[-3, 3]$	1	0.3
b_A	$[0, 12]$	2	0.4
b_D	$[0, 20]$	0.5	0.5
a_P	$[-10, 2]$	-1.5	0.5
d_P	$[0.02, 12]$	2.4	0.6
w_P	$[0.01, 0.99]$	0.22	0.08
a_L	$[-12, 2]$	-4	0.6
d_L	$[0.02, 15]$	3.2	0.8
w_L	$[0.01, 0.99]$	0.1	0.08
h_B	$[-3, 3]$	0.6	0.3
μ_R	$[-4, 4]$	1	0.4
s_R	$[0.01, 5]$	2.2	0.4
v_{30}	$[0.000020, 0.006000]$	0.0015	0.0004
p_v	$[0, 3]$	1.15	0.3
t_L	$[0.01, 0.99]$	0.75	0.08
u_L	$[0.01, 0.99]$	0.2	0.08

Search coordinates. Table 12 gives bounds, initial vector θ^0 and step scales c_k, where a scaled step u_k changes coefficient k by $c_k u_k$. Derivatives and trust radii use these units. All economies start at θ^0. An unoptimized $(\gamma, a_H) = (8.5, 1.1)$ alternative supplies warm starts for the numerical checks without changing targets or portfolios.

Joint search. For the scaled conditions f and their Jacobian J in u, central differences refresh J at accepted centers, using feasible one-sided differences at bounds. A bounded trust-region search compares damped Gauss–Newton, backtracked and projected-gradient candidates against the quadratic model

$$\mathcal{L} + (J'f)'u + \tfrac{1}{2}u'(J'J + S)u.$$

Here a structured secant update estimates the residual curvature S.

The initial search uses a finite-difference step of 0.002 and ends after 107 iterations at $\mathcal{L} = 36.357521$. A restart from that vector uses a step of 0.001 after the larger step fails the initial half-step consistency check, and it retains the objective, bounds and equilibrium tolerances. The restart ends after 41 iterations at $\mathcal{L} = 36.286978$, the lowest retained criterion across both fits, and this vector is the quantitative baseline. Subsequent diagnostic probes do not replace it.

Search checks and interpretation. Outer-search convergence is separate from equilibrium accuracy. At the selected vector, steps of 0.002, 0.001 and 0.000125 give projected-gradient infinity norms of 14.57, 11.01 and 0.376, above the tolerance of 0.01. Changes in participation make the derivatives sensitive to the step size. The feasible probes tested improve $\mathcal{L}$ by at most 2.79×10^{-4}, which bounds only those directions. The selected calibration has no local-stationarity certificate.

Full-state evaluation. The full-state evaluation restores 45 issuer markets, separate risks and measured home-state taxes, with the same seventeen coefficients, quantities and condition definitions. These equilibria supply municipal yields, predictions and policies. Solves take 698–2,006 seconds on six workers, compared with about one second for one market. Across the six blocks, the full-state thirty-year gap minus the one-market gap is $(-0.85, -0.72, 2.48, -0.22, -1.15, -0.93)$ bp, and taxable yields differ by at most 0.52 bp.

Coefficient sensitivity. Table 13 changes one coefficient group per one-market re-solve. Every change raises the criterion. Bond-specific variance moves the gap and taxable yield by less than 0.1 basis points. Lower risk aversion produces the largest change. It lowers the thirty-year taxable yield by 70 basis points and puts term spreads about 20% below targets.

Table 13: Sensitivity of the one-market calibration to the estimated coefficients

Coefficient change	30-year gap (bp)	Change (bp)	Criterion $\mathcal{L}$
Estimated coefficients	102.6		36.29
Risk aversion $\gamma \times 0.8$	89.0	−13.6	55.02
Risk aversion $\gamma = 10$ (upper bound)	104.7	+2.1	36.74
Bond-specific variance $v_{30} \times 0.8$	102.7	0.0	36.61
Bond-specific variance $v_{30} \times 1.2$	102.7	0.0	36.59
Maturity power $p_v - 0.2$	102.6	0.0	36.55
Maturity power $p_v + 0.2$	102.7	0.0	36.47
Household exposure intercept $a_H - 0.25$	99.0	−3.7	78.82
Household exposure intercept $a_H + 0.25$	106.7	+4.1	160.84
Household asset loading $b_A \times 0.8$	100.9	−1.7	62.45
Household asset loading $b_A \times 1.2$	105.2	+2.6	400.24
Household debt loading $b_D \times 0.8$	106.2	+3.5	6,720
Household debt loading $b_D \times 1.2$	103.9	+1.3	82.17
P&C exposures ×0.8	102.2	−0.4	56.35
P&C exposures ×1.2	103.3	+0.6	62.51
Life-insurer exposures ×0.8	101.9	−0.7	59.68
Life-insurer exposures ×1.2	103.4	+0.8	46.24
Bank exposure ×0.8	100.5	−2.1	119.11
Bank exposure ×1.2	104.5	+1.8	241.78
Residual exposure mean ×0.8	102.6	−0.1	538.79
Residual exposure mean ×1.2	101.7	−0.9	65.73
Residual exposure s.d. ×0.8	99.3	−3.3	92.74
Residual exposure s.d. ×1.2	105.8	+3.1	1,687

Notes. The one-market economy is averaged over the six blocks of 2000–2017, with all other coefficients at their estimates. Exposure changes scale every type's exposure in the group. The gap is the municipal yield minus the household after-tax taxable yield at thirty years. The criterion $\mathcal{L}$ is the estimation objective, half the sum of squared scaled deviations of the twenty-two conditions.

Elastic bond supply. I allow borrowing to respond to its cost in both markets, holding the estimated coefficients fixed. For market $a \in \{M, C\}$ and maturity T, supply follows

$$Q_T^a(z) = \overline{Q}_T^a \left(\frac{y_T^a(z)}{\overline{y}_T^a} \right)^{-\chi_{a,T}},$$

where y_T^a is the annual continuously compounded yield. Each schedule passes through the original quantity and fixed-supply yield at the mean rate state. These anchors remain fixed when both yield curves are re-solved. Risky maturities from four to thirty years have elastic supply. The two-year riskless claims retain their baseline treatment.

St. Clair (2024) estimates that a one-percent increase in borrowing costs reduces municipal issuance by approximately 0.3–0.6 percent. I use 0.45 as the midpoint of this rounded range and impose the same elasticity in both markets. The estimate concerns annual borrowing near the bank-qualified threshold. Holding financing terms fixed maps proportional borrowing-cost changes locally to proportional yield changes. Applying that response to outstanding model supply defines the comparative-static exercise.

Table 14 reports the one-market economy over the six calibration blocks. At equal elasticities of 0.3, 0.45, and 0.6, the average thirty-year gap lies between 101.0 and 102.0 basis points, compared with 102.6 under fixed supply. At 0.45, municipal yields fall by 14.1 basis points and the household after-tax taxable benchmark by 13.4, so the gap changes by only 0.7 basis points.

I also consider a larger taxable response. The November 2022 estimates of Siani (2022) imply approximately 1.5% less corporate issuance per ten-basis-point increase in credit spreads, holding the benchmark rate fixed. I match that local slope by setting $\chi_{C,T} = 15\overline{y}_T^C$, while keeping $\chi_{M,T} = 0.45$. The thirty-year taxable elasticity is 0.86–0.97 across blocks. Applying this corporate issuance response to the pooled taxable supply is an additional sensitivity assumption. The average gap is 99.2 basis points, retaining 96.7% of its fixed-supply value. Both yields adjust: municipal yields fall by 19.5 basis points and the after-tax taxable benchmark by 16.1. A larger taxable supply elasticity therefore need not widen the gap.

Table 14: Yield gaps with elastic supply in both markets

Municipal elasticity	Taxable elasticity	10 years	20 years	30 years
0	0	35.7	84.9	102.6
0.30	0.30	36.4	84.1	101.9
0.45	0.45	36.7	84.0	102.0
0.60	0.60	36.8	83.4	101.0
0.45	$15\overline{y}_T^C$	37.2	83.2	99.2

Notes. Gaps are in basis points, averaged over the original rate-state observations within each block and equally across the six blocks of 2000–2017. All other coefficients and inputs remain fixed. Elasticities are reported as positive magnitudes: higher borrowing costs reduce supply. The last row converts a corporate spread semi-elasticity to a yield elasticity at each fixed anchor, with yields in annual decimal units.

C.7 Measured inputs and investor construction

Markets. One model period is two years, and maturities in this subsection are in years. Each of the 43 states with a personal income tax and the District of Columbia forms its own municipal market. The seven states without one (Alaska, Florida, Nevada, South Dakota, Texas, Washington and Wyoming) share a single market because every investor treats their bonds alike: their residents pay no state tax on any municipal bond, and all other investors treat them as out-of-state bonds. Each of the 45 municipal markets has fifteen maturities from two to thirty years, and the taxable market fourteen from four to thirty years, for 689 market–maturity pairs. Cash, the one-period taxable claim, pays the short rate, and the two-year municipal bond is also riskless. All bond yields, including the two-year municipal yield, are determined in equilibrium. The taxable market pools treasury and corporate supply. Its yields are compared with the corporate curve, and its interest is taxed like corporate interest.

Calibration blocks, rates and quantities. The inputs set directly from data come in six three-year calibration blocks, 2000–2002 through 2015–2017, centered on the 2001, 2004, 2007, 2010, 2013 and 2016 Survey of Consumer Finances (SCF) waves, and each block uses its own values. The rate state z of Appendix A.1 follows a first-order autoregression whose annual persistence, 0.854, and innovation standard deviation, 1.63 percentage points, are estimated on 1952–2024 short rates. The mean rate is $\bar{r} = 0.0604$ per two-year period, twice the 2000–2017 average of the two-year corporate zero yield, 3.02% a year. Actual monthly two-year corporate zero yields set the observed rate states (Appendix C.1). Financial Accounts stocks fix sector resources, funding and bond quantities. Mergent outstanding amounts fix the municipal maturity profile and issuer-state shares, and outstanding amounts in the Mergent Fixed Income Securities Database (FISD) fix the corporate maturity profile. Tax rates follow statutes and each SCF tax unit's filing status and income, and households' balance-sheet ratios come from the SCF, as described below. Appendix C.6 gives the remaining details, and Appendix Table 15 lists all measured inputs and their sources.

Investor types. Table 3 lists the six groups. Type i has relative after-tax share ξ_i, resources m_i, funding W_i and outside duration $h_i := -\ell_i$, the negative of the outside exposure ℓ_i in Appendix equation (62). Positive values of h_i add rate risk, the reverse of the sign convention for ℓ_i: they describe outside assets that lose value when rates rise, so long bonds add to the type's rate risk. A negative h_i represents a gain when rates rise, such as a fall in liability value, which long bonds hedge. Total resources m_i, the model's mass of type i, measured in dollars, are the type's debt securities and cash in the Financial Accounts. Funding W_i

Table 15: Measured inputs, estimated coefficients and maintained measurement choices

Input	Source, construction and role
National resources and ownership	Financial Accounts vintage of September 8, 2023, with registered household fund look-through, nonprofit separation and sector exempt-stock adjustments. SCF direct and fund holdings split household municipal ownership. These stocks supply conditions 1–9, and tax-exempt interest flows are not used.
Household heterogeneity	Each SCF wave supplies seven resource-weighted balance-sheet bins per household tax group. Outside financial assets exclude retirement accounts and the modeled resource base of deposits, direct bonds and funds. Debt and outside assets are divided by that resource base before bin averaging. Coefficients a_H, b_A, b_D jointly translate the ratios into exposure.
Household tax products	Dated SCF tax-unit supplements and federal calculations, with annual state-rate snapshots for high-tax households and record-level bracket assignments for lower-tax households. The products are computed before weighting. For lower-tax households, the archived bracket tables lack some filing-status cutoffs or related rules in 209 of 792 state-years. These are imputed from the state's nearest-year thresholds or simple proxies, leaving published rates and top rates unchanged. The repeated national geography is a maintained input.
Insurer portfolios	Annual P&C and life bond-position files, matched to security tax status and remaining maturity. Positive-value positions determine four maturity-dollar and six composition conditions, 10–13 and 17–22. Three annual cross-sectional moments are equally averaged within a block. Portfolio moments inform the estimated exposure mixtures.
Institutional tax treatment	The model applies separate P&C proration, life policyholder-share proration, and bank financing-tax rules, and the residual group is tax-neutral. The measured annual bank funding-cost ratio uses interest expense relative to assets, weighted by municipal holdings. Bank-qualification and coverage assumptions are maintained inputs.
Municipal supply and geography	Dated exempt outstanding Mergent amounts, a within-bin maturity projection and state shares. The seven states without an income tax form one pooled market. National household mass in each bin is allocated by state municipal supply share.
Taxable supply	Treasury and corporate national totals from Financial Accounts, corporate maturity shares from FISD outstanding amounts. Treasury maturity shape is separately tied to the maturities measured in the Monthly Statement of the Public Debt. No national quantity is re-estimated against yields.
Rate process and yield conditions	Annual persistence and innovations from the 1952–2024 TB3MS sample, temporally aggregated to two years. The common mean is the mean of the two-year corporate-like yield over 2000–2017. The monthly corporate-like spot curve supplies conditions 14–16, converted to annual log yields. Broad pooled quantities retain the corporate-like yield and tax approximation.
Joint estimated coefficients	The seventeen coefficients in Table 10 enter the same twenty-two averaged conditions. Ownership informs household and bank exposures. Insurer books, maturities and composition inform exposure mixtures, while taxable yields and joint clearing inform risk curvature, residual exposure and the bond-specific variance profile. Equilibrium prices and allocations link all coefficients to all conditions.

Notes. Six three-year blocks cover 2000–2017. Each averaged condition is scaled by one tenth of its observed six-block mean magnitude, and the estimated coefficients are held fixed in the full-state evaluation and the policy endpoints. Licensed inputs remain subject to their original access restrictions.

Table 16: The twenty-two targeted conditions at the estimated coefficients: signed average deviations in units of 10 percent of the observed magnitude

	Condition	Scaled deviation
1	High-tax households: municipal stock	2.518
2	Lower-tax households: municipal stock	−2.672
3	P&C: municipal stock	−0.199
4	Life: municipal stock	0.140
5	Banks: municipal stock	0.591
6	Households: taxable stock	−0.561
7	P&C: taxable stock	−2.711
8	Life: taxable stock	0.218
9	Banks: taxable stock	−0.277
10	P&C: municipal maturity dollars	1.227
11	P&C: taxable maturity dollars	−1.904
12	Life: municipal maturity dollars	2.269
13	Life: taxable maturity dollars	−0.572
14	Taxable term spread, ten years	0.276
15	Taxable term spread, twenty years	0.165
16	Taxable term spread, thirty years	0.145
17	P&C: municipal-share second moment	2.833
18	P&C: long municipal fraction	−0.341
19	P&C: long taxable fraction	−5.630
20	Life: municipal-share second moment	−0.379
21	Life: long municipal fraction	−0.197
22	Life: long taxable fraction	0.386
	Half sum of squared deviations	36.2870

Notes. Each entry averages six signed dated conditions in original units and divides by one tenth of the observed six-block mean magnitude of the compared quantity, so an entry of one is a residual equal to 10 percent of that condition's specified observed normalizer. Entries are evaluated at the estimated coefficients in the economy with one national municipal market and are not t-statistics. Rows 1–9 are stock differences, scaled by the observed stocks. Rows 10–13 multiply the model stock by its mean-maturity difference and are scaled by the observed book times its mean maturity. Rows 14–16 are annual log-yield differences. Since the model's two-year yield equals the observed one in every month, they are term-spread errors, scaled by the observed mean term spread over the two-year yield. Rows 17 and 20 compare book-weighted second moments. The four long-fraction conditions are $(L - f^{\text{obs}}B)/B^{\text{obs}}_{\text{window}}$, with B and $B^{\text{obs}}_{\text{window}}$ the model and observed books in the greater-than-two-to-thirty-year window and L its greater-than-twenty-year holdings. Rows 17–22 are scaled by their observed values. The two-year municipal stock is part of the ownership and maturity-dollar conditions but not of that distributional window. Conditions are evaluated before averaging. Products of averaged books and ratios generally give a different object.

is the share of these resources available for the modeled bond-and-cash portfolio. Its numerator is the sum of municipal bonds; corporate bonds excluding asset-backed securities; treasury notes and bonds; and cash excluding bank cash. Positions are per dollar of resources, as in Section 5.1. With common risk aversion, each active holder's weight in the tax mix at a municipal maturity is proportional to its resources (Appendix Proposition 9).

Households are grouped by each SCF tax unit's federal rate under that year's law. High-tax households face a federal rate of at least 35%. In 2015–2017 they are about 5% of families and hold 46% of household resources. Lower-tax households, banks and insurers hold over half of measured municipal bonds. Within each household group, tax units are sorted into seven equal-resource bins by outside financial assets less debt, relative to the deposits, bonds and bond funds that fund the portfolio. Bins share group tax rates but have outside duration $h_k = a_H + (b_A Out_k - b_D Debt_k)/2$, where Out_k and $Debt_k$ are bin k's ratios of outside financial assets and of debt to that portfolio base. The three common coefficients a_H, b_A and b_D are estimated, and the factor of one half only normalizes b_A and b_D. For state taxes, high-tax households pay their home state's top rate, and lower-tax households pay each tax unit's bracket rate. The baseline

multiplies federal and state after-tax shares for taxable interest tax unit by tax unit. It gives no federal deduction credit for state taxes on out-of-state municipal interest, and Appendix E.4 varies that convention.

The federal calculation includes a Social Security taxation proxy: the tax induced when exempt interest makes more Social Security benefits taxable. A local increase in ordinary income identifies both the additional taxable benefits and the associated federal liability. Removing the tax on that ordinary-income increment isolates the induced tax from exempt interest. This proxy is part of the household after-tax shares and explains why their absolute home-municipal shares can be slightly below one.

Institutions face the pre-2018 corporate rate of 35% and no modeled state tax. For property and casualty (P&C) insurers, proration taxes 15% of municipal interest at 35%, so $\xi_P = 0.65/0.9475 = 0.686$. Life insurers' policyholders' share, one half before 2018, is taxed at 35%, so $\xi_L = 0.65/0.825 = 0.788$. Banks keep municipal interest and 65% of taxable interest, giving $\xi_B = 0.65$, but cannot deduct the interest expense that funds municipal holdings. Their financing-tax cost, 35% of the ratio of interest expense to assets weighted by municipal holdings, is 1.02% a year in 2000–2002 and 0.11% in 2015–2017. Including it as τ_i^{fin} in the stationarity condition (106) raises the bank's required return by the same amount on every municipal bond. The calibration applies non-bank-qualified treatment (Appendix E.1). The tax-neutral residual group, $\xi = 1$, holds other cash and taxable bonds, mainly through foreign investors, pensions, governments, money market funds and nonprofits. Estimated institutional outside durations comprise two P&C and three life-insurer types with separate resource shares, one bank type and a seven-point Gauss–Hermite approximation to a normal distribution of outside durations for the residual group.

Issuer-state markets. Each household type is replicated in each of the 45 markets, with its national resources allocated by the issuer state's share of municipal supply and its relative after-tax share set at that state's rate. The quantitative assignment applies a home-state exemption in every income-tax jurisdiction and taxes out-of-state municipal interest at the resident state's rate. Institutions stay national and choose across all markets, whose independent bond-specific risks offer geographic diversification. The full economy has $45 \times 14 + 13 = 643$ investor types: fourteen household types in each market and thirteen institutional types.

Estimated coefficients. Seventeen coefficients, common to all six calibration blocks, govern what the data do not measure directly and are estimated jointly (Appendix Table 10). Three set the price and amount of risk. Risk aversion γ prices rate and bond-specific risk for every investor. A bond with n years to maturity has two-year bond-specific return variance $v_{30}(n/30)^{p_v}$ in every municipal market and the taxable market for $n \geq 4$, with the two-year claim riskless (Appendix C.5). Three household coefficients, a_H, b_A and b_D, map measured balance sheets into outside durations. Eight set the outside durations and resource shares of the P&C and life-insurer types, one sets the bank's outside duration, and two set the mean and dispersion of the residual group's outside durations. Outside durations are measured in the units of a bond's rate loading: an outside duration h has the rate risk of investing all of a type's resources in a bond whose log price falls by h per unit rise in the rate state.

The 2018–2020 economy. For the 2018–2020 curve comparison, I hold all seventeen coefficients fixed and update the measured inputs. The extension uses the 2019 SCF composition with annually inflation-adjusted incomes, mean 2018–2020 end-year stocks and dated federal tax rules. Institutions face the 21% corporate rate, P&C proration of 25% and a life-insurer policyholders' share of 30%. The annual bank financing-tax cost is 0.117%, and insurer portfolio fractions are averaged over the available 2018–2019 holdings. This seventh economy supplies the post-2017 curve observations without adding calibration targets. Appendix C.3 reports its equilibrium residuals.

C.8 Bond-specific return risk and its normalization

The independent shock in (31) is a reduced-form component of a bond's holding-period return after conditioning on aggregate rates. For example, an unexpected change in demand for one issue can affect its realized sale proceeds without moving the common rate state. The variance cost grows with the position, making each investor's demand finite and allowing several tax clienteles to hold a maturity. The shock vanishes for the one-period claim.

The shock factor can equivalently be read as part of next-period resale value:

$$\frac{P^a_{T-1}(z')}{P^a_T(z)} e^{\sigma^a_{\eta,T}\eta} = \frac{P^a_{T-1}(z')e^{\sigma^a_{\eta,T}\eta}}{P^a_T(z)}.$$

Both readings give identical realized returns, conditional moments and portfolio conditions. The price function describes the aggregate rate component, and the residual is specified in holding returns. A separate persistent bond-price state would require its own law of motion and a current price that depends on it, and the chosen convention keeps that state out of the term-structure recursion. Bond-specific risk still affects equilibrium yields through risk compensation and the mean of the multiplicative shock.

These effects are quantitatively small, as stated in Sections 4.1 and 4.5. In the one-market calibration, 20 percent changes in the variances move the thirty-year gap by less than 0.1 basis points in either direction (Table 13). The shock's lognormal mean can also be isolated without changing any holding return. Its conditional mean is $\exp\{(\sigma^a_{\eta,T})^2/2\}$. Replacing the shock factor by the mean-one factor $\exp\{\sigma^a_{\eta,T}\eta^a_{T,t+1} - (\sigma^a_{\eta,T})^2/2\}$ gives the corresponding log prices

$$\check{p}^a_T(z) = p^a_T(z) - \frac{1}{2}\sum_{k=1}^{T}(\sigma^a_{\eta,k})^2, \qquad \check{P}^a_T(z) = e^{\check{p}^a_T(z)}.$$

The shifts in adjacent-maturity prices exactly offset the shock normalization:

$$\frac{\check{P}^a_{T-1}(z')e^{\sigma^a_{\eta,T}\eta-(\sigma^a_{\eta,T})^2/2}}{\check{P}^a_T(z)} = \frac{P^a_{T-1}(z')e^{\sigma^a_{\eta,T}\eta}}{P^a_T(z)}.$$

Every realized holding-period return, its conditional mean and covariance and investor optimality are unchanged. With fixed dollar supplies, so are portfolios and market clearing. The one-period boundary is preserved because $\sigma^a_{\eta,1} = 0$.

For T model periods of $\Delta = 2$ years each, the annualized yield shift is

$$\check{y}^a_T - y^a_T = \frac{1}{2T\Delta}\sum_{k=1}^{T}(\sigma^a_{\eta,k})^2.$$

The fitted variance profile is common to municipal and taxable bonds, so both thirty-year continuously compounded yields rise by 0.855 basis points and their difference is unchanged. The corporate-curve normalization of Section 5.4 keeps the model's municipal-to-taxable yield ratio, which rises when both yields rise equally, so applying it raises the average thirty-year model gap from 97.39 to 97.59 basis points, a change of 0.20 basis points. Reading the factor as part of resale value gives exactly the same result.

D Adjustments and empirical tests

This appendix documents the empirical work behind Sections 3 and 6, beginning with the adjustments in Appendix D.1.

D.1 Comparability and bond-side adjustments

The comparability adjustments (call option and issuance premium) define the municipal puzzle, and the bond-side adjustments (capital gains, liquidity and default) leave the residual before the model of Sections 3.7 and 5.4.

D.1.1 Object and conventions

Common unit and sample. Each adjustment is a basis-point contribution to

$$\text{Spread}(T) = y^{M,\text{raw}}_T - \bar{\xi}_H(T)\, y^C_T. \tag{111}$$

The raw municipal zero-coupon curve $y_T^{M,\text{raw}}$ precedes both comparability adjustments and includes callable and noncallable issues, and y_T^C is the corporate curve of Section 3.6. The benchmark $\bar{\xi}_H(T)y_T^C$ averages each bond's household after-tax corporate yield at its dated federal and state tax rates, and $\bar{\xi}_H(T)$ is this average divided by the averaged corporate yield. These rates include the net investment income tax (NIIT) and, from 2018, the cap on state and local tax (SALT) deductions. The household wedge $1 - \bar{\xi}_H(T)$ averages 41.3%.

Both curves and the benchmark are averaged equally within maturity × issue-quarter × half-notch rating cells, then across ratings within quarters and then across quarters, the three-stage averaging used throughout. Figure 14 shows the resulting puzzle in implied-tax units.

Table 27 divides every adjustment by the full-sample, all-bond Spread(T), while each adjustment keeps its own eligible sample and estimator. The credit default swap (CDS) sample used for default starts in 2004 and contains 1,559,971 curve observations. Table 24 reports its spread under the same dated household tax convention, and the default differential estimated there is added to the full-sample accounting.

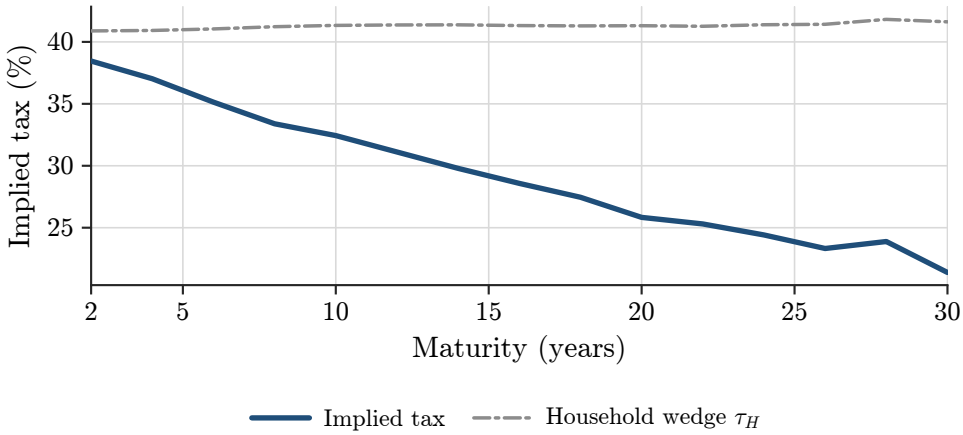


Figure 14: The municipal puzzle in implied-tax units, A-or-above sample, 2000–2020. The implied tax $1 - y_T^M / y_T^C$ uses the call- and issuance-premium-adjusted municipal yields in Figure 2. The household wedge, labeled τ_H in the legend, is $1 - \bar{\xi}_H(T)$.

Bands and additive accounting. The accounting is first order: it estimates each adjustment separately, sets interactions to zero and adds the contributions in common spread units, so the order of removal leaves the sum unchanged.

Appendices D.1.2–D.1.6 give the five estimators in the order of (23), and Appendix D.1.7 combines them.

D.1.2 Adjustment I: embedded call options

Callable yields compensate investors for the issuer's refunding option. More than 95% of bonds beyond twenty years are callable, so dropping them would leave a small long-end subsample.

The call option. Long municipal bonds conventionally allow par calls on coupon dates from year ten. Callable bonds make up 51.1% of curve observations and 51.8% of those at two to thirty years. Investment-grade corporate bonds are usually noncallable bullets or have make-whole strikes that discourage exercise as rates fall. I remove each bond's call value:

$$CV_b \;=\; V_b^{\text{bul}} - V_b^{\text{call}} \;\geq\; 0. \tag{112}$$

The callable and bullet valuations share one fitted option-adjusted spread (OAS) over the model discount curve, which absorbs unpriced credit, liquidity and other factors, so the observed callable price identifies a counterfactual bullet price, conditional on the option model.

Step 1: contractual cash flows. Times are in years under the 30/360 US (NASD) day count, without an end-of-February adjustment or business-day calendar: $\mathrm{DC}(u,v) = [360(Y_v - Y_u) + 30(M_v - M_u) + D_v^* - D_u^*]/360$ for year Y, month M and day D, where $D_u^* = \min(D_u, 30)$, and $D_v^* = 30$ if $D_v = 31$ and $D_u^* = 30$, otherwise $D_v^* = D_v$. A bond with remaining maturity T_b and coupon interval $\Delta = 0.5$ years has $n = \lceil T_b/\Delta - 10^{-9} \rceil$ coupon dates, a first stub $s = T_b - \Delta(n-1)$ and coupon nodes $t_k = s + \Delta(k-1)$, $k = 1, \ldots, n$.

With $f = \mathrm{DC}(\text{settlement}, \text{first coupon})$, the first paying node $\underline{k}$ satisfies $\underline{k} - 1 = \mathrm{round}((f-s)/\Delta)$. Each bond must satisfy $|f - s - \Delta(\underline{k}-1)| \le 0.02$ years, $0 \le \underline{k} - 1 \le 2$ and $\underline{k} < n$. Stubs are rounded to 1/360 year to form groups with common $(n, s, \underline{k})$, which keeps t_n within half a day of T_b. The accrual fraction a_k is zero before $\underline{k}$, equals $s + \Delta(\underline{k}-1)$ at $\underline{k}$ and equals Δ afterward, and the cash flow at node k is $100ca_k + 100\mathbf{1}\{k = n\}$ for annual coupon rate c.

If the dated date, from which interest accrues, precedes settlement, accrued interest $100c\alpha$ is added to the clean target price and to the first paying coupon, with $\alpha = \mathrm{clip}(\mathrm{DC}(\text{dated}, \text{first coupon}) - [s + \Delta(\underline{k}-1)], 0, 1)$. About 29% of observations require this correction.

Step 2: recurring calls. Recorded call times $t^{\mathrm{call}} = \mathrm{DC}(\text{settlement}, \text{call date})$ in $(0, T_b + 0.15)$ are kept, with the lowest strike at ties. Each call maps to node $k = \mathrm{round}((t^{\mathrm{call}} - s)/\Delta) + 1$, which must satisfy $\underline{k} \le k \le n$ and $|t_k - t^{\mathrm{call}}| \le 0.02$ years. The strike is $K_k = \infty$ before the first accepted call date, and thereafter the last recorded strike is carried forward to every coupon date. Special-call provisions are not modeled. A bond marked callable with no accepted call date is excluded, and other bonds without calls have $CV_b = 0$.

Step 3: state-dependent pricing. The issue year selects one of seven dated economies: the six calibration blocks or a 2018–2020 economy using 2018–2019 insurer portfolio fractions and 2018–2020 quantities, rates and taxes. Supply-weighted municipal log prices are interpolated over maturities $\{0, 2, \ldots, 30\}$, including $p_0 = 0$:

$$\log P_T(z) = \mathrm{Interp}_T \left\{ \sum_s w_s p_T^{M_s}(z) \right\}, \qquad w_s = \frac{\text{muni supply in state } s}{\text{total muni supply}}. \tag{113}$$

The OAS is measured relative to this mean curve.

The model's two-year AR(1) in (5) is converted to annual and h-year steps:

$$\rho_y = \sqrt{\rho}, \quad \sigma_y = \frac{\sigma_r}{\sqrt{1+\rho_y^2}}, \quad \rho_h = \rho_y^h, \quad \sigma_h = \sigma_y \sqrt{\frac{1-\rho_y^{2h}}{1-\rho_y^2}}. \tag{114}$$

Pricing yield volatility is measured volatility times a scale m, set before the pricing drift is fitted, by rescaling log prices around $z = 0$:

$$\log P_T^{\mathrm{scaled}}(z) = \log P_T(0) + m\kappa_T[\log P_T(z) - \log P_T(0)], \qquad \kappa_T = \sigma_T^{\mathrm{data}}/\sigma_T^{\mathrm{model}}. \tag{115}$$

Here $\sigma_T^{\mathrm{model}} = T^{-1}|\partial_z \log P_T|_{z=0}\sigma_y\sqrt{2/(1+\rho_y)}$ is the model's standard deviation of twelve-month changes in the T-year yield, and σ_T^{data} is its counterpart in AA-and-above zero-coupon yields, measured in the same quarter of consecutive years after averaging rating cells to maturity–quarter means, with fifty quarters per maturity and a count-weighted $\alpha_0 + \alpha_1 e^{-T/\alpha_2}$ smoothing fit.

The option model fits the conditional mean μ_j of the next rate state at each grid node z_j. This option-pricing mean is distinct from the physical conditional mean and the holding return μ_T^q. The fit uses the model's one-step prices at $T = 1, 1.5, \ldots, 30$, with fifteen-point Gauss–Hermite nodes x_q and weights w_q:

$$\mu_j = \arg\min_{\mu} \sum_T \left[\log P_\Delta(z_j) + \log \sum_q \frac{w_q}{\sqrt{\pi}} P_{T-\Delta}(\mu + \sigma_\Delta \sqrt{2} x_q) - \log P_T(z_j) \right]^2. \tag{116}$$

Prices and drifts are interpolated to 201 nodes spanning the model grid. Transitions Π_{zj} over one coupon interval split quadrature mass between adjacent nodes and clamp tails to the end nodes z_1 and z_N. Over the first stub, $\Pi^{(s)}_{zj}$ uses variance σ_s^2 and drift $\mu^{(s)}(z) = z + (s/\Delta)[\mu(z) - z]$. The median corporate one-year yield y_1, in percent, in the final month of the issue quarter sets $x_0 = \text{clip}(2y_1/100 - \bar{r}, z_1, z_N)$, and contractual dates remain unchanged.

Step 4: callable value and OAS. Starting from $V_n(z) = 100ca_n + 100$, the callable value is computed backward at a flat continuous spread θ:

$$C_k(z) = P_\Delta(z)e^{-\theta\Delta}\sum_j \Pi_{zj}V_{k+1}(z_j), \qquad V_k(z) = 100ca_k + \min\{C_k(z), K_k\}. \tag{117}$$

The issuer calls when the continuation value $C_k(z)$ exceeds the strike K_k, and exercise is ex-coupon. The first stub is discounted using $P_s(z)e^{-\theta s}\sum_j \Pi^{(s)}_{zj}V_1(z_j)$ and interpolated at x_0. The OAS θ^* equates this value to the dirty callable price and is found by 42 bisections on $[-0.20, 0.30]$, which leave an interval of width about 10^{-13} and a repricing error below 10^{-9} per 100 face.

Step 5: bullet value. Removing the call cap in (117), with the OAS and cash flows held fixed, gives the call value $CV_b = \max\{V^{\text{bul}}(\theta^*) - V^{\text{call}}(\theta^*), 0\}$ and the call-adjusted price $P_b^{\text{adj}} = P_b^{\text{dirty}} + CV_b$.

Step 6: zero-coupon yields. On the maturity-interpolated mean discount curve D_0 at $z = 0$, Newton's method solves for a flat continuous spread $\hat{s}$, separately for the raw dirty price treated as a bullet and for the adjusted price:

$$100c\sum_k a_k D_0(t_k)e^{-\hat{s}t_k} + 100D_0(T_b)e^{-\hat{s}T_b} = P_b^{\text{target}}, \tag{118}$$

$$y_b^{\text{target}} = 200\left[\exp\left\{\frac{y_0(T_b) + \hat{s}}{2}\right\} - 1\right], \qquad y_0(T_b) = -\frac{\log D_0(T_b)}{T_b}. \tag{119}$$

These nominal semiannual yields, in percent, give a call spread $y_b^{\text{raw}} - y_b^{\text{adj}}$ that is nonnegative because the adjusted price is no lower.

From bonds to the curve. The 1,828,379 A-or-above pricing bonds are drawn from 1,990,791 issues (Table 1). Bonds must have $T_b \in (1, 30]$, at least two coupons, a complete offering price, coupon and yield, fixed or original-issue-discount (OID) coupons, an offering price of at least $100 - 0.25\,T_b$ per 100 face and consistent state-tax flags. Calendar and price screens exclude 16,295 bonds for irregular first coupons, unsupported conventions, missing accrual dates or unusable call schedules. Eligible bonds retain 97.6%–99.8% of the three-stage weights at each maturity. The three-stage mean of their call spreads is Call(T) in (23), which is subtracted from the observed curve.[23]

The call table uses the screened A-or-above pricing sample, while the main puzzle retains the full observed curve and subtracts the same call adjustment. Screening changes three-stage municipal yields by at most 1.1 bp and the household benchmark by less than 0.1 bp.

Identification of the volatility level. The scale m is fitted to AA-and-above callable and noncallable bonds matched within maturity × issue-quarter × rating × half-point-coupon cells. Their mean callable-minus-noncallable yield difference, averaged across cells and then quarters, is the market gap $\text{gap}_T^{\text{mkt}}$, with across-quarter standard error se_T, and model call spreads in the same cells give the model gap $\text{gap}_T(m)$. If

[23] Premium bonds with early par calls can require a negative OAS and produce extreme call spreads, and they remain in the sample. Under full three-stage weights, call spreads above 100 basis points contribute at most 0.34 basis points to any maturity's three-stage mean, and 0.25 at thirty years.

callability is the only difference, the correct m equates the two gaps. The fitted m is applied to A-or-above bonds without re-estimation. Runs at $m = 1$ and 1.15 give the local elasticity

$$e_T \;=\; \frac{\ln \mathrm{gap}_T(1.15) - \ln \mathrm{gap}_T(1)}{\ln 1.15}, \tag{120}$$

so that $\mathrm{gap}_T(m) \approx \mathrm{gap}_T(1)\, m^{e_T}$ locally. The multiplier m is fitted by weighted least squares at integer maturities six through twenty with twelve or more quarters and positive market and model gaps:

$$\ln \mathrm{gap}_T^{\mathrm{mkt}} \;=\; \ln \mathrm{gap}_T(1) \;+\; e_T \ln m \;+\; \varepsilon_T, \qquad \text{weight } w_T = \left(\mathrm{gap}_T^{\mathrm{mkt}}/\mathrm{se}_T\right)^2, \tag{121}$$

whose solution is

$$\ln \hat{m} \;=\; \frac{\sum_T w_T\, e_T\, \ln\!\left(\mathrm{gap}_T^{\mathrm{mkt}}/\mathrm{gap}_T(1)\right)}{\sum_T w_T\, e_T^2}. \tag{122}$$

The estimate $\hat{m} = 1.089$ has a conditional 95% band of $[0.972, 1.204]$ from 2,000 bootstrap draws of independent quarters, which preserve within-quarter dependence across maturities and fix $\mathrm{gap}_T(1)$, e_T and w_T. The band contains measured volatility ($m = 1$).

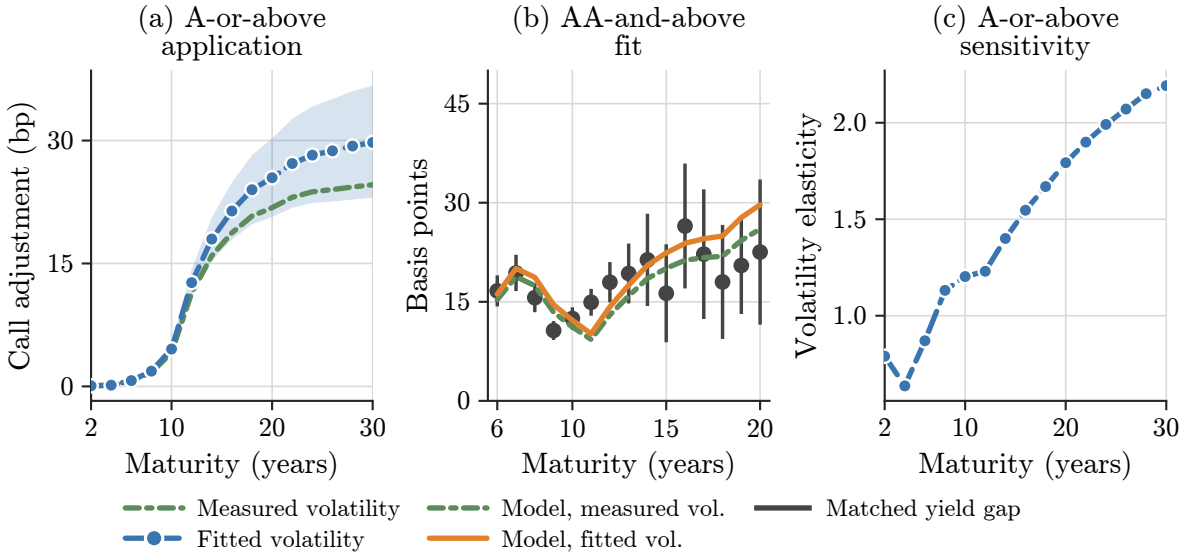


Figure 15: Call adjustment and volatility calibration. (a) A-or-above call adjustments at measured and fitted volatility, with direct-repricing endpoints at the bootstrap band of $\hat{m}$. (b) AA-and-above coupon-matched market gap (±1.96 standard errors) and model gaps at the fitting maturities, 6–20 years. (c) All-bond elasticity e_T^{all} of the A-or-above call adjustment, distinct from the matched-cell elasticity e_T used in the volatility fit.

Result. At $\hat{m}$, the call adjustment for screened A-or-above bonds is small through ten years, 4.6 bp at ten, and reaches 29.8 bp at thirty (Table 17 and panel (a) of Figure 15), with a directly repriced range of $[23.0, 36.7]$ bp. Measured volatility alone gives 24.6 bp.

Direct-repricing validation of the elasticity. At $\hat{m}$ and the two band endpoints, and at all fifteen reporting maturities, the all-bond elasticity approximation is within 0.239 bp of direct repricing, although both endpoints lie outside the pair used to compute the elasticity, below 1.00 and above 1.15.

Validation against the noncallable subsample. The call-adjusted all-bond and noncallable curves in shared cells (Figure 16) differ by less than three basis points through ten years. On the displayed two-year

Table 17: Call adjustment by maturity (bp). The central estimate is a direct repricing at $\hat{m}$, and the range reprices at the 2.5th and 97.5th percentiles of its bootstrap distribution. The measured-volatility column ($m = 1$) and e_T^{all} are diagnostics.

T	measured ($m = 1$)	e_T^{all}	central ($\hat{m}$)	range
2	0.1	0.79	0.1	[0.1, 0.1]
4	0.1	0.64	0.1	[0.1, 0.2]
6	0.7	0.87	0.7	[0.6, 0.8]
8	1.7	1.13	1.9	[1.6, 2.1]
10	4.1	1.20	4.6	[4.0, 5.1]
12	11.4	1.23	12.7	[11.0, 14.3]
14	15.9	1.40	18.0	[15.3, 20.6]
16	18.7	1.55	21.4	[17.9, 24.9]
18	20.8	1.67	24.0	[19.8, 28.2]
20	21.8	1.79	25.5	[20.7, 30.3]
22	23.1	1.90	27.2	[21.8, 32.7]
24	23.7	1.99	28.2	[22.4, 34.2]
26	24.0	2.07	28.7	[22.6, 35.1]
28	24.3	2.15	29.3	[22.8, 36.0]
30	24.6	2.19	29.8	[23.0, 36.7]

maturity grid, the largest absolute difference is eight basis points, at eighteen years. At twenty years the adjusted all-bond yield lies 3.6 basis points below the noncallable yield.

Robustness to option-pricing dynamics. I repeat the valuation with a normal one-factor Hull–White model (Hull and White, 1994), whose time-varying drift fits each dated input discount curve exactly, with separately set mean reversion and volatility. With annual mean reversion fixed at 0.10, fitting volatility to the same AA-and-above coupon-matched gaps at six to twenty years gives diffusion volatility 0.0127, with rates in decimals and time in years. Cash flows, call schedules and the OAS procedure are unchanged. Probability-weighted paired repricings across the seven dated economies give A-or-above adjustments of 4.7, 23.8 and 28.9 basis points at ten, twenty and thirty years, compared with 4.6, 25.5 and 29.8 in the baseline.[24] The root-mean-square error of the Hull–White matched-gap fit is 3.8 basis points, against 4.2 for the baseline model. In the same noncallable-validation cells, the adjusted curve lies 2.1 basis points below the observed noncallable curve at twenty years, compared with 3.6 in the baseline.

D.1.3 Adjustment II: the issuance premium

Spread(T) compares municipal yields at issuance with a fitted curve of seasoned corporate bonds. The adjustment Prem(T) in (23) removes the positive, placement-related difference between offering and secondary-market yields.

Offering and zero-coupon yields. The panel has 1,828,379 A-or-above issues, each observed once at issuance. The premium is measured in offering yields and subtracted one for one from the zero-coupon curve. At five, ten and thirty years, the two yields have correlations 0.998, 0.993 and 0.937, and the conversion from coupon to zero-coupon yields raises mean yields by 2.8, 12.7 and 58.0 bp.

Measurement. Trades come from the Municipal Securities Rulemaking Board (MSRB) dealer-to-customer tape. Both sides use customer purchases, because an offering is an investor purchase, and

[24] The comparison reprices a probability sample of 3,136 draws, covering 2,843 distinct callable contracts, under both models. Weighted mean differences from baseline pricing are added to the baseline population adjustments, which keep the same three-stage weights and zero call values for noncallables. The conditional numerical-integration standard error of the paired thirty-year difference is 0.056 basis points.

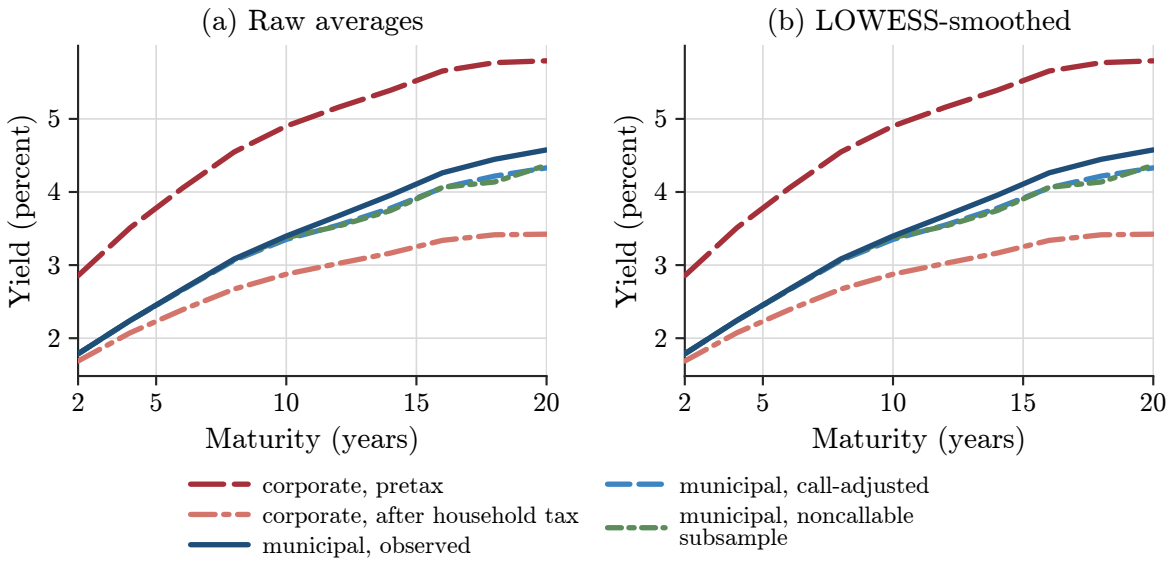


Figure 16: Noncallable validation of Table 17: raw means (a) and locally weighted scatterplot smoothing (LOWESS) (b). Lines show pretax and household after-tax corporate yields alongside observed, call-adjusted and noncallable municipal yields. The call-adjusted and noncallable curves share A-or-above maturity–quarter–rating cells containing noncallables, with fifty or more quarters per maturity and no extrapolation past twenty years. Remaining coupon, credit and other differences prevent identification of individual option values.

comparing it with customer sales would add the bid–ask spread and double-count liquidity (Appendix D.1.5). Same-week seasoned bonds control for market drift:

$$\text{Premium}_b = [y_b^{\text{off}} - \bar{y}_k^{\text{seas}}(w_b^{\text{off}})] - [y_{bj}^{\text{tr}} - \bar{y}_k^{\text{seas}}(w_{bj})]. \tag{123}$$

Here j indexes eligible prints, w weeks and k maturity buckets. Each bond's premium averages over its eligible prints, customer purchases within thirty days of the offering date excluding when-issued trades. The benchmark is the par-weighted yield of same-week customer purchases of bonds in the bucket older than three years, with at least 25 prints per cell. Differencing removes common rate changes and fixed credit differences, while changing credit and call exposure can remain. Early and seasoned prints must have at least $250,000 par, and prints of all sizes give 21.1 bp at thirty years (Panel B of Table 18).

Sample and aggregation. The sample is restricted to the curve's own filtered population. Using all Mergent offering yields would put only 74% of estimation bonds in that universe. Trade age and benchmark weeks are measured from the offering date. The dated date differs from it by a median of fourteen days, by more than a week for 82.7% of bonds and by more than a month for 8.4%, and it precedes the offering for 13.7%.

The three-stage averaging of Appendix D.1.1 is applied within issue-quarter and rating cells and the maturity buckets 1–5, 5–8, 8–12, 12–17, 17–23 and 23–31 years. Estimates and bootstrap endpoints are interpolated from the bucket midpoints $(3, 6.5, 10, 14.5, 20, 27)$ to the two-year grid, with ends held flat. The longest-bucket mean is 11.9 bp with par weights, which measure dollars, and 12.8 with curve weights, which measure curve observations.

Result. The issuance premium rises steeply with maturity, from 2.9 bp at two years to 12.8 bp at thirty (Table 27 and panel (b) of Figure 17). Its range comes from a 10,000-draw issue-quarter bootstrap that samples four-quarter blocks with replacement and endpoint wrapping. Lag-one autocorrelations of quarterly bucket means range from −0.34 (longest bucket) to 0.02. The block bootstrap and a bootstrap of independent quarters give similar short-end intervals, [1.7, 4.2] and [1.8, 4.0] at 1–5 years. The thirty-year interval is [8.1,

17.3] bp, and the intervals exclude zero at every maturity. Intervals condition on the matched sample and seasoned benchmark.

Sensitivity to the trade-age window. In all five trade-age windows of Table 18, premia beyond twelve years exceed those below twelve years. From fourteen days on, the two longest buckets give 9.3–12.8 bp. Three- and seven-day samples have fewer institutional-size prints and are noisier. Panel B shows the same gradient at higher levels.

Table 18: Issuance premium by maturity bucket (bp) under alternative trade-age windows, in days; the baseline uses thirty. Panel A, the baseline, requires prints of at least $250,000 par in both the early print and the seasoned benchmark. Panel B uses prints of all sizes.

maturity bucket (yr)	3	7	14	30	60
Panel A: institutional-size prints (baseline)					
1–5	3.7	3.6	1.4	2.9	3.3
5–8	2.8	5.2	3.0	3.3	4.0
8–12	3.4	5.2	3.9	4.0	4.6
12–17	6.2	9.0	7.7	7.5	7.8
17–23	9.1	7.3	9.3	10.8	12.2
23–31	9.5	7.2	11.9	12.8	12.0
Panel B: prints of all sizes					
1–5	4.5	6.7	5.6	3.6	2.2
5–8	5.9	10.1	7.6	6.1	5.0
8–12	7.2	10.5	9.0	7.9	7.4
12–17	15.5	16.2	13.8	12.8	12.4
17–23	15.0	17.3	16.7	16.6	16.7
23–31	18.1	18.7	20.0	21.1	21.0

Notes. Each cell is the drift-controlled difference-in-differences premium for bonds with at least one print inside the window, averaging each bond's prints inside the window and then bonds in the same three stages as the curve.

Timing, conventions and size. Across the six maturity buckets, premia average 5.8 bp within three days of the offering, 6.4–7.0 at three to thirty days and 8.5 and 11.6 at thirty to sixty and sixty to ninety days. Changing composition across these age bins prevents a within-bond interpretation. For par and discount bonds the two longest buckets give 10.9 and 13.2 bp, and for premium bonds 9.3 and 10.9, so yield-to-worst differences do not explain the gradient. Long-end premia are positive in every trade-size decile of all-size prints and smallest for the largest trades (panel (c) of Figure 17). Small-trade markups also raise the all-size estimates.

Interpretation. The drift-adjusted post-offering yield change captures the mismatch between primary and seasoned pricing that matters for the curve comparison. Changing dealer markups, ongoing distribution, liquidity seasoning and price discovery may also contribute, so I treat the change as a proxy for the issuer's placement cost.

Sample coverage. Only bonds with an institutional-size print within thirty days of the offering contribute, so the estimate covers 4% of the target curve, and the direction of that selection is not bounded here. The trade tape begins in 2005 and the curve spans 2000–2020, so the premium is measured on later issues and applied to the whole curve. It also inherits any staleness in recorded offering yields.

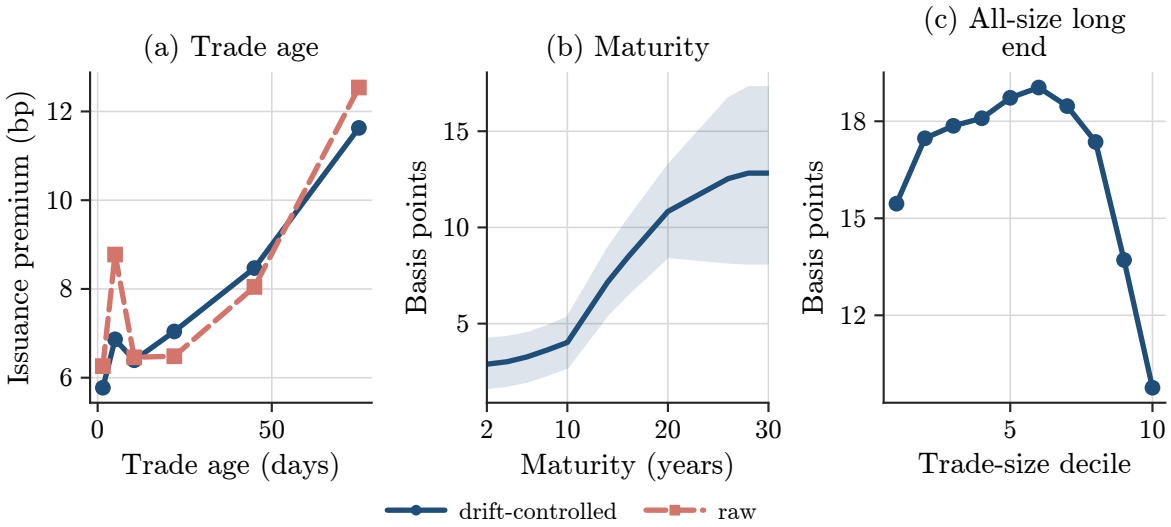


Figure 17: Issuance-premium diagnostics. (a) Institutional-size estimates by trade age, with and without drift controls. (b) Institutional-size maturity profile with block-bootstrap band. (c) All-size long-end estimates by trade-size decile.

D.1.4 Adjustment III: capital-gains taxation

The model's wealth equation (32) in Appendix A.1 taxes the whole holding-period return as interest income. The adjustment $\mathrm{CG}(T)$ in (23) measures how taxing resale gains and losses separately changes the municipal issuance yield, weighting the correction by empirical first-sale horizons.

Taxed amounts and rates. Appendix E.5 gives the accrual and basis rules. Municipal interest and OID are exempt. Market discount and gains are taxable. OID raises basis, original-issue premium (OIP) lowers it without a municipal deduction, and acquisition premium reduces OID. Taxable interest and OID are ordinary income, and premium amortization assumes the Section 171 election. Original buyers use basis A_h (Step 1), and secondary buyers start from the purchase price.

Market discount below a quarter point per full remaining year is de minimis and treated as capital gain. Larger discount is ordinary income at disposition. With n remaining years and face 100, the de minimis threshold is

$$\bar{D}_n = 0.0025 \times 100 \times n. \tag{124}$$

All seven economies use the fixed federal rates $\tau_{oi} = 0.37$ on ordinary income and $\tau_{cg} = 0.20$ on capital gains, excluding NIIT and state tax, unlike the dated household rates in (111). Residual gains and losses are treated symmetrically,

$$\mathcal{T}(G) = \tau_{cg} G, \tag{125}$$

so losses are fully deductible in this calculation.

Benchmark and actual tax rules. For each corporate–municipal bond pair, the benchmark mapping B mirrors the model: it exempts municipal payments and taxes all corporate income, including discount, as ordinary income. The actual mapping A separates interest and capital income under the rules above. Under each mapping, Step 5 solves for the municipal issuance-yield shift δ at which the municipal seller earns the corporate seller's return.

Step 1: the instruments. Each market's par and zero-coupon instruments use its own discount factors d_n, from the model's supply-weighted municipal mean log prices or corporate prices at $z = 0$, interpolated

annually from two-year knots. At even $T = 4, \ldots, 30$ and sale horizons $h = 2, 4, \ldots, T-2$, the two instruments are

$$\text{PAR:}\quad c = \frac{1 - d_T}{\sum_{n=1}^{T} d_n},\quad \text{offer} = 100,\quad A_j \equiv 100; \qquad \text{ZERO:}\quad \text{offer} = 100\, d_T,\quad A_j = \text{offer} \cdot e^{y_0 j}, \tag{126}$$

with $y_0 = -\log(d_T)/T$, where A_j is the original buyer's accreted basis after j years.

For a zero purchased at issue, the interest accruing during year $j+1$ is

$$A_{j+1} - A_j = A_j\big(e^{y_0} - 1\big), \qquad A_T = 100.$$

At a sale in year h, the original buyer's residual capital gain is $P_h - A_h$, so accrued interest is not taxed again as a gain.[25]

Step 2: rate states at the resale date. Sale-date rate states use composite Gauss–Legendre nodes z_i and weights ω_i, split where either market's tax regime changes, and the grid-truncated AR(1) density of the state h years after $z = 0$:

$$w_i \;\propto\; \omega_i \,\frac{1}{\sigma_h}\varphi\Big(\frac{z_i}{\sigma_h}\Big). \tag{127}$$

Here φ is the standard normal density and σ_h is given by (114). Results with twenty-four and with twelve nodes per interval differ by less than 0.001 bp across 2,940 cells, and the residuals of (135), in annualized log return, are at most 10^{-12}.

Step 3: the corporate anchor. The seasoned corporate equilibrium price $P_h^{\text{corp}}(z)$ at the sale date is held fixed. For each mapping $\mathcal{M} \in \{B, A\}$, the remaining annual after-tax cash flows $X_j^{\mathcal{M}}$, contractual payments less the tax due under $\mathcal{M}$, determine a buyer's continuous after-tax return at that price:

$$\sum_{j=1}^{T-h} X_j^{\mathcal{M}}\, e^{-r^*_{\mathcal{M}}(h,z)\, j} \;=\; P_h^{\text{corp}}(z). \tag{128}$$

Flows are annual, with continuous discounting and no day-count or semiannual convention. For a corporate buyer holding to maturity, the two mappings differ only in their de minimis treatment of market discount D:

$$\text{rate}_{\mathcal{M}}(D) = \begin{cases} 0, & D \le 0, \\ \tau_{cg}, & \mathcal{M} = A \text{ and } D < \bar{D}_n, \\ \tau_{oi}, & \text{otherwise.} \end{cases} \tag{129}$$

Step 4: the municipal resale price under each mapping. A municipal buyer holding to maturity under $\mathcal{M}$ must earn $r^*_{\mathcal{M}}$. Benchmark municipal flows CF_k are exempt, giving

$$P^B(h,z) = \sum_{k=1}^{T-h} CF_k\, e^{-r^*_B(h,z)\, k} \;\equiv\; G_B(h,z). \tag{130}$$

The actual price and tax depend jointly on the market discount $A_h - P$. Let G_A be the gross present value at the actual anchor and $q = e^{-r^*_A(h,z)(T-h)}$ the terminal discount factor. Taxing the discount once at maturity at rate τ gives $P = G_A - q\,\tau\,(A_h - P)$, which is linear in P, so

$$P(\tau) = \frac{G_A(h,z) - q\,\tau\, A_h}{1 - q\,\tau}, \tag{131}$$

[25] IRS Publication 1212, *Guide to Original Issue Discount (OID) Instruments*, and Publication 550, *Investment Income and Expenses*, describe accrual and basis. Annual effective yield $e^{y_0} - 1$ corresponds to continuous issuance yield.

and, with $n = T - h$ remaining years, the regime is selected in this order:

$$P^A(h,z) = \begin{cases} G_A, & G_A \geq A_h \quad \text{(no discount: premium, untaxed)}, \\ P(\tau_{cg}), & A_h - P(\tau_{cg}) < \bar{D}_n \quad \text{(de minimis)}, \\ P(\tau_{oi}), & \text{otherwise.} \end{cases} \tag{132}$$

When both regime-specific prices satisfy their conditions, this order selects the de minimis price, the higher of the two.

Step 5: the two shifts. A scalar δ shifts the municipal log-discount curve in parallel:

$$\hat{p}_n^{\delta}(z) = \hat{p}_n(z) - \delta n \quad \Longleftrightarrow \quad y_n^{\delta} = y_n + \delta. \tag{133}$$

Both sellers are compared on the annualized log return

$$R \;=\; \frac{1}{h} \log \frac{\sum_i w_i \, \mathcal{W}(z_i)}{P_0}, \tag{134}$$

where $\mathcal{W}(z_i)$ is terminal wealth at rate state z_i and P_0 is the offering price. Both sellers reinvest interim coupons to h on the unshifted municipal curve. Corporate wealth carries forward coupons net of τ_{oi} and the annual OID tax, and the residual gain $P_h^{\text{corp}} - A_h^C$ is taxed at τ_{oi} under B and at τ_{cg} under A. Municipal wealth equals the resale price plus exempt coupons, less τ_{cg} on its residual gain under A. Bisection solves for each mapping's shift,

$$R^{\text{muni}}(\delta_B; \text{ benchmark}) = R_B^{\text{corp}}, \qquad R^{\text{muni}}(\delta_A; \text{ actual}) = R_A^{\text{corp}}, \tag{135}$$

and the conditional adjustment is their difference,

$$\Delta_{\text{cg}}(h,T) = 10^4 \big(\delta_A - \delta_B\big) \text{ basis points.} \tag{136}$$

The adjustment averages across instrument pairs with issuance weights, in continuously compounded annual yields. The difference $\delta_A - \delta_B$ isolates capital-income taxation from curve mismatch between the markets. Each shift recomputes only the municipal issuance coupon or zero price and basis, together with its resale price. Rate states, weights, corporate prices, cash flows and targets, anchors and the reinvestment curve stay fixed.

Instrument types and issuance weights. Issues with positive coupons and prices of at least 90 are PAR-like, including premium issues above 110, and issues priced below 90 or with zero coupons are ZERO-like. In the Mergent Fixed Income Securities Database (FISD), nonpositive prices denote missing values.

Pooled 1998–2021 amount-weighted ZERO shares are 1.3% municipal and 2.2% corporate (Table 19). The baseline shares match original maturities within three years of each node, pooling when offering amounts are insufficient, and are 0.97% and 0.31% at thirty years. Each of the four pairs in $\{\text{PAR}, \text{ZERO}\}^2$ receives the product of its corporate and municipal shares. The PAR/PAR pair receives 98.7% of the weight at thirty years and drives the adjustment.

First-sale weights. The conditional adjustment $\Delta_{\text{cg}}(h,T)$ is weighted by the original buyer's probability of a first sale at age h. Only first sales count: Step 4 already prices the next buyer's tax under hold-to-maturity, so turnover weights would double-count later dispositions.

Let t_a be seasoned traded par over outstanding par at age a, measured from the MSRB dealer-to-customer tape (71.6 million records, 2005–2021) and pooling original maturities within ± 4 years of T. To exclude primary distribution, I omit when-issued trades and the first quarter after the dated date. A trading

Table 19: Four instrument pairs at $T = 30$: pooled issuance weights and each pair's maximum and mean adjustment over sale horizons h, averaged across the seven dated economies. The baseline uses maturity-matched shares, which put PAR/PAR at 98.7% at thirty years.

corp / muni	weight (%)	max over h (bp)	mean over h (bp)
PAR / PAR	96.5	6.51	5.49
ZERO / PAR	2.2	20.07	14.90
PAR / ZERO	1.3	7.71	5.74
ZERO / ZERO	0.0	15.75	10.91

segment, a share $\phi(T)$ of par, carries all resales, and the remainder is held to maturity. With aggregate turnover fixed, the first-sale hazard at age a is

$$\lambda_a = \min\big(t_a/\phi,\, 0.99\big), \tag{137}$$

where the cap binds only at the first two ages of thirty-year bonds in the central calibration, at which t_a/ϕ would otherwise reach 1.003 and 0.991. The first-sale density and held-to-maturity mass are

$$f_\phi(a) = \phi\,\lambda_a \prod_{j<a}(1-\lambda_j), \qquad S_\phi(T) = 1 - \phi\Big(1 - \prod_{a<T}(1-\lambda_a)\Big), \qquad \sum_a f_\phi(a) + S_\phi(T) = 1, \tag{138}$$

and the unconditional adjustment is $\mathrm{CG}(T) = \sum_h \pi_h(T)\,\Delta_{\mathrm{cg}}(h,T)$, where $\pi_h(T)$ sums f_ϕ over the ages assigned to sale horizon h and the held-to-maturity branch carries a zero adjustment. Each h receives ages $h-1$ and h. The first node also receives age 0, and the last node, with two years remaining, also receives the final pre-maturity age $T-1$.

Calibrating the trading segment. Repeated trades in a narrow segment can generate high turnover without nearly universal resale. Cumulative seasoned volume in each CUSIP (bond identifier), capped at its outstanding par, bounds the par ever sold (Table 20). The parameter ϕ is fitted over the observed window and then extended to full life. At thirty years, par weights over original maturities within four years give a ceiling of 0.602 (single-bin 0.617), so $\phi = 0.602$ and $S_\phi = 0.398$, with first sales concentrated at young ages.

Table 20: The measured ceiling on the fraction of par ever sold, for selected one-year bins of original maturity. Cumulative traded par counts repeat trades of the same dollar, so the ceiling is an upper bound; the band's 0.5 and 0.75 segment multipliers allow for less. Never traded: par share of CUSIPs with no seasoned trade in the window. Entries are par-weighted over issues dated from 2005.

orig. maturity	ceiling on P(ever sold)	never traded	obs. window (y)	CUSIPs
4	0.466	0.195	3.9	124520
6	0.571	0.155	5.6	121412
8	0.556	0.227	7.0	118265
10	0.625	0.132	8.0	110693
14	0.558	0.171	9.4	73189
18	0.546	0.216	9.7	46382
22	0.586	0.219	10.0	13056
26	0.612	0.193	10.6	6551
30	0.617	0.159	10.3	22151

Result. The thirty-year adjustment is 2.3 bp, with range [1.2, 2.5] bp (Table 21 and panel (b) of Figure 18). With (125), the expected tax on the residual gain depends only on its mean.

The range is the envelope of eighteen variants, each averaged equally over the seven 2000–2020 economies: three tape windows (the full window and its two halves), three segment multipliers (0.5, 0.75 and 1.0) and two weighting schemes (pooled and maturity-matched). For multipliers below one, the segment share is

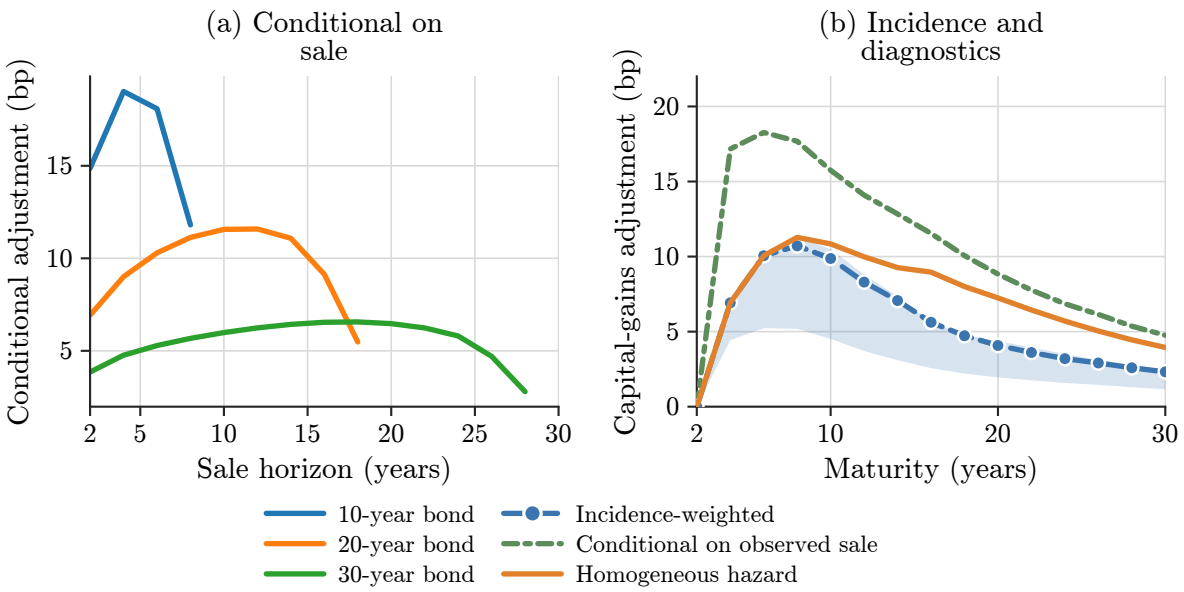


Figure 18: Capital-gains adjustment, averaged over the seven dated economies. (a) Pair-weighted conditional adjustment $\Delta_{\text{cg}}(h, T)$ by sale horizon for ten-, twenty- and thirty-year bonds. (b) Adjustment weighted by first-sale probabilities, with the range of its eighteen variants, and the conditional-on-observed-sale and homogeneous-hazard diagnostics.

refitted on the full window before first-sale weights are measured on each half window. The central estimate uses the ceiling, and the effects of the window and the weights have either sign.

With the same economies, tax rates and trade tape, the thirty-year adjustment is 4.8 bp conditional on an observed sale and 3.9 bp with a homogeneous hazard ($\phi = 1$). At long maturities the homogeneous hazard exceeds the measured resale ceiling, and its adjustment lies above the band. At short maturities the fitted segment share is $\phi = 1$, so the reported and homogeneous-hazard estimates coincide.

D.1.5 Adjustment IV: liquidity

Construction. The imputed round-trip cost (RTC) is $(\max - \min)/\max$ price within CUSIP $\times$ day $\times$ size cells that contain both customer buys and customer sells. The extreme prices need not belong to one dealer's round trip. The comparison uses 2005–2020 MSRB municipal trades and Trade Reporting and Compliance Engine (TRACE) corporate trades of at least \$250,000, made more than three months after issue and more than one year before maturity. Remaining maturity is rounded down to whole years and assigned to the six five-year bins in Table 22. Costs are averaged to bond-months before matching. Trades at the TRACE size caps of 1MM+ and 5MM+ are retained at their lower bounds, and trade size is not otherwise used.

Cells are matched by month $\times$ rating family $\times$ maturity bin, with ten or more bonds per market. Agency letter grades map AAA to notch one, AA to two–four and A to five–seven on the twenty-two-notch scale. Averages weight ratings equally within months, then months within years, then years.

Let c_m and c_c be the matched municipal and corporate costs in percent of price. One purchase and resale over a holding period of h years requires annual compensation of c_m/h and c_c/h in the two markets, giving the yield differential in basis points

$$\Delta y^{\text{liq}} = \underbrace{\frac{100\, c_m}{h}}_{\text{municipal cost per year}} - \underbrace{\frac{100\, c_c}{h}}_{\text{corporate cost per year}} = \frac{100(c_m - c_c)}{h}.$$

Table 21: Capital gains by maturity (bp). The reported column uses first-sale probabilities, with the eighteen-variant envelope as its range. The cond. on sale and homogeneous columns are diagnostics. The maximum is over sale horizons of the dated-average, pair-weighted conditional curve. Columns ϕ and P(hold) give the trading-segment share and the held-to-maturity probability. At two years no sale horizon precedes maturity on the two-year grid, so the adjustment is zero.

T	cond. on sale	max over h	ϕ	P(hold)	homogeneous	reported	range
2	0.0	0.0	–	1.00	0.0	0.0	[0.0, 0.0]
4	17.2	17.2	1.00	0.60	6.9	6.9	[4.4, 7.0]
6	18.3	19.2	1.00	0.45	10.1	10.1	[5.2, 10.3]
8	17.7	19.9	0.88	0.40	11.3	10.7	[5.2, 11.3]
10	15.7	19.0	0.82	0.37	10.8	9.9	[4.5, 10.5]
12	14.1	17.8	0.73	0.39	10.0	8.3	[3.7, 8.8]
14	12.8	16.2	0.67	0.40	9.3	7.1	[3.1, 7.3]
16	11.5	14.6	0.58	0.44	9.0	5.6	[2.6, 5.6]
18	10.1	13.0	0.57	0.44	8.0	4.7	[2.2, 4.8]
20	8.8	11.6	0.57	0.43	7.2	4.1	[2.0, 4.4]
22	7.8	10.4	0.58	0.42	6.4	3.6	[1.8, 3.9]
24	6.9	9.3	0.59	0.41	5.7	3.2	[1.6, 3.5]
26	6.1	8.5	0.60	0.40	5.0	2.9	[1.5, 3.1]
28	5.4	7.4	0.60	0.40	4.5	2.6	[1.3, 2.8]
30	4.8	6.6	0.60	0.40	3.9	2.3	[1.2, 2.5]

Both markets use $h = 5$. Trading costs reduce after-tax wealth in both markets, so the differential is not multiplied by an interest-income after-tax share. A positive difference raises the yields municipal holders require and is subtracted from the puzzle.

Regressing A-or-above offering yields on bond-average round-trip costs, with maturity-bin × rating × offer-year effects, gives 15.4 bp per percentage point of price in round-trip cost, compared with 20 at $h = 5$, so this regression map scales $h = 5$ estimates by 0.772. Standard errors are the standard deviation of sixteen annual differences divided by $\sqrt{16}$. The band spans $h = 2$, $h = 10$, the regression map and the central estimate ± 1.96 standard errors.

The six bin estimates at $3, 8, 13, 18, 23, 28$ years are interpolated linearly to $2, 4, \ldots, 30$, with end values held flat, to give Liq(T) in (23). Thirty years uses the 26–30 bin, and the interpolation avoids artificial jumps across maturities.

Result. Institutional-size municipal round trips are slightly cheaper than corporate ones before twenty years and costlier afterward (Table 22 and Figure 19). At 26–30 years, costs of 0.5335% and 0.5310% imply 0.050 bp annually at $h = 5$, which is reported as 0.1 bp, with a band of $[-2.2, 2.3]$ bp. The higher-cost 21–25 bin produces the peak adjustment of 2.9 bp at twenty-four years.

At $h = 5$, standard errors are 0.8–2.3 bp. The 21–25 estimate is about 1.5 standard errors from zero and depends on the holding period. Appendix D.2 gives separate issuer-level regressions on trading frequency and trading costs, which are not part of the band.

Table 22: Liquidity by maturity bin. Round-trip costs (RTC) in percent of price, and adjustments in bp under three holding periods h and the regression map.

bin	muni RTC	corp RTC	difference	$h = 2$	$h = 5$	$h = 10$	regr.
1–5	0.225	0.246	-0.022	-1.08	-0.43	-0.22	-0.33
6–10	0.382	0.403	-0.021	-1.05	-0.42	-0.21	-0.33
11–15	0.582	0.588	-0.006	-0.30	-0.12	-0.06	-0.09
16–20	0.609	0.619	-0.010	-0.49	-0.20	-0.10	-0.15
21–25	0.684	0.506	0.178	8.92	3.57	1.78	2.75
26–30	0.533	0.531	0.003	0.13	0.05	0.03	0.04

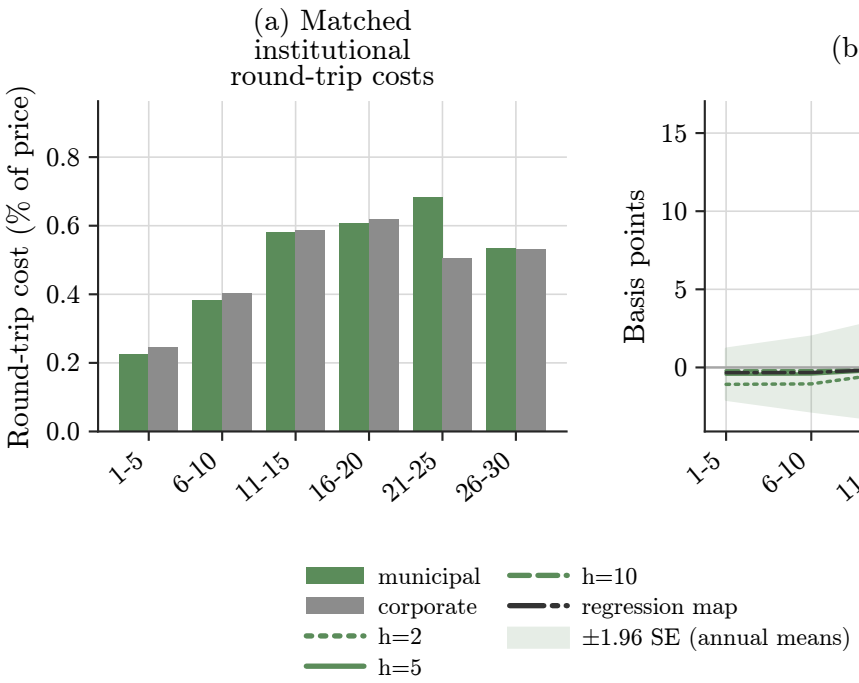


Figure 19: Matched institutional-size customer trades of at least $250,000, 2005–2020. Panel (a): imputed round-trip costs in common month, rating and maturity cells, percent of price. Panel (b): the municipal-minus-corporate cost difference annualized under three holding periods and under the regression map. Shading is ±1.96 standard errors calculated across annual means at $h = 5$.

Sensitivity to the cell rule. Table 23 varies the minimum number of bonds in each market's cells, using the same underlying trades and size screen. A one-bond minimum admits thin cells and changes the 26–30 year estimate from 0.05 to −6.53 bp. A twenty-bond minimum leaves only 19 common months at 11–15 years and changes the estimate there from −0.12 to −2.99 bp. The ten-bond minimum balances coverage against the influence of thin cells.

Table 23: Liquidity adjustment at $h = 5$ (bp) under alternative minimum bond counts per market cell.

bin	≥ 1	≥ 5	≥ 10 (used)	≥ 20
1–5	-0.59	-0.50	-0.43	-0.39
6–10	-0.79	-0.49	-0.42	-0.27
11–15	-0.12	0.42	-0.12	-2.99
16–20	-0.09	-0.30	-0.20	-0.31
21–25	2.05	2.20	3.57	1.81
26–30	-6.53	-5.81	0.05	0.30

D.1.6 Adjustment V: default

Method. For $\mathrm{Def}(T)$ in (23), I use the simple taxable-equivalent convention in Schwert (2017), which divides municipal interest by the household after-tax share before comparing it with taxable interest. State-government and pooled AA corporate Markit CDS spreads serve as proxies for credit compensation on this common scale. Let s_b^M and s_b^C denote the matched CDS spreads for curve observation b, and $\xi_{H,b}$ its dated household after-tax share. The default differential is

$$\mathrm{Def}(T) = \mathrm{Avg}_T[\xi_{H,b} s_b^M] - \mathrm{Avg}_T[\xi_{H,b} s_b^C]. \tag{139}$$

Both CDS quotes are pretax. Multiplying each by the dated household after-tax share once puts their difference on the scale of $y_T^M - \bar{\xi}_H(T) y_T^C$, which adapts Schwert's mapping to these proxies. The after-tax share includes federal and state rates, NIIT from 2013 and the 2018 change in state-tax deductibility. No further corporate-tax or loss-deduction factor is applied.

Matching and extrapolation. State general-obligation (GO) CDS are matched to A-or-above bonds by rating, maturity and quarter, and pooled quarterly AA corporate CDS form the corporate leg. After-tax shares are applied before Avg_T averages within maturity–quarter–whole-notch-rating cells, then across ratings and quarters, retaining each leg's nonmissing support. The AA-and-above sample of Appendix D.12.1 uses the same construction. Beyond quoted tenors, the cumulative spread $T\,\text{CDS}(T)$ is extrapolated linearly, with negative filled spreads floored at zero and nonnegative quotes left unchanged.

Coverage and sensitivity. In the A-or-above CDS panel, exactly observed cells cover 49% at 1–10 years, 47% at 11–20 and 13% at 21–30, and the remaining cells use filled curves (panel (c) of Figure 20). The sensitivity range is the pointwise minimum and maximum of five variants, each with the same household tax mapping: the baseline, cells with at least half of their source observations quoted or interpolated (quality-capped), the 2007-onward sample and CDS–cash basis factors 0.75 and 0.50 (Table 25).

Result. The thirty-year default adjustment is 16.5 bp, with a sensitivity range of $[8.2, 20.8]$ bp. It equals 9.9% of the CDS-matched spread and 10.3% of the full-sample denominator $\text{Spread}(T)$, and it remains positive at every maturity (Table 24).

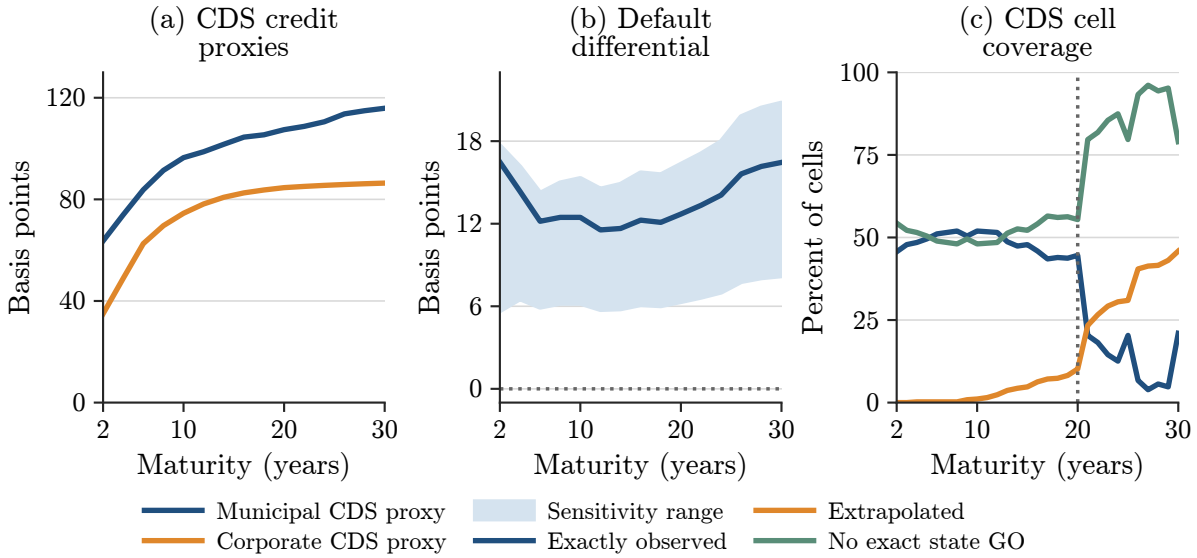


Figure 20: Default adjustment on the A-or-above CDS panel. (a) Municipal and corporate CDS proxies in taxable-equivalent units. (b) Their differential after the dated household tax mapping in (139), with the five-variant sensitivity range. (c) Exactly observed, extrapolated and no-exact-state-GO shares of rating-matched CDS cells. The last two overlap, so the three shares are not a partition. The vertical rule marks twenty years.

D.1.7 The combined effect

Table 26 reports the accounting at selected maturities, and Table 27 gives every maturity with each adjustment's band.

The call and issuance-premium adjustments grow with the spread. The three bond-side adjustments together remain near 17–20 bp from twelve years onward. Capital gains peak at intermediate maturities

Table 24: Default by maturity. Yields in percent, spread and component (the default adjustment) in bp. The CDS-matched spread uses each observation's dated federal and state household after-tax share, as does the full-sample Spread(T), on a different sample.

T	muni yield	corp after-tax	spread (bp)	component (bp)	band
2	1.56	1.51	5.7	16.48	[5.69, 17.64]
4	1.99	1.85	14.6	14.35	[6.55, 16.10]
6	2.39	2.15	24.8	12.18	[5.94, 14.20]
8	2.79	2.42	37.0	12.46	[6.23, 14.95]
10	3.12	2.63	49.5	12.46	[6.23, 15.27]
12	3.47	2.79	68.1	11.55	[5.77, 14.50]
14	3.73	2.89	84.0	11.65	[5.83, 14.85]
16	3.93	2.96	97.1	12.26	[6.13, 15.68]
18	4.09	3.01	108.4	12.10	[6.05, 15.54]
20	4.26	3.04	122.2	12.70	[6.35, 16.30]
22	4.36	3.07	129.0	13.34	[6.67, 17.06]
24	4.47	3.08	138.3	14.08	[7.04, 17.94]
26	4.59	3.09	149.3	15.62	[7.81, 19.77]
28	4.58	3.09	149.5	16.16	[8.08, 20.39]
30	4.77	3.11	165.7	16.45	[8.23, 20.75]

Table 25: Default sensitivity in basis points. Every column uses the dated household tax mapping of (139). The basis columns scale both CDS proxies by the stated factor. Quality-capped cells have at least half of their source observations quoted or interpolated.

T	2007+	basis 0.50	basis 0.75	baseline	quality-capped
2	17.64	8.24	12.36	16.48	5.69
6	14.20	6.09	9.13	12.18	5.94
10	15.27	6.23	9.35	12.46	8.35
14	14.85	5.83	8.74	11.65	8.02
18	15.54	6.05	9.07	12.10	8.76
22	17.06	6.67	10.00	13.34	11.24
26	19.77	7.81	11.71	15.62	17.47
30	20.75	8.23	12.34	16.45	18.66

because the resale-tax wedge is amortized over the remaining life and the de minimis threshold rises with it. At thirty years, the residual before the model is 98.4 bp, and the normalized model leaves an unexplained residual of 1.0 bp (Figure 8). Negative residuals before the model at the shortest maturities refer to the full curve, and an adjustment's own matched sample can give a different sign.

Model and observed curves. Figure 8 and Table 4 report the normalized and unnormalized model comparisons of Section 5.4. The normalization (22) preserves the empirical household benchmark.

Bars denote sample averages, and superscripts m and d distinguish model and data. With model municipal and taxable yields $\bar{y}_T^{M,m}$ and $\bar{y}_T^{C,m}$, observed corporate yield $\bar{y}_T^{C,d}$ and observed household benchmark B_T^d, the normalization is

$$\widetilde{y}_T^M = \frac{\bar{y}_T^{M,m}}{\bar{y}_T^{C,m}} \bar{y}_T^{C,d}, \qquad \widetilde{\mathcal{G}}_T = \widetilde{y}_T^M - B_T^d.$$

Here $B_T^d = \bar{\xi}_H(T)\bar{y}_T^{C,d}$, with $\bar{\xi}_H(T)$ as in (111). At thirty years the model's taxable yield exceeds the observed corporate yield by 39 basis points, mostly reflecting 2018–2020. Model and data yields are converted to nominal semiannual units before date-by-date after-tax products are formed and averaged with the empirical bond, rating and issue-quarter weights of Appendix D.1.1. Simply rescaling the unnormalized model gap would also carry over the model's own averaged household benchmark, whereas the benchmark here uses

Table 26: What explains the municipal spread over the household benchmark: contributions in basis points to Spread(T) of equation (111). The two comparability columns remove the call option and the issuance premium from the municipal yield itself, giving the puzzle of Figure 2. The three bond-side columns, headed differentials, are municipal-minus-corporate differentials and do not map to a single adjusted curve.

		comparability			differentials					
T	Spread	Call	Issuance premium	Puzzle	Capital gains	Liquidity	Default	Total differential	Share of puzzle	Residual
2	9.9	0.1	2.9	6.9	0.0	−0.4	16.5	16.0		−9.1
6	27.9	0.7	3.3	23.9	10.1	−0.4	12.2	21.8	91%	2.1
10	52.2	4.6	4.0	43.6	9.9	−0.3	12.5	22.0	51%	21.6
14	87.0	18.0	7.2	61.8	7.1	−0.1	11.7	18.6	30%	43.2
18	110.3	24.0	9.6	76.7	4.7	−0.2	12.1	16.6	22%	60.1
22	128.8	27.2	11.4	90.2	3.6	2.8	13.3	19.8	22%	70.5
26	145.0	28.7	12.5	103.7	2.9	1.5	15.6	20.0	19%	83.7
30	159.8	29.8	12.8	117.2	2.3	0.1	16.5	18.8	16%	98.4

Notes. The puzzle is the spread less the call and issuance-premium adjustments. The total differential sums the three bond-side adjustments, the share divides that sum by the puzzle (omitted where the puzzle is below 20 basis points), and the residual is the residual before the model, the puzzle less the sum. Thirty-year bands, in basis points: call $[23.0, 36.7]$, issuance premium $[8.1, 17.3]$, capital gains $[1.2, 2.5]$, liquidity $[-2.2, 2.3]$, default $[8.2, 20.8]$. Using unrounded band endpoints gives a thirty-year residual before the model of $[80.4, 121.4]$ bp, an envelope with no joint coverage level. Appendix D.1 gives the construction. The issuance-premium estimate uses the 4% of curve bonds with an institutional-size trade within one month of issuance.

Table 27: All five adjustments by maturity, in bp of Spread(T) (equation (111)). Brackets give each adjustment's band. The total band adds the band endpoints and has no joint coverage level. The final column divides the total by Spread(T) and is suppressed below 20 bp of observed spread. Table 26 divides the three bond-side adjustments by the puzzle.

T	call pt band	capital gains pt band	liquidity pt band	default pt band	issuance premium pt band	total pt band	obs. Spread(T)	% expl.
2	0.1 [0.1, 0.1]	0.0 [0.0, 0.0]	−0.4 [−2.1, 1.2]	16.5 [5.7, 17.6]	2.9 [1.7, 4.2]	19.0 [5.4, 23.1]	9.9	
4	0.1 [0.1, 0.2]	6.9 [4.4, 7.0]	−0.4 [−2.2, 1.3]	14.3 [6.6, 16.1]	3.0 [1.8, 4.3]	24.0 [10.7, 28.9]	16.9	
6	0.7 [0.6, 0.8]	10.1 [5.2, 10.3]	−0.4 [−2.5, 1.7]	12.2 [5.9, 14.2]	3.3 [2.0, 4.5]	25.8 [11.3, 31.4]	27.9	92
8	1.9 [1.6, 2.1]	10.7 [5.2, 11.3]	−0.4 [−2.8, 2.0]	12.5 [6.2, 15.0]	3.6 [2.3, 4.9]	28.2 [12.6, 35.2]	41.1	69
10	4.6 [4.0, 5.1]	9.9 [4.5, 10.5]	−0.3 [−3.1, 2.5]	12.5 [6.2, 15.3]	4.0 [2.7, 5.3]	30.6 [14.4, 38.7]	52.2	59
12	12.7 [11.0, 14.3]	8.3 [3.7, 8.8]	−0.2 [−3.3, 3.0]	11.5 [5.8, 14.5]	5.6 [4.1, 7.1]	37.9 [21.2, 47.7]	71.2	53
14	18.0 [15.3, 20.6]	7.1 [3.1, 7.3]	−0.1 [−3.3, 3.0]	11.7 [5.8, 14.8]	7.2 [5.5, 8.9]	43.7 [26.4, 54.7]	87.0	50
16	21.4 [17.9, 24.9]	5.6 [2.6, 5.6]	−0.2 [−3.0, 2.6]	12.3 [6.1, 15.7]	8.4 [6.5, 10.4]	47.6 [30.2, 59.2]	99.4	48
18	24.0 [19.8, 28.2]	4.7 [2.2, 4.8]	−0.2 [−2.6, 2.2]	12.1 [6.0, 15.5]	9.6 [7.5, 11.8]	50.3 [32.9, 62.5]	110.3	46
20	25.5 [20.7, 30.3]	4.1 [2.0, 4.4]	1.3 [−2.0, 4.6]	12.7 [6.4, 16.3]	10.8 [8.5, 13.2]	54.4 [35.5, 68.8]	123.1	44
22	27.2 [21.8, 32.7]	3.6 [1.8, 3.9]	2.8 [−1.4, 7.1]	13.3 [6.7, 17.1]	11.4 [8.4, 14.4]	58.4 [37.2, 75.1]	128.8	45
24	28.2 [22.4, 34.2]	3.2 [1.6, 3.5]	2.9 [−1.3, 7.2]	14.1 [7.0, 17.9]	12.0 [8.3, 15.5]	60.3 [38.0, 78.3]	136.9	44
26	28.7 [22.6, 35.1]	2.9 [1.5, 3.1]	1.5 [−1.7, 4.6]	15.6 [7.8, 19.8]	12.5 [8.2, 16.7]	61.3 [38.3, 79.3]	145.0	42
28	29.3 [22.8, 36.0]	2.6 [1.3, 2.8]	0.1 [−2.2, 2.3]	16.2 [8.1, 20.4]	12.8 [8.1, 17.3]	61.0 [38.1, 78.7]	145.6	42
30	29.8 [23.0, 36.7]	2.3 [1.2, 2.5]	0.1 [−2.2, 2.3]	16.5 [8.2, 20.8]	12.8 [8.1, 17.3]	61.4 [38.4, 79.5]	159.8	38

observed taxable yields. Date-by-date household after-tax shares covary differently with model and observed yields, so the two calculations differ.

The normalized root-mean-square unexplained residual is 11.21 bp (16.97 bp unnormalized), and the maximum absolute residual is 14.87 bp at sixteen years. Five of fifteen absolute residuals are below 10 bp. The signed residuals are −12.03 bp at twelve years, −11.26 at twenty and +1.01 at thirty.

D.2 Issuer-level liquidity and offering-yield regressions

These issuer-level regressions are separate diagnostics for the liquidity differential of Section 3.7. The combined adjustment uses the matched institutional-size round-trip costs of Appendix D.1.5.

Liquidity measures. Table 28 associates offering yields with full-life issuer liquidity proxies. Because the proxies are not predetermined at issuance, the coefficients are neither causal ex-ante premia nor formal bounds. The trading-frequency measure is the issuer mean of each bond's average daily trades reported to the Municipal Securities Rulemaking Board (MSRB), with days counted from the bond's dated date to maturity. The spread measure is the corresponding issuer mean of each bond's average imputed round-trip cost (Appendix D.1.5), in bond-price units (0.01 is 1% of price, or 100 basis points).

Table 28: Issuer liquidity and offering yield, by maturity bin.

	Trading frequency		Transaction-cost spread	
	(1) Above corp median	(2) Top 25% trade	(3) Issuer-avg spread (cont.)	(4) Lowest-spread quartile
Maturity 1–5y	−0.046***	−0.028***	13.446***	−0.057***
	(0.009)	(0.007)	(0.489)	(0.001)
Maturity 6–10y	−0.007**	0.021***	11.624***	−0.048***
	(0.003)	(0.005)	(0.216)	(0.002)
Maturity 11–15y	−0.011***	0.041***	11.860***	−0.040***
	(0.004)	(0.001)	(0.355)	(0.001)
Maturity 16–20y	−0.035***	0.024***	13.447***	−0.043***
	(0.003)	(0.006)	(0.557)	(0.003)
Maturity 21–25y	−0.091***	−0.041**	20.427***	−0.104***
	(0.009)	(0.016)	(1.225)	(0.011)
Maturity 26–30y	−0.093***	−0.082***	27.262***	−0.146***
	(0.030)	(0.018)	(4.815)	(0.037)
Fixed effects	Yes	Yes	Yes	Yes
Maturity cluster	Yes	Yes	Yes	Yes
Observations	785,256	785,256	779,988	785,256

Notes. AA-and-above federally exempt issues, 2000–2021, offering yields in percent. Regressors are issuer averages: (1) daily MSRB trades per bond above the median in the Trade Reporting and Compliance Engine (TRACE) for plain-vanilla debentures and medium-term notes; (2) top quartile of municipal trade frequency; (3) continuous MSRB spread; (4) lowest-spread quartile. Thus −0.046 is −4.6 bp, and column (3)'s 13.4 implies 13.4 bp per 0.01 spread change. Fixed effects interact rounded maturity, rating rounded down, state, issue year–month and call status. Standard errors cluster by rounded maturity, and the sample has no bond-count restriction. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Translation to long-end magnitudes. At 26–30 years, the indicator coefficients in Columns (1), (2) and (4) are about 8–15 bp in absolute value, and the continuous Column (3) coefficient of 27.262 implies 27.3 bp for a 0.01 spread contrast. At these maturities, more liquid issuers have lower offering yields on all four measures.

D.3 Paired issues: Build America Bonds (BABs) and tax-exempt bonds

Scaling BAB offering yields to $0.65 \times y^{BAB}$ puts them on the household after-tax scale.

Sample and construction. The issue-level Mergent sample, one observation per bond identifier (CUSIP), includes bonds issued in 2009Q2–2010Q4 and not subject to the alternative minimum tax (AMT). Table 29 uses AA-and-above issues (agency score ≤ 3) and A-or-above issues (≤ 7). Figure 3 uses the former and averages yields within cells of integer maturity $m \in \{1, \ldots, 30\}$ and bond type.

Regression specification. In the notation of equation (4), with m_b the remaining maturity at issuance, I estimate

$$y_b \;=\; \alpha_1 \, EXMP_b \;+\; \beta \, (EXMP_b \times m_b) \;+\; \gamma \, (0.65 \, y^{C}_{m_b,t_b}) \;+\; \lambda_{j(b)\times m_b \times q_b} \;+\; \varepsilon_b. \tag{140}$$

The corporate control $0.65 \, y^{C}_{m_b,t_b}$ varies across issue months within matched quarters, identifying γ. Equation (4) uses the unscaled control, with $\gamma_C = 0.65\gamma$, which changes only the control's coefficient and standard error.

Results. Table 29 shows a positive maturity gradient under both rating cuts and call matching, with all maturity interactions significant at one percent and Panel B bin coefficients rising with maturity. Call matching lowers the longest-bin coefficient by about fifteen percent in the AA-and-above sample but leaves the $[5, 10)$ coefficient unchanged.

Call features. BAB call flags occur on 49 percent of bonds, versus 39 percent of tax-exempts. Call-matched columns match these flags only, so exercise terms, coupons, liquidity and the treatment of original-issue discount (OID) or premium can still differ.

Original-issue premium and discount. Exact after-tax internal rates of return would treat BAB and exempt OID, acquisition premium, interest income and basis separately, so I check the 0.65 yield approximation on bonds issued near par. Restricting the call-matched AA-and-above sample to offering prices of 99–101 leaves a maturity gradient of 2.1 bp per year (s.e. 0.6), against 3.1 in the full call-matched sample. The 98–102 restriction gives 2.7 bp (s.e. 0.4). Both remain significant at one percent.

Interpretation. The paired comparison of Section 3.8 holds issuer and credit fixed, with the BAB yield replacing the corporate benchmark. The widening gap supports the participation mechanism under the 0.65 approximation and comparable security design and issuance premia. The 35% top federal rate behind the scaling is below the 41.1% average household wedge of the AA-and-above curve (Appendix D.12.1), and a lower after-tax share would widen the gap. Against that curve, this short program's comparison agrees most closely at the endpoints and least at $[15, 20)$ years.

D.4 The municipal puzzle on unlimited-tax general-obligation bonds

Section 3.6 states that the declining implied tax also appears among unlimited-tax general-obligation (GO) bonds.

Sample. Unlimited-tax GO bonds pledge full faith, credit and unlimited taxing power, while revenue bonds pledge project revenues. Figure 21 restricts the comparison to AA-and-above unlimited-tax GO bonds[26] passing the main call-pricing and calendar screens (Appendix D.1.2). No default adjustment is applied, because local obligations' credit differs from the state-government credit default swap (CDS) proxy (Appendix D.1.6).

[26] Mergent security code "K".

Table 29: Paired issues: the tax-exempt-versus-BAB gap widens with maturity.

	AA and above (score ≤ 3)		A− and above (score ≤ 7)	
	(1) base	(2) call-matched	(3) base	(4) call-matched
Panel A. Continuous maturity				
Tax-exempt	-0.102^{***}	-0.091^{**}	-0.062^{**}	-0.043
	(0.034)	(0.036)	(0.027)	(0.030)
Tax-exempt × maturity	0.034^{***}	0.031^{***}	0.033^{***}	0.030^{***}
	(0.003)	(0.003)	(0.002)	(0.003)
After-tax corporate yield	0.379^{***}	0.376^{***}	0.264^{*}	0.261^{*}
	(0.138)	(0.135)	(0.134)	(0.130)
Panel B. Five-year maturity bins				
Tax-exempt	-0.048	-0.050	-0.007	-0.008
	(0.033)	(0.033)	(0.024)	(0.024)
Tax-exempt × $[5, 10)$ y	0.192^{***}	0.191^{***}	0.192^{***}	0.190^{***}
	(0.022)	(0.022)	(0.017)	(0.017)
Tax-exempt × $[10, 15)$ y	0.396^{***}	0.364^{***}	0.368^{***}	0.342^{***}
	(0.033)	(0.033)	(0.026)	(0.030)
Tax-exempt × $[15, 20)$ y	0.450^{***}	0.390^{***}	0.419^{***}	0.403^{***}
	(0.041)	(0.039)	(0.029)	(0.039)
Tax-exempt × $[20, 25)$ y	0.576^{***}	0.540^{***}	0.593^{***}	0.510^{***}
	(0.061)	(0.066)	(0.071)	(0.066)
Tax-exempt × $[25, 30]$ y	0.755^{***}	0.639^{***}	0.996^{***}	0.804^{***}
	(0.079)	(0.039)	(0.074)	(0.060)
After-tax corporate yield	0.380^{***}	0.378^{***}	0.264^{*}	0.262^{**}
	(0.138)	(0.135)	(0.134)	(0.130)
Issuer × maturity × quarter FE	Yes	Yes	Yes	Yes
× call status	No	Yes	No	Yes
State cluster	Yes	Yes	Yes	Yes
Observations	29,839	28,890	61,816	59,886

Notes. Offering yields in percent, 2009Q2–2010Q4, excluding AMT bonds. BAB yields are multiplied by 0.65. The corporate-yield row reports γ, the coefficient on the after-tax control in equation (140). The omitted bin is $[1, 5)$ years. Standard errors are clustered by issuing jurisdiction (51 in columns (1)–(2), 52 in columns (3)–(4)). Call-matched columns add the optional-call flag to the absorbed issuer-by-maturity-by-quarter cell. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Result. After call and issuance-premium adjustments, GO implied tax falls from 43.1% at two years to 26.0% at thirty, matching the 17.1-point decline of the A-or-above full sample. The screens exclude 4,029 GO bonds (0.86%). Relative to the unscreened GO call-adjusted series, with the issuance premium held fixed, they change the decline by 0.49 points, almost entirely at the short end. The result agrees with earlier evidence that the long-end residual persists after default and liquidity adjustments and under high-grade or GO restrictions (Longstaff, 2011; Fleckenstein and Longstaff, 2025; Chalmers, 1998).

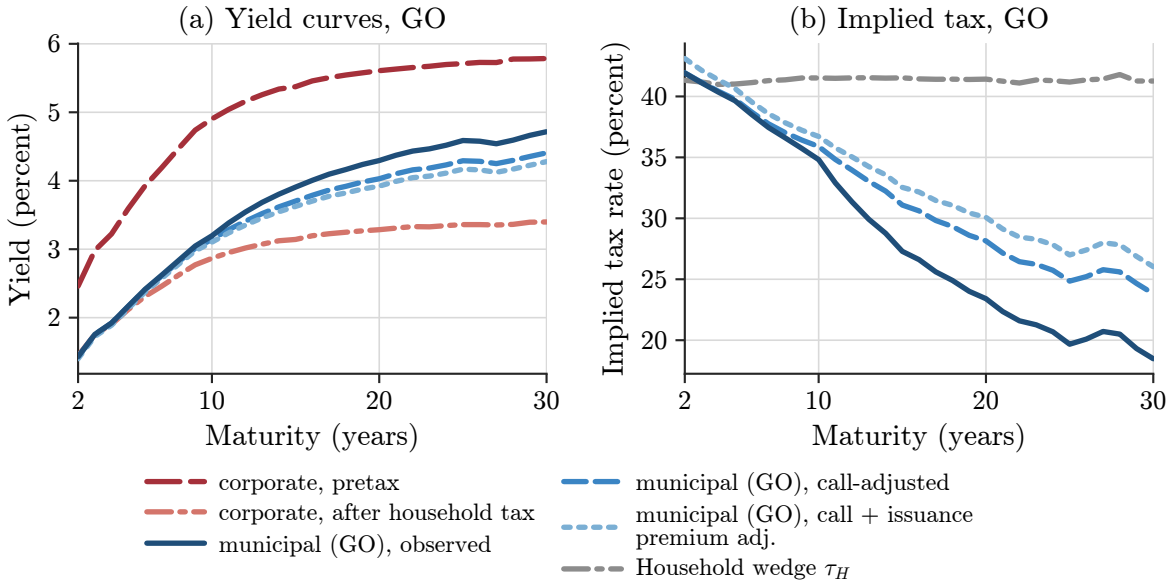


Figure 21: AA-and-above unlimited-tax GO bonds: 465,258 bonds pass the main pricing and calendar screens, and curves use the sample's own weights and household benchmark. Calls use the calibrated model. Because GO offering-week matches are scarce, the issuance-premium adjustment uses the all-bond institutional estimate (Appendix D.1.3). Adjusted implied tax falls 17.1 points from two to thirty years, matching the A-or-above full sample.

D.5 The single-tax-rate benchmark of Green (1993)

Sections 2 and 3.7 refer to the single-tax-rate explanation of Green (1993).

Green's mechanism. Green explains rising ratios of municipal to taxable yields through taxable-bond tax timing. Buying two five-percent market-discount bonds and shorting one ten-percent par bond of equal maturity produces no net cash flow before maturity and one dollar of principal then, deferring tax. At rate τ, the package costs P_T and pays $1-\tau(1-P_T)$ after tax at maturity. With d_T the price of a dollar paid at T, it is therefore worth $P_T = (1-\tau)d_T/(1-\tau d_T)$. The limiting friction is the asymmetric taxation of long and short positions, because borrowing and short-sale deductions cannot exceed the investment income they support. Green's high-tax investor prices both municipal bonds and the package, and dealers, who face no such limit, link taxable par prices to it, lowering long taxable yields below fully taxed returns. Thus one investor can price municipal bonds at every maturity as the yield ratio rises, without the different price-setting investors across maturities that Fama (1977) requires.

Relation to participation. Deep market-discount bonds may be scarce, annual OID accrual removes the deferral and shorts consume deduction capacity. Households forgoing the package can accept the household benchmark $\xi_H y_T^C$, below Green's relation. With sufficient capacity, these higher-benefit households set municipal prices, and investors using the strategy do not. Raising municipal yields to Green's relation would require shorting municipal bonds, subject to the frictions in Appendix E.6. His relation therefore need not

bind. Nor does it account for observed long yields: his own 1987–1990 estimates leave thirty-year municipal yields about twenty taxable-equivalent basis points above his relation, a gap that increases with maturity.

Green's critique of clientele explanations. Green rejects earlier clientele explanations because term premia persisted after mid-1980s reforms disrupted bank holdings. Here participation depends on exemption benefits and the duration benefit (Section 4.3), so the clientele that prices each maturity can change. Households alone hold short maturities and set their yields. Institutions with smaller exemption benefits also hold long maturities, whose yields reflect their tax treatment too, and implied taxes respond to supply and risk capacity. Section 3.4 documents ownership across benefits. I retain $\xi_H y_T^C$, measure tax timing separately in the capital-gains adjustment (Appendix D.1.4) and test household state-tax pass-through in P1 and bank and insurer benefits in P2 (Sections 6.1 and 6.2), distinguishing participation from tax timing alone.

D.6 State-tax pass-through: full regression tables

These tables support the state-tax pass-through test (P1) of Section 6.1.

Sample. The main P1 regressions use federally exempt AA-and-above issuance (agency score ≤ 3) from 2000–2017. They exclude zero-maturity bonds and group maturities into $[0, 5)$, $[5, 10)$, $[10, 15)$, $[15, 20)$ and $[20, 30]$ years. The zero-yield columns have 681,839 complete-control observations. Six missing offering yields leave 681,833 in the offering-yield columns.

Call-adjusted outcome. The main outcome uses call-adjusted zero-coupon yields. The P1 call adjustment prices each bond on the fifteenth of its issue quarter's final month, while fixed effects use actual issue months. The option-adjusted spread is fitted on the calibrated model's supply-weighted curve for the issue date, with a regular semiannual cash-flow grid. Observed call dates map to the nearest grid date within 0.27 years, and the call is exercisable on every grid date from the first mapped date, unlike the contractual calendar of the main adjustment (Appendix D.1.2). Callable bonds are excluded when the first call date lies more than 0.27 years before the first semiannual grid date or the call schedule lacks a usable date and price.

Controls and fixed effects. All P1 tables retain equation (25)'s six lagged state controls and the rounded average rating, each interacted with all five maturity bins, plus log offering par and security-type indicators. Stock and issuance controls use all observed federally exempt bonds, regardless of rating. Outstanding par equals the offering amount until the reported outstanding date and the reported outstanding amount afterward. Fixed effects are cells defined by issue year, issue month, rounded maturity and callability. Observations with missing controls are excluded without imputation, and singleton cells are then removed.

Estimation and inference. Unweighted ordinary least squares (OLS) is the main estimator. For the instrumental-variables (IV) comparison, the 1990 rate interacted with bins instruments the current-rate interactions, giving a Kleibergen–Paap (KP) rk Wald F of 10.4 and Sanderson–Windmeijer (SW) conditional statistics of 76.1, 115.9, 104.4, 155.7 and 230.4. Standard errors are state-clustered, with $t(G-1)$ critical values for G states ($t(46)$ in the main sample). The endpoint test of $\beta_1 = \beta_5$ gives each table's decline p-value.

D.6.1 Comparison across yield constructions

Table 30 compares Mergent offering yields, which may be yields to worst for premium callable bonds, with zero yields from offering prices, with and without call adjustment. Both OLS and IV reject endpoint equality at one percent for every construction.

Table 30: P1 across yield constructions, 2000–2017.

	Offering yield		Call-adjusted zero		Unadjusted zero	
	OLS	IV	OLS	IV	OLS	IV
β_1: $[0, 5)$ years	0.985^{***}	1.084^{***}	1.072^{***}	1.092^{***}	1.080^{***}	1.102^{***}
	(0.238)	(0.249)	(0.195)	(0.224)	(0.196)	(0.227)
β_2: $[5, 10)$ years	0.635^{***}	0.648^{***}	0.682^{***}	0.705^{***}	0.683^{***}	0.693^{***}
	(0.075)	(0.093)	(0.071)	(0.088)	(0.069)	(0.087)
β_3: $[10, 15)$ years	0.414^{***}	0.410^{***}	0.484^{***}	0.448^{***}	0.515^{***}	0.479^{***}
	(0.053)	(0.075)	(0.098)	(0.121)	(0.100)	(0.117)
β_4: $[15, 20)$ years	0.268^{***}	0.257^{***}	0.386^{***}	0.325^{***}	0.428^{***}	0.360^{***}
	(0.060)	(0.067)	(0.102)	(0.117)	(0.108)	(0.121)
β_5: $[20, 30]$ years	0.193^{***}	0.152^{**}	0.298^{***}	0.227^{***}	0.335^{***}	0.255^{***}
	(0.052)	(0.057)	(0.075)	(0.081)	(0.078)	(0.085)
Decline $\beta_1 - \beta_5$	0.791^{***}	0.933^{***}	0.774^{***}	0.865^{***}	0.745^{***}	0.847^{***}
p-value of decline	0.004	< 0.001	0.002	< 0.001	0.003	0.001
KP F	—	10.4	—	10.4	—	10.4
Observations	681,833	681,833	681,839	681,839	681,839	681,839
States	47	47	47	47	47	47
Full main controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Full controls, fixed effects and state-clustered inference as in Appendix D.6, 2000–2017. IV instruments each bin's current rate with its 1990 rate. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D.6.2 State credit-risk controls

Table 31 restricts the pre-2018 sample to observations with a preceding-month five-year state-government CDS quote and adds that quote interacted with the maturity bins. On this common sample of 345,878 bonds from 25 states, the CDS controls raise the longest-bin OLS estimate from 0.323 to 0.401 and IV from 0.309 to 0.353. Both estimators retain a declining profile and reject endpoint equality at five percent in every row.

Table 31: P1 sample and credit controls, 2000–2017.

Sample	Obs.	States	β_1, OLS	β_5, OLS	Decline p	β_1, IV	β_5, IV	Decline p
Main sample	681,839	47	1.072^{***}	0.298^{***}	0.002	1.092^{***}	0.227^{***}	< 0.001
				(0.075)			(0.081)	
Observed CDS sample	345,878	25	1.210^{***}	0.323^{***}	0.008	1.350^{***}	0.309^{***}	0.008
				(0.082)			(0.098)	
CDS sample + CDS × bin	345,878	25	1.400^{***}	0.401^{***}	0.010	1.392^{***}	0.353^{***}	0.022
				(0.070)			(0.088)	

Notes. Full controls, fixed effects and state-clustered inference as in Appendix D.6, 2000–2017. IV uses bin-interacted 1990 rates. Parentheses give standard errors of β_5. CDS is the prior-month five-year state-government quote, without imputation. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D.7 State-tax pass-through: instrument vintages, weighting and further checks

These checks for P1 (Section 6.1) keep the controls, fixed effects and state-clustered inference of Appendix D.6.

Instruments. The IV comparisons use the state's 1990 top income-tax rate as the baseline instrument. The maintained exclusion restriction is that the historical rate affects current implied taxes only through the current rate, conditional on the controls and fixed effects. Tables 32 and 33 compare 1945–2000 vintages on the same pre-2018 sample.

Continuous-maturity specification. Table 32 replaces the five current-rate bin interactions in (25) with the current rate and its interaction with maturity in years, m_b. The interaction coefficient is the pass-through gradient per year, and IV uses the historical rate and its maturity interaction. All instrument vintages give negative IV gradients, unweighted and with equal total weight per state, as does OLS.

Table 32: Continuous-maturity P1 checks, 2000–2017.

		Unweighted				Equal per state	
Vintage	Corr.	Gradient	s.e.	KP F	SW F	Gradient	KP F
OLS	—	-0.0399^{***}	(0.0131)	—	—	-0.0330^{**}	—
1945	0.40	-0.0548^{**}	(0.0251)	8.0	19.2	-0.0454	2.8
1950	0.31	-0.0622^{**}	(0.0274)	7.8	18.4	-0.0904	1.8
1960	0.38	-0.0636^{**}	(0.0256)	9.0	28.8	-0.1022^{**}	2.5
1970	0.51	-0.0588^{***}	(0.0203)	11.5	45.2	-0.0818^{***}	7.3
1980	0.76	-0.0434^{**}	(0.0164)	10.6	103.4	-0.0396^{**}	26.9
1990	0.82	-0.0447^{***}	(0.0129)	30.1	404.1	-0.0310^{**}	25.9
2000	0.88	-0.0405^{***}	(0.0125)	65.1	998.5	-0.0262^{*}	36.5

Notes. Full controls, fixed effects and state-clustered inference as in Appendix D.6, 2000–2017. Gradients are per year. IV uses the indicated historical rate and its maturity interaction. SW is the interaction's conditional statistic. Corr. is the correlation across states between each state's mean current top rate and its top rate in the vintage year, using state means over the main estimation sample. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Binned profiles. Table 33 compares the five-bin IV profiles with OLS, and Figure 22 plots them. Every vintage rejects endpoint equality at five percent.

Table 33: P1 instrument-vintage checks, 2000–2017.

Estimator/vintage	β_1	β_2	β_3	β_4	β_5	Decline p	KP F
OLS (main)	1.072^{***}	0.682^{***}	0.484^{***}	0.386^{***}	0.298^{***}	0.002	—
	(0.195)	(0.071)	(0.098)	(0.102)	(0.075)		
1945	1.267^{***}	0.886^{***}	0.546^{***}	0.359^{**}	0.246^{**}	0.025	3.1
	(0.422)	(0.168)	(0.161)	(0.142)	(0.104)		
1950	1.464^{***}	0.959^{***}	0.613^{***}	0.423^{***}	0.311^{**}	0.022	3.1
	(0.475)	(0.192)	(0.158)	(0.143)	(0.118)		
1960	1.494^{***}	0.954^{***}	0.615^{***}	0.426^{***}	0.327^{***}	0.013	3.6
	(0.441)	(0.176)	(0.152)	(0.141)	(0.115)		
1970	1.404^{***}	0.864^{***}	0.551^{***}	0.392^{***}	0.317^{***}	0.005	4.4
	(0.352)	(0.132)	(0.148)	(0.143)	(0.115)		
1980	1.092^{***}	0.759^{***}	0.520^{***}	0.367^{***}	0.273^{***}	0.009	3.8
	(0.273)	(0.085)	(0.119)	(0.118)	(0.083)		
1990	1.092^{***}	0.705^{***}	0.448^{***}	0.325^{***}	0.227^{***}	< 0.001	10.4
	(0.224)	(0.088)	(0.121)	(0.117)	(0.081)		
2000	1.075^{***}	0.684^{***}	0.482^{***}	0.375^{***}	0.275^{***}	0.001	23.5
	(0.215)	(0.080)	(0.097)	(0.096)	(0.069)		

Notes. Full controls, fixed effects and state-clustered inference as in Appendix D.6, 2000–2017. First row: main OLS. IV instruments each bin's current rate with its indicated historical top rate. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Weighting. Table 34 compares the unweighted main estimates with equal total weight per state, equal total weight per state-year and weights of $1/\sqrt{n_{\text{state}}}$, with n_{state} the observation count of the bond's state. Every weighting retains a declining OLS profile, with endpoint differences significant at one percent.

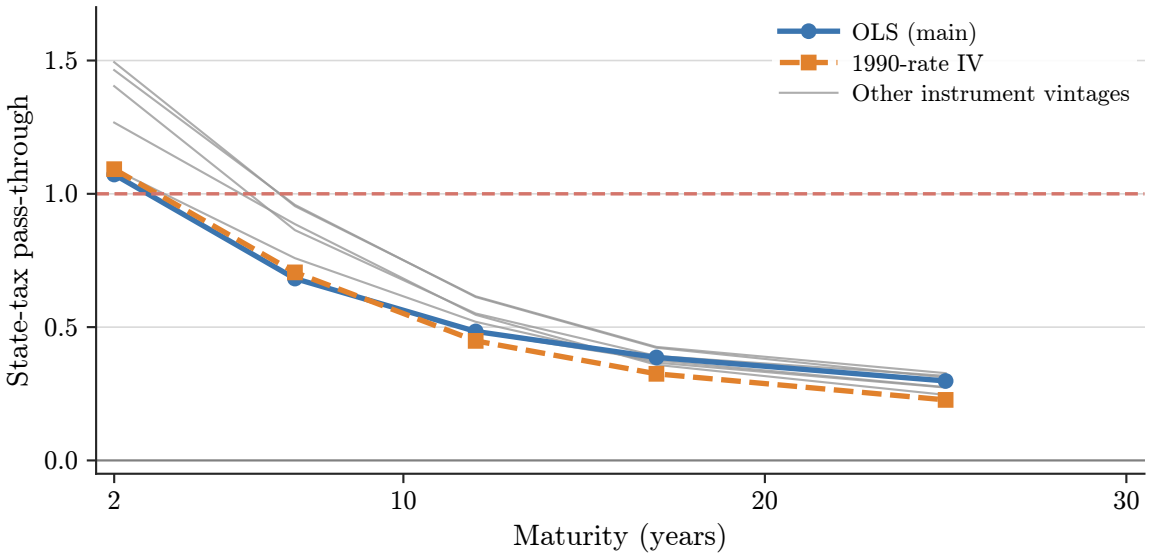


Figure 22: State-tax pass-through using historical tax-rate instruments, 2000–2017. Blue shows the main OLS profile. Orange shows IV using the 1990 rate. Gray lines use the 1945, 1950, 1960, 1970, 1980 and 2000 rates. The dashed red line is the representative-tax benchmark. Table 33 reports state-clustered uncertainty.

Table 34: P1 OLS estimates under four weighting schemes, 2000–2017.

Weighting	β_1	β_2	β_3	β_4	β_5	Decline p
Unweighted (main)	1.072^{***}	0.682^{***}	0.484^{***}	0.386^{***}	0.298^{***}	0.002
	(0.195)	(0.071)	(0.098)	(0.102)	(0.075)	
Equal per state	0.869^{***}	0.562^{***}	0.384^{***}	0.314^{***}	0.240^{***}	0.007
	(0.191)	(0.084)	(0.109)	(0.115)	(0.089)	
Equal per state-year	0.915^{***}	0.519^{***}	0.319^{***}	0.261^{***}	0.211^{***}	0.002
	(0.190)	(0.069)	(0.083)	(0.090)	(0.073)	
$1/\sqrt{n_{\text{state}}}$	0.996^{***}	0.630^{***}	0.425^{***}	0.345^{***}	0.281^{***}	0.005
	(0.202)	(0.082)	(0.113)	(0.122)	(0.092)	

Notes. Full controls, fixed effects and state-clustered inference as in Appendix D.6, 2000–2017. Weight counts follow complete-case and singleton restrictions. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

State fixed effects. Adding state fixed effects to the pre-2018 OLS specification gives short and long coefficients of 0.658 (s.e. 0.205) and −0.124 (0.211), with a decline of 0.782 (0.220, $p = 0.001$), on 681,839 bonds. State effects absorb fixed tax levels. The five excluded instruments, the 1990 rate interacted with each maturity bin, sum to the state's 1990 rate, which is collinear with the state effects, so IV is not identified.

Leave-one-state-out. All 47 single-state omissions retain a positive short-to-long decline under both OLS and IV (Figure 23). OLS long-bin estimates span 0.224–0.469 and declines 0.467–0.898. IV long-bin estimates span 0.113–0.309 and declines 0.678–1.007. The largest endpoint-test p-values are 0.054 for OLS and 0.039 for IV.

The two tax regimes. The main P1 sample ends in 2017, matching the compounded federal and state after-tax shares of the six model economies in Figure 9. From 2018, the Tax Cuts and Jobs Act capped state and local tax (SALT) deductions at $10,000 per return. When the cap binds, an additional dollar of

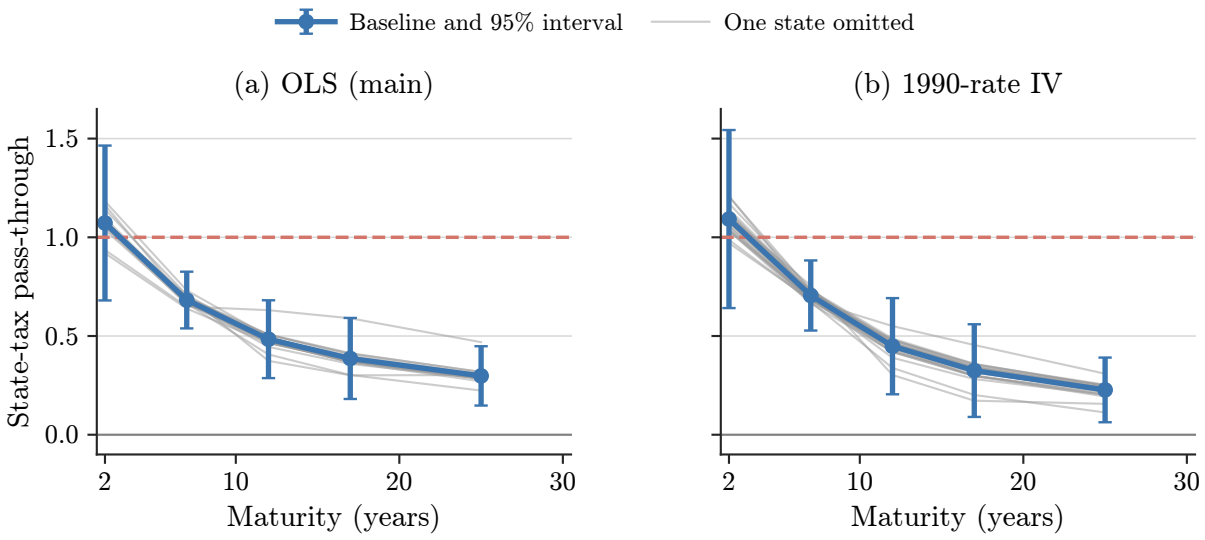


Figure 23: Leave-one-state-out P1 profiles for 2000–2017. Gray lines show each of the 47 state omissions, and blue shows the baseline estimates with state-clustered 95% intervals. Panel (a) uses OLS and panel (b) uses the 1990-rate instruments. The dashed line is the representative-tax benchmark.

state tax is no longer federally deductible, and the combined after-tax share on taxable interest is $\xi^{\text{fed}} - \tau_s^{\text{st}}$ (notation of Section 6.1). The implied state tax for this regime is

$$\mathit{ImpliedStateTax}_{s,t}^{\text{cap}}(T) \;\equiv\; \xi^{\text{fed}} - \frac{y_{s,t}^{M}(T)}{y_t^{C}(T)}. \tag{141}$$

It recovers the statutory state rate under representative-tax indifference for a household with a binding cap.[27] Table 35 estimates each regime separately. OLS declines from 1.072 to 0.298 in 2000–2017 and from 0.952 to 0.091 in 2018–2020. IV declines from 1.092 to 0.227 and from 0.912 to 0.082. Both estimators reject endpoint equality at one percent in each period.

[27] See the IRS, 2017 Schedule A instructions, and its summary of the 2018 changes.

Table 35: P1 under the two federal/state tax regimes.

	2000–2017 (main)		2018–2020	
	OLS	IV	OLS	IV
β_1: $[0, 5)$ years	1.072^{***}	1.092^{***}	0.952^{***}	0.912^{***}
	(0.195)	(0.224)	(0.181)	(0.246)
β_2: $[5, 10)$ years	0.682^{***}	0.705^{***}	0.842^{***}	0.900^{***}
	(0.071)	(0.088)	(0.126)	(0.154)
β_3: $[10, 15)$ years	0.484^{***}	0.448^{***}	0.450^{***}	0.466^{***}
	(0.098)	(0.121)	(0.065)	(0.076)
β_4: $[15, 20)$ years	0.386^{***}	0.325^{***}	0.318^{***}	0.326^{***}
	(0.102)	(0.117)	(0.081)	(0.107)
β_5: $[20, 30]$ years	0.298^{***}	0.227^{***}	0.091	0.082
	(0.075)	(0.081)	(0.091)	(0.105)
Decline $\beta_1 - \beta_5$	0.774^{***}	0.865^{***}	0.861^{***}	0.830^{***}
p-value of decline	0.002	< 0.001	< 0.001	0.007
KP F	—	10.4	—	7.9
Observations	681,839	681,839	146,174	146,174
States	47	47	47	47
Full main controls	Yes	Yes	Yes	Yes

Notes. Full controls, fixed effects and state-clustered inference as in Appendix D.6. IV uses bin-interacted 1990 rates. Outcomes follow equation (24) before 2018 and equation (141) afterward, for a binding SALT cap. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D.8 The bank-qualified discontinuity: construction, robustness and placebos

For the P2 test of Section 6.2, Table 36 reports the full five-bin two-stage least squares (2SLS) estimates of (27), and Figure 24 extends Figure 10 to all five maturity bins.

D.8.1 Benchmarks and the designation premium

The two benchmarks. Both benchmarks of (26) compare annual after-tax carry per dollar of basis on matched bonds, in the notation of Section 6.2. I suppress dates, and $y_k^C > 0$ is the taxable yield at maturity k.

The representative-tax benchmark has $y_k^N = (1 - \tau_H) y_k^C$, where τ_H is the top household rate. At this yield a designated bond earns the bank $y_k^N - 0.2\,\tau_B c$ after the disallowed funding cost, against $(1 - \tau_B) y_k^C$ on a taxable bond, an advantage of

$$\underbrace{(1 - \tau_B) y_k^C}_{\text{taxable alternative}} - \underbrace{\left[(1 - \tau_H) y_k^C - 0.2\,\tau_B c\right]}_{\text{qualified bond at the household yield}} = (\tau_H - \tau_B) y_k^C + 0.2\,\tau_B c > 0. \tag{142}$$

It is positive whenever $\tau_H > \tau_B - 0.2\,\tau_B c / y_k^C$, in particular when household rates are at least banks' and funding costs are positive. Banks still prefer taxable bonds and designation leaves household taxes unchanged, so $y_k^Q = y_k^N$ and $\varpi_k = 0$ at every maturity.

In the bank-only benchmark, both municipal bonds satisfy the bank's after-tax indifference conditions with taxable bonds:

$$\begin{aligned} y_k^N - \underbrace{\tau_B c}_{\text{full disallowance}} &= (1 - \tau_B) y_k^C, \\ y_k^Q - \underbrace{0.2\,\tau_B c}_{\text{qualified disallowance}} &= (1 - \tau_B) y_k^C. \end{aligned} \tag{143}$$

Table 36: Bank-qualified designation and yield premia by maturity.

	(1)	(2)	(3)
Dependent variable	BQ designation	Offering yield (bp)	Yield / expected benefit
Estimator	OLS	2SLS	2SLS
Regressor, interacted with each maturity bin	Below-cap indicator	BQ designation	BQ designation
Reported effect	Designation increase	Yield reduction	Scaled yield reduction
1–2 years	0.42***	1.8	0.17
	(0.015)	(2.2)	(0.11)
2–5 years	0.42***	8.7***	0.31***
	(0.015)	(2.2)	(0.08)
5–10 years	0.44***	13.8***	0.41***
	(0.015)	(2.2)	(0.06)
10–20 years	0.45***	18.4***	0.52***
	(0.017)	(2.3)	(0.07)
Over 20 years	0.43***	18.2***	0.47**
	(0.027)	(6.3)	(0.19)
10–20 minus 1–2 years		16.5***	0.34***
		(2.7)	(0.11)
Observations (bonds)	355,372	355,372	355,372
Standard-error clustering	Issuer	Issuer	Issuer
Number of issuers	15,889	15,889	15,889

Notes. Each column is a separate pooled regression containing all five maturity interactions. Column (1) reports the designation increase associated with below-cap status, estimated by ordinary least squares (OLS). Columns (2)–(3) instrument designation by below-cap status within each bin, as in (27), and report $-\beta_k$, so positive numbers mean lower yields on designated bonds. Column (3) divides each bond's yield by its expected annual benefit $X_{tm} = 0.8\,\tau_{B,t}\hat{c}(t,m)$ before estimation (Appendix D.8.3). Contrasts use unrounded estimates. Parentheses give standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Fixed effects in every column. Maturity bucket × year, state × year, state × offering year–month, rating category × year, security type × year, use of proceeds, coupon bin × maturity bucket and callability × maturity bucket × year. Other controls are log tranche par, maturity-bin indicators and separate linear issuance slopes on each side of the cap within each bin.

Sample. Tax-exempt new issues in 2000–2021 within \$0.5–\$4 million of the statutory cap (\$30 million in 2009–10, \$10 million otherwise). Maturity is measured at issuance. Appendix D.8.2 defines the fixed-effect buckets.

Subtracting gives $\varpi_k = 0.8\,\tau_B c = X$, because the taxable yields cancel within each maturity pair, so constant funding costs imply a premium flat in maturity in basis points. The expected annual benefit X_{tm} (Appendix D.8.3) evaluates this benefit at expected annual funding costs.

Derivation of the designation premium. Take matched bonds of maturity k in supply $q_k > 0$, with outside opportunities and non-tax characteristics fixed. The designation benefit u is zero for a non-designated bond and X for a designated one. At yield y, household demand is $d_k^H(y)$ and bank demand $d_k^B(y+u)$, both reflecting baseline funding costs, taxes and risk compensation. For each $u \in [0, X]$, assume that a clearing yield exists, that demands are continuous and nondecreasing with finitely many continuously differentiable regimes and that total demand has a strictly positive slope at clearing. Then

$$d_k^H(y_k(u)) + d_k^B(y_k(u)+u) = q_k, \qquad y_k^N = y_k(0), \quad y_k^Q = y_k(X). \tag{144}$$

Differentiating (144) on a smooth segment gives

$$-\frac{dy_k(u)}{du} = \frac{\partial d_k^B/\partial y}{\partial d_k^H/\partial y + \partial d_k^B/\partial y} \equiv w_{Bk}(u) \in [0,1], \tag{145}$$

with demand slopes at the clearing net returns, so $w_{Bk}(u)$ is the banks' share of the total demand response. A larger benefit raises bank demand, and the yield falls to offset it. Integrating continuously across participation boundaries gives

$$\varpi_k = y_k(0) - y_k(X) = \int_0^X w_{Bk}(u)\,du = \theta_k X, \qquad \theta_k = \frac{1}{X}\int_0^X w_{Bk}(u)\,du. \tag{146}$$

If banks never participate, $\partial d_k^B/\partial y = 0$ and $\varpi_k = 0$, the representative-tax benchmark. If only banks hold the bond, $\partial d_k^H/\partial y = 0$ and $\varpi_k = X$, the bank-only benchmark. When both respond, capitalization lies between them. The capitalization share θ_k, the yield reduction as a fraction of the benefit, is distinct from banks' share of holdings.

The flat-X prediction concerns annual yield-equivalent carry and makes no claim about a fixed price premium. The benefit applies per dollar of tax basis, so mapping it into offering yields uses comparable cash flows and the ratio of basis to price.

D.8.2 Design

Running variable. The running variable R_{jt} of Section 6.2 measures issuance as all tax-exempt par within each six-character CUSIP issuer prefix and offering calendar year, before sample restrictions. Statutory issuance aggregates subordinate entities, includes bank loans and leases, mostly absent from Mergent, and excludes refundings of designated bonds (deemed designated under Section 265(b)(3)(D)) and private-activity bonds other than 501(c)(3) bonds. This measurement noise motivates the fuzzy design.

Among bonds \$0.5–4 million above the cap, 21% are designated. In that window, 4% of single-sale issuer-years designate, and issuer-years with multiple sales designate 44% of their issues. The statute tests reasonably anticipated issuance at each sale, so a March designation stands even when an unanticipated October sale takes the year over the cap. Refundings are 46% of the bonds in that window, and measuring issuance net of refundings reduces designation in the corresponding window to 9%. Designation falls from 24% at \$0.5–1 million above the cap to 15–16% at \$4–8 million and 5% beyond \$10 million, consistent with measurement noise. Designation, refunding status, security type, bond insurance and credit enhancement come from the Mergent bond master files.

Controls. In (27), V_b holds a linear function of R_{jt} on each side of the cap, separately by maturity bin, the log par amount of the maturity tranche (floored at \$1,000) and the fixed effects listed under Table 36. The seven maturity buckets, used only in these fixed effects and distinct from the five maturity bins $\mathbf{1}_k$ of the coefficients, have upper limits of 2, 5, 10, 15, 20 and 30 years, with a seventh bucket beyond thirty.

The window. The baseline window keeps issuer-years with $0.5 \le |R_{jt}| \le 4$, and its half-million donut excludes managed bunching (2,081 issuer-years within \$0.25 million below the cap versus 572 above) and 1,034 issuer-years with 12,076 bonds exactly at the \$10 million cap (\$30 million in 2009–10). At-cap issuers are eligible (Section 6.2) and designate 79.6% of their bonds, the below-cap rate, against 27% at \$1–50 thousand above the cap. The below-cap indicator is always $E = \mathbf{1}\{R \le 0\}$; a strict inequality would misclassify at-cap issuers, lowering the no-donut first stage from 0.39 to 0.30 but barely changing 2SLS, because dividing by the first stage offsets the attenuation.

Balance at the cap. Local linear covariate regressions in the baseline window, with maturity-bucket-by-year and state-by-year effects, find callability and the number of maturities per issue balanced in every bin. Ratings, observed for 62%, differ only at 1–2 years (0.17 notch, $t = 2.1$). Coupons, set at pricing and covered by coupon-bin-by-maturity-bucket fixed effects, are 0.09–0.14 percentage points lower below the cap in the four bins beyond two years. Log par, always controlled, is lower below the cap at 1–2 years (-0.26) and higher at 10–20 years ($+0.12$).

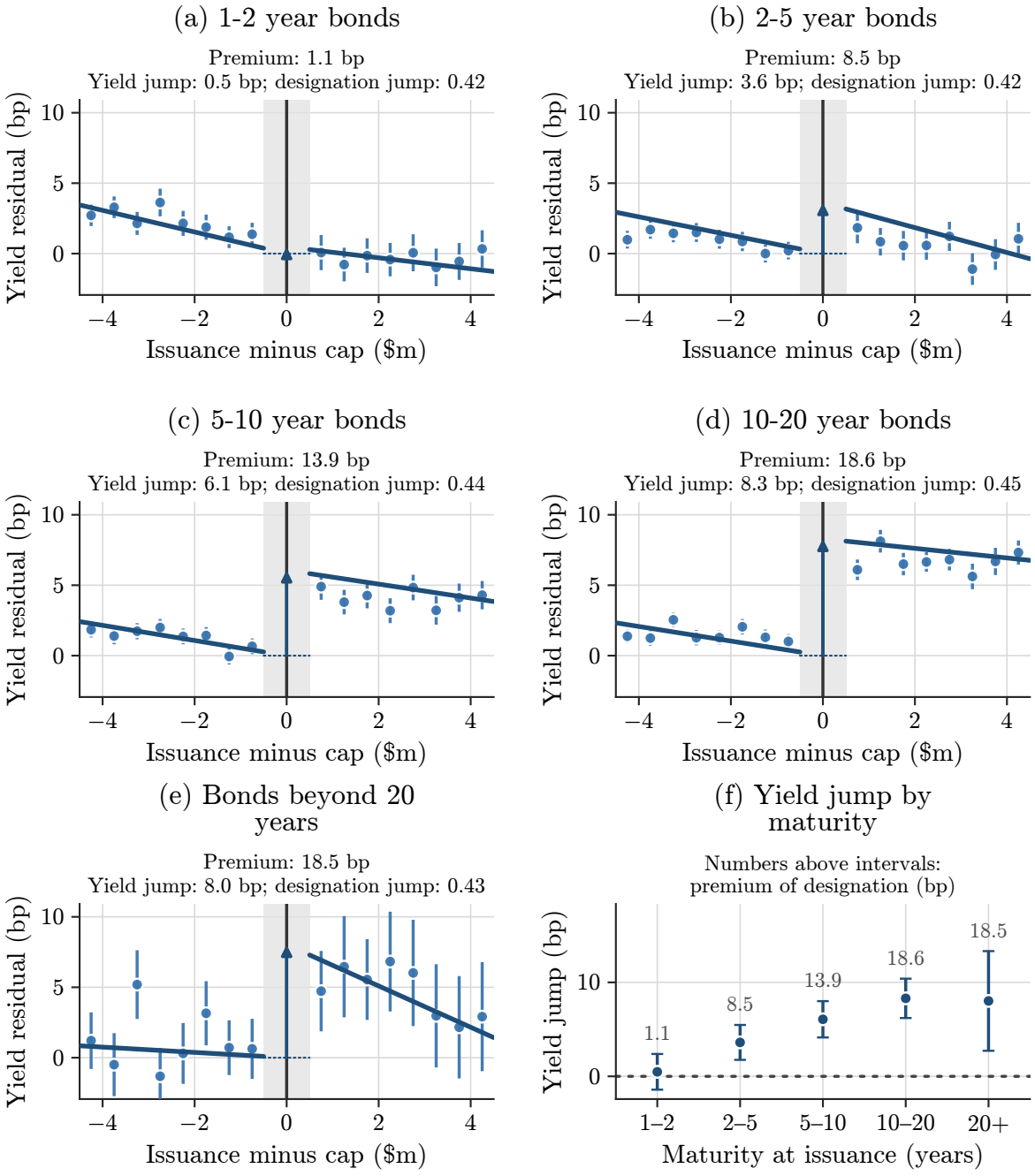


Figure 24: Offering-yield discontinuities by maturity. Panels (a)–(e) show residualized $0.5 million issuance-bin means relative to the below-cap fitted value at the cap, with local linear fits, donut shading and 95 percent bars. Arrows show above-cap-minus-below-cap jumps, and headers divide each yield jump by the designation jump to obtain the premium. Panel (f) shows jumps and premia, the binwise counterparts to Table 36. The common scale covers the central 96 percent of binned means.

D.8.3 Funding costs and the benefit measures

Funding costs. Section 265(b) allocates a bank's interest expense across its assets in proportion to their average adjusted bases, without matching each bond to borrowing of the same maturity. With bank n's annual interest expense I_{nt} and average asset basis $\bar{A}_{nt}$, a dollar of municipal basis held throughout the year is allocated $c_{nt} = I_{nt}/\bar{A}_{nt}$ of interest expense. Designation reduces the nondeductible fraction from one to 0.2, saving $X_{nt} = 0.8\,\tau_{B,t}c_{nt}$ at the bank's federal tax rate $\tau_{B,t}$, 35% before 2018 and 21% thereafter. Year-to-date interest expense from banks' December reports in the BankFind database of the Federal Deposit Insurance Corporation (FDIC) is divided, without multiplying by four, by average assets. This expense-to-assets ratio is neither a deposit rate nor a maturity-matched funding cost.

Banks are weighted by their municipal-security holdings: state and local securities held to maturity at amortized cost plus those available for sale at fair value, covering all municipal securities, bank-qualified or not, excluding loans and with no bank-size cutoff.[28] With M_{nt} these holdings and $\mathcal{B}_t$ the retained banks with positive holdings, the holdings-weighted funding cost and the current-year benefit are

$$c_t = \frac{\sum_{n\in\mathcal{B}_t} M_{nt}(I_{nt}/\bar{A}_{nt})}{\sum_{n\in\mathcal{B}_t} M_{nt}}, \qquad X_t = 0.8\,\tau_{B,t}c_t. \tag{147}$$

The retained banks have positive average assets, nonmissing interest expense and an expense-to-assets ratio from zero to below 15%. Mean c_t is 2.34% in 2000–04, 2.31% in 2005–08, 1.09% in 2009–10, 0.51% in 2011–13, 0.31% in 2014–17 and 0.66% in 2018–19. The corresponding current-year benefits X_t are approximately 66, 65, 30, 14, 9 and 11 bp, with the 2009–10 value under the ordinary rule.

Scope of the measure. The Call Report measure approximates the tax allocation with book assets and expense, while the statute uses adjusted tax bases and otherwise-deductible interest, unavailable taxpayer by taxpayer, and the IRS permits specified book-asset averaging conventions. X_t is an observable C-corporation scale of the ordinary-rule benefit and does not replicate any bank's tax-return calculation.[29] The weights include Subchapter S banks' municipal holdings. In 2009–10, the Section 265(b)(7) exception for holdings up to two percent of assets could reduce the disallowance on non-designated bonds and with it designation's incremental benefit. Treating 2009–10 through the \$30 million cap alone omits this provision, and those years are reported separately.

Bank holdings of municipal securities rose from \$97 billion in 2000 to \$370 billion in 2017 and fell after the 2018 rate cut.

The expected annual benefit. Because a bond's benefit accrues over its life, the expected annual benefit is $X_{tm} = 0.8\,\tau_{B,t}\,\hat{c}(t,m)$ for issue month t and maturity m, where $\hat{c}(t,m)$ is the expected annual funding cost over that maturity. Fitting annual holdings-weighted funding costs to one-year expected rates over 2000–2021 ($R^2 = 0.75$) gives $\hat{c}(t,m) = 0.38 + 0.55\,r_{tm}$. Here r_{tm} is the issue-month Kim–Wright expected-average-short-rate component at the rounded whole-year maturity, capped at ten years. It is the fitted yield less its term premium, because raw zero yields would put a term premium rising with maturity into the denominator of the scaled outcome. Bin means of X_{tm} are 33.9, 41.5, 50.5, 56.8 and 53.3 bp.

D.8.4 Scaled and interacted specifications

Premia and the current-year benefit. The current-year benefit X_t falls from 104 bp in 2000 to an average of 9 bp in 2013–17, but premia do not track it. Year terciles with average benefits of 8.5, 26.7 and 70.2 bp give 5–20 year premia of 20.8–21.1, 12.3–23.2 and 6.5–13.2 bp, each with a significant long-minus-short contrast. In two- to four-year blocks, ratios of the 5–20 year premium to X_t exceed two in 2013–17

[28] Interest expense is BankFind series `EINTEXP` (Schedule RI, item 2.e, `RIAD4073`), average assets `ASSET5` and municipal holdings `SCMUNI` (Schedule RC-B, item 3, columns A/D, 8496/8499 with applicable prefixes), as defined in the FDIC's Financial API definitions and the FFIEC December 2017 Call Report, Schedules RI and RC-B. BankFind harmonizes the historical regulatory-report series.

[29] See the IRS, Bank Qualified Bonds: Section 265, pp. 13-11–13-12, for the allocation and 20% disallowance. The annual, per-dollar-of-basis benefit is distinct from the present value of deductions over a bond's remaining life.

and 2020–21, while the pre-2009 period estimates (Appendix D.8.6) are at most about 0.3 of X_t, so pooled ratios depend on how years are weighted.

Outcome scaled by the expected benefit. Dividing each offering yield by its own X_{tm} before 2SLS estimation of (27), bond by bond and before any pooling, gives the premia in expected-benefit units in column (3) of Table 36: 0.17, 0.31, 0.41, 0.52 and 0.47, with a 10–20 minus 1–2 year contrast of 0.34 (s.e. 0.11). The Adrian–Crump–Moench expected-short-rate component subtracts larger term premia and gives 0.17, 0.35, 0.46, 0.55 and 0.48 (contrast 0.39).

Within-regime long-minus-short contrasts are 0.57 in low-benefit years, 0.37 in middle years and 0.27 in the high-benefit years 2000–02 and 2005–08, when expected benefits were nearly flat in maturity (62.8–71.2 bp) and yield curves often inverted, so the benefit's maturity structure does not drive the gradient. Excluding the strong bank-demand years 2020–21 gives 0.18, 0.25, 0.35, 0.39 and 0.36 (contrast 0.21).

Yield-level benefit interactions. A second specification keeps offering yields in basis points and interacts designation with the expected annual benefit:

$$\begin{aligned} y_b &= \sum_{k=1}^{5} \beta_k B_b X_{t_b m_b} \mathbf{1}_k + \widetilde{V}_b' \delta + u_b, \\ B_b X_{t_b m_b} \mathbf{1}_k &\quad \text{instrumented by } E_{jt} X_{t_b m_b} \mathbf{1}_k, \qquad k = 1, \ldots, 5. \end{aligned} \tag{148}$$

Here $\widetilde{V}_b$ adds X_{tm} by bin to the fixed effects and running-variable controls of (27), and u_b is the regression error. Table 37 reports the yield reductions $-\beta_k$ and the joint first-stage strength. Table 36 divides the outcome by X_{tm}, whereas this specification multiplies designation and its instruments by X_{tm} and adds benefit controls, so their levels need not coincide. Both give rising profiles. The statement of Section 6.2 that about half of the benefit is priced at 10–20 years refers to the outcome-scaled specification. Excluding 2009–2010 leaves the interacted 10–20 minus 1–2 year contrast at 0.113 (s.e. 0.032).

Table 37: Bank designation interacted with the expected annual benefit.

Maturity (years)	1–2	2–5	5–10	10–20	> 20
Yield reduction per benefit unit	0.058**	0.096***	0.154***	0.175***	0.233***
	(0.024)	(0.025)	(0.027)	(0.031)	(0.089)
10–20 minus 1–2	0.117***				
	(0.032)				

Notes. Offering yield and expected annual benefit X_{tm} are in basis points. The table reports $-\beta_k$ in equation (148). Each designation–benefit interaction is instrumented by the below-cap indicator times the benefit in that bin. Controls include the benefit separately by bin and the matched fixed effects and running-variable slopes of equation (27). The window is 0.5–4 million dollars from the cap of the year. Observations: 355,372, issuer clusters: 15,889, joint Kleibergen–Paap F: 122.0. Issuer-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D.8.5 Robustness

Donut, bandwidth and window. Donuts from zero to \$1.5 million and bandwidths from \$2 million to \$8 million (Table 38) give 10–20 minus 1–2 year contrasts of 13.5–18.4 bp, and the 1–2 year premia are never significant. In the \$5–\$15 million window of Dagostino (2025), without a donut, the contrast is 15.4 (1.8) bp, one of the most precise estimates.

The last row of Table 38 drops the region her bunching estimation treats as contaminated, \$8.5–\$13.5 million at the \$10 million cap, so within $|R| \leq 8$ it compares issuers at least \$1.5 million below the cap with issuers at least \$3.5 million above it. First stages rise to 0.47–0.54, and the more distant extrapolation gives premia of 7.6 and 10.0 bp at 5–10 and 10–20 years and a contrast of 9.8 (4.2).

Table 38: Bank-qualified discontinuity: sensitivity to the donut, the bandwidth and the window.

	first stage	1–2y	2–5y	5–10y	10–20y	20y+	10–20 minus 1–2
Donut half-width, bandwidth $4 million							
none	0.39***	1.9 (1.8)	7.5*** (1.8)	13.9*** (1.8)	17.7*** (1.8)	17.7*** (4.5)	15.7*** (2.0)
$0.25 million	0.40***	1.6 (2.0)	7.6*** (2.0)	13.4*** (2.0)	16.7*** (2.2)	16.3*** (5.5)	15.1*** (2.4)
$0.5 million (baseline)	0.42***	1.8 (2.2)	8.7*** (2.2)	13.8*** (2.2)	18.4*** (2.3)	18.2*** (6.3)	16.5*** (2.7)
$1 million	0.44***	1.2 (2.9)	8.2*** (2.7)	13.3*** (2.8)	19.7*** (3.1)	20.6*** (8.0)	18.4*** (3.5)
$1.5 million	0.44***	2.0 (4.4)	7.3* (3.8)	11.8*** (4.0)	15.5*** (4.2)	19.5* (10.2)	13.5*** (5.2)
Bandwidth, donut $0.5 million							
$2 million	0.38***	0.8 (4.7)	7.9 (4.9)	13.3*** (4.8)	16.9*** (5.0)	21.0 (14.1)	16.1*** (4.8)
$3 million	0.40***	2.7 (3.0)	8.9*** (3.0)	15.2*** (2.9)	19.2*** (3.1)	19.9*** (7.6)	16.5*** (3.3)
$5 million	0.42***	−0.9 (1.9)	5.7*** (1.8)	11.4*** (1.8)	14.9*** (1.9)	15.9*** (4.9)	15.8*** (2.3)
$6 million	0.43***	−0.5 (1.7)	5.4*** (1.5)	10.8*** (1.6)	13.2*** (1.6)	11.4*** (4.1)	13.6*** (2.1)
$8 million	0.45***	0.0 (1.5)	6.3*** (1.2)	10.1*** (1.3)	13.5*** (1.3)	12.3*** (3.5)	13.5*** (1.8)
Other windows							
$5–15 million, no donut	0.40***	0.1 (1.6)	5.6*** (1.5)	12.1*** (1.6)	15.5*** (1.5)	16.7*** (3.8)	15.4*** (1.8)
$8.5–13.5 million dropped, $\|R\| \leq 8$	0.47***	0.2 (3.5)	6.0** (3.0)	7.6** (3.1)	10.0*** (3.4)	−1.8 (9.5)	9.8** (4.2)

Notes. Cells report the 2SLS premium of designation in basis points by maturity bin, pooled over 2000–2021 at each year's cap, with matched fixed effects and issuer-clustered standard errors in parentheses. The first-stage column refers to the 1–2 year bin. The $5–15 million window is the one used in the spread comparison of Dagostino (2025). The last row drops the region her bunching estimation excludes. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Construction and the relation to Dagostino (2025). Dagostino (2025) measures issuer-year issuance with Ipreo MuniIC data and documents the same bunching and first stage at the $30 million cap in 2009–10. Her regressions on $1 million issuance bins control for size, maturity, insurance, rating and use of proceeds, with year and issuer effects and standard errors clustered by year. In that study, tax-adjusted spreads over treasuries jump about 10 bp at the cap for general-obligation (GO) bonds issued before 2008. In her Call Report data for 2000–2010, designated bonds are about 4% of bank assets, rising to 5% by 2010, about nine times the non-designated share. Table 39 compares her construction choices with mine and nets refunding par from the running variable. Every row retains a larger long-maturity premium: issuer effects steepen it, while netting refundings shifts issuer-years and widens confidence intervals.

Table 39: Bank-qualified discontinuity: construction checks.

	1–2y	2–5y	5–10y	10–20y	20y+	10–20 minus 1–2
Baseline (Table 36)	1.8 (2.2)	8.7*** (2.2)	13.8*** (2.2)	18.4*** (2.3)	18.2*** (6.3)	16.5*** (2.7)
Bond insurance and enhancement × year	0.9 (2.2)	8.0*** (2.2)	13.0*** (2.2)	17.7*** (2.3)	17.3*** (6.2)	16.8*** (2.6)
General-obligation bonds only	1.7 (2.3)	7.6*** (2.2)	13.3*** (2.3)	15.2*** (2.5)	15.8** (7.3)	13.5*** (2.9)
Issuer fixed effects	−5.0* (2.6)	1.8 (2.5)	7.5*** (2.6)	17.3*** (2.8)	16.8*** (4.1)	22.3*** (2.4)
Tax-adjusted spread over treasuries	5.3 (3.3)	12.6*** (3.2)	21.4*** (3.2)	26.9*** (3.5)	22.8*** (8.6)	21.6*** (3.8)
Insurance controls, year clusters (22)	0.9 (2.9)	8.0*** (2.6)	13.0*** (3.6)	17.7*** (3.4)	17.3** (7.6)	16.8*** (3.7)
Running variable net of refundings	0.8 (3.1)	4.9 (3.1)	8.1*** (3.1)	15.7*** (3.6)	11.0 (8.2)	14.8*** (3.7)
and refunding bonds excluded	0.9 (3.1)	6.3* (3.2)	8.9*** (3.2)	16.0*** (3.6)	10.9 (7.5)	15.1*** (3.6)

Notes. Cells report the 2SLS premium of designation in basis points, pooled over 2000–2021 at the year's cap in the window $0.5 to $4 million, with matched fixed effects and issuer-clustered standard errors unless stated. First stages are 0.40 to 0.51 in every bin of every row. Insurance and enhancement: adds bond insurance × year and credit enhancement × year (47% of window bonds are insured). The general-obligation, issuer-fixed-effect, tax-adjusted-spread and year-clustered rows also include these controls. General obligation: unlimited-tax, limited and double-barreled general-obligation bonds, 70% of the window. Issuer fixed effects: identified from issuers that cross the cap across years. Tax-adjusted spread: $100\,[y/(1-\tau_{B,t}) - y^{\text{Tsy}}]$, the outcome of Dagostino (2025). Refunding-net: the running variable excludes refunding par, and the last row also drops refunding bonds from the estimation sample. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Applying her $1 million issuance-bin estimator to her window, with maturity-bucket, state and rating effects by year and use-of-proceeds controls, gives flat below-cap bins and a step above the cap, matching her Figure 3. Pooled across maturities, the step is 4.2 bp in the offering yield and 5.6 bp in the tax-adjusted spread, 1.1 and 1.4 bp at 1–5 years and 7.8 and 10.2 bp at 10–20 years. In her pooled-spread setting for GO bonds issued before 2008, with her controls, state fixed effects and issuer clustering, the reduced-form jump

at the cap is 10.4 bp (s.e. 2.5), against her approximately 10. With her issuer fixed effects and clustering by year, which leaves eight clusters, the jump is 6.3 (3.8), because within-issuer variation across the cap is thin before 2008.

D.8.6 Placebos and periods

Placebo cutoffs and the ARRA cap. At a placebo cutoff of $5 million, first stages are 0.00–0.05 and all yield effects are within two bp of zero. At $15 million, first stages are −0.01 to 0.02, and yield effects are zero except in one 20+ year cell. Around the $30 million cap of the American Recovery and Reinvestment Act (ARRA), with matched effects and a $0.5–$8 million window, the 2009–10 first stages are 0.19, 0.20, 0.19, 0.16 and 0.26 on 16 thousand bonds from issuers with $22–$38 million of issuance, jointly significant at five percent. At that cutoff, first stages are 0.01–0.04 in 2004–08 and −0.05 to 0.01 in 2011–15. At $10 million in 2009–10, they are 0.01–0.06 and not jointly significant. The 2009–10 reduced-form yield effects at the $30 million cap are −0.5, −3.2, −1.6, −10.9 and −21.5 bp, and the average beyond ten years minus the 1–2 year effect is −15.7 (6.7), significant at five percent.

By period. Period estimates for 2000–04, 2005–08, 2009–10, 2011–13, 2014–17 and 2018–19 use matched fixed effects and issuer clustering, with the $30 million cap and a $0.5–8 million window in the ARRA period and the $10 million cap and $0.5–4 million window otherwise. Before 2009, with funding costs c_t between 2.3 and 2.4% and X_t of 65–66 bp, premia are at most about 0.3 of X_t at every maturity: the deduction is weakly capitalized when the carrying penalty on non-designated bonds approaches a percentage point.

The 2009–10 $30 million cap has a weaker first stage than the $10 million cap's 0.42–0.45, reflecting limited use. With an ordinary-rule X_t of 30 bp, the 2009–10 premia are 11 (24), 22 (23), 17 (23), 50 (28) and 60 (29) bp. The contrast between the average beyond ten years and the 1–2 year premium is 43 (20), and its reduced-form counterpart above is significant at five percent.

In 2011–17, with funding costs of 0.31–0.51% and X_t of 9–14 bp, premia are 2.4–2.5 bp at 1–2 years and 14–28 bp at 5–20 years, at or above X_t.

After the 2018 cut in the bank tax rate, premia are 6.9 (4.3), 6.4 (5.0), 7.7 (5.7), 14.9 (7.7) and 3.7 (24.2) bp. The difference-in-discontinuities estimate of Section 6.2, which pools 2014–2019 and interacts designation with a post-2018 indicator, gives post-2018 changes of 3.1 (6.5), −9.1 (6.7), −19.2 (7.6), 3.6 (10.0) and −1.3 (28.7) bp, with unchanged first stages. Subchapter S banks, whose income is taxed to their shareholders and which the 2017 Tax Cuts and Jobs Act (TCJA) affected less, hold 6–14% of banks' municipal holdings over 2000–2021.

Annual reduced forms at the $10 million cap, pooled across maturities, are largest in 2011–13, 2019 and 2021, at 16–22 bp, and within about 5 bp of zero in 2001, 2004–05, 2017 and 2020. Premia do not simply track the current-year benefit, and the pattern is consistent with changing bank participation.

D.9 Investor holdings by tax status

Portfolio shares vary with exemption benefits (Section 3.4 and Table 2). In 2000–2017, exempt municipal bonds account for 34–48% of the average property and casualty (P&C) insurer's bond portfolio, compared with 6–9% for life insurers (Figure 25). Public pensions' exempt and taxable municipal shares never exceed 0.13% and 0.78%, respectively (Figure 26). After the corporate-rate cut, the P&C exempt share fell from 38.0% in 2017 to 28.6% in 2019, while the other-taxable share rose from 50.2% to 57.5%. This reallocation is the quantity counterpart of the repricing in the insurer test of P2 (Appendix D.10).

D.10 The 2017 Tax Cuts and Jobs Act and insurer exposure: difference-in-differences and event study

This appendix gives the difference-in-differences (DiD) test of P2 summarized in Section 6.2, with inference in Appendices D.10.1 and D.10.2.

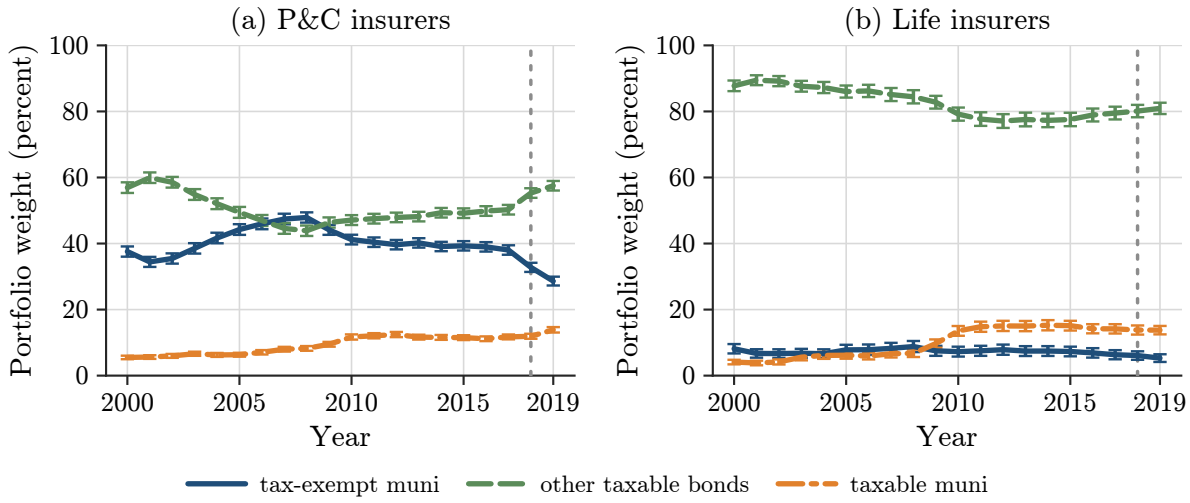


Figure 25: Tax-exempt municipals, taxable municipals and other taxable bonds as shares of insurers' total bond holdings: P&C in panel (a) and life in panel (b), 2000–2019, NAIC Schedule D. The dotted vertical line marks the TCJA corporate-rate cut. Bars are 95 percent intervals for yearly mean insurer portfolio shares, with standard errors clustered by insurer.

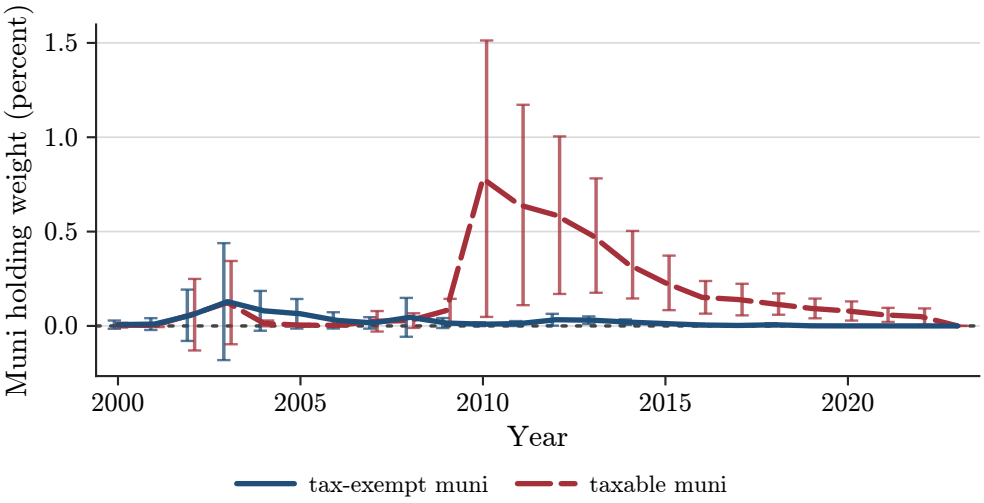


Figure 26: Municipal holdings by federal tax status as shares of total bond portfolios in the public-pension sample obtained by Freedom of Information Act (FOIA) requests (Section 3.1), 2000–2023. All thirty plans are covered from 2015. Bars are 95 percent intervals for yearly mean plan portfolio shares, with standard errors clustered by plan. Other taxable bonds are omitted. Taxable municipals include Build America Bonds.

Prediction. Appendix Proposition 6 gives zero pass-through of the lower-tax investor's exemption benefit at the riskless short end, where the household's after-tax cash return anchors the yield, and positive pass-through at the long end. Through portfolio adjustment, a change in that benefit also reaches risky maturities shorter than those the lower-tax investor holds. P&C insurers are lower-tax institutions in this sense. Since the TCJA lowered their exemption benefit, P2 predicts that after the reform the implied tax fell more at longer maturities in states whose municipal debt P&C insurers held more heavily.

The 2017 tax change. The TCJA cut the corporate rate from 35% to 21% effective 2018Q1, and the §832 proration rule amplified the resulting decline in P&C insurers' exemption benefit. The disallowed fraction of their tax-exempt interest rose from 15 to 25 percent, so the effective tax on municipal interest stayed at 5.25% ($35\% \times 15\%$ before, $21\% \times 25\%$ after), while the after-tax share of taxable interest rose from 0.65 to 0.79. The P&C relative after-tax share ξ_P rose from 0.686 to 0.834, and the exemption benefit $1-\xi_P$ narrowed from 0.31 to 0.17.

The top household ordinary rate also fell, from 39.6% to 37%, and the cap on state and local tax (SALT) deductions changed the value of in-state exemptions. Life insurers' relative after-tax share rose from $0.65/0.825 \simeq 0.788$ to $0.79/0.937 \simeq 0.843$ (Appendix E.1), and banks' taxable after-tax share and the tax cost of their financing changed. The design compares states with different P&C exposure, so tax changes common to all states drop out.

Treatment and sample. The treatment is a state's exposure to P&C insurers, $PCShare_s$: the market value of P&C insurers' holdings of state s's municipal bonds at the end of 2015 relative to the state's par outstanding, before the 2016Q3 start of the estimation window and two years before the reform. Across the 50 states and the District of Columbia in the sample, it averages 0.103 with a standard deviation of 0.030, and each bond b takes the exposure of its issuing state, $PCShare_{s(b)}$. The sample is the 1,733,881 customer trades reported to the Municipal Securities Rulemaking Board (MSRB) over 2016Q3–2019Q1 in non-callable municipal bonds rated AA and above and not subject to the alternative minimum tax (AMT). Restricting to non-callable bonds rules out direct repricing of embedded options. The outcome is the implied tax of (2) for each trade, $ImpliedTax_{b,t} = 1 - y^M_{b,t}/y^C_t(m_b)$, computed from the trade yield and the corporate zero yield for the last month of the trade's calendar quarter at the bond's whole-year maturity, its maturity year minus the trade year.

Specification. I estimate

$$\begin{aligned} ImpliedTax_{b,t} = \beta_0\big(PostTCJA_t \times PCShare_{s(b)}\big) + \beta_1\big(PostTCJA_t \times PCShare_{s(b)} \times m_b\big) \\ + \alpha_{\lfloor m_b \rfloor \times Purchase} + \alpha_{issuer \times Purchase} + \alpha_t + \Gamma' Z_{b,t} + \varepsilon_{b,t}, \end{aligned} \tag{149}$$

where $PostTCJA_t = \mathbf{1}\{t \geq 2017Q4\}$ starts the post period at the enactment quarter and m_b is the bond's remaining maturity in years. The fixed effects compare trades of the same issuer and of the same whole-year maturity, purchases with purchases and sales with sales ($Purchase_{b,t}$ is the MSRB dealer-side indicator for a customer purchase), and α_t absorbs each quarter's level. The controls $Z_{b,t}$ are $PCShare_{s(b)} \times m_b$; the issuing state's resident tax rate, the pre-period life-insurer holding share and the bond rating, each alone and interacted with maturity; and interactions of event time with maturity and with the state tax rate. The event-time-by-maturity interactions let the common maturity gradient change each quarter, so β_1 is identified from cross-state differences in $PCShare$ within a quarter.

The differential effect of exposure at maturity m_b is $\beta_0 + \beta_1 m_b$. The model predicts $\beta_1 < 0$, and the representative-tax benchmark predicts $\beta_0 = \beta_1 = 0$.

Instruments. Pre-reform P&C ownership may be correlated with state conditions that also move the maturity profile of implied taxes, so the instrumental-variables (IV) specifications instrument $PCShare_s$ with the municipal capacity of the P&C insurers domiciled in the state:

$$IV1_s = \log\Big(\frac{\sum_{i \in D_s} \text{Muni Assets}_i}{\text{Muni Outstanding}_s}\Big), \qquad IV2_s = \log\Big(\frac{\sum_{i \in D_s^{\text{act}}} \text{Bond Assets}_i}{\text{Muni Outstanding}_s}\Big), \tag{150}$$

where D_s is the set of P&C insurers domiciled in state s and D_s^{act} the subset holding any municipal bonds. Muni Assets$_i$ and Bond Assets$_i$ are insurer i's par holdings of municipal bonds and of all bonds in its Schedule D filings with the National Association of Insurance Commissioners (NAIC), and Muni Outstanding$_s$ is the state's municipal par outstanding. $IV2_s$ uses total bond holdings, so it does not depend on which municipal bonds the domiciled insurers chose to hold. For each instrument, its interactions with m_b, with $PostTCJA_t$ and with their product serve as instruments for the three corresponding exposure terms. Relevance comes from insurer home bias: locally domiciled insurers disproportionately hold the state's bonds. The Kleibergen–Paap first-stage F statistics are 25.7 for $IV1_s$ and 17.5 for $IV2_s$.

The exclusion restriction is that, conditional on the fixed effects and controls, the domiciled insurers' capacity affects the post-TCJA change in the maturity gradient of implied taxes only through P&C insurers' exposure to the state's bonds. Insurer domicile largely predates the reform. Quarter effects and event-time-by-maturity controls absorb the common component of the federal tax changes. Event-time-by-state-tax interactions absorb level changes associated with state tax rates, including SALT-induced ones, and state-tax-by-maturity controls capture a persistent maturity gradient, but neither absorbs a post-reform change in the gradient that varies with state tax rates. Appendix D.10.2 states the covariance conditions under which omitting such a SALT gradient biases the OLS estimate toward zero.

Results. Table 40 reports the static DiD estimates. The maturity gradient β_1 is -0.017 under OLS and -0.038 and -0.040 under the two IV specifications, with issuer-clustered standard errors of 0.005, 0.010 and 0.015. For two states whose pre-reform P&C shares differ by one standard deviation, 3.0 percentage points, the implied tax in the more exposed state falls after the reform by an additional 0.05 percentage point per year of maturity under OLS and 0.11 under the first IV specification. Between a five- and a twenty-five-year bond, the first IV estimate cumulates to about 2.3 percentage points of implied tax. At a corporate yield of four percent, that is about 9 basis points of municipal yield relative to the taxable yield.

Table 40: TCJA difference-in-differences: implied-tax response to insurer exposure. The sample is MSRB trades in non-callable, AA-and-above, non-AMT municipal bonds in 2016Q3–2019Q1. The outcome is the implied tax of equation (2), and *PCShare* is the end-2015 P&C ownership share. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	(1) OLS	(2) IV1	(3) IV2
$PostTCJA \times PCShare$	-0.122^{***}	-0.129	-0.017
	(0.029)	(0.084)	(0.100)
$PostTCJA \times PCShare \times m_b$	-0.017^{***}	-0.038^{***}	-0.040^{***}
	(0.005)	(0.010)	(0.015)
$\lfloor m_b \rfloor \times Purchase$ FE	Yes	Yes	Yes
$Issuer \times Purchase$ FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Issuer cluster	Yes	Yes	Yes
Observations	1,733,881	1,733,881	1,733,881
First-stage F (Kleibergen–Paap)		25.7	17.5

Notes. Columns (2) and (3) instrument *PCShare* with $IV1_s$ and $IV2_s$ of equation (150). All columns include the controls of equation (149), among them the rating and its interaction with m_b. Standard errors are clustered by issuer (six-digit CUSIP, 9,903 issuers) and reported in parentheses.

The maturity gradient persists on the wider A-or-above sample (Appendix D.12.2). With the rating control omitted in both samples (Table 47), widening the sample to A or above roughly doubles the OLS gradient, from -0.018 to -0.036, leaves the IV1 gradient essentially unchanged, from -0.037 to -0.039, and roughly doubles the IV1 level term.

Event study. The event study replaces the post-reform indicator in (149) with quarter indicators, omitting 2017Q3, so $\hat{\beta}_{1,q}$ measures the maturity gradient in quarter q relative to 2017Q3. The IV versions use the corresponding quarter interactions of the instruments, with the fixed effects and clustering of Table 40.

These estimates locate the change at the reform. Across all three estimators, eleven of the twelve pre-TCJA coefficients are individually indistinguishable from zero at the five percent level, while the coefficients become significantly negative at 2018Q1 in every column and remain negative thereafter. Five of the six post-reform quarters exclude zero under IV1 and four under IV2 (Figure 27; Table 41 reports each quarter).

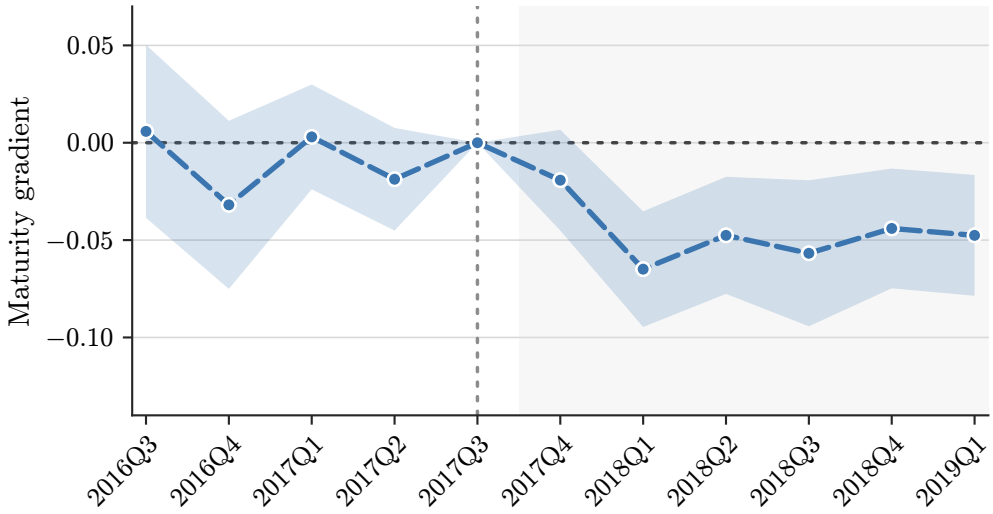


Figure 27: IV1 event-study estimates of the quarter-specific maturity gradient, the coefficient on $\mathbf{1}\{t = q\} \times PCShare \times m_b$, relative to 2017Q3 (dashed line). Shading marks the post-reform quarters from 2017Q4. Bands are 95% intervals clustered by six-digit CUSIP issuer. Appendix D.10.1 reports further inference checks.

Table 41: TCJA dynamic event study: full triple-interaction coefficients $\hat{\beta}_{1,q}$ on $\mathbf{1}\{t = q\} \times PCShare \times m_b$, by quarter. Standard errors in parentheses are clustered by issuer (six-digit CUSIP) as in Table 40, and 2017Q3 is the omitted base quarter. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Quarter q	(1) OLS	(2) IV1	(3) IV2
2016Q3	0.019**	0.006	−0.020
	(0.010)	(0.023)	(0.036)
2016Q4	−0.014	−0.032	−0.029
	(0.009)	(0.022)	(0.036)
2017Q1	0.001	0.003	0.009
	(0.007)	(0.014)	(0.020)
2017Q2	−0.005	−0.019	−0.021
	(0.006)	(0.013)	(0.020)
2017Q4	−0.001	−0.019	−0.021
	(0.006)	(0.013)	(0.020)
2018Q1	−0.028***	−0.065***	−0.063***
	(0.007)	(0.015)	(0.023)
2018Q2	−0.027***	−0.048***	−0.039*
	(0.007)	(0.015)	(0.023)
2018Q3	−0.031***	−0.057***	−0.063**
	(0.009)	(0.019)	(0.027)
2018Q4	−0.011	−0.044***	−0.051**
	(0.008)	(0.016)	(0.023)
2019Q1	−0.014*	−0.048***	−0.065***
	(0.007)	(0.016)	(0.024)
$\lfloor m_b \rfloor \times$ *Purchase* FE	Yes	Yes	Yes
Issuer × *Purchase* FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Issuer cluster	Yes	Yes	Yes
Observations	1,733,881	1,733,881	1,733,881

D.10.1 State-clustered inference

As a check for shocks shared within states, I also cluster the insurer regressions above by issuing state (51 clusters). Point estimates are unchanged. State-clustered standard errors are 0.0129, 0.0245 and 0.0342 for the OLS, IV1 and IV2 maturity coefficients, and their 95% intervals include zero. The post-reform level coefficient under OLS remains negative and significant: −0.122 (s.e. 0.061), with $p = 0.044$ from a state-level wild-cluster bootstrap (9,999 replications).

D.10.2 SALT-induced household demand and the insurer-channel regression

A binding SALT cap can raise high-tax households' exemption benefit and lower municipal yields where they are active. The conditions below determine when this household channel attenuates the OLS insurer coefficient.

Table 42: P3: observed coupon-yield response to treasury par yield by remaining maturity.

	(1) Continuous $\Delta y^M_{b,t}$	(2) Bin $\Delta y^M_{b,t}$
$\Delta y^{\text{Tsy}}_{m_b,t}$	0.346^{***}	0.351^{***}
	(0.016) [0.012]	(0.013) [0.006]
$\Delta y^{\text{Tsy}}_{m_b,t} \times m_b$	0.007^{***}	
	(0.002) [0.002]	
$\Delta y^{\text{Tsy}}_{m_b,t} \times Long_b$		0.075^{***}
		(0.012) [0.009]
$Long_b$		0.016^{***}
		(0.002) [0.002]
m_b	0.002^{***}	
	(0.000) [0.000]	
Trade-date cluster	Yes	Yes
Observations	952,874	952,874
Weights	Inverse pair count	Inverse pair count

Notes. The sample is pairs of consecutive MSRB trades within 90 days in federally tax-exempt, non-callable bonds rated AA and above (agency score ≤ 3), January 2005 to August 2020, over which the maturity-matched treasury yield moved at least 10 basis points in either direction. The treasury yield is interpolated on the daily GSW curve at the bond's exact remaining maturity on each transaction date. Observations are weighted by the inverse count of their CUSIP's base-sample trade pairs, computed before observations with missing or trimmed outcomes are dropped. The dependent variable is trimmed at its first and ninety-ninth percentiles. $Long_b = \mathbf{1}\{m_b \geq 5\}$. First differencing absorbs time-invariant bond characteristics. Parentheses give trade-date-clustered standard errors (3,864 clusters). Brackets give state-clustered standard errors (54 clusters). Stars are identical under either clustering. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Write $Y_{b,t} = 1 - y^M_{b,t}/y^C_t(m_b)$ for the implied tax of equation (2), $X_{b,t} = PostTCJA_t PCShare_{s(b)} m_b$ for the insurer interaction and $W_{b,t} = PostTCJA_t\, Tax_{s(b)} m_b$ for a household-SALT interaction, where the state tax rate Tax_s proxies SALT exposure. Here $X_{b,t}$ is unrelated to the bank benefit X of Appendix D.8. Consider

$$Y_{b,t} = \beta_1 X_{b,t} + \beta_2 PostTCJA_t\, Tax_{s(b)} + \beta_3 W_{b,t} + \text{controls} + \text{FE} + \varepsilon_{b,t}. \tag{151}$$

The channels under study imply $\beta_1 < 0$ for the insurer mechanism, and $\beta_2 > 0$ and $\beta_3 < 0$ for stronger household demand at the short end.

Suppose that the SALT level term is controlled for, that W is omitted and that the remaining disturbance is orthogonal to residualized X, with tildes denoting residuals after projecting on all included controls and fixed effects. The omitted-variable formula is

$$\operatorname{plim} \widetilde{\beta}_1 = \beta_1 + \beta_3 \frac{\operatorname{Cov}(\widetilde{X}, \widetilde{W})}{\operatorname{Var}(\widetilde{X})}. \tag{152}$$

Consequently,

$$\beta_3 < 0, \quad \operatorname{Cov}(\widetilde{X}, \widetilde{W}) < 0 \quad \Longrightarrow \quad \operatorname{plim} \widetilde{\beta}_1 > \beta_1. \tag{153}$$

Equation (152) uses residualized variables, so it accounts for controls, fixed effects and weights. An upward bias attenuates a negative coefficient as long as the estimate keeps its sign.

IV identifies the insurer coefficient when the instruments are relevant and orthogonal to the remaining household-demand term, under the exclusion restriction stated in Appendix D.10.

D.11 P3: municipal-yield response to treasury yields

Transaction sample. Each MSRB customer trade records the CUSIP, trade date, par, dollar price, implied yield to maturity and dealer side. Estimation trades, from January 2005 to August 2020, are in non-callable, federally exempt bonds rated AA and above (agency score ≤ 3, Section 6.3). Noncallability removes option contamination. I exclude trades less than 92 days after issue, when offering yields still anchor prices, and trades less than 365 days before maturity, where the treasury substitute for the short rate is erratic.

Matched treasury changes. At each trade, the daily treasury par-yield curve of Gürkaynak, Sack and Wright (GSW) is interpolated linearly between integer maturities at the bond's remaining maturity $m_b(t) = d_{b,t}/365$, where $d_{b,t}$ is its days to maturity, and clamped above thirty years. Within each CUSIP and dealer side, consecutive trades form the municipal yield change $\Delta y^{M}_{b,t}$ and the matched treasury change $\Delta y^{\text{Tsy}}_{m_b,t} = y^{\text{Tsy}}_{m_b(t),t} - y^{\text{Tsy}}_{m_b(t-1),t-1}$, where $t-1$ denotes the trade preceding t. Comparing purchases with purchases and sales with sales limits bid-ask and intraday contamination. Pairs with gaps of 1–90 days and treasury moves of at least 10 bp are kept, with the weights and trimming of Table 42. Over Δt elapsed days, aging contributes $-(\Delta t/365)\partial_m y^{\text{Tsy}}_{m,t} + O((\Delta t/365)^2)$ to the matched change, an effect measured below.

Specifications. Column (1) of Table 42 estimates the continuous interaction in (28), including maturity but no fixed effects. Column (2) replaces maturity by $Long_b = \mathbf{1}\{m_b \geq 5\}$, so its short and long responses are β_0 and $\beta_0 + \beta_1$.

Holding the benchmark maturity fixed. Slide, $y^{\text{Tsy}}_{m_b(t),t} - y^{\text{Tsy}}_{m_b(t-1),t}$, is the part of the matched change due only to the bond's shorter remaining maturity at the later trade. Fixing the benchmark at the initial maturity, $\Delta y^{\text{Tsy,fixed}}_{b,t} = y^{\text{Tsy}}_{m_b(t-1),t} - y^{\text{Tsy}}_{m_b(t-1),t-1}$, removes it. On upward-sloping curves slide is negative, averaging -2.6 bp below five years and -1.6 bp beyond, against absolute changes near 26 bp. Removing it lowers the short response more and increases the gradient (Table 43). Because municipal and treasury curves have different slopes, municipal aging does not offset the treasury slide. The third row also absorbs linear municipal aging through elapsed time and its maturity interaction, which further increases the gradient. The headline specification keeps the benchmark matched at each trade, the more conservative choice.

Table 43: P3 observed coupon-yield responses and treasury-curve aging: equation (28) on Table 42's 952,874 pairs, inverse-CUSIP-base-pair weighted and trade-date clustered.

	Bin specification		Interaction	
Treasury benchmark	< 5y	≥ 5y	Bin	Continuous
Benchmark matched at each trade's maturity (Table 42)	0.351^{***}	0.426^{***}	0.0749^{***}	0.0070^{***}
Benchmark held at the first trade's maturity	0.340^{***}	0.423^{***}	0.0825^{***}	0.0084^{***}
plus elapsed time and elapsed time × maturity	0.334^{***}	0.419^{***}	0.0850^{***}	0.0086^{***}

Notes. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Sensitivity to the filter thresholds. Across the treasury-move cutoffs and trade windows of Table 44, all observed maturity-interaction estimates are positive. Every binned cell is significant at one percent, and every continuous cell is significant at five percent, with twenty-seven of the thirty significant at one percent. Appendix D.11.1 applies the coupon benchmark to each cell.

State clustering. Brackets in Table 42 give standard errors clustered by issuing state, which allow correlation among a state's bonds across dates, while the headline trade-date clustering allows common shocks across states on the same date. Both headline interactions remain significant at one percent, and binned-grid significance levels are unchanged. In the continuous grid of Table 44, the 0.05, 0.075 and 0.10 cutoffs give $p < 0.05$ at thirty days and $p < 0.10$ at sixty days, and the 0.15 cutoff gives $p < 0.05$ at sixty

Table 44: P3 sensitivity: observed maturity interaction across the treasury-move cutoff and trade-gap window (weighted by inverse CUSIP base-pair counts).

Panel A. Continuous interaction $\Delta y^{\text{Tsy}}_{m_b,t} \times m_b$					
Cutoff \ window	15d	30d	60d	90d	120d
≥ 0.05	0.0053^{***}	0.0057^{***}	0.0051^{**}	0.0071^{***}	0.0090^{***}
≥ 0.075	0.0061^{***}	0.0057^{***}	0.0047^{**}	0.0068^{***}	0.0087^{***}
≥ 0.10	0.0065^{***}	0.0060^{***}	0.0049^{**}	0.0070^{***}	0.0087^{***}
≥ 0.15	0.0080^{***}	0.0071^{***}	0.0059^{***}	0.0077^{***}	0.0091^{***}
≥ 0.20	0.0088^{***}	0.0082^{***}	0.0067^{***}	0.0081^{***}	0.0094^{***}
≥ 0.25	0.0104^{***}	0.0083^{***}	0.0074^{***}	0.0087^{***}	0.0097^{***}
Panel B. Binned interaction $\Delta y^{\text{Tsy}}_{m_b,t} \times Long_b$					
Cutoff \ window	15d	30d	60d	90d	120d
≥ 0.05	0.049^{***}	0.061^{***}	0.070^{***}	0.074^{***}	0.088^{***}
≥ 0.075	0.056^{***}	0.063^{***}	0.068^{***}	0.074^{***}	0.086^{***}
≥ 0.10	0.062^{***}	0.066^{***}	0.069^{***}	0.075^{***}	0.086^{***}
≥ 0.15	0.079^{***}	0.077^{***}	0.079^{***}	0.083^{***}	0.092^{***}
≥ 0.20	0.094^{***}	0.090^{***}	0.085^{***}	0.088^{***}	0.097^{***}
≥ 0.25	0.111^{***}	0.100^{***}	0.096^{***}	0.096^{***}	0.102^{***}

Notes. Each cell re-estimates the headline specification on pairs whose maturity-matched treasury move is at least the cutoff (percentage points) and whose trades are within the window. Standard errors are clustered by trade date as in Table 42. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

days. All other significance levels are unchanged, so twenty-seven of the thirty continuous cells and all binned cells remain significant at five percent.

Taxable-market comparison. I also compare municipal changes with monthly corporate-curve changes. Both reference curves are measured at monthly averages and the bond's remaining maturity in each trade month. Retaining cross-month pairs whose absolute monthly treasury change is at least 10 bp gives 517,864 pairs, with the original inverse-CUSIP weights. The municipal-to-corporate maturity interaction is 0.0132 (s.e. 0.0026), while the municipal-to-treasury interaction on these pairs is 0.0072 (s.e. 0.0025), both with trade-date clustering. The cash-flow comparison below translates each reference market's zero curve into municipal coupon yields on a common sample.

D.11.1 Coupon cash flows and the zero-yield benchmark

P3 concerns zero-coupon responses, while the data report coupon yields and treasury par yields. Even when zero yields satisfy $y^{M,0}(u,t) = \xi_H y^{\text{Tsy},0}(u,t)$ at every horizon u, coupon yields need not have a constant response ratio, because their cash flows span horizons with different yield movements. The coupon benchmark measures the response that coupon conventions alone produce.

I fix the after-tax share at 0.65 and price each bond's regular coupon and redemption payments at municipal zero yields of 0.65 times the GSW treasury zero yields:

$$P^{M,0}_{b,t} = \sum_k CF_{b,k} \exp\left[-0.65\, y^{\text{Tsy},0}(u_{b,k,t}, t)\, u_{b,k,t}\right],$$

where $CF_{b,k}$ is payment k and $u_{b,k,t}$ its time to payment, and zero yields are continuously compounded fractions. I then recover the nominal-semiannual yield to maturity, $y^{M,0}_{b,t}$, from this price at each trade date, keeping the bond's coupon and maturity, both transaction dates and the original par-yield regressor.

Semiannual payments are reconstructed backward from maturity with US 30/360 day-count fractions. The comparison retains zero-coupon bonds and fixed-rate semiannual bonds whose first coupon precedes the

first trade, and excludes other payment conventions, maturities over thirty years and pairs with incomplete curves. Zero yields are interpolated linearly, with the one-year value held below one year. The 918,350 common pairs retain 94.9% of the original weight, with the original trimming and inverse-CUSIP weights. Simulated outcomes are not trimmed.

Table 45 applies the same regression to the observed changes, the benchmark changes and their difference, whose standard error retains their covariance. The coupon benchmark accounts for 30.0% of the observed daily gradient, leaving a difference of 0.0056 (s.e. 0.0021). With after-tax shares from 0.60 to 0.80, the benchmark accounts for 27.6%–37.0%, and the difference remains positive throughout. The binned difference is 0.0623 (s.e. 0.0123). Benchmarks that hold treasury maturity fixed, with or without elapsed-time controls, also leave a positive interaction.

The monthly comparisons keep the cross-month pairs and monthly reference regressors of the taxable-market comparison, restricted to the 500,006 of its 517,864 pairs with regular coupon schedules. Treasury comparisons price at daily proportional GSW zero curves on trade dates, corporate comparisons at monthly proportional corporate zero curves converted to continuous yields. Both differences remain positive and significant at five percent (Table 45).

Table 45: P3: response gradient after matching coupon cash flows.

	Observed	Coupon benchmark	Difference	Pairs
Daily, continuous	0.0080***	0.0024***	0.0056***	918,350
	(0.0021)	(0.0001)	(0.0021)	
Daily, long minus short	0.0738***	0.0115***	0.0623***	918,350
	(0.0122)	(0.0003)	(0.0123)	
Fixed initial maturity	0.0095***	0.0034***	0.0061***	918,350
	(0.0021)	(0.0001)	(0.0021)	
Fixed maturity + gap controls	0.0098***	0.0036***	0.0062***	918,350
	(0.0021)	(0.0001)	(0.0021)	
Monthly treasury	0.0082***	0.0019**	0.0063**	500,006
	(0.0026)	(0.0009)	(0.0026)	
Monthly corporate	0.0141***	−0.0006***	0.0147***	500,006
	(0.0027)	(0.0001)	(0.0026)	

Notes. The observed and benchmark coupon changes of Appendix D.11.1 use identical pairs, regressors and inverse-CUSIP-base-pair weights, and the difference is estimated jointly. Interactions are per year of remaining maturity, except the five-year-bin contrast. Fixed-maturity rows retain the first trade's treasury maturity. Gap controls are elapsed days and their maturity interaction. Monthly rows use cross-month pairs and monthly regressors. Parentheses give second-trade-date-clustered standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Applying the coupon benchmark separately in each cell of Table 44, on that cell's common coupon pairs, leaves all thirty continuous differences positive, from 0.0034 to 0.0104, with twenty-four significant at five percent. All thirty benchmark-adjusted binned differences are positive and significant at five percent, ranging from 0.0393 to 0.1016.

D.12 Robustness to the rating cut

Both samples apply their rating cutoffs to the unrounded mean agency score on the twenty-two-notch scale, on which 1 = AAA, 3 = AA and 7 = A−. The curves in Sections 3 and 5 use scores of seven or better, the A-or-above sample, matching the corporate A band (A+, A and A−). The Build America Bond (BAB) comparison in Section 3 and the P1, Tax Cuts and Jobs Act (TCJA) and P3 tests in Section 6 use scores of three or better, the AA-and-above sample. Each exercise otherwise keeps its own rating cells and rating controls.

The curve pattern survives the narrower cut, and the BAB, P1, TCJA and P3 tests retain their gradient signs on the wider cut. On both samples, BAB interactions are significant at one percent and the bins rise monotonically. P1's pre-2018 state-tax pass-through declines across all five bins under OLS and IV

(Table 46), with endpoint differences significant at five percent under OLS and one percent under IV. In the TCJA insurer test, the maturity gradients are negative and significant under all three estimators with issuer clustering, and Appendix D.10.1 reports state-clustered inference. P3's observed maturity interactions in the coupon-yield response to par yields are positive and significant at one percent. The wider sample gives larger P3 interactions and a larger OLS TCJA gradient, with similar BAB, P1 and IV TCJA gradients.

D.12.1 Curve construction at AA and above

Figure 28 uses the AA-and-above sample and its own bond, rating and month weights. Each implied tax is one minus the ratio of mean municipal to mean taxable yields, averaged with these weights. Net of calls and issuance premia, the implied tax falls from 40.1% at two years to 25.5% at thirty, a decline of 14.6 points against 17.1 on the A-or-above sample.

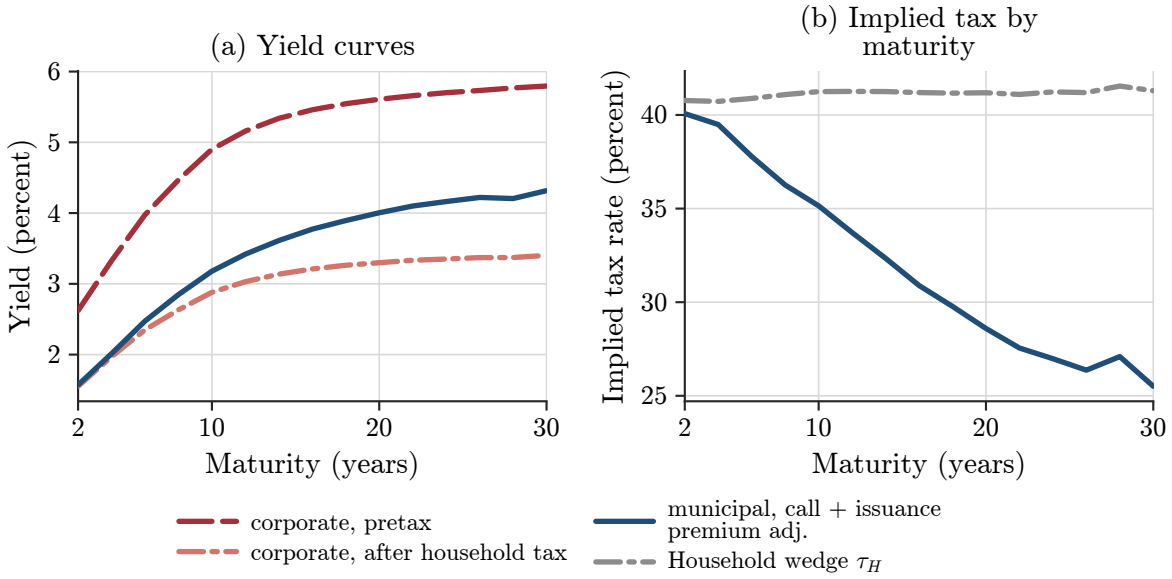


Figure 28: The AA-and-above municipal puzzle, 2000–2020, with this sample's own weights and household benchmark. Panel (a) shows pretax and household after-tax corporate zero-coupon yields and the municipal yield after removing callability and the issuance premium, and panel (b) shows the implied tax $1 - y_T^M / y_T^C$ and the household wedge τ_H. Calls use dated calibrated prices, and the sample's all-size issuance premium is scaled by the full sample's institutional-to-all-size ratio (see text).

The model comparison repeats the accounting of Section 5.4, applying this sample's model ratio of municipal to taxable mean yields to observed corporate yields (the corporate normalization of (22)). Call, liquidity and default estimates are sample-specific, and capital gains use the common method. The issuance adjustment scales the sample's all-size premium by the full-sample institutional-to-all-size ratio, giving 11.5 bp at thirty years. The normalized model gap exceeds the residual before the model at every maturity, 96.2 against 85.2 basis points at thirty years, so all unexplained residuals are negative. In absolute value, 7/15 are below ten basis points, the thirty-year residual is −11.0 and the root-mean-square residual is 11.1 basis points.

D.12.2 Empirical tests on the A-or-above sample

Table 29 (Appendix D.3) compares the two BAB rating cuts in columns (1) and (3), with call matching in columns (2) and (4). Table 46 repeats the P1 specification of Appendix D.6, so its AA-and-above columns reproduce the headline P1 estimates. Table 47 repeats the TCJA difference-in-differences of Table 40 with issuer clustering but omits the headline rating and rating-by-maturity controls in both samples, so its AA-and-above columns differ slightly. Tables 48 and 49 repeat the P3 specification of Table 42 and the binned

grid of Table 44 with state-clustered standard errors, where the headline P3 estimates cluster by trade date. Their AA-and-above coefficients equal the headline ones.

Table 46: P1 by rating cut, 2000–2017.

	OLS		IV	
	Score ≤ 3	Score ≤ 7	Score ≤ 3	Score ≤ 7
β_1: $[0, 5)$ years	1.072***	0.990***	1.092***	1.084***
	(0.195)	(0.256)	(0.224)	(0.234)
β_2: $[5, 10)$ years	0.682***	0.618***	0.705***	0.680***
	(0.071)	(0.106)	(0.088)	(0.122)
β_3: $[10, 15)$ years	0.484***	0.379***	0.448***	0.382***
	(0.098)	(0.084)	(0.121)	(0.101)
β_4: $[15, 20)$ years	0.386***	0.284***	0.325***	0.262***
	(0.102)	(0.087)	(0.117)	(0.097)
β_5: $[20, 30]$ years	0.298***	0.239***	0.227***	0.216***
	(0.075)	(0.077)	(0.081)	(0.080)
Decline $\beta_1 - \beta_5$	0.774***	0.751**	0.865***	0.868***
p-value of decline	0.002	0.010	< 0.001	< 0.001
KP F	—	—	10.4	12.6
Observations	681,839	1,397,769	681,839	1,397,769
States	47	47	47	47
Full main controls	Yes	Yes	Yes	Yes

Notes. Full controls, fixed effects and state-clustered inference as in Appendix D.6, 2000–2017. IV uses bin-interacted 1990 rates. All columns use call-adjusted zeros. Pricing availability differs across rating bands. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 47: P2: TCJA insurer difference-in-differences, by rating cut

	OLS		IV1		IV2	
	≤ 3	≤ 7	≤ 3	≤ 7	≤ 3	≤ 7
Post × *PCShare*	−0.118***	−0.161***	−0.124	−0.255**	−0.015	−0.175
	(0.029)	(0.035)	(0.085)	(0.110)	(0.102)	(0.131)
Post × *PCShare* × m_b	−0.018***	−0.036***	−0.037***	−0.039***	−0.038**	−0.037***
	(0.005)	(0.005)	(0.011)	(0.009)	(0.015)	(0.012)
$\lfloor m_b \rfloor$ × *Purchase* FE	Yes	Yes	Yes	Yes	Yes	Yes
Issuer × *Purchase* FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Issuer cluster	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,733,881	3,043,035	1,733,881	3,043,035	1,733,881	3,043,035
First-stage F			25.6	45.5	17.5	34.6

Notes. *Post* is the post-reform indicator *PostTCJA* of equation (149). Standard errors are clustered by issuer (six-digit CUSIP), with 9,903 issuers at score ≤ 3 and 17,634 at score ≤ 7. First-stage F is Kleibergen–Paap. Both samples omit Table 40's rating and rating-by-maturity controls. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 48: P3: observed coupon-yield response to treasury par yields, by rating cut

	(1) continuous		(2) bin <5y vs ≥5y	
	score ≤ 3	score ≤ 7	score ≤ 3	score ≤ 7
$\Delta y^{\text{Tsy}}_{m_b,t}$	0.346***	0.281***		
	(0.012)	(0.011)		
$\Delta y^{\text{Tsy}}_{m_b,t}\times$ maturity	0.007***	0.013***		
	(0.002)	(0.002)		
$\Delta y^{\text{Tsy}}_{m_b,t}\times$ (≥5y)			0.075***	0.113***
			(0.009)	(0.006)
State cluster	Yes	Yes	Yes	Yes
Observations	952,874	2,023,637	952,874	2,023,637
Weights	Inverse pair count			

Notes. The estimates use Table 42's first-differenced specification for coupon yields and par yields, weighted by inverse CUSIP pair counts and clustered by state. The wider-sample continuous interaction roughly doubles and the binned one rises by about half. Appendix D.11.1 maps the zero-coupon benchmark into the observed yield conventions on the main sample. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 49: P3 observed-response sensitivity grid, binned specification, by rating cut

	AA and above (score ≤ 3)					A− and above (score ≤ 7)				
cutoff	15d	30d	60d	90d	120d	15d	30d	60d	90d	120d
≥ 0.05	0.049***	0.061***	0.070***	0.074***	0.088***	0.078***	0.099***	0.098***	0.114***	0.121***
≥ 0.075	0.056***	0.063***	0.068***	0.074***	0.086***	0.080***	0.098***	0.096***	0.112***	0.118***
≥ 0.1	0.062***	0.066***	0.069***	0.075***	0.086***	0.088***	0.101***	0.097***	0.113***	0.119***
≥ 0.15	0.079***	0.077***	0.079***	0.083***	0.092***	0.095***	0.110***	0.105***	0.117***	0.123***
≥ 0.2	0.094***	0.090***	0.085***	0.088***	0.097***	0.108***	0.118***	0.111***	0.122***	0.128***
≥ 0.25	0.111***	0.100***	0.096***	0.096***	0.102***	0.120***	0.126***	0.119***	0.128***	0.134***

Notes. Cells report the observed interaction Δtreasury × (≥5y) in the regression of coupon-yield changes on par-yield changes, with state-clustered standard errors. Rows give minimum absolute treasury moves in percentage points. Columns give maximum days between consecutive trades, as in Table 44. All thirty cells are significant at one percent on both samples. Every wider-sample coefficient is larger, with the largest proportional gains at the lowest cutoffs. Appendix D.11.1 reports the corresponding coupon-benchmark contrasts on the main sample. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

E Institutional detail and data

This appendix documents the tax rules and trading restrictions summarized in Section 3, separating statutory treatment from the assumptions of the quantitative model.

E.1 Investor-by-investor federal tax treatment since 2000

Investor i keeps the absolute after-tax shares $\xi_i^C = 1 - \tau_i^C$ of taxable and $\xi_i^M = 1 - \tau_i^M$ of municipal interest. Pricing depends on the relative after-tax share $\xi_i = \xi_i^C / \xi_i^M$ (Section 4.2). The exemption benefit $1 - \xi_i = (\tau_i^C - \tau_i^M)/(1 - \tau_i^M)$ equals τ_i^C only when municipal interest is fully exempt. Deduction rules for interest that finances bond holdings add a further margin (Appendices E.1 and E.6). Table 50 restates Table 2 for the 2018–2020 sample.

Households. Municipal interest is generally federally exempt and taxable bond interest is taxed as ordinary income, so a household's federal after-tax shares are approximately $(\xi_i^M, \xi_i^C) = (1, 1 - \tau_{oi})$, with τ_{oi} its ordinary-income tax rate. Exempt interest is reported in both high- and lower-income brackets (Figure 1), so the exemption benefit varies within the household clientele.

Table 50: Investor heterogeneity in the municipal market after the 2017 Act

Investor type	Federal treatment	State treatment	Net exemption benefit
High-tax household	Interest exempt	Exempt on in-state bonds only	0.35–0.41
Middle-tax household	Interest exempt	Exempt on in-state bonds only	0.22–0.32
Low-tax household	Interest exempt	Exempt on in-state bonds only	0.10–0.12
P&C insurer	Exempt, less proration	Premium taxes, no distinction	0.17
Life insurer	Exempt, less proration	Premium taxes, no distinction	0.16
Bank	Exempt, funding cost disallowed	Varies by state	0.17–0.20
Other (pensions, etc.)	Varies	Varies	< 0.05

Notes. Household values are federal marginal tax rates in three groups (35% and above, 22% to 32%, below 22%), and the top value includes the 3.8% net investment income tax. The $10,000 cap on the state and local tax deduction makes the home-state exemption more valuable to households whose cap binds. Institutions face the 21% corporate rate. P&C proration rises to 25% of exempt interest, giving $\xi = 0.79/0.9475$, and the life insurer's policyholders' share is set at 30%, giving $\xi = 0.79/0.937$. The bank value is the break-even implied tax of Table 2 at the 21% rate with the 2015–2017 interest-expense cost and taxable yields. The Act narrows the difference between the P&C and life-insurer exemption benefits.

Table 51 gives top federal rates. The 2001 phase-down, accelerated in 2003, cut the top ordinary rate to 35% in 2003. The rate rose to 39.6% in 2013 and fell to 37% in 2018. The top long-term capital-gains rate fell to 15% in 2003 and rose in 2013. Since 2013, the 3.8% net investment income tax (NIIT) has applied to taxable interest and many gains, but not to exempt interest. The calibration computes federal and state rates for each Survey of Consumer Finances (SCF) tax unit under each year's law (Appendix C.7).

Table 51: Top federal tax rates relevant to municipal pricing, 2000–2020

Period	Top ordinary rate	+ NIIT (on taxable interest)	Top corporate rate
2000	39.6%	39.6%	35%
2001	39.1%	39.1%	35%
2002	38.6%	38.6%	35%
2003–2012	35.0%	35.0%	35%
2013–2017	39.6%	43.4%	35%
2018–2020	37.0%	40.8%	21%

Banks. With c and τ_B as in Section 6.2, interest expense that finances non-bank-qualified exempt bonds is generally nondeductible, so it costs c after tax, compared with $c(1-\tau_B)$ if deductible, a penalty of $\tau_B c$. Section 265(b)(3) excepts bank-qualified bonds, but Section 291 cuts their deduction by 20%, leaving 80% and an after-tax cost of $c(1-0.8\tau_B)$. Interest expense is allocated to exempt bonds in proportion to adjusted tax basis.[30]

The 15% disallowance under the Tax Equity and Fiscal Responsibility Act became 20% in 1984. The 1986 Act generally denied financial institutions' interest deductions allocable to exempt obligations acquired after August 7, 1986. The ordinary bank-qualified exception covers designated small-issuer obligations under the $10 million annual limit and other conditions, and temporary 2009–2010 rules expanded eligibility. In the model, each dollar of bank municipal holdings bears a financing-tax cost of τ_B times the measured ratio of interest expense to assets times $(1-0.8q_{BQ})$, where q_{BQ} is the bank-qualified share of those holdings. The aggregate calibration sets $q_{BQ} = 0$. Appendix D.8.1 derives how much of the deduction on designated bonds is capitalized in yields.

[30] See the IRS, Bank Qualified Bonds: Section 265, pp. 13-9 and 13-11. Special rules apply to certain 2009–2010 issues and grandfathered obligations.

Property and casualty insurers. P&C insurers pay the corporate rate τ_c on taxable bond interest. Proration cuts their losses-incurred deduction by a percentage of exempt interest, so that percentage of municipal interest is in effect taxed at τ_c. Before the 2017 Tax Cuts and Jobs Act, the percentage was 15%, giving an effective municipal rate of $0.15 \times 0.35 = 0.0525$. The Act replaced it with 5.25% divided by the highest corporate rate, or 25% at $\tau_c = 21\%$. The municipal after-tax share thus stays at 0.9475 while the taxable share rises from 0.65 to 0.79. The relative after-tax share rises from 0.686 to 0.834 and the exemption benefit falls from 31.4% to 16.6%. Cutting the corporate rate while keeping 15% proration would not reproduce the 2018 law.

The calibration applies the top corporate rate to tax-paying insurers. Before 2018, marginal corporate rates were 15–34% below \$100,000 and 34–39% from \$100,000 to \$18,333,333, including bubble rates that recapture the lower brackets' benefit, and a flat 35% above. Since 2018 the rate is 21% at every taxable-income level.

Life insurers and other institutions. Section 812 divides a life insurer's net investment income into company and policyholder shares, and exempt interest reduces its reserve deductions. In the baseline, a one-half policyholders' share taxed at 35% gives an effective municipal rate of 17.5%, absolute after-tax shares of 0.825 on municipal and 0.65 on taxable interest and a relative after-tax share of 0.788. The post-2017 calibration's 30% policyholders' share and 21% corporate rate give an effective municipal rate of 6.3%, after-tax shares of 0.937 on municipal and 0.79 on taxable interest and a relative after-tax share of 0.843. Tax-exempt institutions and tax-deferred accounts gain little from the exemption.

E.2 Relative after-tax shares, absolute tax liabilities and fiscal scope

Pricing needs only relative after-tax shares, but the tax receipts of Section 7 require absolute tax rates. The pre-2018 P&C relative after-tax share of 0.6860 reflects a 5.25% tax on municipal interest and a 35% tax on taxable interest. Treating municipal interest as untaxed and taxing taxable interest at $1 - 0.6860$, or 31.4%, would give the same relative share but misstate both receipts. When a reform changes absolute after-tax shares, risk preferences and liability hedges stay in each investor's baseline payoff units (Section 4.2), preserving its absolute objective. Household tax assignments retain the dated federal tax responses, the state-tax approximations and the Social Security taxation proxy of Appendix C.7. State receipts are assigned by state of residence. Before 2018, a group's federal tax per dollar of taxable interest is one minus its combined federal-state after-tax share, less its state tax rate, a residual consistent with the multiplicative after-tax share that needs no new tax-unit calculation.

The fiscal ledger covers bond and taxable cash interest. Each investor's cash balance is its funding less its bond holdings, and cash uses the same absolute tax rates as taxable bond interest. Fixed aggregate funding, debt supplies and the short rate imply unchanged aggregate cash interest. Changes in cash tax receipts therefore offset changes in investors' after-tax cash income, preserving identity (29). Changes in the bank financing-tax cost are separate federal receipts, deducted from banks' income without double-counting interest expense or issuers' payments. Other liability flows remain outside the ledger. Financing costs use $y_T Q_T$ at $z = 0$, summed over maturities, without resetting existing coupons, OID schedules or redemption payments. Nominal semiannual yields are a sensitivity check. Equal-treatment averages give each calendar year the same weight and divide mean dollar costs by mean receipts. Table 53 reports the individual periods.

Portfolio substitution and interest receipts. After home-state repeal, both home- and out-of-state municipal interest are state-taxable, so switching issuers does not avoid tax. Residents' cash share rises by 2.42–2.78 percentage points across the seven states. The smaller rise in their total riskless share in Table 52 reflects sales of riskless municipal bonds. In California, applying reform yields to baseline holdings raises annual state receipts by \$135.5 million relative to unchanged prices and holdings. Reallocation then reduces bond-interest receipts by \$200.7 million, comprising \$177.6 million from a smaller bond portfolio and \$23.1 million from its composition. The quantity term rescales each type's initial bond basket to its new total holdings, and the composition term is the remainder. Added cash-interest receipts of \$244.8 million more than offset this bond-income reduction. Total added state receipts are \$1.432 billion, compared with \$1.253 billion at unchanged prices and holdings.

Table 52: Portfolio allocation under the two tax reforms

Panel A. Home-state exemption repeal: resident households			
	Home-state municipal	All risky	Riskless
California	16.47 → 6.02	36.01 → 35.45	63.99 → 64.55
New York	16.42 → 5.02	35.06 → 34.69	64.94 → 65.31
New Jersey	15.49 → 2.05	34.95 → 34.69	65.05 → 65.31
Massachusetts	15.60 → 1.81	34.37 → 34.27	65.63 → 65.73
Ohio	14.84 → 1.70	34.23 → 34.13	65.77 → 65.87
Pennsylvania	14.52 → 2.24	33.80 → 33.77	66.20 → 66.23
Minnesota	17.52 → 0.97	35.44 → 35.19	64.56 → 64.81
Panel B. Equal treatment: average investor allocations, 2000–2017			
	Risky municipal	Risky taxable	Riskless
High-tax households	24.69 → 4.44	0.00 → 18.60	75.31 → 76.96
Lower-tax households	10.18 → 6.64	23.75 → 28.42	66.07 → 64.95
Banks	33.45 → 22.01	66.55 → 77.99	0.00 → 0.00
P&C insurers	40.20 → 16.16	38.11 → 83.19	21.70 → 0.65
Life insurers	2.76 → 23.45	96.86 → 75.87	0.38 → 0.68
Tax-neutral investors	0.00 → 7.66	72.10 → 62.78	27.90 → 29.56

Notes: Entries are baseline → reform percentages of each population's modeled financial assets, including cash and excluding outside assets and liabilities. Panel A aggregates resident households in each state of the 2015–2017 economy. Each row is a separate repeal. Panel B averages each investor group's portfolio shares across the six 2000–2017 economies under 99.9% equal treatment. Both panels use investor masses and $z = 0$. Riskless holdings comprise cash and one-period municipal bonds (two years on the numerical grid). All longer bonds are risky. In Panel A, home-state municipal holdings overlap the other columns, and the last two columns sum to 100%. In Panel B, the three columns sum to 100%, subject to rounding. These portfolio shares differ from shares of market supply.

Table 53: Equal-treatment reform across the six pre-2018 economies

	Thirty-year yield increase (bp)		Costs per receipt dollar	
Period	Municipal	Taxable	All issuers	Taxable issuers
2000–2002	199.8	38.2	1.434	0.282
2003–2005	136.9	9.3	1.125	0.089
2006–2008	146.5	11.4	1.077	0.073
2009–2011	152.7	11.2	1.113	0.043
2012–2014	178.9	33.9	1.245	0.170
2015–2017	208.6	34.7	1.242	0.174
Average 2000–2017	170.6	23.1	1.196	0.133

Notes. Each row removes 99.9% of the municipal–taxable tax distinction from its own calibrated baseline. Yield changes average monthly rate states. Fiscal ratios use continuous yields and holdings at $z = 0$, with bond and taxable cash interest in receipts. The average row divides calendar-year-weighted mean costs by mean receipts.

E.3 State tax exemption and who receives it

Resident households. For a household resident in state s, with state tax rate τ_s^{st}, the baseline normalizes the home-state municipal after-tax share to one and sets the out-of-state municipal share to $1 - \tau_s^{\text{st}}$, with no additional federal deduction credit (varied in Appendix E.4). Through 2017, the after-tax share on taxable interest multiplies the federal relative after-tax share by $1 - \tau_s^{\text{st}}$, tax unit by tax unit. The post-2017 mapping treats the cap on the state and local tax deduction as binding, so the combined share on taxable interest is $\xi^{\text{fed}} - \tau_s^{\text{st}}$, as in equation (141).

Funds and institutions. Municipal funds generally pass the federal exemption through to shareholders as exempt-interest dividends. State treatment depends on the shareholder's residence and the share of the

fund's interest from that state's bonds: a single-state fund preserves the home-state exemption for resident shareholders, while a national fund passes it through only on that share, so funds can attenuate the household state incentive. Among institutions, tax-deferred accounts and many pensions owe no current state tax on interest. P&C insurers commonly pay state premium taxes, which do not distinguish municipal from taxable interest, and the state taxes of banks and other corporations depend on the tax base and business apportionment. None of these is the exemption that depends on both the issuer's and the holder's state, which P1 exploits (Section 6.1), so the model assigns that wedge only to households and, for insurers and banks, emphasizes federal proration and financing deductions.

E.4 Federal deduction credit for out-of-state municipal interest

Before 2018, itemizers could deduct state income taxes on federally exempt interest against ordinary federal income tax. The deduction's full benefit is not available to every household: taking the standard deduction, paying the alternative minimum tax or choosing a larger sales-tax deduction can eliminate the incremental saving.[31]

To test sensitivity, I grant full credit, without reducing it for deduction eligibility, at the ordinary-income component of the baseline federal tax response, excluding NIIT, forming products tax unit by tax unit before aggregation. I re-solve the six pre-2018 economies, holding all other inputs and calibrated coefficients fixed. The post-2017 binding-cap convention is unchanged.

For the 2015–2017 California high-tax group, the effective after-tax shares with the credit are 0.498 for taxable interest and 0.919 for out-of-state municipal interest. At a riskless taxable yield of 5%, a municipal yield of 2.71% equates the two after-tax returns, compared with 2.87% for the high-tax group in states without an income tax. The taxable alternative already incorporates the baseline deduction treatment, and the added municipal credit lowers the yield needed to match it. Residents of high-tax states can consequently accept lower out-of-state yields than local residents, compressing cross-state yield differences, and so state-tax pass-through, even at the riskless maturity.

Using the main sample's 2000–2020 weights and corporate normalization, the thirty-year model gap falls only 0.23 basis points, from 97.39 to 97.16. The model's P1 pass-through, averaged over the six blocks with its original taxable-yield normalization, falls from 1.001 to 0.904 at the two-year riskless maturity and from 0.458 to 0.412 at twenty-five years (interpolated on the two-year grid). The credit lowers the level of pass-through but preserves its declining profile and the large long-maturity gap.

E.5 Accrual, price changes and secondary-market taxation

Equation (32) applies the investor's interest-income after-tax share to the whole model return $\mathcal{R} = \mu + u$, approximating the federal rules below for accrual, basis and gains.[32]

Gains and basis. For individuals, federal law separates interest accrual from gains realized at disposition (Sections 171, 1272–1278 and 1288). For a bond with face value 100, annual coupon payment c in dollars, original price A_0 and annual effective issue yield y_0, the adjusted issue price satisfies

$$A_t = (1 + y_0)A_{t-1} - c, \qquad t = 1, \ldots, n, \qquad A_n = 100. \tag{154}$$

[31] IRS Publication 550 (2017), "State income taxes"; 2017 Instructions for Form 6251, line 3; and 2017 Instructions for Schedule A, line 5. 2016 Instructions for Form 8960, line 9b, restrict NIIT deductions to taxes attributable to net investment income, which excludes exempt interest.

[32] The after-tax share multiplies both the log of the expected gross return, $\log \mathbb{E}[1+R]$, and the centered log-return innovation. Taxing a simple return R gives the after-tax gross return $1+\xi R$, with $\log \log(1+\xi R)$. For a deterministic two-year simple return of 5.4 percent and $\xi = 0.62$, this log differs from $\xi \log(1+R)$ by 3.2 basis points per two-year period (1.6 a year) and zero for a fully exempt leg. Appendix B.1 reports the effect of taxing simple returns exactly in the solved model. The return convention matches the empirical implied tax and the after-tax yield mappings of Appendix D.1.4.

Here A_t is a dollar price, unrelated to the rate loading A_T^a. A discount, $A_0 < 100$, is OID, and a premium, $A_0 > 100$, is original-issue premium (OIP). Annual interest $y_0 A_{t-1} = c + (A_t - A_{t-1})$, including the noncash accrual, is taxed at the ordinary-income rate τ_{oi}:

$$\text{tax}_t = \begin{cases} \tau_{oi} y_0 A_{t-1}, & \text{taxable bond,} \\ 0, & \text{tax-exempt bond.} \end{cases} \tag{155}$$

The basis change $A_t - A_{t-1}$ prevents taxing accrued interest again at redemption. OID raises basis, including for zero-coupon bonds, and OIP lowers it. Amortizing premium on a taxable bond requires a Section 171 election, which the calculation assumes, and amortization on an exempt bond gives no deduction. At maturity, basis equals face, so there is no capital gain. For taxable bonds, de minimis OID below a quarter point per full year can be treated as zero, with the discount recognized as capital gain at maturity. This OID exception generally does not apply to tax-exempt obligations.[33]

A sale at price P_h at date h realizes the gain $G_h = P_h - \text{basis}_h$, long-term after one year and then taxed at the capital-gains rate τ_{cg}. A secondary buyer's basis starts at the purchase price P. If P exceeds face, the premium amortizes at the buyer's yield, mandatorily for exempt bonds and by election for taxable bonds. On an OID bond bought at date s for $A_s < P \leq 100$, the acquisition premium $P - A_s$ reduces the remaining OID by the fraction $(P - A_s)/(100 - A_s)$, leaving basis 100 at maturity.

Market discount is the shortfall of the purchase price below A_s for OID bonds or below face for par and premium bonds. It accrues ratably, or at constant yield by election. Absent annual inclusion, accrued market discount D_h is taxed as ordinary income at sale or redemption, up to any positive gain. This rule also covers exempt bonds bought after April 30, 1993. Below the de minimis threshold (124), the discount is treated as zero and the gain is capital income. Thus

$$\text{tax at sale} = \tau_{oi} \min\{\max(G_h, 0), D_h\} + \tau_{cg}[G_h - \min\{\max(G_h, 0), D_h\}], \qquad G_h = P_h - \text{basis}_h. \tag{156}$$

Individual capital losses offset gains and up to $3,000 of ordinary income a year, with carryforwards (Sections 1211–1212). The capital-gains adjustment assumes full offsets at the capital-gains rate, as in (125). Corporations tax gains at their ordinary rate but offset capital losses only against gains. Banks treat gains and losses on debt as ordinary income (Section 582(c)). The model's institutional after-tax shares apply one rate to the full return. Appendix D.1.4 applies the household accrual and basis mappings above, with its stated federal tax rates and loss offsets.

E.6 Institutional foundations of the no-short-sale constraint

The model rules out short sales and borrowing (Sections 3.5 and 4.1). Tax asymmetries, financing rules and residual risk limit short arbitrage, and the nine cases below show how they sustain the different post-hedge duration valuations required by (166).

Household tax rules matter because high-income households hold much of the market. In custodial data, municipal bonds are wealthy households' second-largest liquid risky asset, and their portfolio share rises with wealth (Gabaix et al., 2026). In tax years 2008, 2016 and 2019, returns with adjusted gross income (AGI) of $1 million or more, about 0.3 percent of all returns, reported 26–33 percent of tax-exempt interest.[34]

Define the after-tax coupon gap $g = c_M - (1 - \tau)c_C$, where c_M and c_C are municipal and corporate par coupons of the same maturity and τ is the investor's marginal tax rate. This coupon measure differs from the 117-basis-point thirty-year puzzle in zero-coupon yields (Section 5.4) and from the model's one-period return gap g_T. Carry is computed for equal principal in each leg, before fees, margin and collateral costs, and the legs' duration and price risk need not match.

(i) Financing municipals with debt or a corporate short. Let c_F be the interest on a dollar borrowed to buy a dollar of municipal principal. If that interest were deductible, carry would be $c_M - (1-\tau)c_F$.

[33] IRS Publication 550 (2017), "Original Issue Discount (OID)"; Publication 1212, "De minimis rule." The market-discount threshold in (124) applies separately.

[34] IRS Statistics of Income, Individual Income Tax Returns, Table 1.4, all returns by size of adjusted gross income, tax years 2008, 2016 and 2019. Tax-exempt interest includes exempt-interest dividends from funds.

Section 265(a)(2) denies deductions for interest on debt incurred or continued to purchase or carry tax-exempt obligations, so carry becomes

$$c_M - c_F = \{c_M - (1-\tau)c_F\} - \tau c_F. \tag{157}$$

At after-tax parity, $c_M = (1-\tau)c_F$, the denial turns zero carry into a loss of τc_F. Financing the position with a corporate short does not restore the tax shield, because Section 265(a)(5) treats the short seller's compensatory coupon payments c_C as interest under Section 265(a)(2), leaving carry $c_M - c_C$. This carry is negative at every maturity because the sample-average municipal zero curve, and hence the implied par yield, lies below its corporate counterpart. At thirty years with annual compounding, the municipal and corporate par yields are 4.498% and 5.443%.[35] Municipal fund disclosures tell investors that interest on debt allocable to exempt-interest dividends is nondeductible.[36] While the deduction was available, depository institutions financed municipal holdings at scale, holding 51.8% of municipal bonds in 1972 and $235.1 billion (27.3%) in 1985. After the 1986 denial (Appendix E.1), their holdings fell to $120.4 billion in 1990 and $95.4 billion (7.5%) in 1995, while municipal debt outstanding grew from $862 billion in 1985 to $1.27 trillion in 1995.[37]

(ii) Pairing corporate positions. If the proceeds of a corporate short buy a corporate bond, the investor's own capital finances the municipal bond, and the remaining trade is the hedge in (iii). If they finance the municipal bond, the denial in (i) applies. Revenue Procedure 72-18 tests the purpose of borrowing through tracing of the proceeds, exempt collateral or the facts and circumstances. Its presumption that business borrowing is not incurred to carry exempt bonds lapses when the borrowing exceeds business needs. If the positions form an offsetting straddle, Section 263(g) also capitalizes interest and carrying charges, net of the positions' taxable income, deferring the deductions.[38]

(iii) Hedging duration with a corporate short. An unlevered hedge of a municipal bond's duration shorts a long-maturity corporate bond and buys a short-maturity corporate bond with the proceeds. Let c_ℓ and c_s be the long and short corporate coupons, so that $g = c_M - (1-\tau)c_\ell$, and let $\delta \in [0,1]$ be the currently deductible fraction of the short payment. With equal principal in each leg, annual carry relative to holding the short corporate bond alone is

$$\Delta = \{c_M + (1-\tau)c_s - (1-\delta\tau)c_\ell\} - (1-\tau)c_s = g - (1-\delta)\tau c_\ell. \tag{158}$$

Denial under Section 265 gives $\delta = 0$ and $\Delta = c_M - c_\ell$, negative at every maturity and a loss of 95 basis points a year at thirty years. Section 263(g) capitalization and Section 163(d) carryforwards defer deductions, so current carry approaches $c_M - c_\ell$ and later benefits carry a timing and liquidity cost.

Even where neither Section 265 nor Section 263(g) applies, Section 163(d) lets individuals deduct investment interest only if they itemize and up to net investment income, carrying the excess forward. Coupons on the short-maturity corporate bond add to this deduction capacity, while wages and exempt coupons do not. With principal q on both corporate legs, coupon difference $k = c_\ell - c_s > 0$, unused deduction capacity $C \geq 0$ and no prior carryforwards, the hedge's current cash flow is

$$\Pi(q) = -(1-\tau)qk - \tau[qk - C]_+, \tag{159}$$

[35] 26 U.S.C. §265(a)(2), (5) and (b), 2018 U.S. Code. Section 265(a)(5)(B) excepts specified short-sale payments when the taxpayer posts cash collateral that earns no material return. That exception does not remove the opportunity cost of the collateral. See also Hines and Horton (2024) on banks' financing-tax restrictions.

[36] BlackRock Municipal Bond Fund, Inc. (2025), Part II, "Dividends and Taxes," p. II-131; iShares Trust (2026), "Taxes," p. 210; Nuveen Municipal Trust (2026), "Tax Matters," p. S-80.

[37] Board of Governors of the Federal Reserve System, Financial Accounts of the United States (Z.1), release of September 8, 2023, Table L.212, year-end levels. Depository institutions comprise U.S.-chartered depository institutions, foreign banking offices in the U.S. and banks in U.S.-affiliated areas.

[38] Rev. Proc. 72-18, 1972-1 C.B. 740, §§3.01–3.04 and 6.01, as summarized in Revenue Ruling 2004-47 and in IRS Field Service Advice 200129009, which is nonprecedential. On straddles, see 26 U.S.C. §263(g)(1)–(2) and §1092(c)(1)–(2). Positions are offsetting when holding one substantially diminishes the risk of loss from holding the other.

where $[x]_+ = \max\{x, 0\}$, so $\delta < 1$ once qk exceeds C.[39]

Even with $\delta = 1$, the hedge removes only common rate risk. Bond-specific risk (Appendix A.1) and credit, liquidity and call differences (Appendix D) remain. Shorting requires available bonds and incurs fees, recall risk and margin demands against an illiquid callable long position.[40] Fees for borrowing corporate bonds are 10–20 bp in the major-lender sample of Asquith et al. (2013).

(iv) Hedging with swaps. A pay-fixed, receive-floating swap hedges duration without borrowing or shorting a bond, but household access is restricted. A person that is not an eligible contract participant may trade a swap only on, or subject to the rules of, a designated contract market. An individual qualifies only with more than $10 million invested on a discretionary basis, or more than $5 million when the swap manages the risk of an asset owned or a liability incurred.[41] Ordinary swap payments are not automatically interest on indebtedness, so Section 265(a)(2) alone does not deny their deduction. A household's Section 212 investment expenses attributable to exempt interest fall under Section 265(a)(1).[42] Independently of that classification, let X be a net taxable amount and $d \in [0, 1]$ the denied fraction of its loss tax shield, so that its after-tax value is

$$A_d(X) = (1 - \tau)X - d\tau X^-, \qquad X^- := \max\{-X, 0\}. \tag{160}$$

Symmetric offsets give $d = 0$. Taxing gains and denying losses gives $d = 1$, which lowers expected proceeds by $\tau\mathbb{E}[X^-]$.

Netting gives the swap a tax advantage when the corporate short payment is disallowed. With matched periodic receipts $F \geq 0$ and payments $K \geq 0$, the bond trade's after-tax cash is $A_C = (1 - \tau)F - K$ when the corporate short payment is wholly nondeductible, and the swap's is $A_S = F - K - \tau(F - K)^+$, with $(F - K)^+$ the positive part of $F - K$, under the rule that taxes net gains and denies losses. Hence

$$A_S - A_C = \tau \min\{F, K\} \geq 0, \tag{161}$$

and the two coincide when the corporate payment is deductible against F. Additional deduction capacity favors the corporate short. A pay-fixed swap has the rate exposure of a short fixed-rate bond and a long floating-rate bond, without the corporate short's credit and borrowing exposures but with its own collateral and counterparty exposure. It weakly dominates a corporate hedge in the mean–variance objective whenever the latter delivers the same exposure with only an additional nonnegative cost and uncompensated risk orthogonal to the existing portfolio and the swap.[43] An outside dealer can also absorb swap risk without trading domestic bonds, whereas physical shorts change the gross long positions required to clear those markets.

Because the wealthiest households can meet the eligibility thresholds, the equilibrium test of Appendix E.6.1 grants every household a thirty-year pay-fixed swap from an outside dealer with no fee, margin or notional limit and taxes swap gains while denying losses ($d = 1$). Physical bond shorts remain excluded, institutions' hedging is unchanged and both bond markets are re-solved. The model's thirty-year gap for 2015–2017 falls only from 104.3 to 94.8 basis points, retaining 90.9% of its baseline value.

[39] 26 U.S.C. §163(d)(1)–(3), which includes otherwise deductible short-sale expenses in investment interest; IRS Publication 550 (2016), Chapter 3, "Interest Expenses" and "Interest and carrying charges on straddles"; IRS Form 4952 and instructions (2016). Net capital gain and qualified dividends count as investment income only through an election that relinquishes their preferential rates to that extent.

[40] FINRA Rule 4210, paragraph (e)(2), sets margin requirements for bond shorts.

[41] Commodity Exchange Act §2(e) and §1a(18)(A)(xi), 7 U.S.C. §§2(e), 1a(18)(A)(xi); see also the CFTC's 2012 final-rule summary.

[42] Treasury Regulations §1.446-3 and §1.265-1(a)(2), (c). IRS Field Service Advice 200037034 treats the ordinary swap payments before it as non-interest in the absence of a loan. The advice concerns a corporate taxpayer and is nonprecedential.

[43] In common payoff units, let the corporate hedge be $B = \alpha S - \varsigma + \varepsilon$, where S is the swap payoff, $\alpha > 0$, $\varsigma \geq 0$, $\mathbb{E}\varepsilon = 0$, and $\mathrm{Cov}(\varepsilon, W) = \mathrm{Cov}(\varepsilon, S) = 0$ for the investor's existing risk index W. With $\mathrm{CE}(Y) = \mathbb{E}Y - \gamma\,\mathrm{Var}(Y)/2$, one has $\mathrm{CE}(W + qB) - \mathrm{CE}(W + q\alpha S) = -q\varsigma - \gamma q^2\,\mathrm{Var}(\varepsilon)/2 \leq 0$ for $q \geq 0$, provided the replacement is feasible at no greater cost of financing the position.

(v) Treasury futures. Regulated futures avoid the swap eligibility rules but require cash margin and annual marking to market, with gains and losses treated as 60% long-term and 40% short-term. Individuals offset these losses fully against capital gains, deduct at most $3,000 against ordinary income and carry the excess forward. Net losses on these Section 1256 contracts may be carried back three years against net Section 1256 gains. Section 1092 defers losses to the extent of unrecognized gains on offsetting bonds. Gains can be taxed before losses yield tax benefits, a timing and liquidity cost equivalent to partial loss denial in present value, $0 \le d < 1$ in (160).[44]

(vi) Shorting municipals. The substitute payment a short seller makes for the exempt coupon loses its exemption, so a lender with tax rate τ_L who requires the after-tax amount of the coupon c_M must receive L satisfying

$$(1-\tau_L)L = c_M, \qquad L - c_M = \frac{\tau_L}{1-\tau_L} c_M. \tag{162}$$

A short also requires a willing lender of the specific issue and satisfaction of the contract's collateral terms.[45] For the reverse trade, which shorts a municipal bond and buys an equal-principal corporate bond, let τ_B be the short seller's tax rate and δ the currently deductible fraction of the substitute payment, as in (iii). Its coupon carry under this lender compensation is

$$\pi_{M\to C} = (1-\tau_B)c_C - (1-\delta\tau_B)\frac{c_M}{1-\tau_L}. \tag{163}$$

Even with full deduction, $\delta = 1$, borrower and lender rates equal to τ give $\pi_{M\to C} = (1-\tau)c_C - c_M$. This carry is nonpositive when $c_M \ge (1-\tau)c_C$, and the carry $c_M - c_C$ of (i) when $c_M \le c_C$, so both coupon trades have nonpositive carry throughout the band

$$(1-\tau)c_C \le c_M \le c_C, \tag{164}$$

and long-end municipal coupons lie in it: above the household after-tax benchmark and, on the sample-average curves, below the corporate coupon. A municipal curve trade, short long-maturity and long short-maturity municipal bonds, pays the gross-up in (162) on its short leg. To the extent its proceeds buy the short-maturity municipal bond, that payment is nondeductible as in (i).

(vii) Tax-exempt floaters. A tender-option-bond partnership divides a long municipal bond into variable-rate equity and an inverse interest that receives the residual. Exempt income passes through to both classes, so the structure needs no interest deduction. The inverse holder receives the municipal curve's slope and bears its duration, so capturing g still requires one of the hedges in (iii)–(v). Demand for the floaters limits the scale of the strategy.[46]

(viii) The de minimis presumption. For insubstantial holdings, Revenue Procedure 72-18 ordinarily presumes, absent tracing or exempt collateral, that borrowing is not incurred to carry exempt bonds. A corporation qualifies only up to 2% of average business assets. Direct evidence overrides the presumption, and larger holdings face the facts-and-circumstances test. Sections 163(d), 263(g), 1092 and 1211(b) still constrain deductions and losses.[47]

(ix) Other entities. Financial institutions outside the bank-qualified exception face the pro rata disallowance of Appendix E.1, without tracing. Revenue Procedure 72-18 applies a similar ratio to dealers' business borrowing that cannot be traced. Insurer proration reduces the exemption regardless of financing, through P&C insurers' loss deductions (Section 832(b)(5)(B)) and life insurers' reserve deductions (Sections 807 and 812). Partnerships pass the exempt character of municipal interest through to partners under

[44] 26 U.S.C. §1256(a)(1), (a)(3), (b) and (g)(1); §1211(b); §1212(b)–(c); §1092(a)(1)(A). Section 1256(a)(4) removes Sections 1092 and 263(g) only for straddles consisting entirely of Section 1256 contracts.

[45] IRS Publication 550 (2016), "Reporting Substitute Payments"; Treasury Regulation §1.6045-2(a)(4).

[46] Revenue Procedure 2003-84, Section 2; 26 U.S.C. §702(b).

[47] Rev. Proc. 72-18, §3.05, as summarized in Revenue Ruling 2004-47 and Field Service Advice 200129009.

Section 702(b), but their interest expense remains subject to Section 265(a)(2) under Section 703(a). Over 2000–2020, dealers held on average 0.85% of municipal bonds, with a maximum of 1.94%, and hedge funds, reported from 2012, held at most 0.36%.[48]

The model constraint. With a concave objective and linear shorting costs, a zero short position is optimal when the risk-adjusted benefit of the first dollar shorted does not exceed the tax, borrowing and financing cost. The quantitative model imposes this corner, with nonnegative bond positions financed from funding W_i, and charges banks the measured financing-tax cost, which excludes unconstrained after-tax arbitrage as a benchmark. The measured costs do not establish the corner for every investor, and some hedges remain profitable.

Residual risk and participation after hedging. The mechanism requires only that valuations of additional duration differ after available hedges, which allows continued derivative trading. To isolate the rate-hedging tradeoff, suppose bond funding is nonbinding and set bond-specific risk to zero. Let a hedge remove q_i units of rate exposure at an expected charge $\kappa_i q_i$ and a differentiable convex additional cost $\psi_i(q_i)$, all in the investor's normalized payoff units. With residual exposure $\widehat{D}_i^{\text{hedge}} = D_i - \ell_i - q_i$, where $D_i - \ell_i$ is the net rate exposure (10) before hedging, the hedge-dependent part of the objective is $-\kappa_i q_i - \psi_i(q_i) - \gamma_i \sigma_r^2 (\widehat{D}_i^{\text{hedge}})^2/2$, so an interior hedge choice satisfies

$$\lambda_i := \gamma_i \sigma_r^2 \widehat{D}_i^{\text{hedge}} = \kappa_i + \psi_i'(q_i). \tag{165}$$

Here λ_i is the unit cost of rate risk at the post-hedge exposure. A binding hedge limit replaces the equality by the appropriate one-sided condition. At a household-only municipal allocation, with hedges and other holdings optimized, the lower-tax investor's municipal slack, its required return on the T-period municipal bond minus the offered return, is

$$\nu_{I,T}^{M,\text{hh}} = (\xi_I - \xi_H) r - A_{T-1}^{M} (\lambda_H^{\text{hh}} - \lambda_I^{\text{hh}}). \tag{166}$$

This is the participation comparison (44) applied to residual, post-hedge risk costs. Differences in liabilities sustain unequal duration valuations through unequal hedge access, binding limits or hedging costs that rise with the hedge, even when both investors hedge.

If unlimited hedging in either direction spans rate risk, $\psi_i = 0$ and both investors face a common normalized charge κ, then $\lambda_H = \lambda_I = \kappa$. With equal unit costs of rate risk, the slack reduces to the positive tax difference $(\xi_I - \xi_H) r$, which excludes lower-tax municipal holdings. This conclusion requires a common after-tax hedging opportunity; zero dealer fees alone do not suffice. Observed insurer holdings (Figure 25) therefore require a departure from this benchmark, which any of the documented frictions can sustain. The calibrated outside exposures are not measured directly net of derivative positions.

Insurers already hedge: at year-end 2015, interest-rate swaps accounted for 64% of the notional value of U.S. insurers' interest-rate hedges, and derivative users held 87% of life-insurer assets but 19% of P&C assets.[49]

E.6.1 Household swaps: incentives and equilibrium

The model gives households no rate hedge, and off-exchange swaps face eligibility restrictions (Section 3.5). Using the pay-fixed swap of Appendix E.6, I first evaluate the incentive to add a swap at fixed baseline bond portfolios, the 2015–2017 mean rate state and all 630 household types (14 types in each of 45 markets). I then let households choose swaps and bonds jointly and re-solve both bond markets.

[48] Rev. Proc. 72-18, §§5.03 and 7, as summarized in Revenue Ruling 2004-47; 26 U.S.C. §265(b)(1)–(2), §832(b)(5)(B)–(C), §807(a)–(b) and §812. Holdings: Financial Accounts of the United States (Z.1), release of September 8, 2023, Tables L.212 and B.101.f, quarterly levels.

[49] NAIC, "Update on the Insurance Industry's Use of Derivatives and Exposure Trends," year-end 2015, Tables 1 and 8 and the accompanying discussion. Notional shares measure instrument use and are silent on how much liability risk is eliminated.

Quotes, pricing and taxes. Swap quotes come from the 95 dates with a two-year rate within 10 bp of the model's 3.0206%, all in 2001–2008, outside the comparison block. Fixed rates at ten, twenty and thirty years are 4.8217%, 5.0976% and 5.3736%, and the twenty-year rate is interpolated.[50]

Floating payments are quarterly and fixed payments semiannual, and liquidation after two years uses an approximate calendar. One discount and forward curve with Gaussian pricing drifts fits quotes and liquidation values. Quarterly physical dynamics preserve the calibrated two-year persistence and variance. Annual net gains and the second-year liquidation payoff are taxed as ordinary income. Losses are denied in the baseline and offset symmetrically in a sensitivity check. Municipal funds likewise distribute swap gains as taxable ordinary income or capital gains and allocate net payments pro rata against taxable and exempt income.[51] Segregated swap cash earns taxable interest when positive and pays nondeductible interest when negative. Existing cash is not reoptimized.

Objective and tax units. Let $\widetilde{W}_i$ be the calibrated log-return risk index, including outside exposure $-h_i z'$, and M_i^0 its fixed baseline mean, where $h_i = -\ell_i$ is the outside duration of Appendix C.7 (positive values add rate risk). The absolute home-municipal after-tax share $\xi^M_{i,\mathrm{abs}}$ is ξ_i^M of Appendix E.1 on home-state bonds; multiplied by the relative after-tax share ξ_i, it gives the swap's absolute after-tax share. High- and lower-tax home-municipal after-tax shares are 0.999905 and 0.990429, and swap tax rates span 21.49%–50.25%. After-tax cash A_i per unit notional (unrelated to the rate loading A_T^a) changes normalized wealth by $qA_i/\xi^M_{i,\mathrm{abs}}$. The objective and the score b_i, its annualized slope at zero notional in basis points, are

$$\mathcal{W}_i(q) = \xi^M_{i,\mathrm{abs}} \left[M_i^0 + \frac{q\mathbb{E}A_i}{\xi^M_{i,\mathrm{abs}}} - \frac{\gamma_i}{2} \operatorname{Var}\left(\widetilde{W}_i + \frac{qA_i}{\xi^M_{i,\mathrm{abs}}} \right) \right], \tag{167}$$

$$b_i = \frac{10^4}{2} \mathcal{W}_i'(0+) = \frac{10^4}{2} \left[\mathbb{E}A_i - \gamma_i \operatorname{Cov}(\widetilde{W}_i, A_i) \right]. \tag{168}$$

The score uses the model's log-return approximation. Bond-specific shocks remain in baseline risk but, by independence, have zero covariance with a rate-only swap.

Table 54: Fixed-portfolio swap benefits with gains taxed and losses denied.

Household group	Years	Cash	Risk benefit	Total	Median	Positive
Higher tax	10	−408.6	852.8	444.3	−14.4	135/315
Higher tax	20	−473.5	996.0	522.6	−13.1	135/315
Higher tax	30	−473.9	988.2	514.2	−17.2	135/315
Lower tax	10	−387.9	675.6	287.7	−28.8	135/315
Lower tax	20	−450.0	787.6	337.7	−31.5	135/315
Lower tax	30	−449.5	783.3	333.9	−33.3	135/315

Notes. Annualized scores b_i of equation (168), in basis points per unit notional. Cash and risk benefit are its two terms; total is their sum. These columns use calibrated household-mass weights within each group; identical budgets make resource weights equivalent. Medians and positive counts weight types equally, and entries are rounded separately. "Higher tax" denotes the high-tax group.

Results. At every tenor (Table 54), 360 types have negative and 270 positive scores. The positive types, the top three outside-exposure bins of both groups, hold 42.9% of household resources and 9.8% of household municipal holdings. Individual scores reach 3,100 bp, but a finite trade also bears the quadratic term $-\gamma_i q^2 \operatorname{Var}(A_i)/(2\xi^M_{i,\mathrm{abs}})$. Dispersion across types gives negative medians but positive weighted means.

Relative to symmetric offsets, loss denial lowers expected swap cash for every type and, through the covariance term, lowers the score for 450 types and raises it for 180; on net it lowers 585 of the 630 scores.

[50] Federal Reserve H.15 DSWP10 and DSWP30 are mid-market fixed rates against three-month London Interbank Offered Rate (LIBOR) and differ from household quotes.

[51] BlackRock Municipal Bond Fund, Inc. (2025), Part II, pp. II-20, II-126 and II-132; iShares Trust (2026), "Taxes," p. 213.

Doubling pricing resolution and draws preserves signs (maximum tail change 2.57 bp), as does using terminal dollar payoffs and their covariance. Widening the quote window to 25 bp gives 285, 302 and 275 positive scores at ten, twenty and thirty years, some near zero.

Equilibrium with household hedge access. With the 2015–2017 calibration fixed, households jointly choose bonds and nonnegative thirty-year pay-fixed swap notionals. Outside dealers quote swaps without trading the modeled bonds. A new swap has zero upfront value at the current rate state's par rate, and liquidation uses the same pricing rule and the historical quote family for the contract's age. Both markets' bond prices and all portfolios adjust across rate states, retaining nonnegative bonds and cash, gains taxation, loss denial and segregated interim financing.

Swap access reduces the thirty-year gap by 9.48 bp, or 9.1% (Table 55), with thirty-year municipal and taxable yields falling 15.95 and 12.04 bp. At the mean rate state, 270 types hedge, but over 99.99% of their municipal holdings remain riskless two-year bonds, since they mainly hedge outside exposure. Portfolio responses at other rate states affect continuation prices.

Table 55: Household swap access and the equilibrium municipal gap.

Maturity	Baseline	Swap access	Change
2 years	0.00	0.00	0.00
10 years	23.68	22.24	−1.44
20 years	72.63	65.42	−7.20
30 years	104.27	94.79	−9.48

Notes. Entries are the supply-weighted municipal yield minus the household benchmark, in basis points, averaged over the 36 monthly rate states of 2015–2017 with continuous yields. After-tax shares, parameters and dollar supplies are fixed; both yields adjust. Changes are computed before rounding. Refining the simulation fourfold moves either yield curve or the gap by less than 0.003 bp, and price and optimality residuals are below 10^{-10}.

Hedging costs and interpretation. The test includes asymmetric taxes and nondeductible interest on negative swap cash but excludes dealer fees, bid–ask spreads, margin, collateral opportunity costs and notional limits. The largest mean-state notional is 6.88 times the type's budget, so the test substantially expands household access.

At fixed bond prices and holdings, let $c_i \geq 0$ be any additional two-year certainty-equivalent cost per unit notional, in A_i's absolute units. Maximizing (167) net of this cost gives

$$q_i^*(x; c_i) = \frac{\xi_{i,\mathrm{abs}}^M}{\gamma_i \operatorname{Var}(A_i)} \left[\mathbb{E}A_i - \gamma_i \operatorname{Cov}\left(\widetilde{W}_i(x), A_i \right) - c_i \right]_+ . \tag{169}$$

Here $[u]_+ = \max\{u, 0\}$, as in (159), and $\widetilde{W}_i(x)$ is the risk index of bond portfolio x. Costs reduce hedge demand at given bond holdings, and margin and position limits restrict feasible hedges. These frictions oppose risk transfer and are expected to attenuate yield responses. The equilibrium gap depends on both yields, and the 9.1% reduction excludes these costs.

E.7 Data sources

Table 56 lists the data sources of Section 3.1 and the model inputs. FISD is Mergent's Fixed Income Securities Database, FDIC the Federal Deposit Insurance Corporation, FOIA the Freedom of Information Act, CAFR/ACFR the plans' comprehensive annual financial reports, SOI the IRS Statistics of Income, MSRB RTRS the Municipal Securities Rulemaking Board's Real-Time Transaction Reporting System, FINRA TRACE the Financial Industry Regulatory Authority's Trade Reporting and Compliance Engine, GSW the treasury curves of Gürkaynak, Sack and Wright, FRED the Federal Reserve Bank of St. Louis economic database, HQM the U.S. Treasury's High Quality Market corporate yield curve, CDS credit default swaps and BEA the Bureau of Economic Analysis.

Table 56: Data sources, variables and coverage.

Source	Variables	Coverage
Municipal issues: Mergent Municipal Bond Database (FTSE Russell)	CUSIP, issue & maturity dates, coupon, callability, tax status (federal/state), issuer state, purpose	1986–2023
Corporate issues: Mergent FISD (FTSE Russell)	CUSIP, issue & maturity dates, coupon, callability, rating, issuer industry, seniority	1986–2023
Insurers: NAIC Schedule D (annual, Q1–Q3 supplements)	CUSIP-level par, book, and market values for U.S. P&C and life insurers	1991–2023
Banks: FDIC BankFind (Call Reports RI, RC-B)	Annual interest expense, average assets and municipal securities holdings by bank	2000–2021
Public pensions: FOIA, CAFR/ACFR investment schedules	CUSIP-level muni/taxable par and market values, 30 state/local plans (coverage broadens from 2015)	2000–2023
Sector holdings: Financial Accounts (Z.1, L.212, B.101.f)	Quarterly munis: households, funds, banks, P&C, life, state/local pensions, other	1972–2022
Households: SCF, Federal Reserve	Tax-unit income and filing status, financial assets and debt	2001–2019 waves
Household taxes: IRS SOI	Filer counts, taxable and tax-exempt interest, effective federal rates by AGI bracket	2000–2022
Municipal trades: MSRB RTRS	Trade-level price, par volume, customer-side flag, trade-condition codes	2005–2023
Corporate trades: FINRA TRACE Enhanced (Academic)	Trade-level price, par volume, customer-side flag, condition codes	2002–2023
Treasury curve: GSW, Federal Reserve	Daily par and zero yields, 1–30y. Par yields enter P3 and zeros price its hypothetical coupon cash flows	2000–2023
Short rates: TB3MS, FRED	Monthly three-month treasury bill rates for the rate process	1952–2024
Treasury supply: Monthly Statement of the Public Debt	Security-level amounts and maturities for the treasury maturity profile	2001–2020
Corporate zeros: HQM (FRED), Mergent FISD robustness	Monthly A-or-above zero curve, 1–30y	2000–2023
Credit risk: Markit/IHS	State-government and AA corporate CDS at 1–30y tenors, coverage of the matched adjustment sample	2004–2020
Expected rates: Kim–Wright, Federal Reserve	Monthly fitted yields and term premia, expected-average-short-rate component capped at 10y	2000–2021
Swaps: Federal Reserve H.15 (FRED)	Par interest-rate swap rates for household swap pricing	2001–2008
State taxes: Tax Foundation	Top marginal rate, bracket thresholds, treatment of out-of-state muni interest	2000–2022
State income: BEA	Quarterly state-level personal income	2000–2022
Historical state rates: Institute on Taxation and Economic Policy (ITEP)	Top statutory rate by state, the P1 instruments	1945–2000 vintages